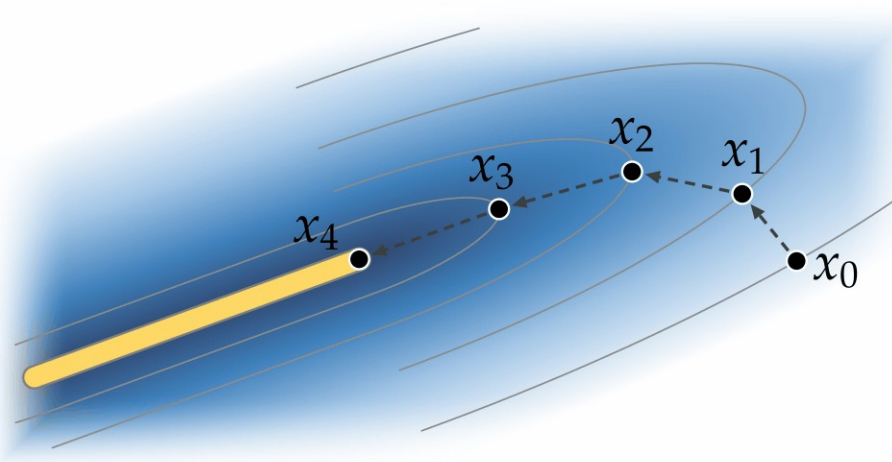


How to Think About Optimization

The Proximal Point Method and Its Impact on Optimization Theory and Machine Learning

Peter Richtárik

King Abdullah University of Science and Technology
Kingdom of Saudi Arabia



Optimization & Machine Learning Lab



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KAUST (King Abdullah University of Science and ...
Oct 23, 2025 • Edited

#KAUST PhD student Kaja Gruntkowska has been awarded a Google PhD Fellowship, earning the 2025 fellowship in Algorithms & Optimization.

Guided by her advisor, Prof. [Peter Richtarik](#), Kaja's innovative research bridges mathematics, statistics, and AI, advancing both the theory and practice of optimization for machine learning to make AI training faster, more cost-effective, and more resource-efficient.

She shares: "My goal is to bridge rigorous mathematical theory with practical solutions that scale to today's trillion-parameter AI models. This fellowship will help turn these ideas into tools that keep AI breakthroughs within reach."

At KAUST, we are proud to empower the next generation of AI leaders and drive innovation from Saudi Arabia to the world.



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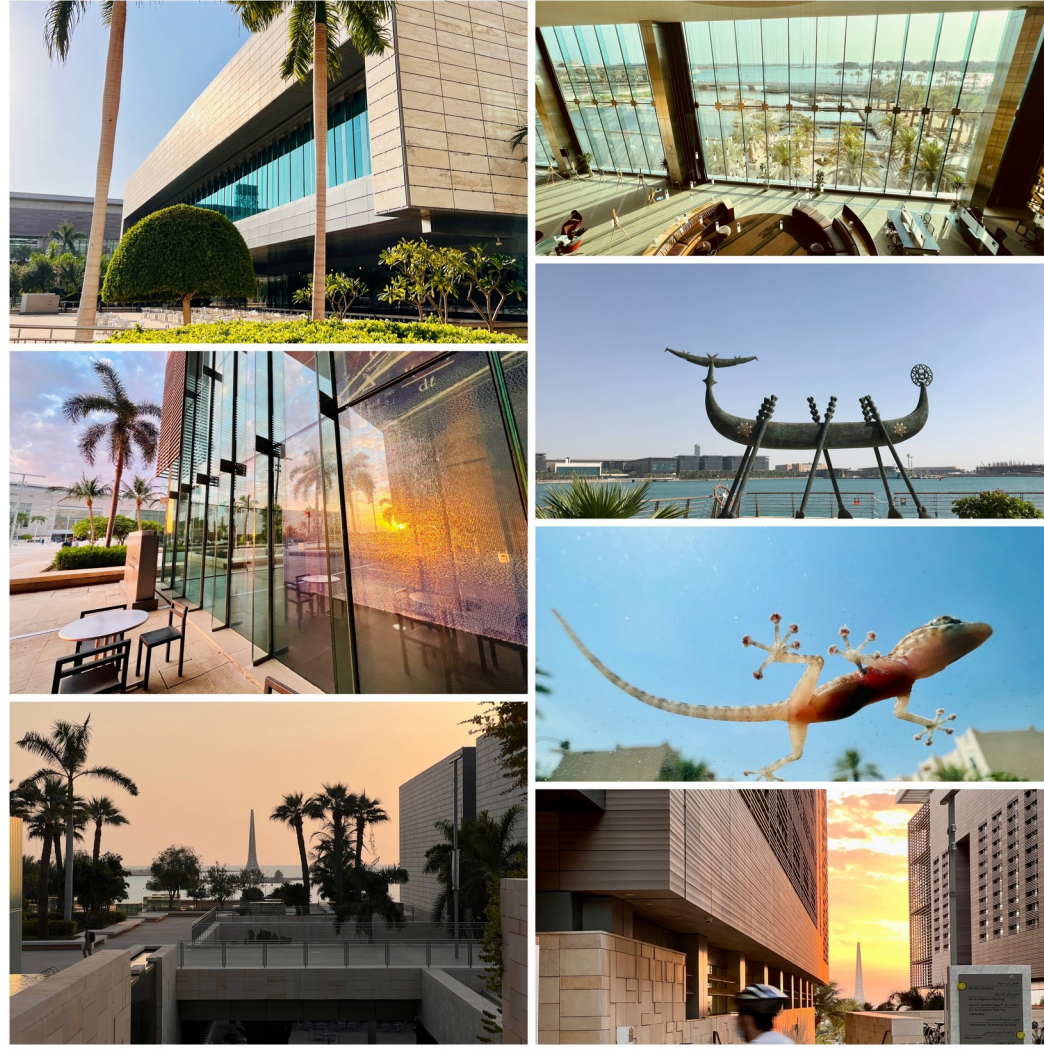
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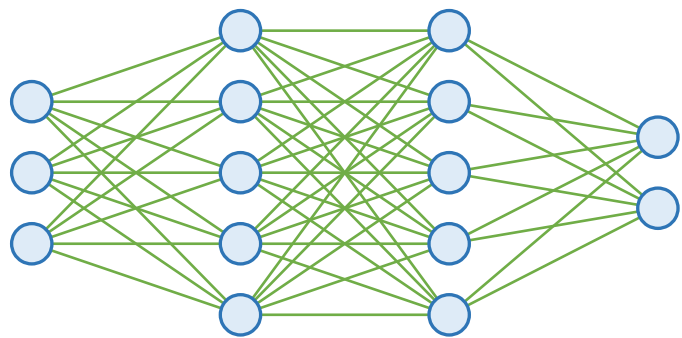
A Few Random Photos of KAUST



Optimization

$$\min_{x \in \mathbb{R}^d} f(x)$$

Optimization



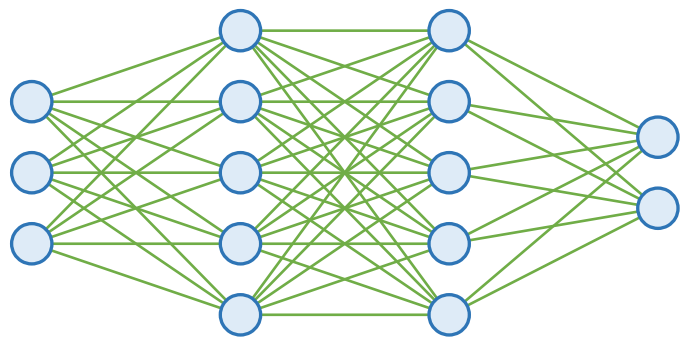
x = weights of a neural network

GPT-3: $d \approx 1.75 \cdot 10^{11}$

$$\min_{x \in \mathbb{R}^d} f(x)$$

number of
trainable
parameters

Optimization



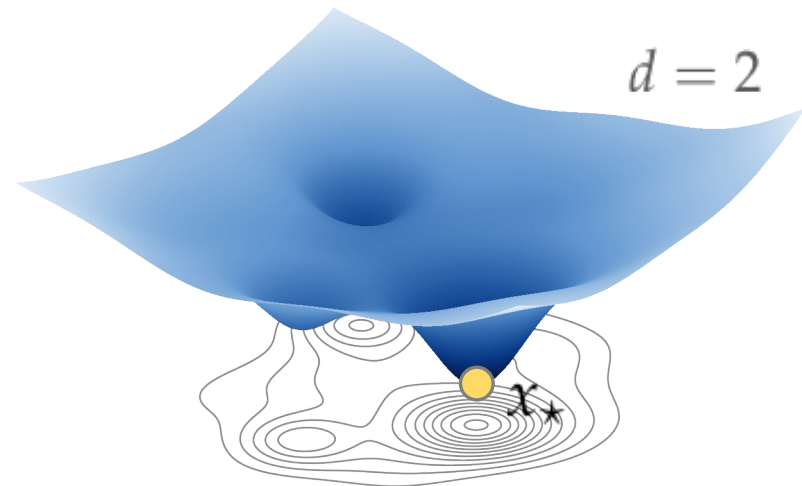
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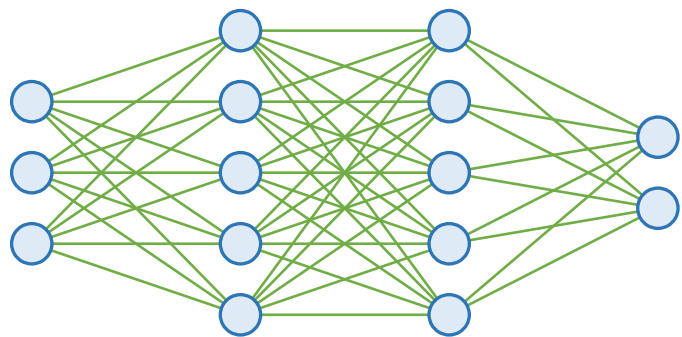
$$\min_{x \in \mathbb{R}^d} f(x)$$

number of
trainable
parameters

$$f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$$



Optimization



x = weights of a neural network

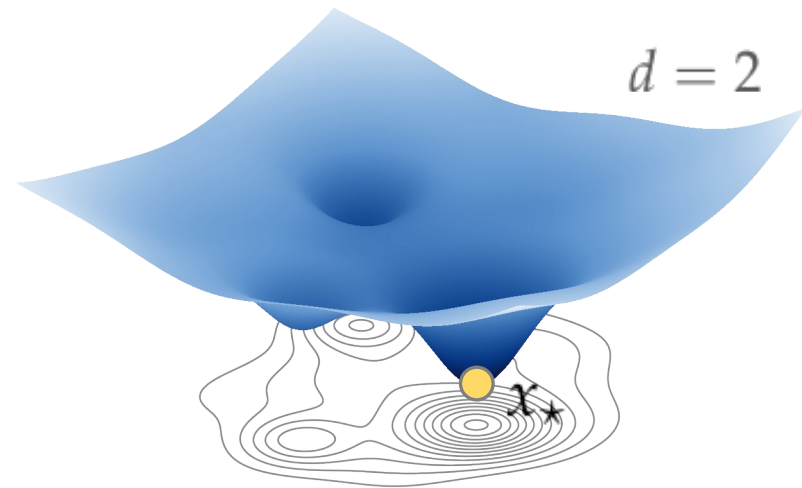
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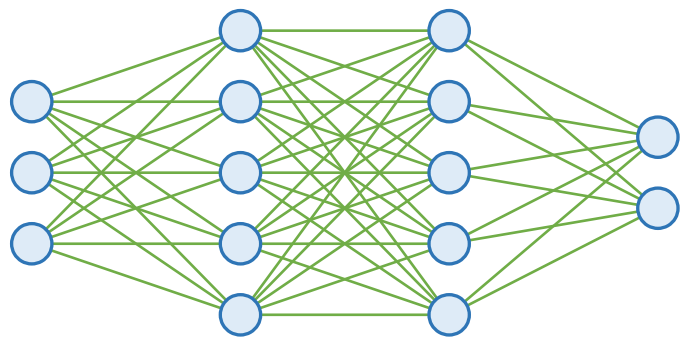
number of
trainable
parameters

$$f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$$

f is proper



Optimization

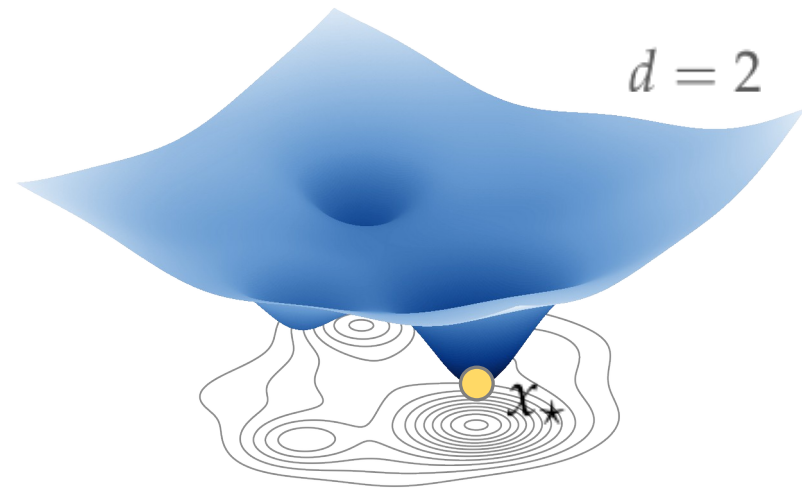


x = weights of a neural network

GPT-3: $d \approx 1.75 \cdot 10^{11}$

$$\min_{x \in \mathbb{R}^d} f(x)$$

number of
trainable
parameters



$$f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$$

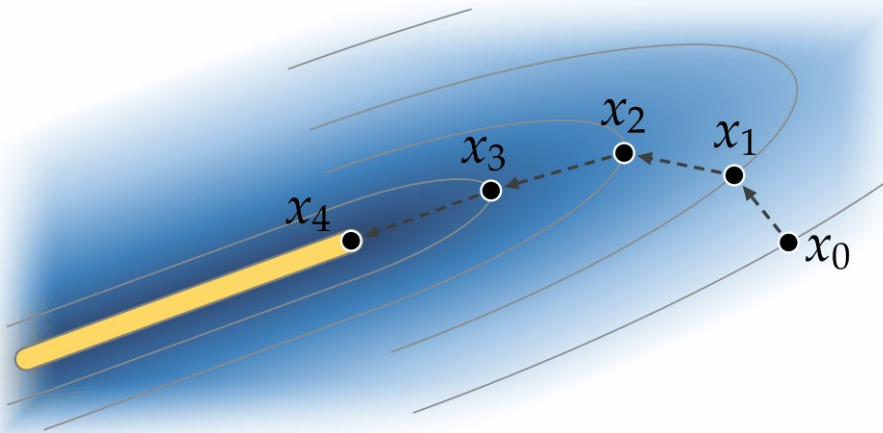
f is proper

f is closed (= lower semi-continuous)

Outline

- Part 1 BPM: Convex Regime
- Part 2 BPM: Convergence Theory
- Part 3 BPM: Nonconvex Regime
- Part 4 BPM: Practical Approximations
- Part 5 BPM: Four Formulations

$$\min_{x \in \mathbb{R}^d} f(x)$$



BPM = Broximal Point Method

arXiv:2003.08078v1 [math.OA] 18 Mar 2020

Acceleration with a Ball Optimization Oracle

Yair Caramon^{*} Arun Jambulapati[†] Qijia Jiang^{*} Yujia Jin^{*} Yin Tai Lee[‡]
Aaron Sidford^{*} Kevin Tian^{*}

Abstract

Consider an oracle which takes a point x and returns the minimizer of a convex function f in an ℓ_2 ball of radius r around x . It is straightforward to show that roughly $r^{-1} \log \frac{1}{\epsilon}$ calls to the oracle suffice to find an ϵ -approximate minimizer of f in an ℓ_2 unit ball. Perhaps surprisingly, this is not optimal: we design an accelerated algorithm which utilizes an ϵ -approximate minimizer with roughly $r^{-1/3} \log \frac{1}{\epsilon}$ oracle queries, and gives a matching lower bound. Further, we implement ball optimization oracles for functions with locally stable Hessians using a variant of Newton's method. The resulting algorithms provide to a number of problems of practical and theoretical import, improving upon previous results for logistic and ℓ_{∞} regression and selecting guarantees comparable to the state-of-the-art for ℓ_2 regression.

1 Introduction

We study unconstrained minimization of a smooth convex objective $f: \mathbb{R}^d \rightarrow \mathbb{R}$, which we access through a ball optimization oracle $\mathcal{O}_{\text{ball}}$ that when queried at any point x , returns the minimizer¹ of f restricted to a ball of radius r around x , i.e.,

$$\mathcal{O}_{\text{ball}}(x) := \arg \min_{y \in B(x, r)} f(y),$$

$$r = \frac{1}{\epsilon}, \frac{1}{\epsilon^2}.$$

Such oracles underlie trust region methods [14] and, as we demonstrate via applications, encapsulate problems with local stability. Iteratively $\mathcal{O}_{\text{ball}, \epsilon+1} \leftarrow \mathcal{O}_{\text{ball}, \epsilon}(x)$ minimizes f in $\tilde{O}(d/r)$ iterations (see Appendix A), where R is the initial distance to the minimizer, x^* , and $\tilde{O}(\cdot)$ hides polylogarithmic factors in problem parameters, including the desired accuracy.

Given the fundamental geometric nature of the ball optimization abstraction, the central question motivating our work is whether it is possible to improve upon this $\tilde{O}(d/r)$ query complexity. It is natural to conjecture that the answer is negative; we require R/r oracle calls to observe the entire line from x_0 to the optimum, and therefore finding a solution using less queries would require jumping into completely unexplored regions. Nevertheless, we prove that the optimal query complexity scales as $\tilde{O}(d/r)^{1/3}$. This result has positive implications for the complexity for several key regression tasks, for which we can efficiently implement the ball optimization oracles.

1.1 Our contributions

Here we overview the main contributions of our paper: accelerating ball optimization oracles (with a matching lower bound), implementing them under Hessian stability, and applying the resulting techniques to regression problems.

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[†]University of Washington, ystatis@uw.edu.

[‡]In this introduction we discuss exact oracles for simplicity, but our results account for inexactness.

arXiv:2502.02002v1 [math.OA] 1 Feb 2025

The Ball-Proximal (or “Broximal”) Point Method: a New Algorithm, Convergence Theory, and Applications

Kaja Gruntkowska¹ Hammin Li¹ Anil Rao^{1,2} Peter Richtárik¹

Abstract

Non-smooth and non-convex global optimization poses significant challenges across various applications, where standard gradient-based methods often struggle. We propose the *Ball-Proximal Point Method*, *Broximal Point Method*, or *Ball-Point Method* (BPM) for short – a novel algorithmic framework inspired by the classical Proximal Point Method (PPM) (Rockafellar, 1976), which, as we show, sheds new light on several foundational optimization paradigms and phenomena, including non-convex and non-smooth optimization, acceleration, smoothing, adaptive stepsize selection, and trust-region methods. At the core of BPM lies the ball-proximal (“broximal”) operator, which arises from the classical proximal operator by replacing the quadratic distance penalty by a ball constraint. Surprisingly, and in sharp contrast with the sublinear rate of PPM in the non-smooth convex regime, we prove that BPM converges linearly and in a finite number of steps in the same regime. Furthermore, by introducing the concept of ball-convexity, we prove that BPM retains the same global convergence guarantees under weaker assumptions, making it a powerful tool for a broad class of proximal-type optimization problems. Just like PPM plays the role of a conceptual method inspiring the development of practically efficient algorithms and algorithmic elements, e.g., gradient descent, adaptive step sizes, acceleration (Arai & Iwata, 2020), and “W” in Iterative (Zhang et al., 2022), we believe that BPM should be understood in the same manner: as a blueprint and inspiration for further development.

Non-smooth and non-convex global optimization poses significant challenges across various applications, where standard gradient-based methods often struggle. We propose the *Ball-Proximal Point Method*, *Broximal Point Method*, or *Ball-Point Method* (BPM) for short – a novel algorithmic framework inspired by the classical Proximal Point Method (PPM) (Rockafellar, 1976), which, as we show, sheds new light on several foundational optimization paradigms and phenomena, including non-convex and non-smooth optimization, acceleration, smoothing, adaptive stepsize selection, and trust-region methods. At the core of BPM lies the ball-proximal (“broximal”) operator, which arises from the classical proximal operator by replacing the quadratic distance penalty by a ball constraint. Surprisingly, and in sharp contrast with the sublinear rate of PPM in the non-smooth convex regime, we prove that BPM converges linearly and in a finite number of steps in the same regime. Furthermore, by introducing the concept of ball-convexity, we prove that BPM retains the same global convergence guarantees under weaker assumptions, making it a powerful tool for a broad class of proximal-type optimization problems. Just like PPM plays the role of a conceptual method inspiring the development of practically efficient algorithms and algorithmic elements, e.g., gradient descent, adaptive step sizes, acceleration (Arai & Iwata, 2020), and “W” in Iterative (Zhang et al., 2022), we believe that BPM should be understood in the same manner: as a blueprint and inspiration for further development.

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³In this introduction we discuss exact oracles for simplicity, but our results account for inexactness.

arXiv:2510.00823v1 [math.OA] 1 Oct 2025

NON-EUCLIDEAN BROXIMAL POINT METHOD: A BLUEPRINT FOR GEOMETRY-AWARE OPTIMIZATION

Kaja Gruntkowska¹ KAUST²
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Thuwal, Saudi Arabia

Peter Richtárik¹ KAUST²
Center of Excellence for Cognitive AI
Thuwal, Saudi Arabia

Abstract

The recently proposed Broximal Point Method (BPM) (Gruntkowska et al., 2023) offers an idealized optimization framework based on iteratively minimizing the objective function over norm balls centered at the current iterate. It enjoys striking global convergence guarantees, converging linearly and in a finite number of steps for proper, closed and convex functions. However, its theoretical analysis has so far been confined to the Euclidean geometry. At the same time, emerging trends in deep learning optimization, exemplified by algorithms such as Moen (Jordan et al., 2024) and Stein (Petrich et al., 2025), demonstrate the practical advantages of minimizing over balls defined via non-Euclidean norms which better align with the underlying geometry of the associated loss landscapes. In this note, we ask whether the convergence theory of BPM can be extended to this more general, non-Euclidean setting. We give a positive answer, showing that most of the elegant guarantees of the original method carry over to arbitrary norm geometries. Along the way, we clarify which properties are preserved and which necessarily break down when leaving the Euclidean realm. Our analysis positions Non-Euclidean BPM as a conceptual blueprint for understanding a broad class of geometry-aware optimization algorithms, shedding light on the principles behind their practical effectiveness.

1 Introduction

Optimization stands as a cornerstone of modern machine learning, powering the success of virtually everything, from deep neural networks to state-of-the-art large language models. As models grow ever larger and more complex, the demand for better optimization algorithms evolves in tandem. What is needed now is a new class of methods—one that are not only computationally efficient and scalable, but also inherently robust to the nonconvex, high-dimensional landscapes that characterize contemporary deep learning.

The field of deep learning optimization has long relied on Adam and related algorithms (Kingma and Ba, 2014; Loshchilov and Hutter, 2017), whose core innovation lies in adaptive moment estimation. While these methods have achieved significant empirical success, their theoretical behavior, particularly in nonconvex landscapes, remains only partially understood. Recently, however, a new class of optimization methods has begun to challenge this long-standing dominance. Algorithms such as Moen (Jordan et al., 2024), Stein (Petrich et al., 2025) and Quon (Fridman et al., 2023) break away from the adaptive moment paradigm. Instead, they adopt a different principle: structured updates derived by minimizing a linear approximation of the loss function over a carefully chosen norm ball. Crucially, these norm balls are non-Euclidean, aiming to reflect the intrinsic geometry of the optimization problem (see Section 3.1). Operating at the individual layer level, these algorithms combine simplicity, scalability, and promising empirical performance, with several studies suggesting their potential to supersede AdamW in large-scale deep learning tasks (Jin et al., 2025; Petrich et al., 2025; Shao et al., 2025; Thorelli et al., 2025; Viet et al., 2025).

While Moen and Stein—both of which utilize spectral norm balls—are among the most prominent and actively studied examples in this emerging line of work, the underlying principle extends well beyond these specific instances. By changing the geometry through altering the norms, one can recover a range of familiar optimization methods. For

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arXiv:2604.13483v1 [math.OA] 15 Apr 2026

Broximal Alignment for Global Non-Convex Optimization

Kaja Gruntkowska¹ Hammin Li¹ Xun Qian¹ Peter Richtárik¹

Abstract

Most non-convex optimization theory is built around gradient dynamics, leaving global convergence largely unexplored. The dominant paradigm focuses on stationarity, certifying only that the gradient norm vanishes, which is often a weak proxy for actual optimization success. In practice, gradient norms can stagnate or even increase during training, and stationary points may be far from global solutions. In this work, we propose a new framework for global non-convex optimization that avoids gradient-based reasoning altogether. We revisit the Ball-Proximal Point Method (BPM), a trust-region-style algorithm introduced by Gruntkowska et al. (2023), and propose a novel structural condition – *Broximal Alignment* – under which BPM provably converges to a global minimizer. Our condition requires convexity, smoothness, or Lipschitz assumptions, and it permits multiple and disconnected global minima as well as non-optimal local minima. We show that this class generalizes standard non-convex frameworks such as quasi-convexity, star-convexity, quasi-convexity, and the aiming condition. Our results provide a new conceptual foundation for the global non-convex optimization beyond stationarity.

1. Introduction

Non-convex optimization lies at the heart of modern machine learning (ML). Matrix factorization (Koren et al., 2009), phase retrieval (Pillay, 1994; Candes et al., 2013), representation learning, and most prominently deep neural networks (Lecun et al., 2015; Goodfellow et al., 2016) all lead to objective functions with highly non-convex landscapes. Although such functions provide the expressive capacity needed to model complex learning tasks, they often pose significant challenges for optimizer designers. The non-convex landscape is often characterized by a large number of local minima and saddle points that can trap optimization algorithms (Poupart et al., 2014; Vidal et al., 2017; Saffari & Shamsi, 2018; Jin et al., 2021).

We study the most Euclidean ball-proximal point method for proper, closed, convex functions, where each iteration minimizes the objective over a ball centered at the current point. Retaining the algorithm’s key features of rapid descent yields sharper bounds on objective values, stationarity, and the symmetric Bregman distance to a minimizer. For constant radius $r > 0$ and initial distance $D_0 > 0$ to the solution set, the objective gap after k iterations is at most its initial value multiplied by $\exp(-k/r^2 D_0^2)$.

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Preprint, April 16, 2026.

deriving landscape can be highly irregular, with many poor local minima and saddle points that can trap optimization algorithms (Poupart et al., 2014; Vidal et al., 2017; Saffari & Shamsi, 2018; Jin et al., 2021).

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We characterize convergence for arbitrary positive radius sequences. If their sum diverges, the method reaches a minimizer in finitely many iterations whenever one enters otherwise, the objective values converge to the infimum and the iterates escape every bounded set. If the radii are summable, the iterates converge to a possibly nonoptimal point. Self-contraction gives finite trajectory length for bounded iterates and a constant-radius termination bound $O(k) = D_0/r$, whose implicit constant depends only on the dimension. A polished family shows that the dimension-independent quadratic bound remains asymptotically sharp.

1.1 Limitations of Stationarity-Based Non-Convex Optimization Theory

To fix notation, we consider solving the generic (potentially non-convex) optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper (that is, the set $\text{dom } f := \{x \in \mathbb{R}^d : f(x) < +\infty\}$ is nonempty), closed, and has at least one minimizer. We denote the set of all minimizers by X_f and write $f_* := \min_x f(x)$.

The dominant theoretical paradigm for such tasks is built around stationarity. Convergence guarantees are typically stated in terms of the norm of the gradient, asserting that an algorithm reaches a point where $\|\nabla f(x)\|$ is small. This framework underlies much of the modern theory of stochastic gradient methods.

From the modeling perspective, however, stationarity-based guarantees are fundamentally misaligned with the goals of learning. A stationary point can be a global minimizer, a poor local minimizer, or a saddle point; first-order information alone does not distinguish among these possibilities. Therefore, a small gradient provides little guidance on whether an algorithm meaningfully solves the under-

A Sharper Theory of Ball-Proximal Optimization: Convergence and Radius Selection

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Thuwal 23955-6900, Saudi Arabia

28 September 2026

Abstract

We study the most Euclidean ball-proximal point method for proper, closed, convex functions, where each iteration minimizes the objective over a ball centered at the current point. Retaining the algorithm’s key features of rapid descent yields sharper bounds on objective values, stationarity, and the symmetric Bregman distance to a minimizer. For constant radius $r > 0$ and initial distance $D_0 > 0$ to the solution set, the objective gap after k iterations is at most its initial value multiplied by $\exp(-k/r^2 D_0^2)$.

We characterize convergence for arbitrary positive radius sequences. If their sum diverges, the method reaches a minimizer in finitely many iterations whenever one enters otherwise, the objective values converge to the infimum and the iterates escape every bounded set. If the radii are summable, the iterates converge to a possibly nonoptimal point. Self-contraction gives finite trajectory length for bounded iterates and a constant-radius termination bound $O(k) = D_0/r$, whose implicit constant depends only on the dimension. A polished family shows that the dimension-independent quadratic bound remains asymptotically sharp.

We also identify a uniform nonvacuous geometric lower bound and analyze adaptive radius rules based on subgradients or objective gaps, eigengap reformulation, and related updates. Together, these results strengthen the foundation and convergence guarantees of the method without assuming smoothness or strong convexity.

Keywords: ball-proximal point method; convex optimization; finite termination; self-contracted sequences; radius selection; oracle complexity.

Mathematics Subject Classification (2020): Primary 90C25; Secondary 90C30, 65R05.

1 Introduction

We study the convex optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, closed, and convex. Unless stated otherwise, we assume neither that the infimum is attained nor that f is smooth. We focus on the exact ball-oracle iteration considered by Caramon et al. (2020, Appendix A) and subsequently developed as the ball-proximal (broximal) point framework by Gruntkowska et al. (2023). Starting from $x^0 \in \text{dom } f$, the ball-proximal point method (BPM) minimizes f over a Euclidean ball centered at the current point:

$$x^{k+1} \in \arg \min_{x \in B(x^k, r)} f(x), \quad t_k > 0, \quad (\text{BPM})$$

Part 1

Proximal Point Method: Convex Regime



Kaja Grunkowska, Hanmin Li, Aadi Rane, P.R.

The ball-proximal (= "proximal") point method: a new algorithm, convergence theory, and applications

arXiv:2502.02002, 2025



Kaja Grunkowska, P.R.

Non-Euclidean proximal point method: a blueprint for geometry-aware optimization

arXiv:2510.00823, 2025

BPM with ℓ_2 Balls

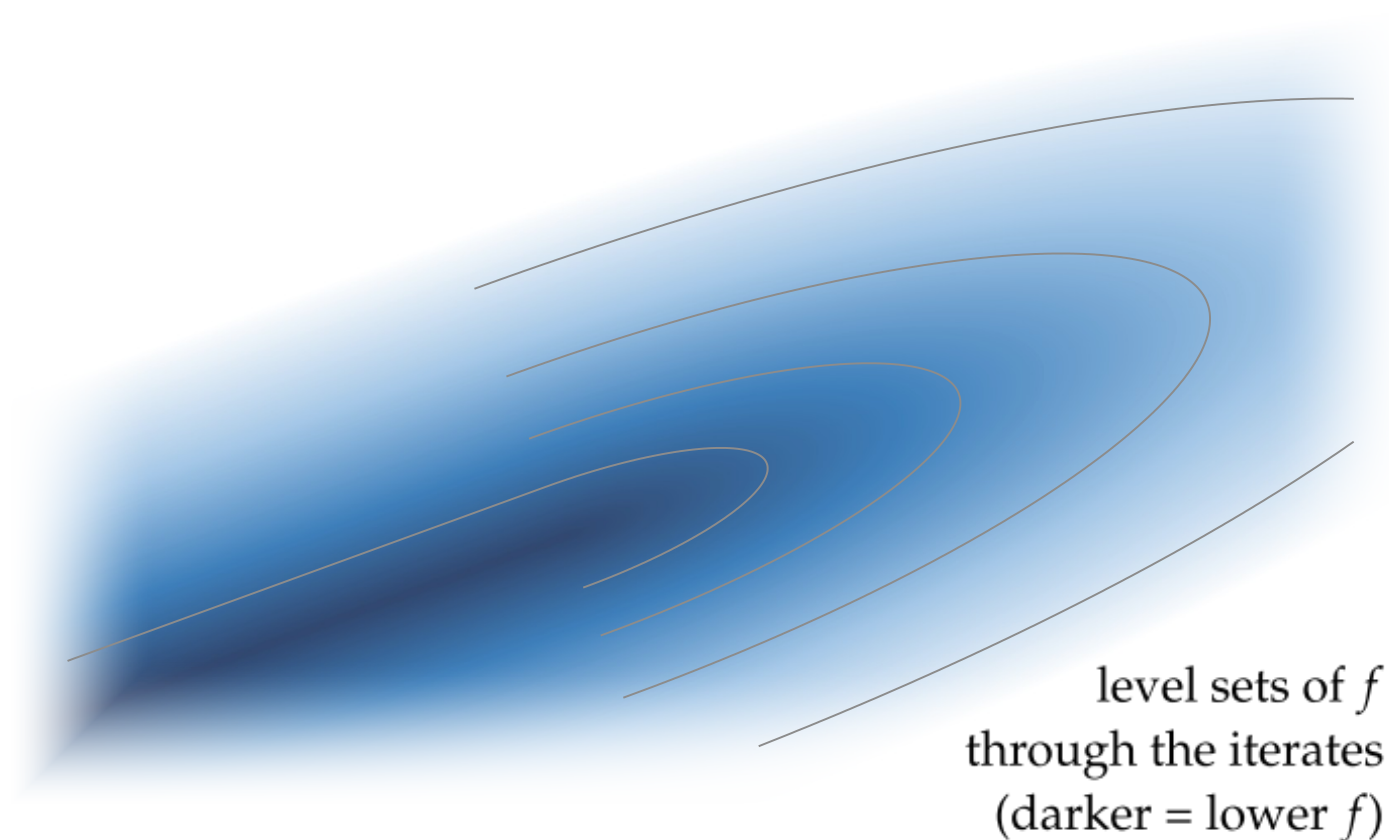
$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_2 \leq t_k\}$$

Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

BPM with ℓ_2 Balls

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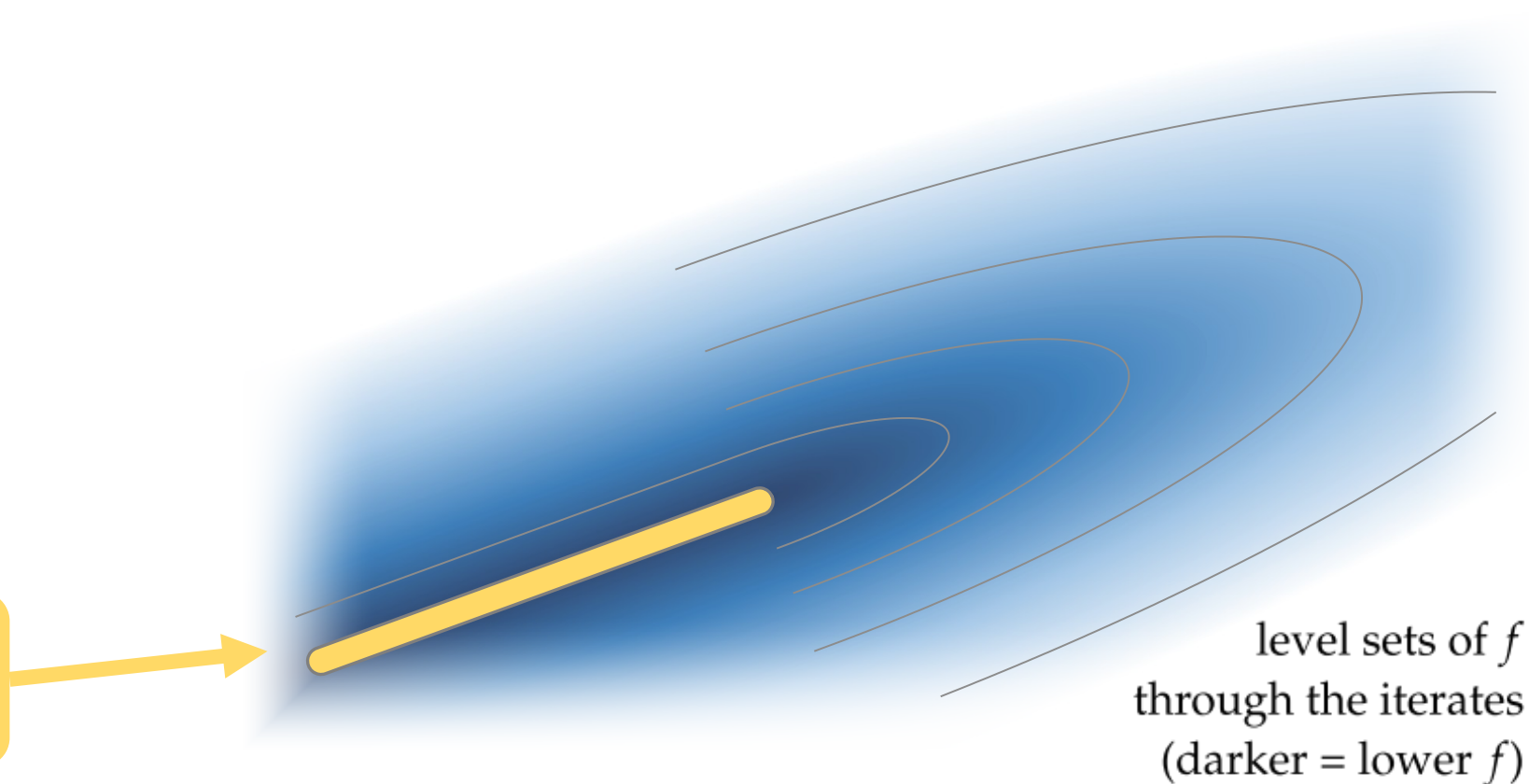


BPM with ℓ_2 Balls

$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_2 \leq t_k\}$$

Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

$\operatorname{Argmin} f = [-2a, 2a]$
 $f_\star := \min f = 0$



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BPM run

$$d_k = \operatorname{dist}_2(x_k, \operatorname{Argmin} f)$$

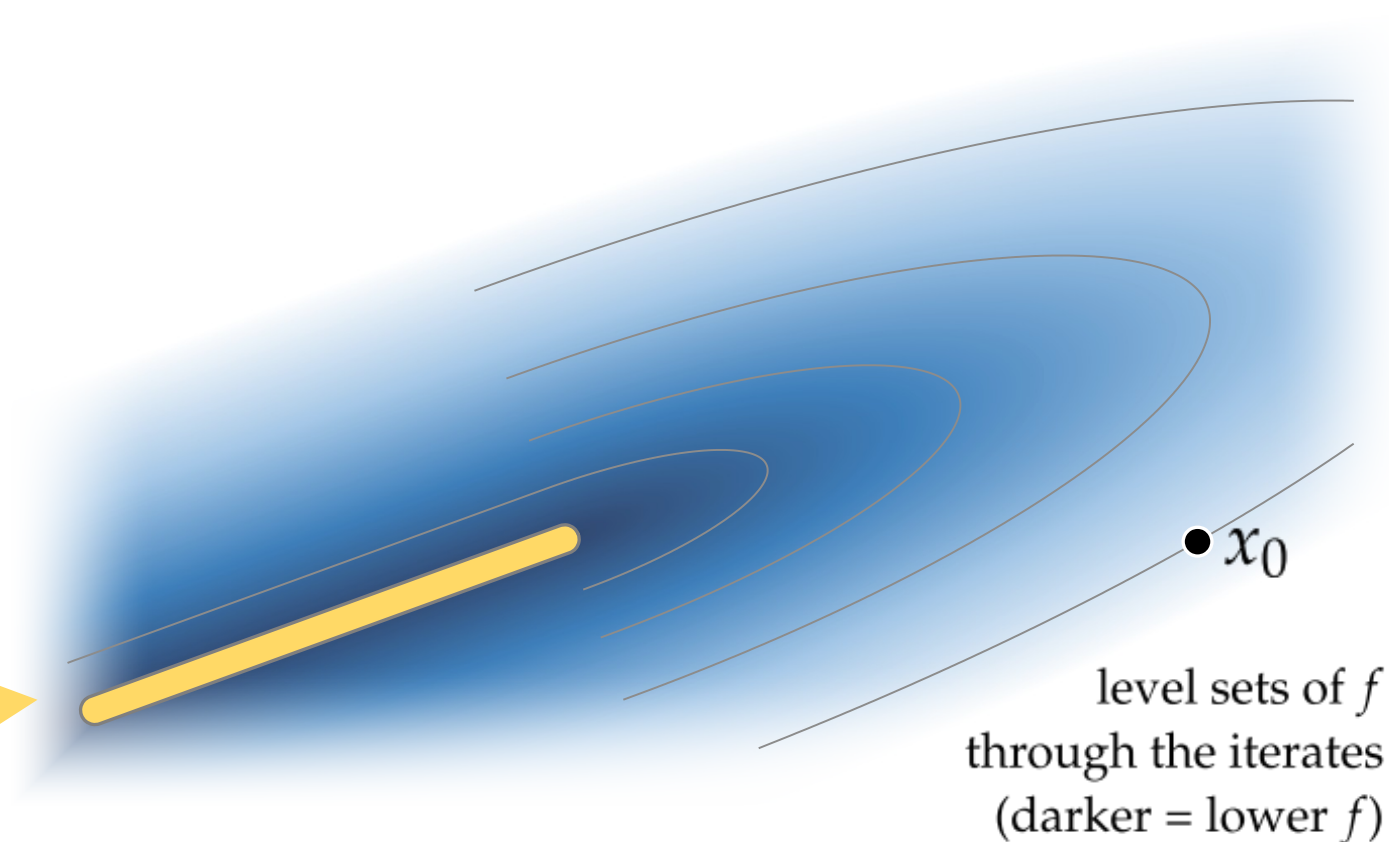
$$f_k = f(x_k) - f_\star$$

$$g_k = \|\nabla f(x_k)\|_2$$

$$d_0 = 5.06 \quad f_0 = 35.78 \quad g_0 = 28.40$$

$$\operatorname{Argmin} f = [-2a, 2a]$$

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BPM with ℓ_2 Balls

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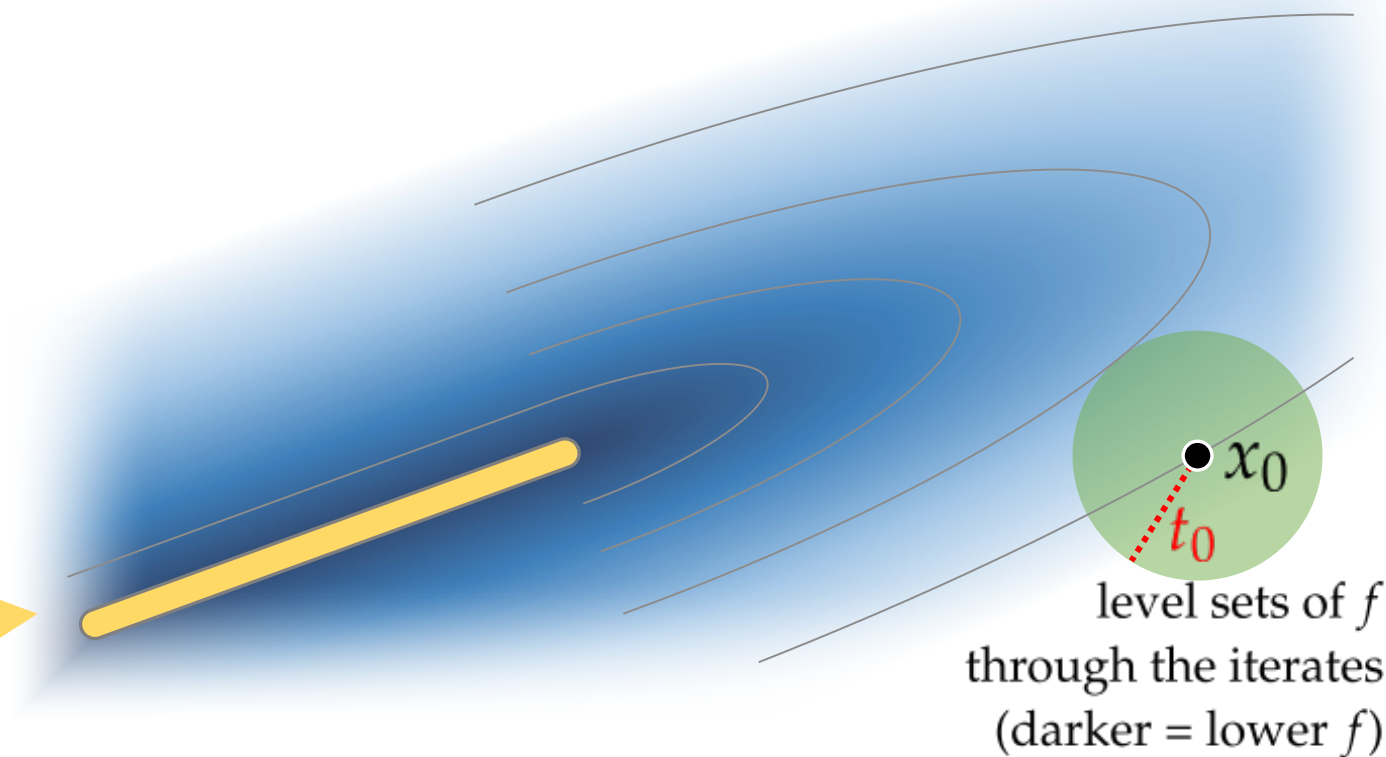
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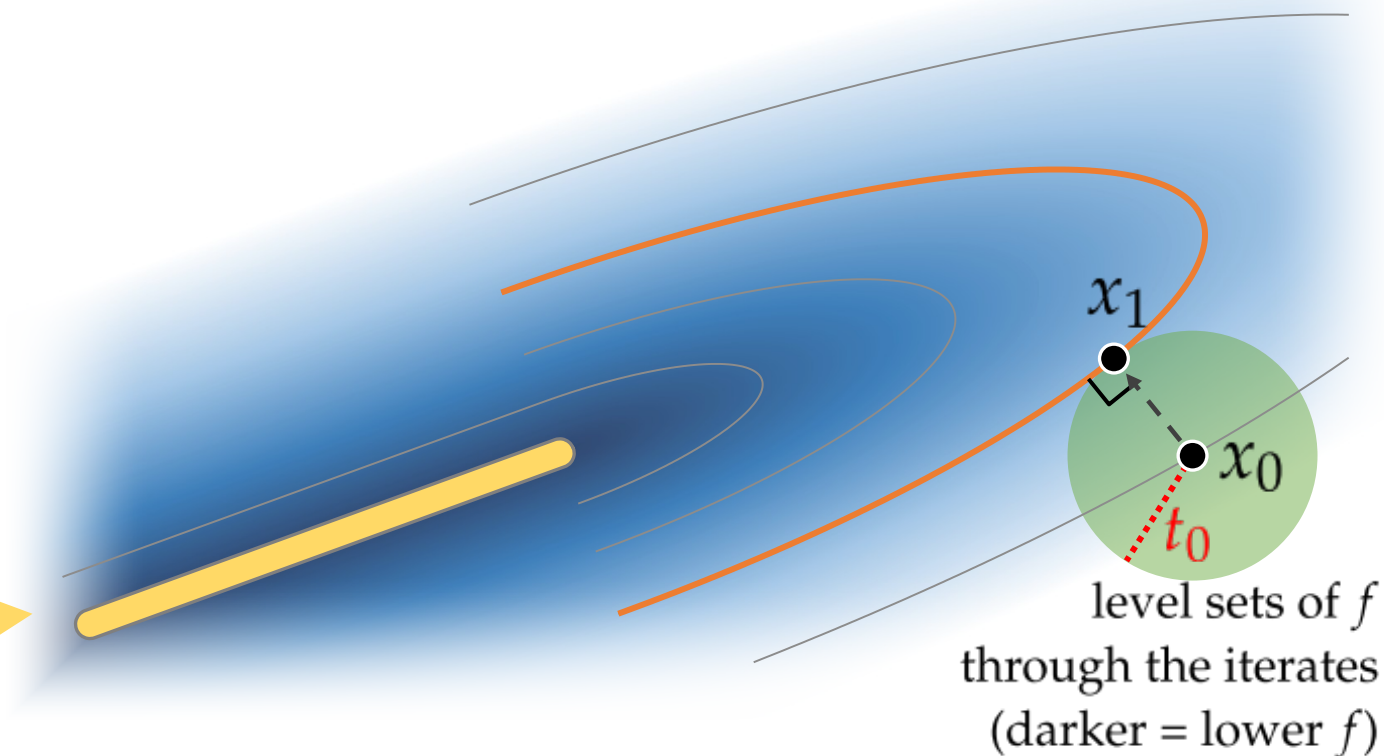
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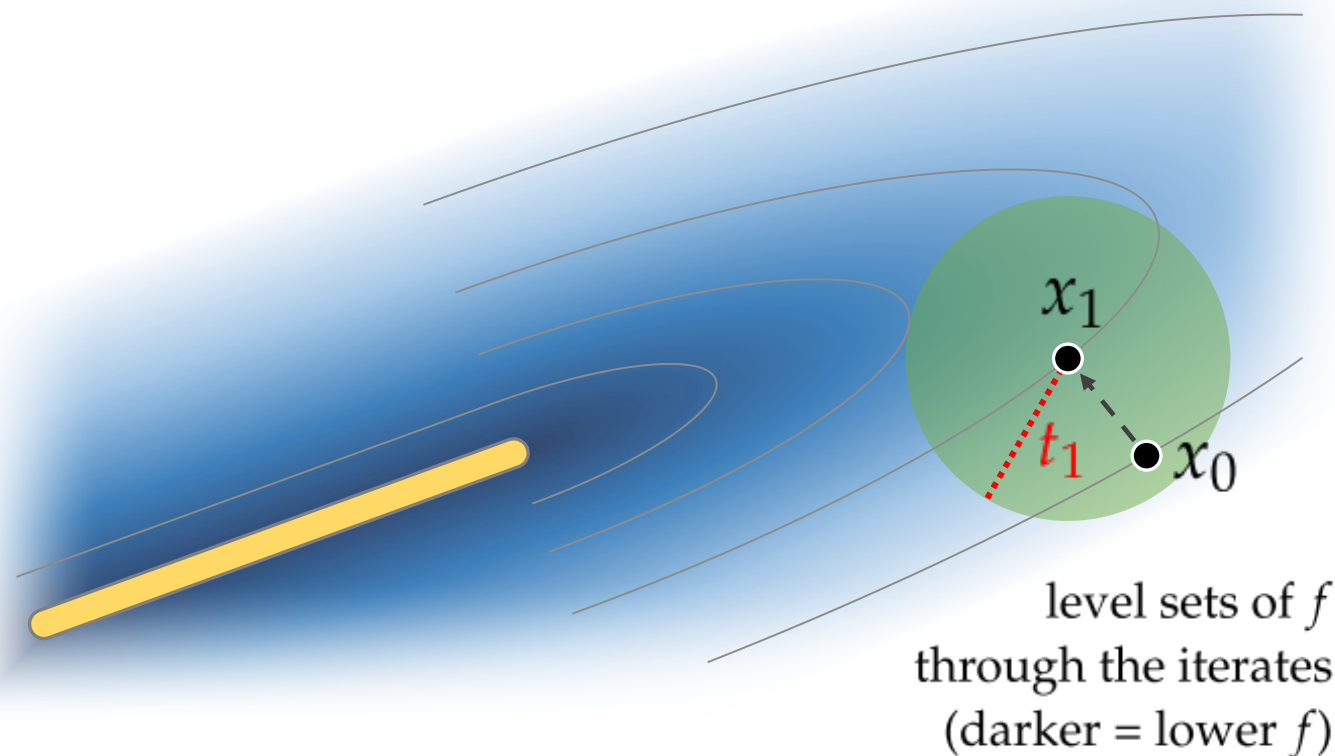
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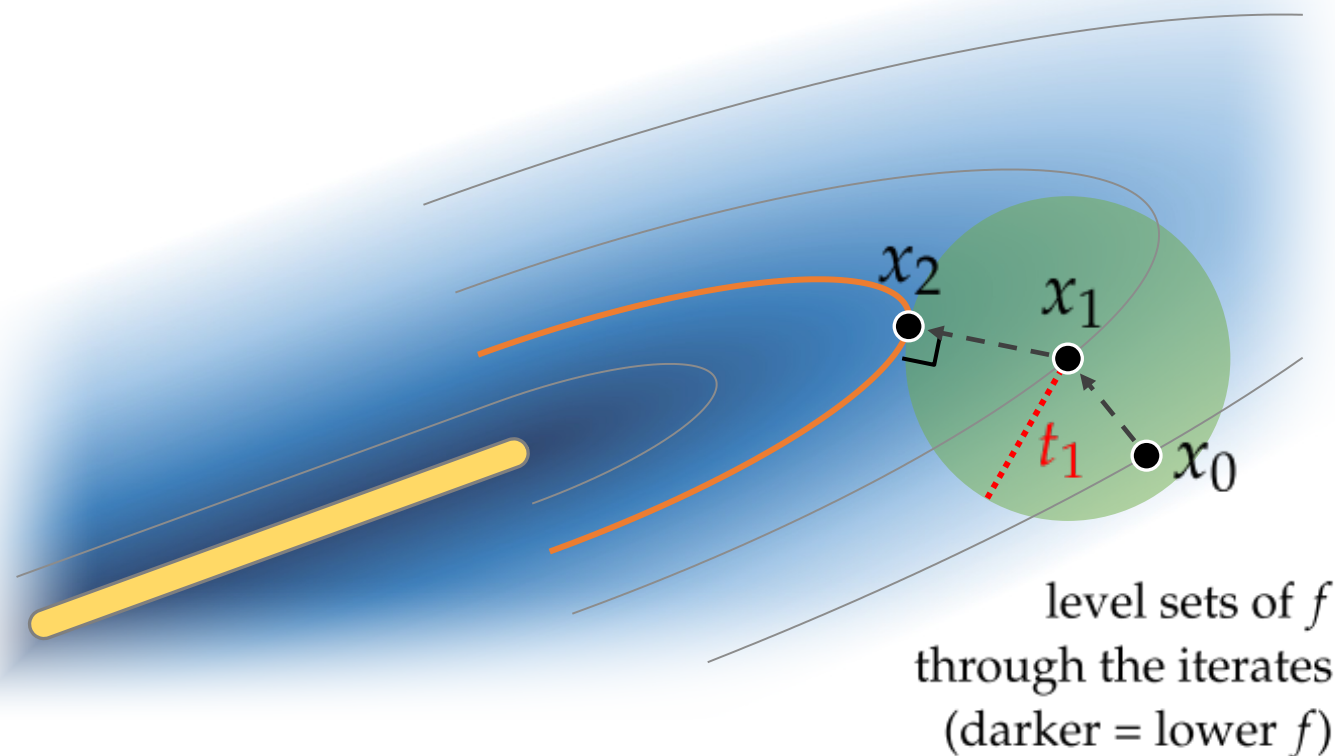
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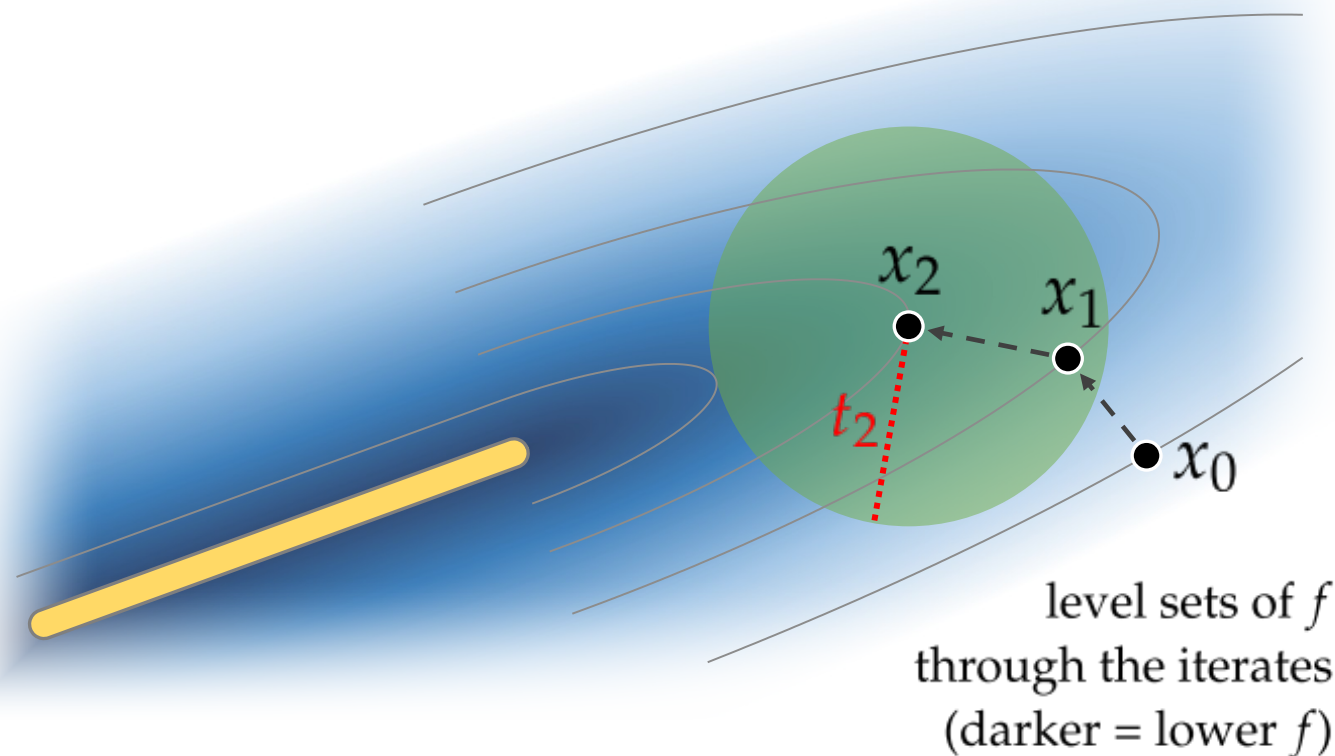
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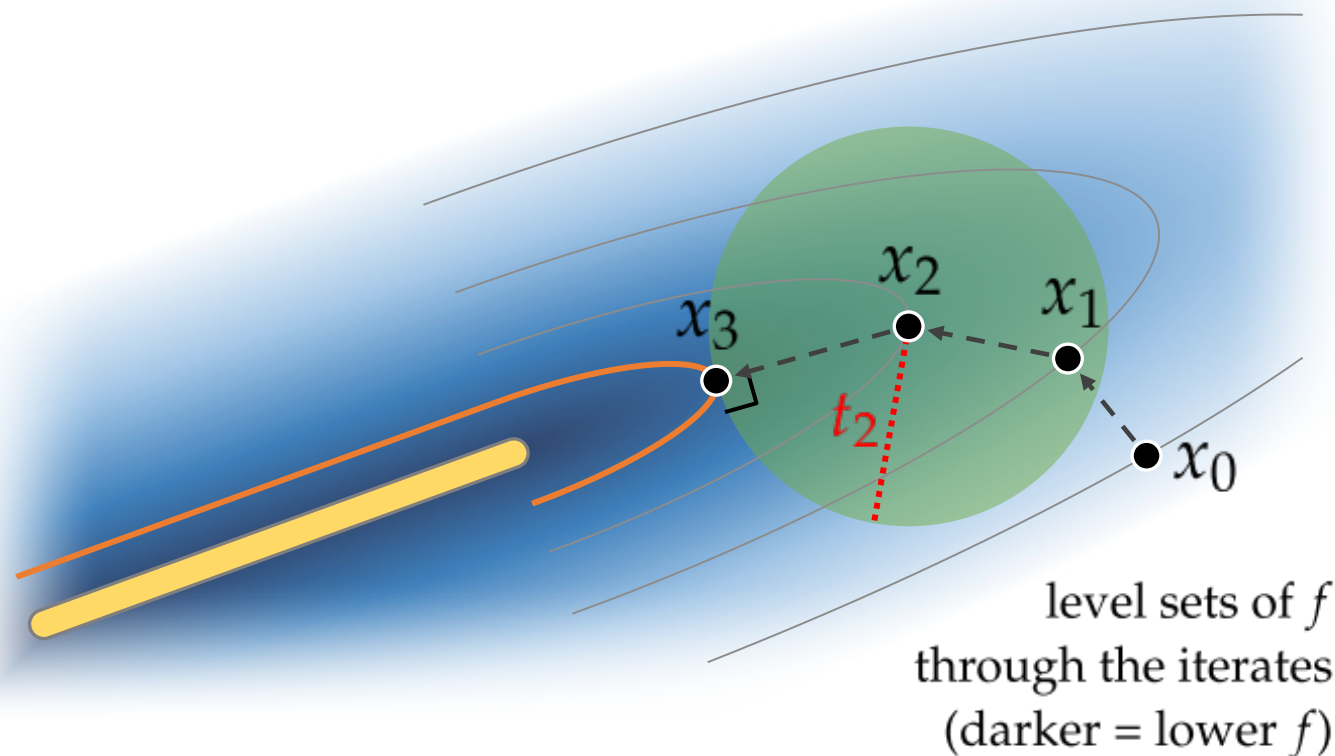
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$$t_2 = 1.6 \quad d_3 = 1.72 \quad f_3 = 1.48 \quad g_3 = 1.72$$

$$\operatorname{Argmin} f = [-2a, 2a]$$

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BPM with ℓ_2 Balls

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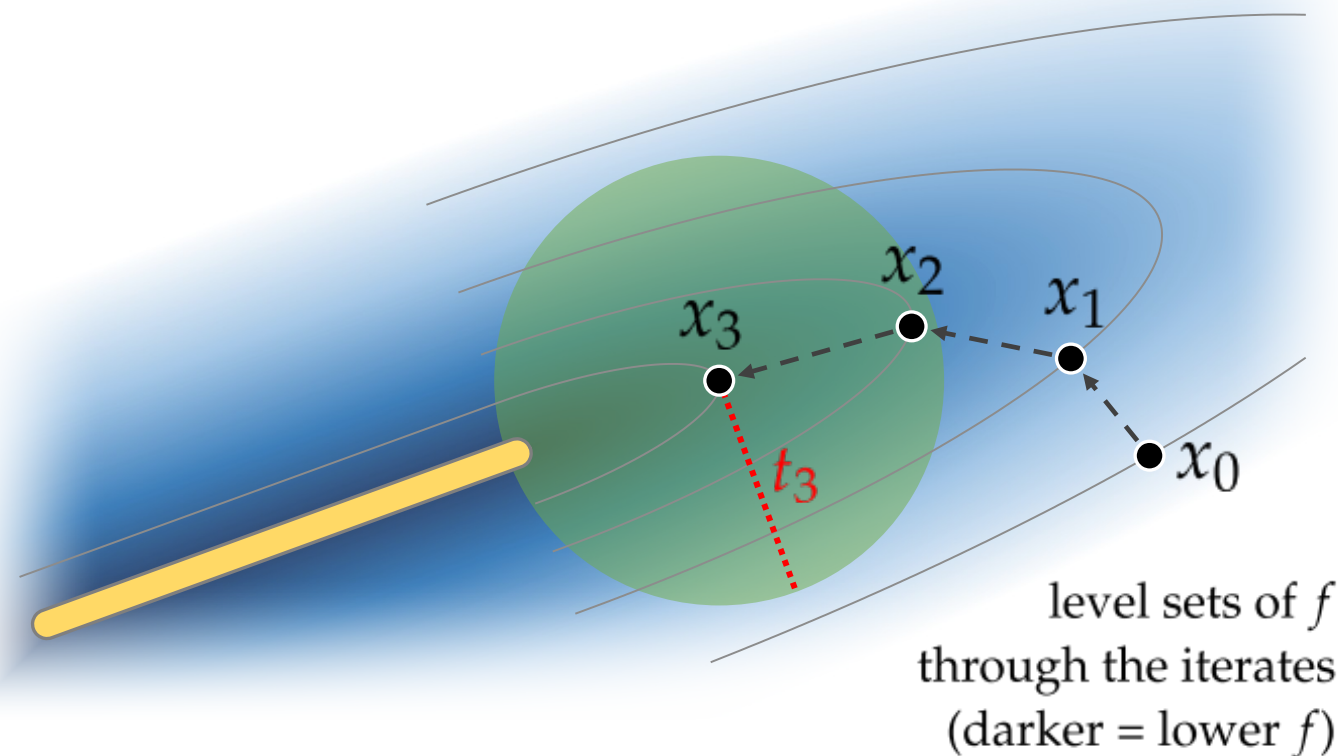
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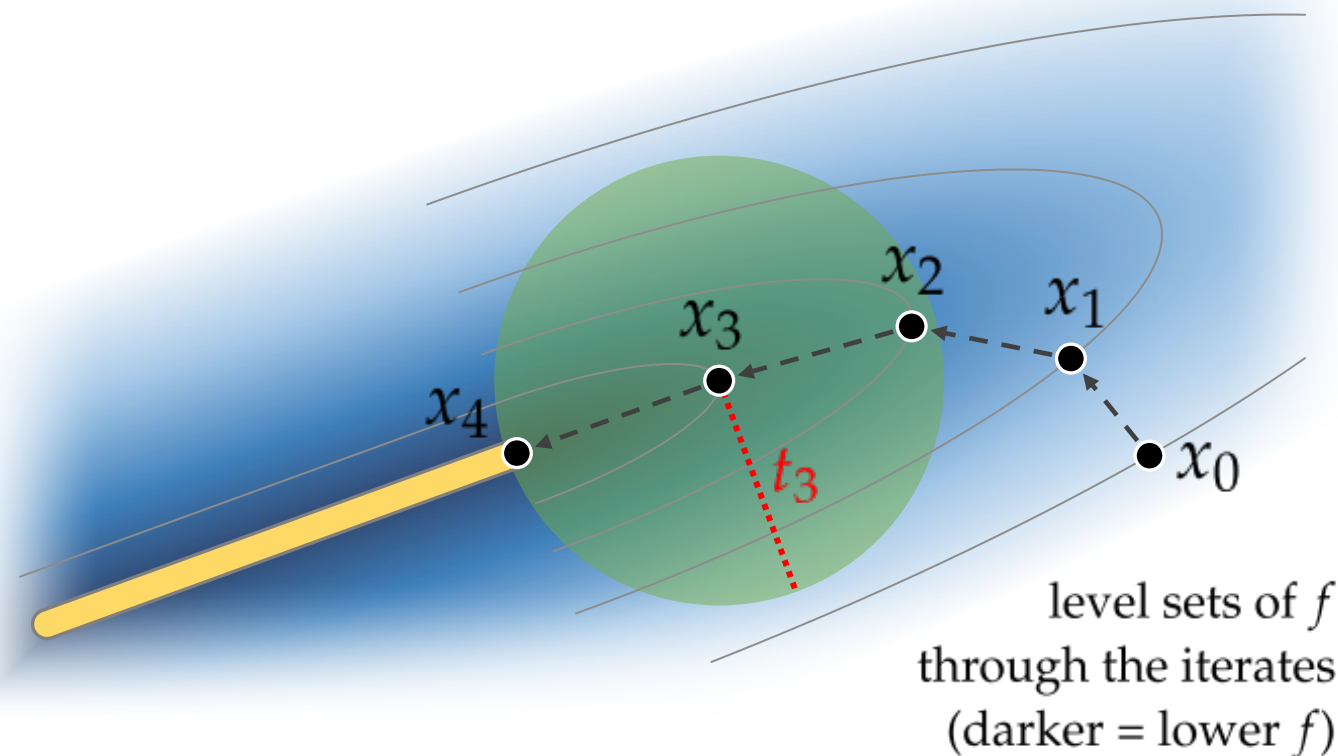
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$$t_3 = 1.8 \quad d_4 = 0.00 \quad f_4 = 0.00 \quad g_4 = 0.00$$

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_\star := \min f = 0$$



BPM with ℓ_2 Balls

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Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

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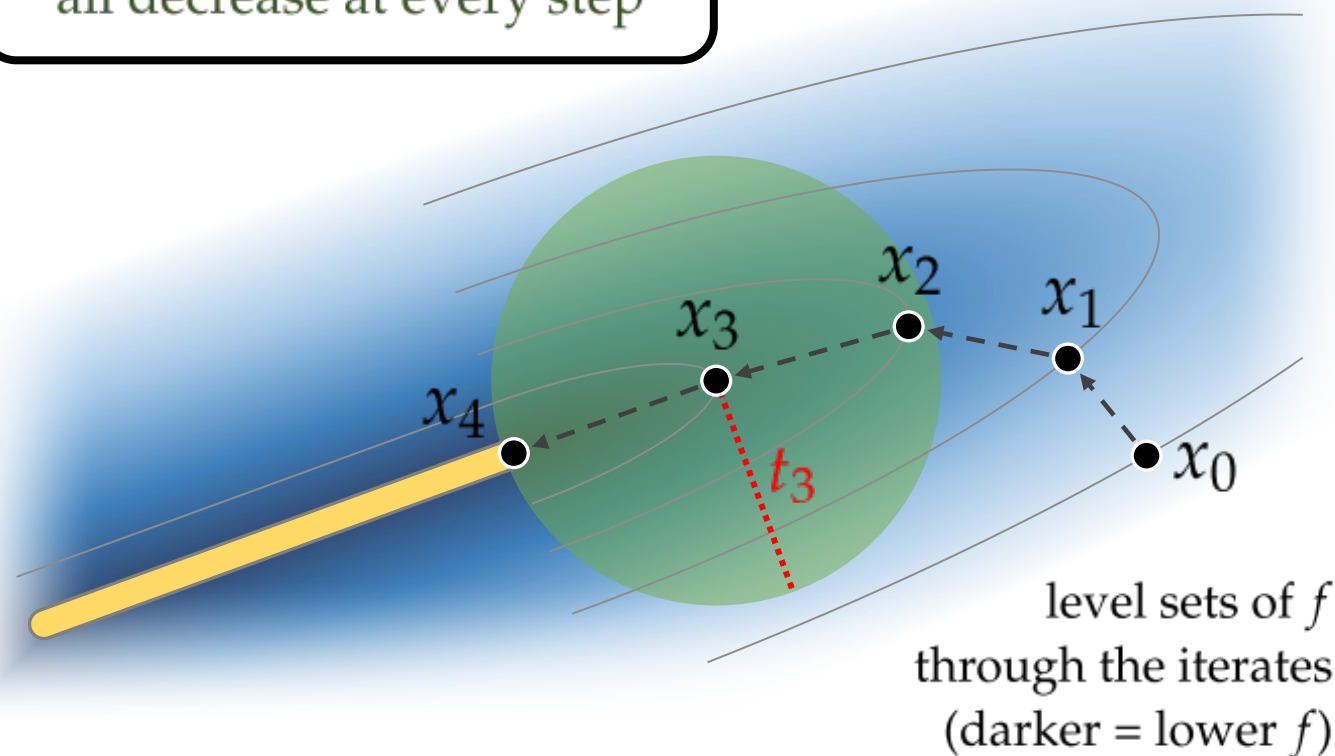
$$t_3 = 1.8 \quad d_4 = 0.00 \quad f_4 = 0.00 \quad g_4 = 0.00$$

d_k, f_k and g_k

all decrease at every step

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_\star := \min f = 0$$



Proximal Point vs. Broximal Point

PPM: Proximal Point Method (Martinet, 1970)

$$x_{k+1} = \operatorname{argmin}_y \left\{ f(y) + \frac{1}{2\gamma} \|y - x_k\|_2^2 \right\}$$

$$\|x_{k+1} - x_k\|_2 = \gamma \|\nabla f(x_{k+1})\|_2 \rightarrow 0$$

f differentiable: $x_k \notin \operatorname{Argmin} f$ for every k



$\operatorname{Argmin} f$

• x_0

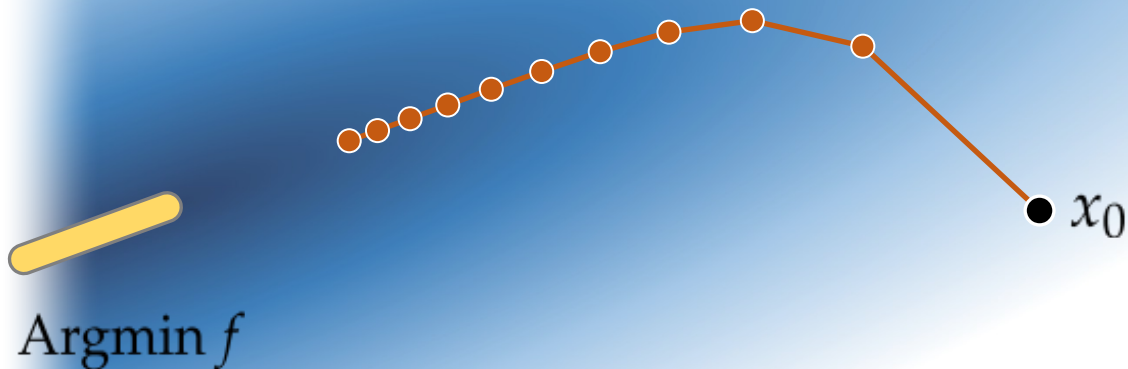
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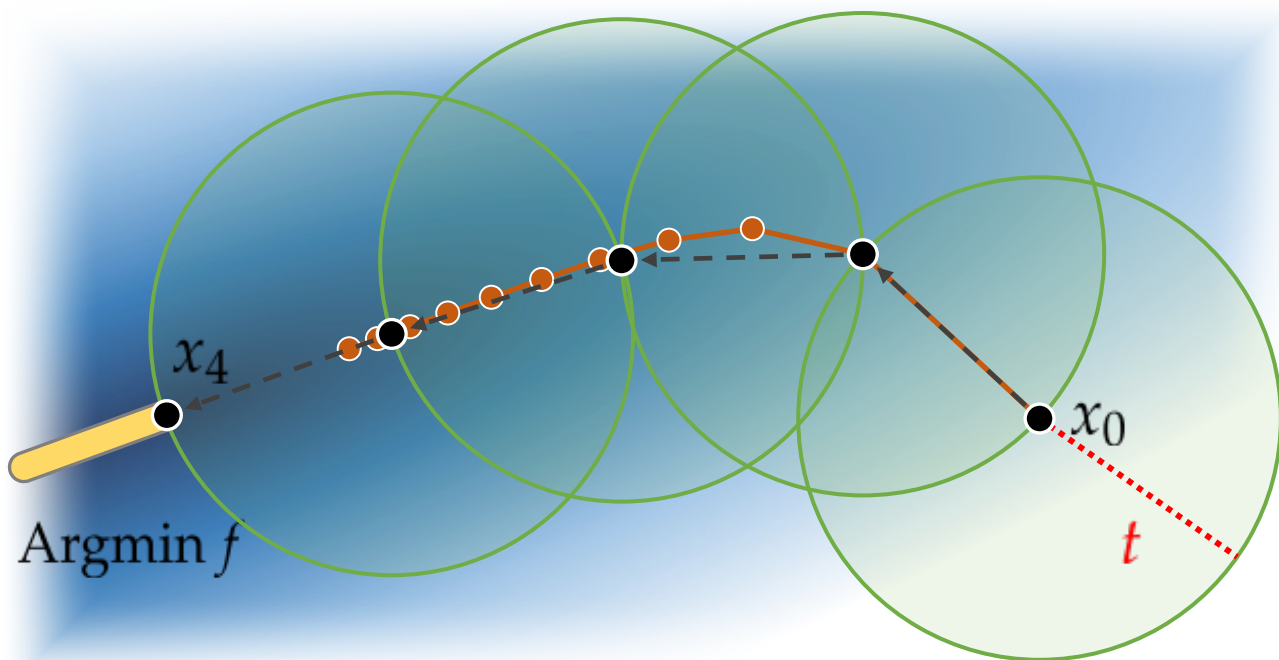
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BPM: Broximal (= ball-proximal) Point Method

$$x_{k+1} \in \operatorname{Argmin}_y \left\{ f(y) : \|y - x_k\|_2 \leq t \right\}$$

$\|x_{k+1} - x_k\|_2 = t$ until the ball meets $\operatorname{Argmin} f$
exact after $\lceil d_0^2/t^2 \rceil$ steps; no condition number



• PPM, $\gamma = 0.155$ • BPM, $t = 1.4$

same x_0 , same first step

after 4 steps: $d_4 = 0$ vs. $d_4 = 2.67$

$d_k := \operatorname{dist}_2(x_k, \operatorname{Argmin} f)$

Proximal Point vs. Broximal Point

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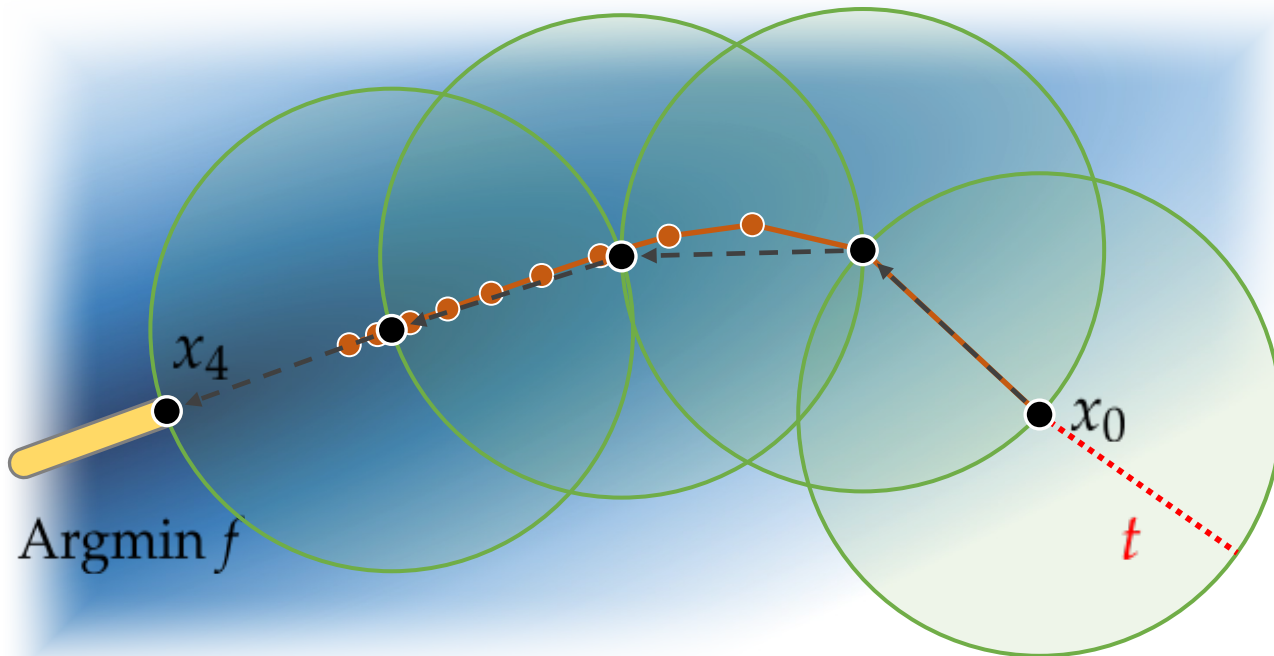
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$d_k := \operatorname{dist}_2(x_k, \operatorname{Argmin} f)$

Penalty $\rightarrow$ constraint: the steps
no longer shrink, so BPM reaches
 $\operatorname{Argmin} f$ in finitely many steps.

BPM with ℓ_1 Balls

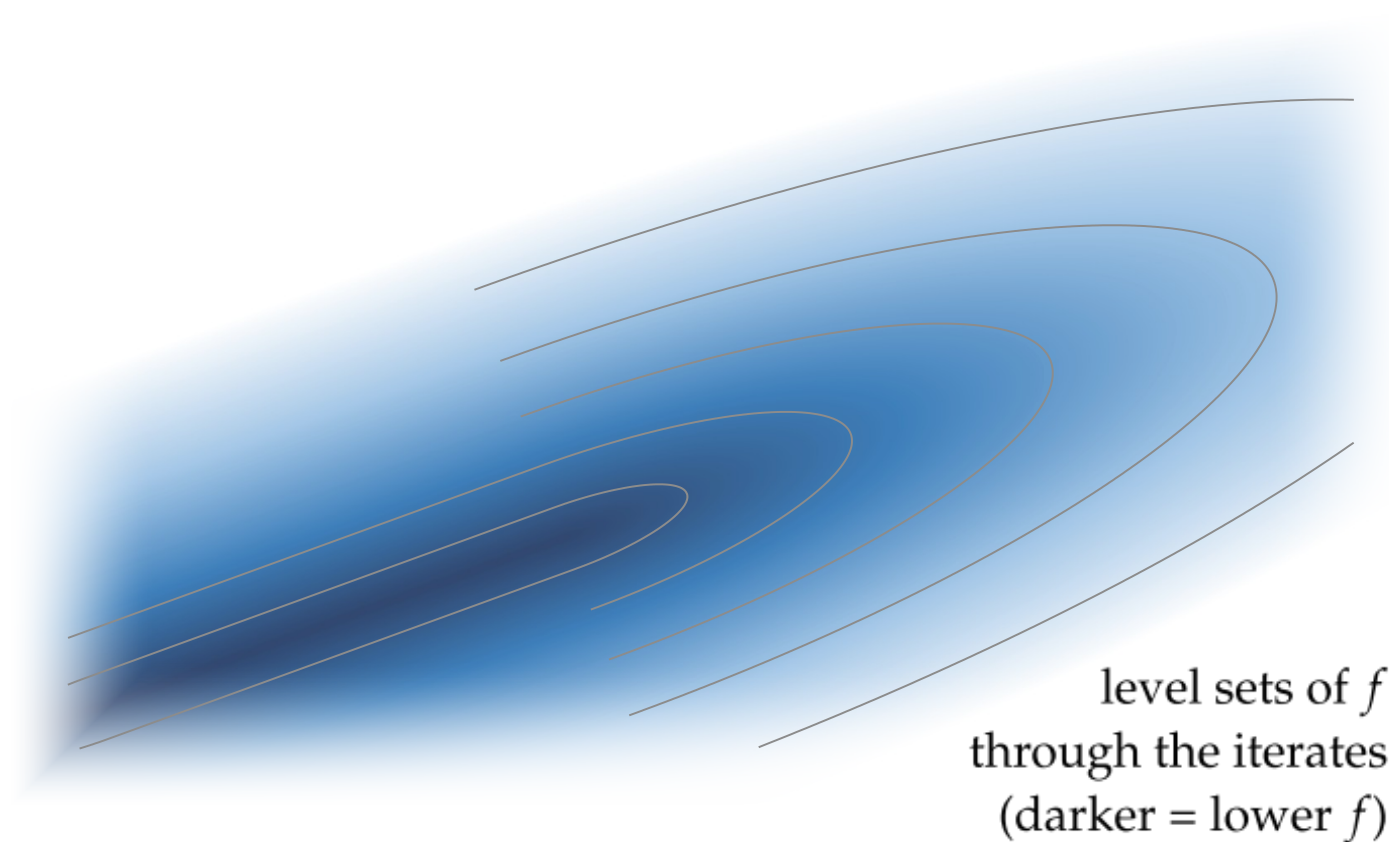
$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_1 \leq t_k\}$$

Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

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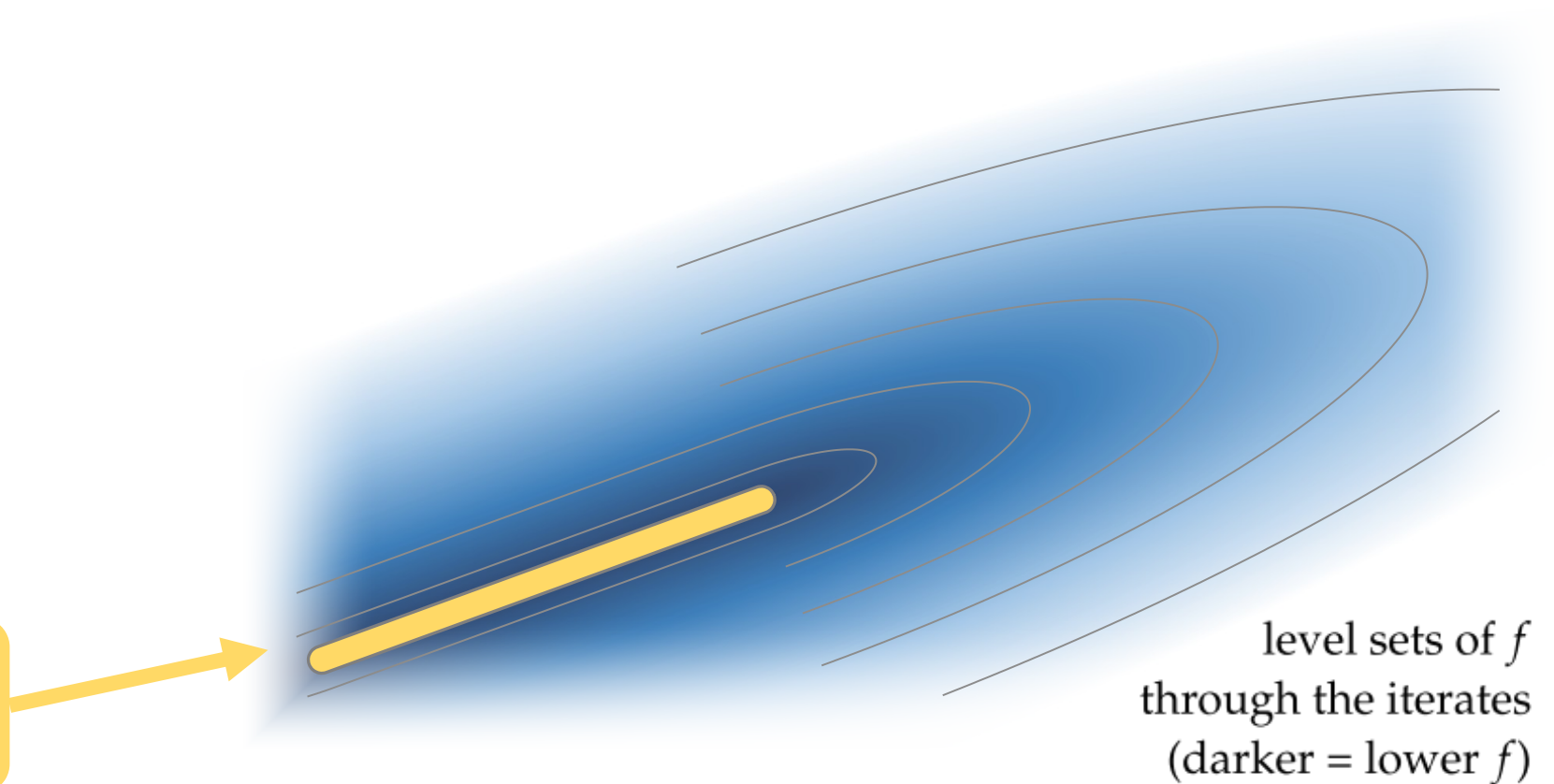


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BPM run

$$d_k = \operatorname{dist}_1(x_k, \operatorname{Argmin} f)$$

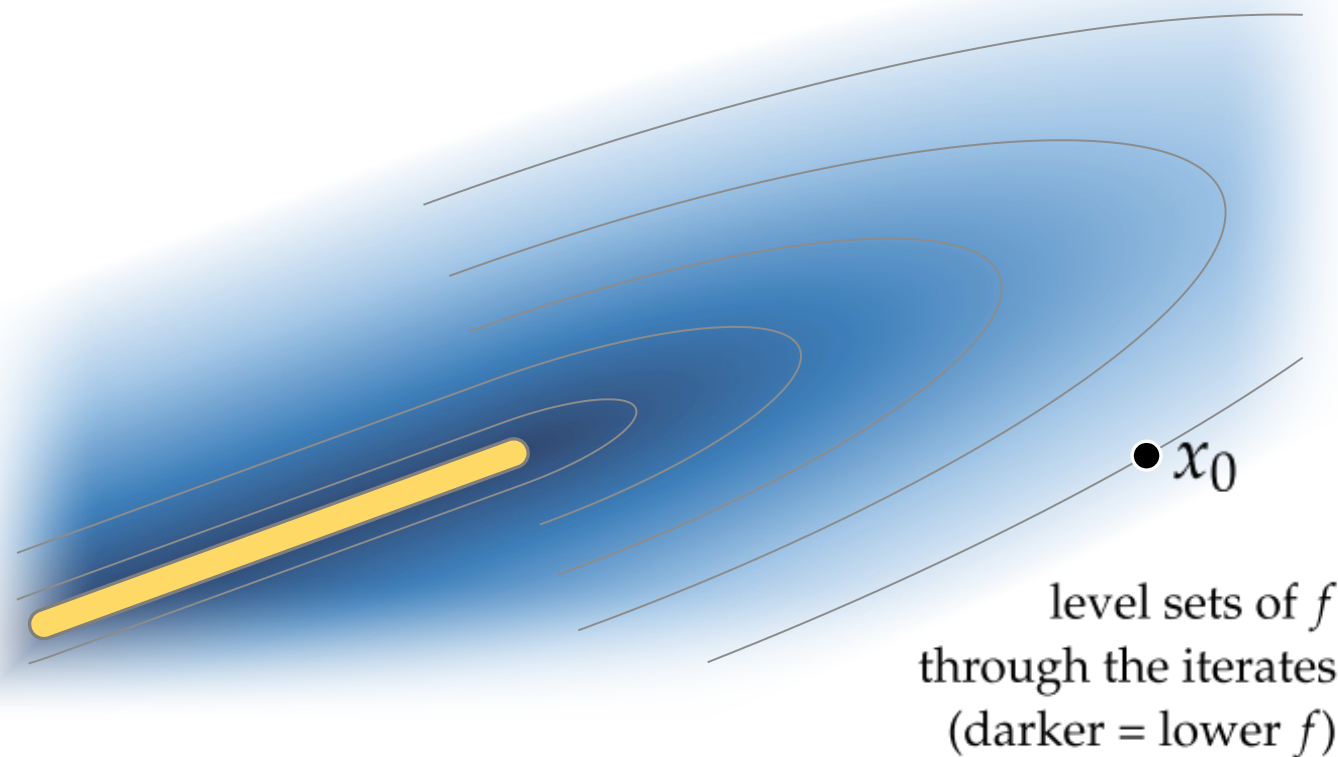
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$$g_k = \|\nabla f(x_k)\|_\infty \quad (\text{dual norm})$$

$$d_0 = 5.08 \quad f_0 = 35.78 \quad g_0 = 24.69$$

$$\operatorname{Argmin} f = [-2a, 2a]$$

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BPM with ℓ_1 Balls

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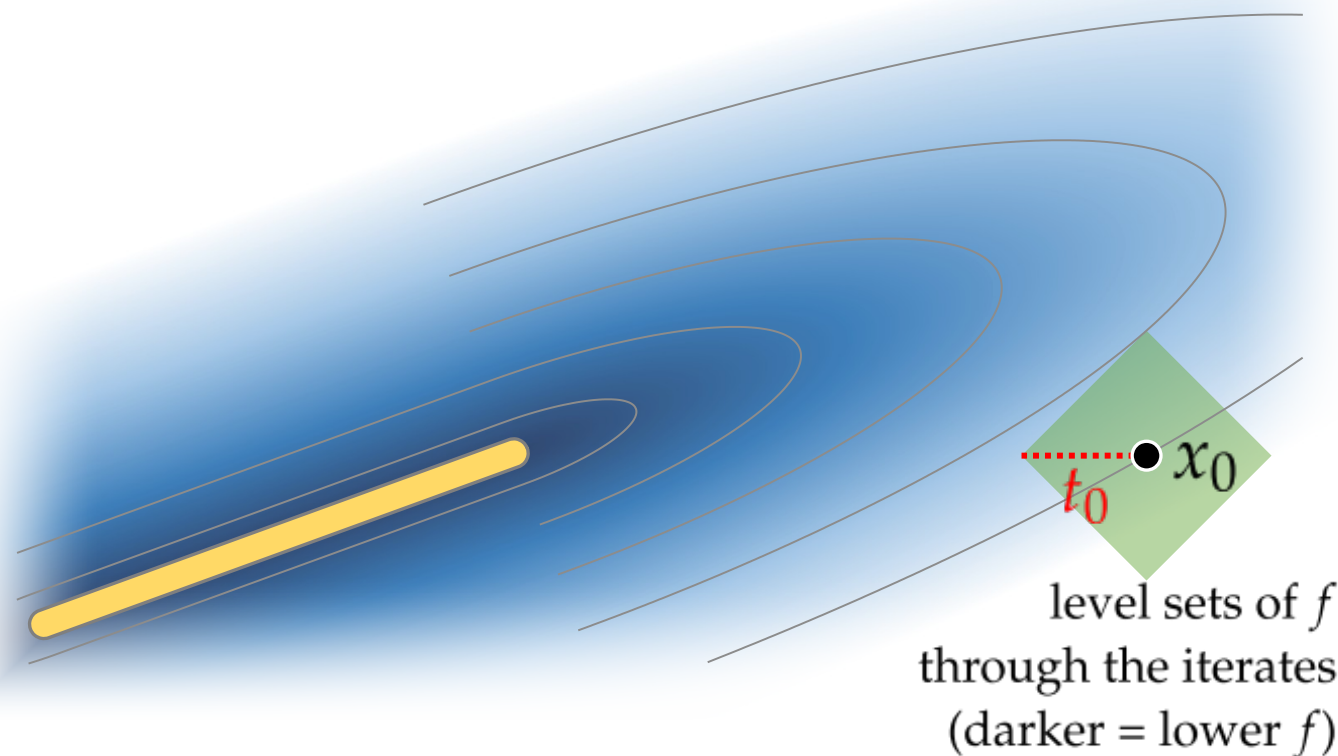
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level sets of f
through the iterates
(darker = lower f)

BPM with ℓ_1 Balls

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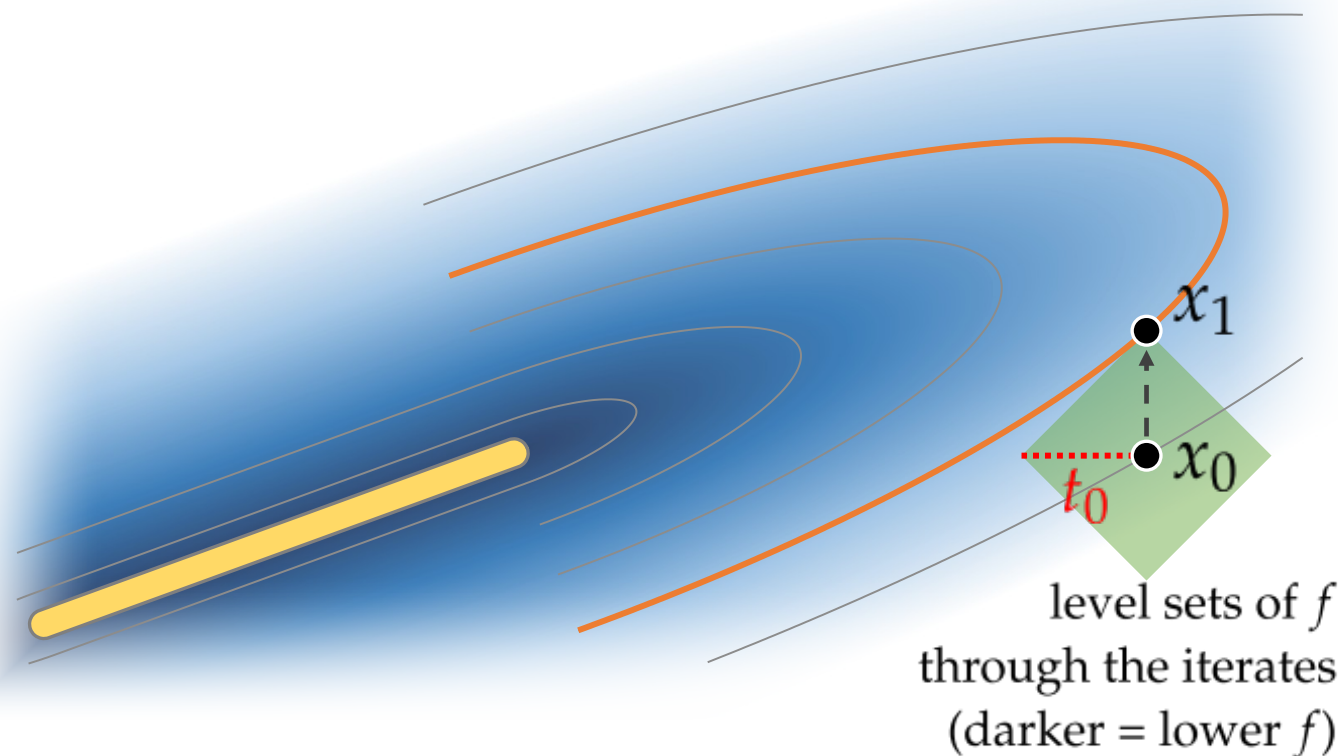
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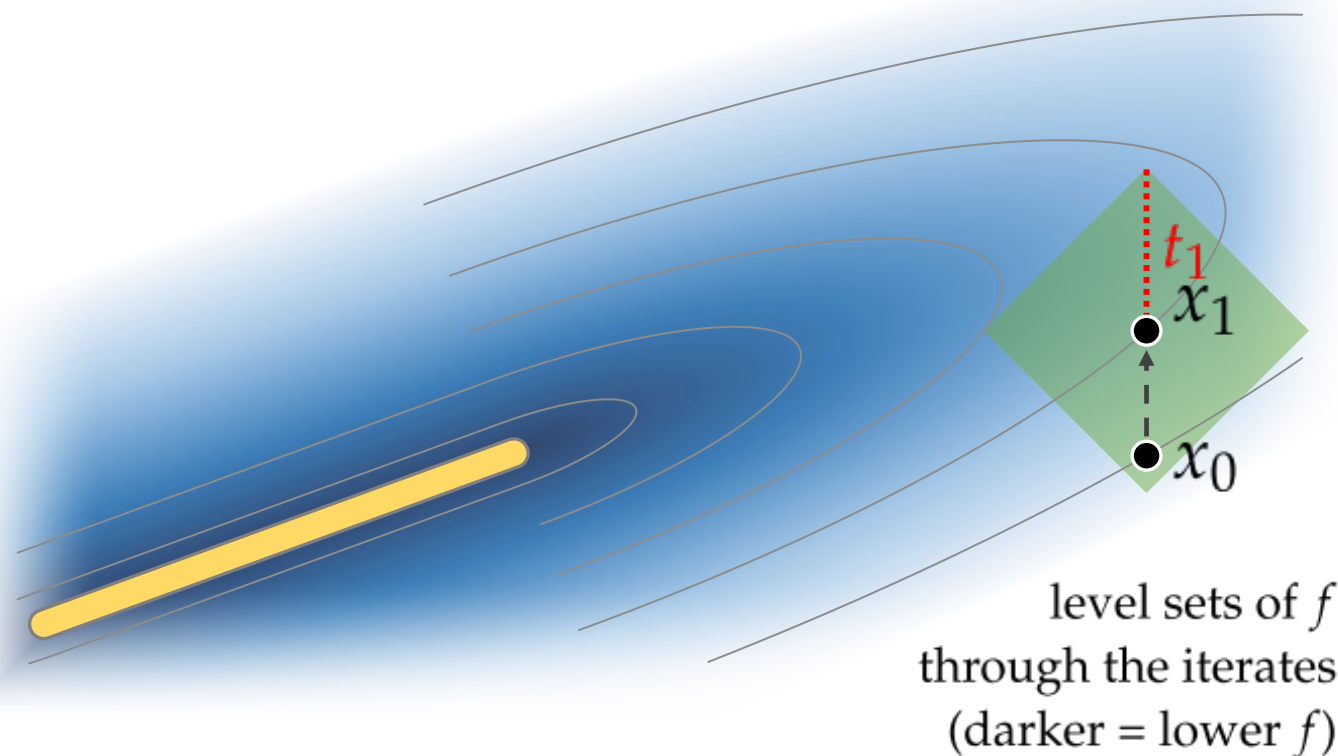
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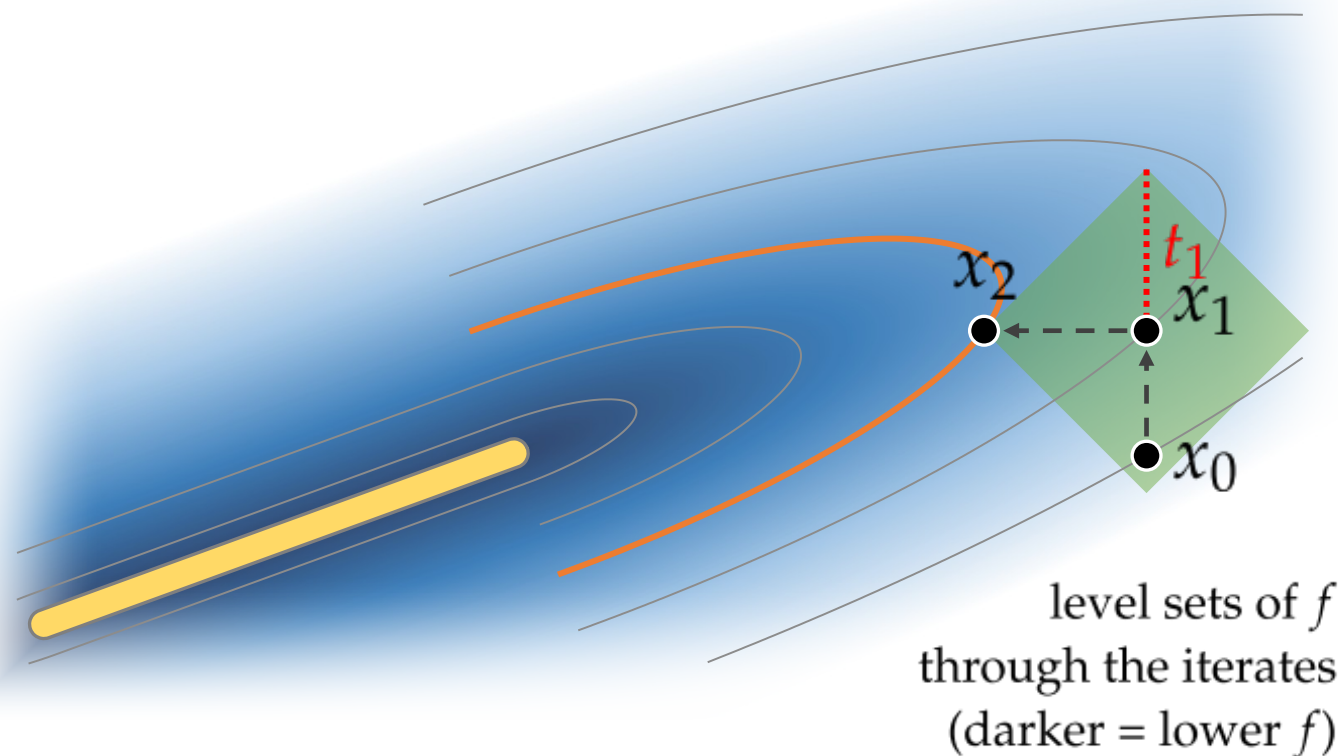
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$t_0 = 1.0$	$d_1 = 6.04 \uparrow$	$f_1 = 18.22$	$g_1 = 10.44$
$t_1 = 1.3$	$d_2 = 4.74$	$f_2 = 8.56$	$g_2 = 5.64$

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_\star := \min f = 0$$



BPM with ℓ_1 Balls

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BPM run

$$d_k = \operatorname{dist}_1(x_k, \operatorname{Argmin} f)$$

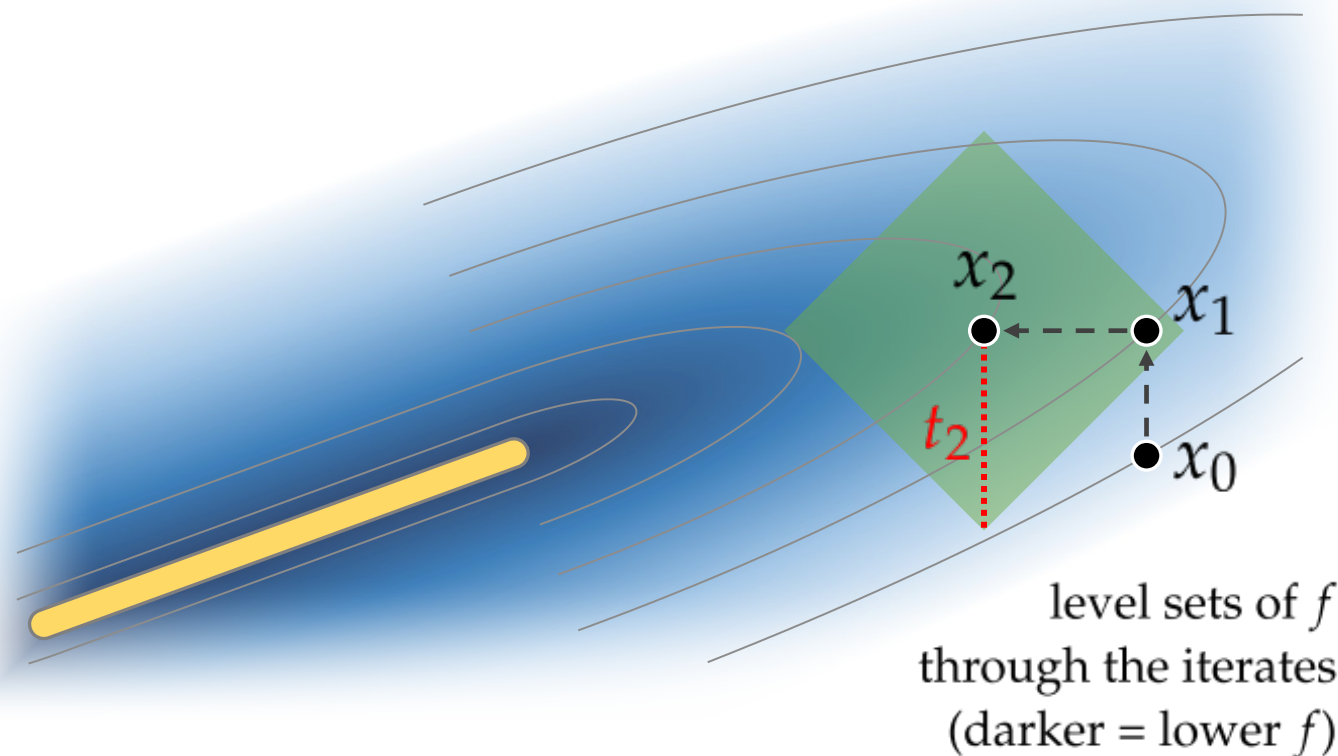
$$f_k = f(x_k) - f_\star$$

$$g_k = \|\nabla f(x_k)\|_\infty \quad (\text{dual norm})$$

	$d_0 = 5.08$	$f_0 = 35.78$	$g_0 = 24.69$
$t_0 = 1.0$	$d_1 = 6.04 \uparrow$	$f_1 = 18.22$	$g_1 = 10.44$
$t_1 = 1.3$	$d_2 = 4.74$	$f_2 = 8.56$	$g_2 = 5.64$

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_\star := \min f = 0$$



BPM with ℓ_1 Balls

$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_1 \leq t_k\}$$

Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

BPM run

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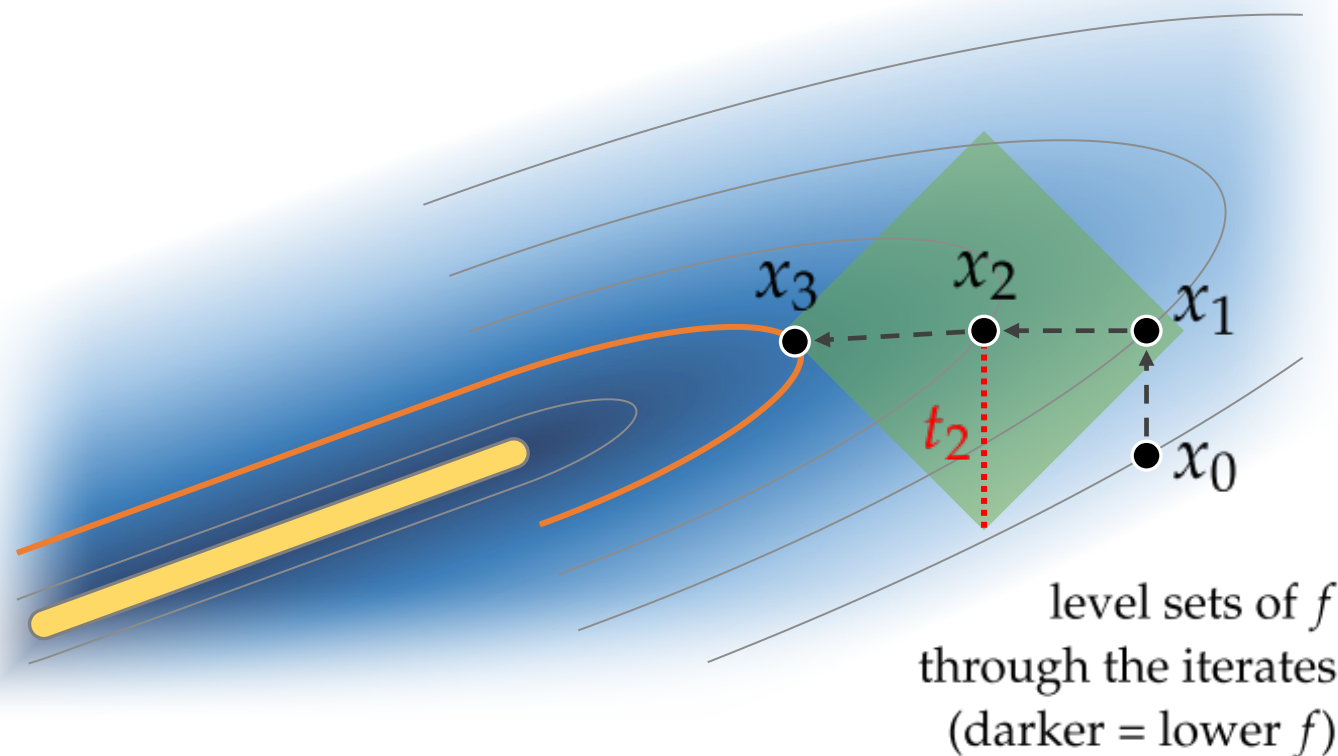
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$t_2 = 1.6$	$d_3 = 3.14$	$f_3 = 2.96$	$g_3 = 1.89$

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BPM run

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$$d_0 = 5.08 \quad f_0 = 35.78 \quad g_0 = 24.69$$

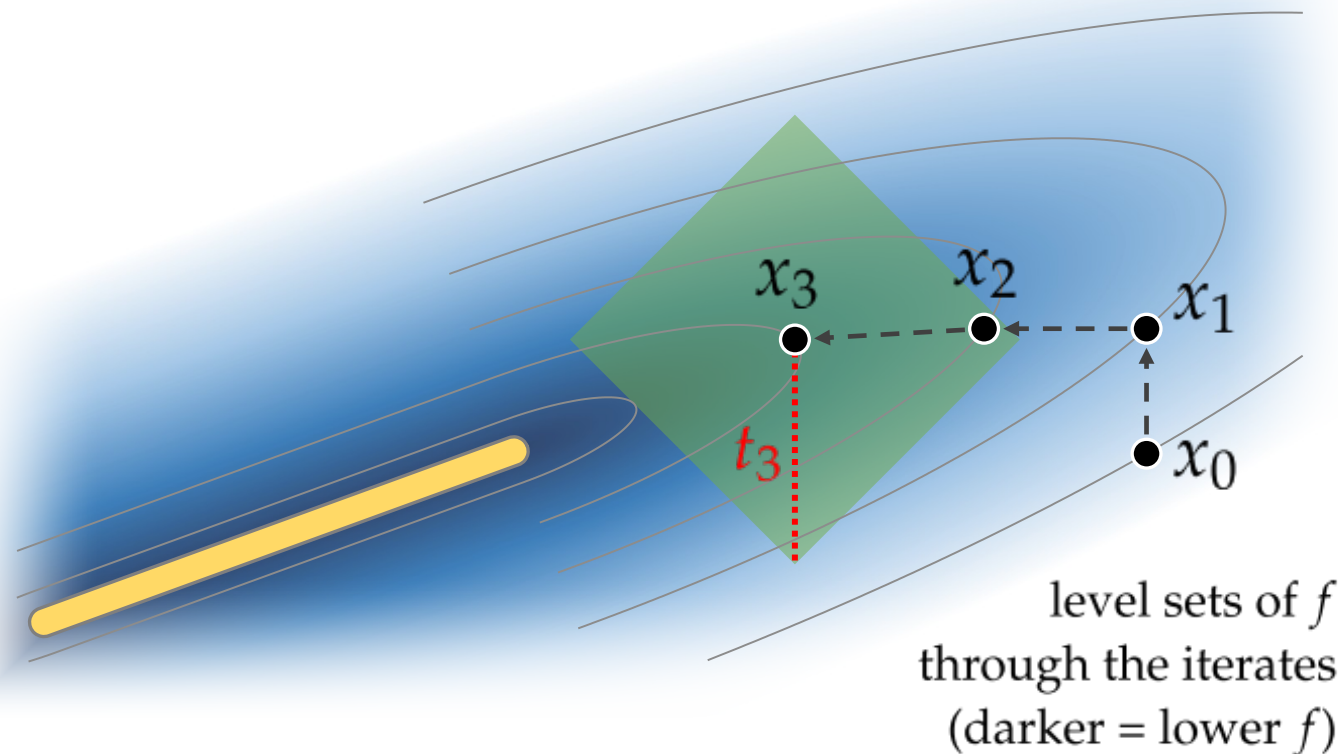
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BPM run

$$d_k = \operatorname{dist}_1(x_k, \operatorname{Argmin} f)$$

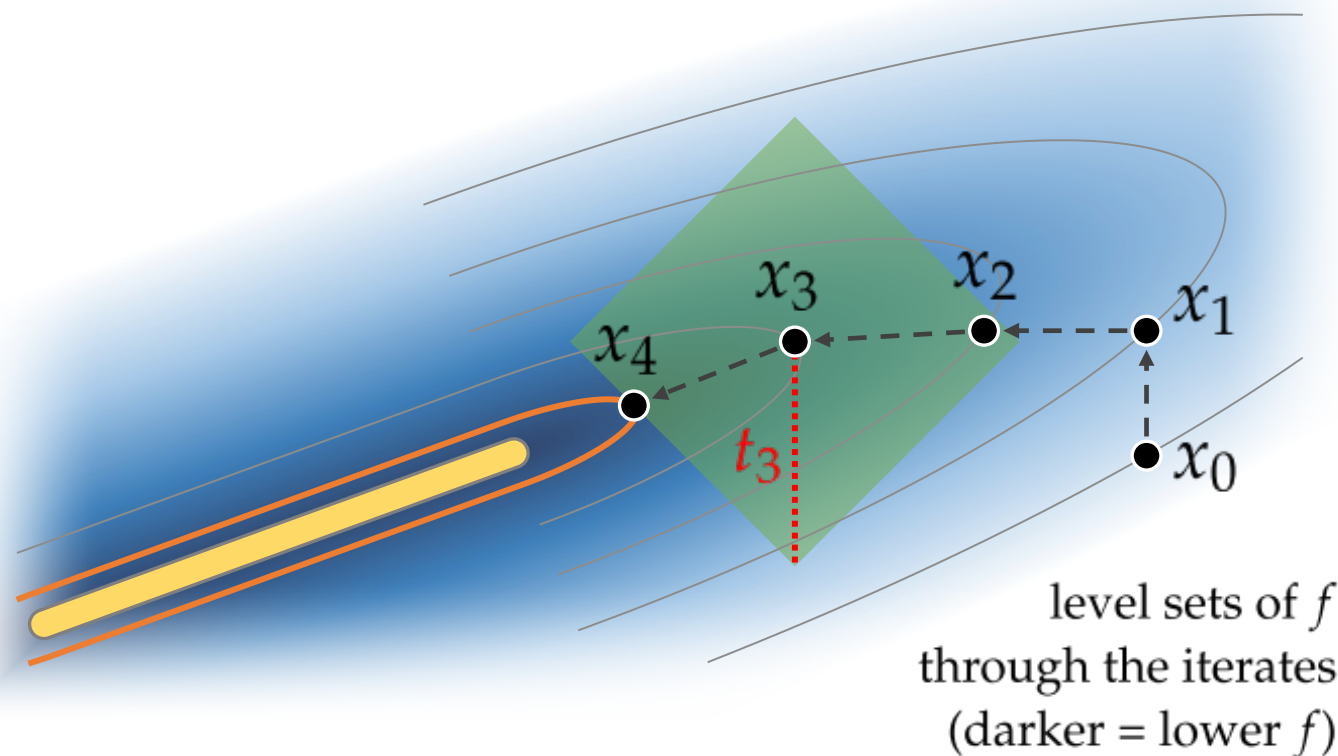
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BPM with ℓ_1 Balls

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BPM run

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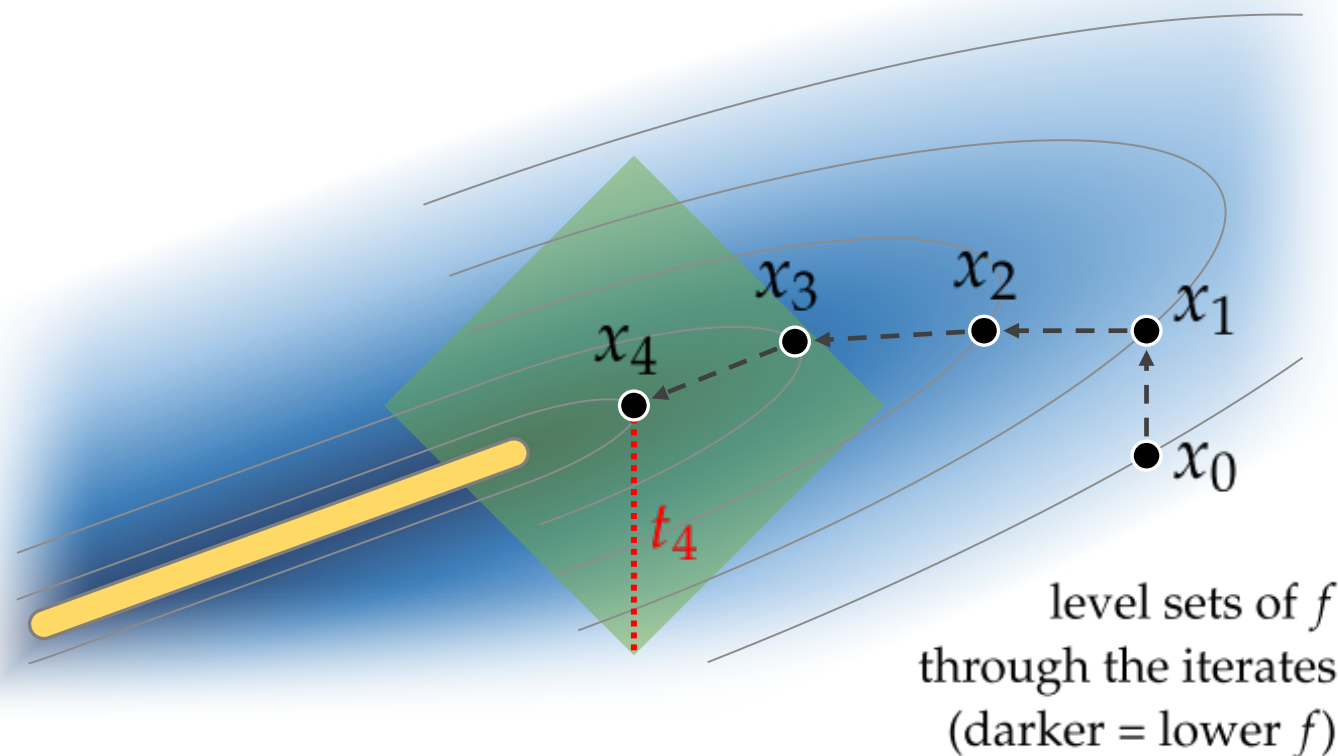
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BPM with ℓ_1 Balls

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BPM run

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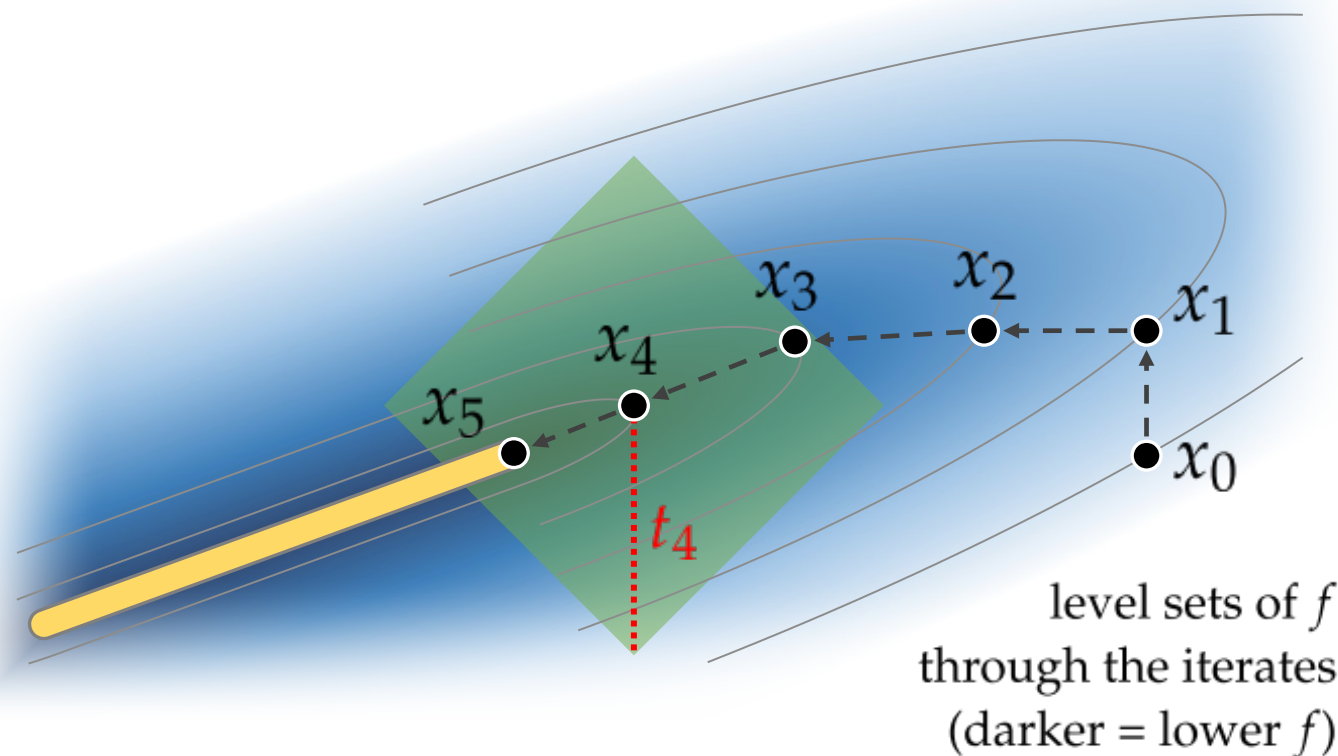
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$t_2 = 1.6$	$d_3 = 3.14$	$f_3 = 2.96$	$g_3 = 1.89$
$t_3 = 1.8$	$d_4 = 1.34$	$f_4 = 0.54$	$g_4 = 0.81$
$t_4 = 2.0$	$d_5 = 0.00$	$f_5 = 0.00$	$g_5 = 0.00$

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_\star := \min f = 0$$



BPM with ℓ_1 Balls

$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_1 \leq t_k\}$$

Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

BPM run

$$d_k = \operatorname{dist}_1(x_k, \operatorname{Argmin} f)$$

$$f_k = f(x_k) - f_*$$

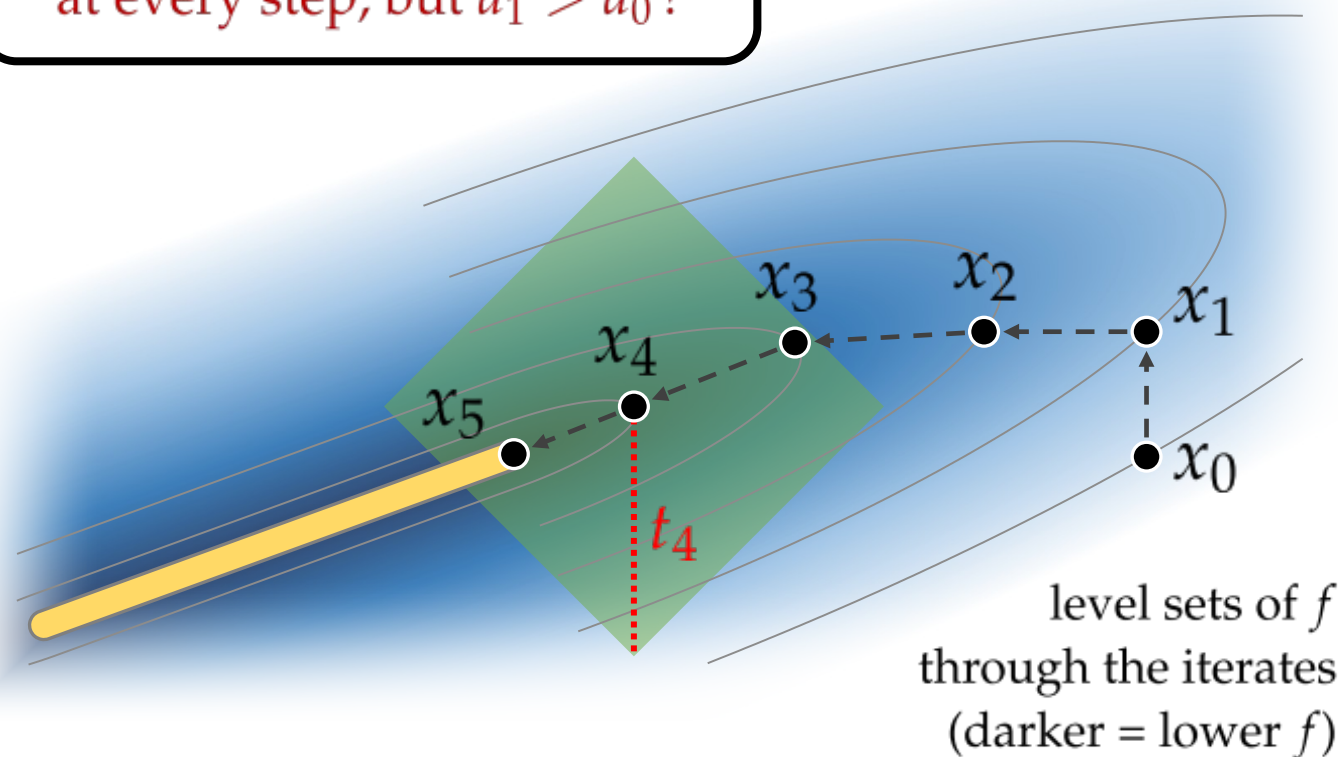
$$g_k = \|\nabla f(x_k)\|_\infty \quad (\text{dual norm})$$

	$d_0 = 5.08$	$f_0 = 35.78$	$g_0 = 24.69$
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$t_4 = 2.0$	$d_5 = 0.00$	$f_5 = 0.00$	$g_5 = 0.00$

f_k and g_k decrease
at every step, but $d_1 > d_0$!

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_* := \min f = 0$$



BPM with ℓ_∞ Balls

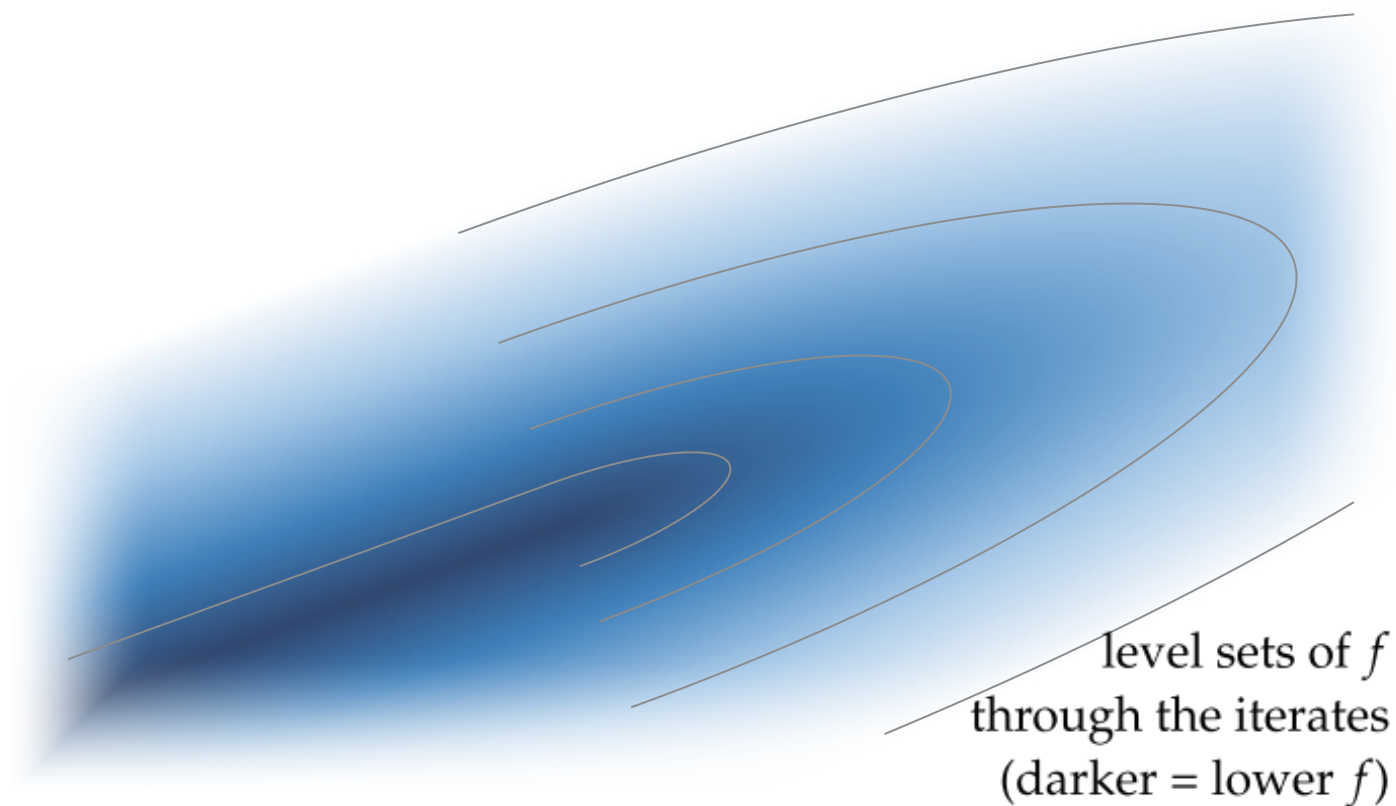
$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_\infty \leq t_k\}$$

Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

BPM with ℓ_∞ Balls

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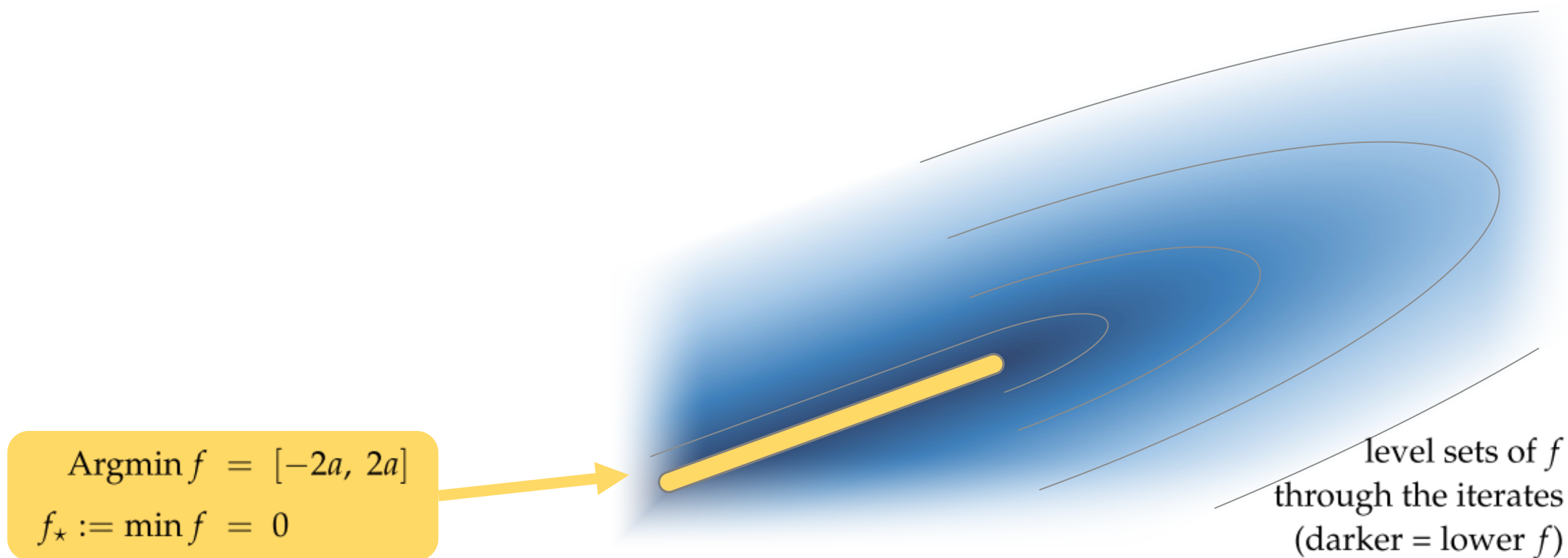
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BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

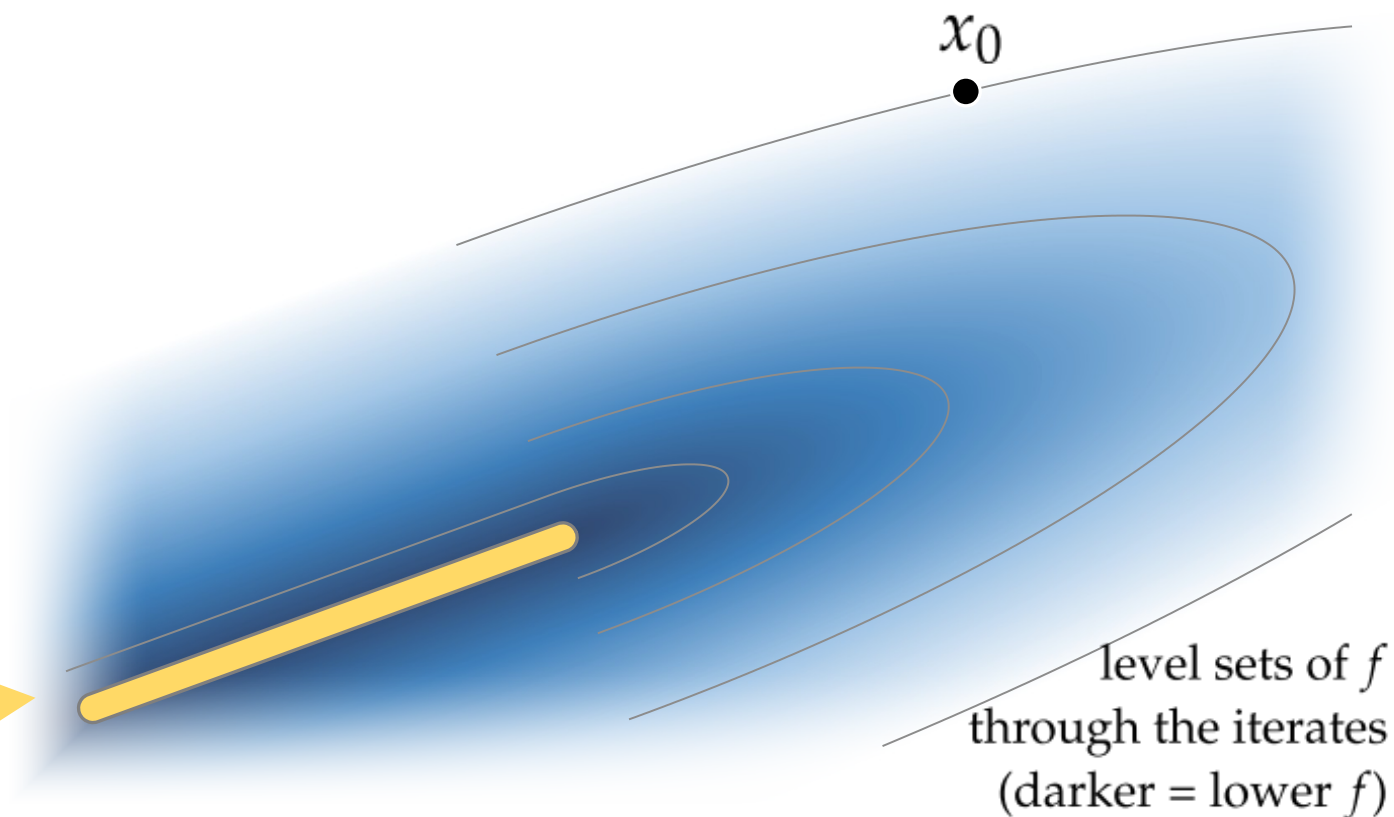
$$f_k = f(x_k) - f_\star$$

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$$d_0 = 3.57 \quad f_0 = 49.53 \quad g_0 = 43.60$$

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_\star := \min f = 0$$



BPM with ℓ_∞ Balls

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BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

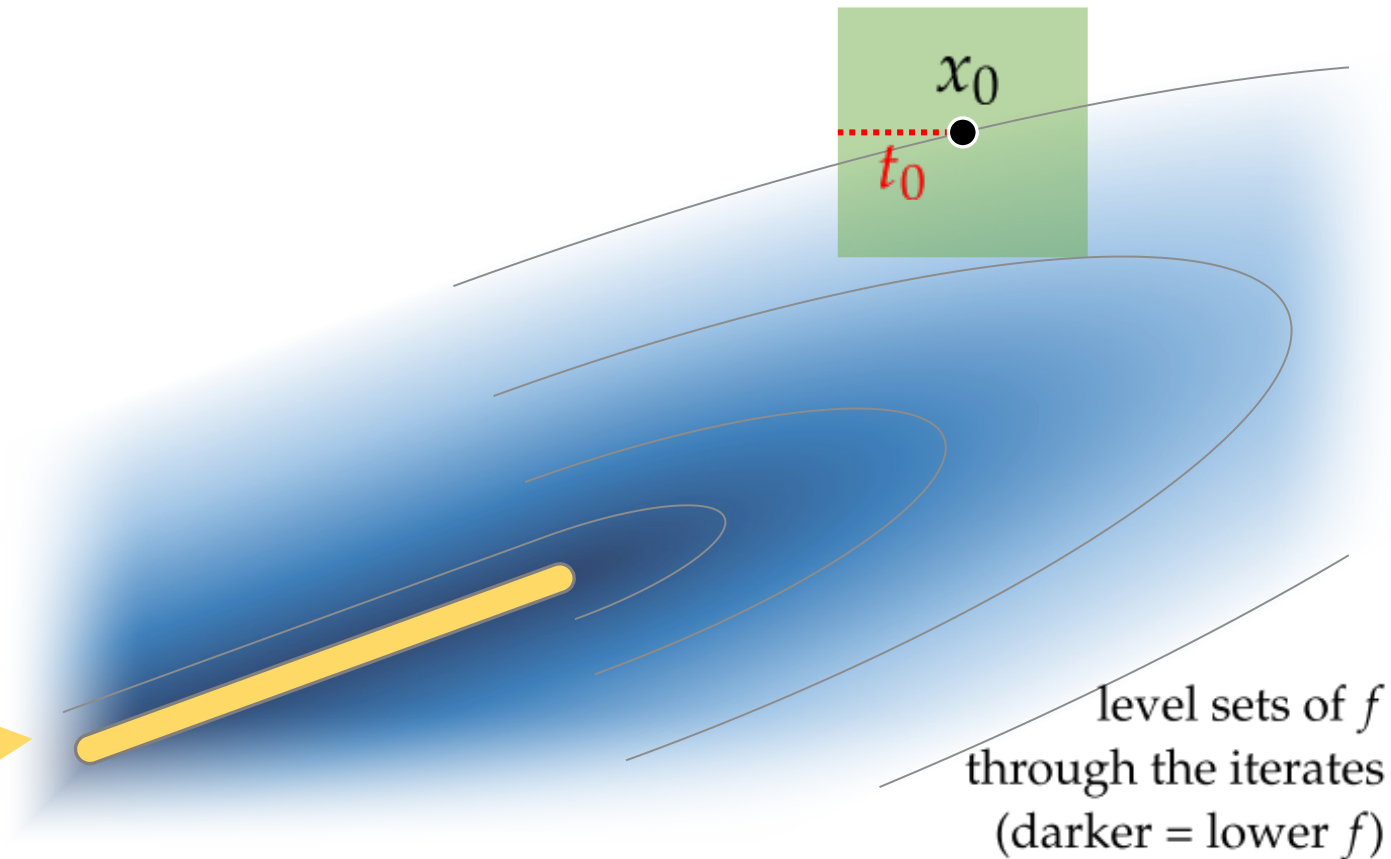
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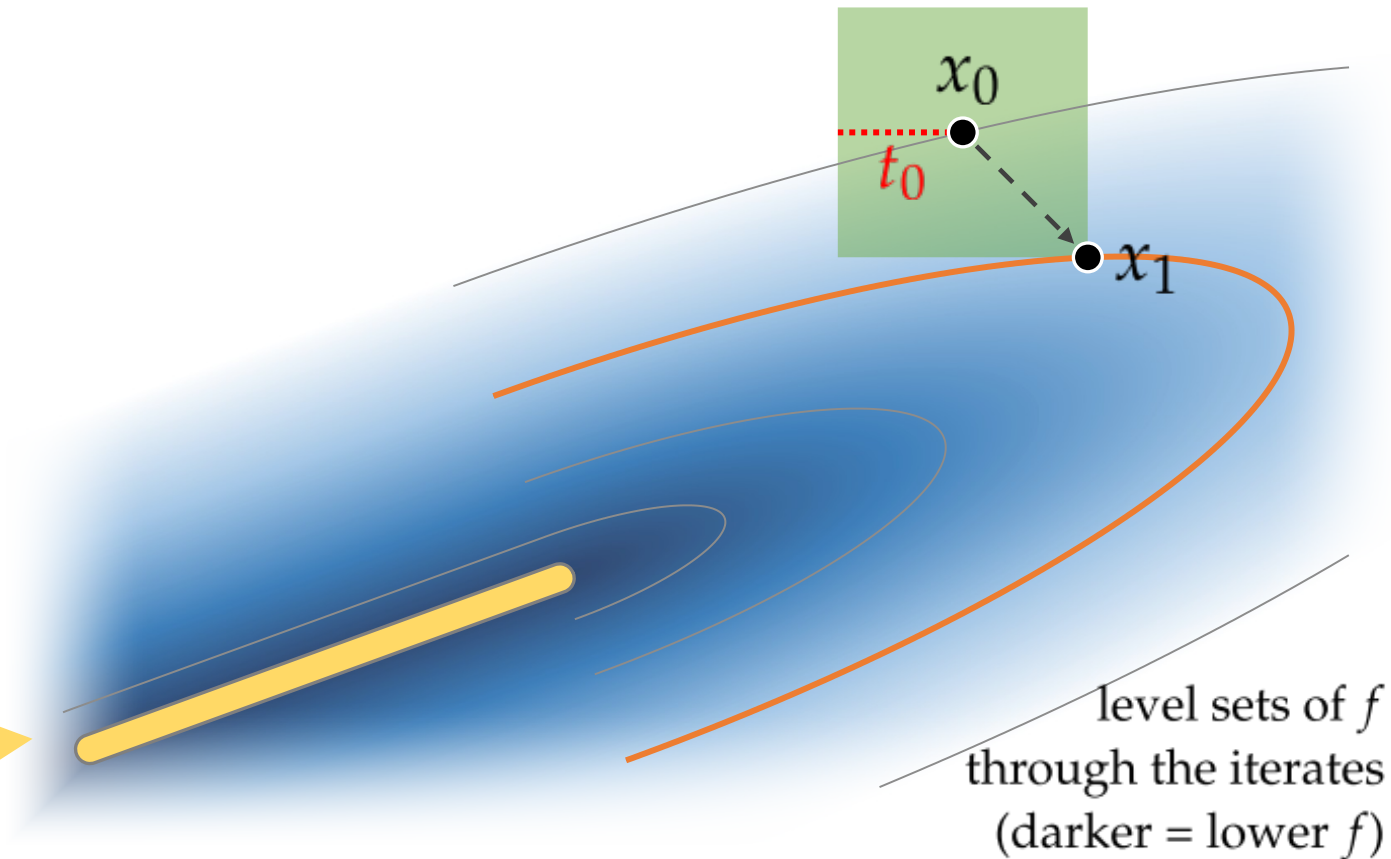
$$g_k = \|\nabla f(x_k)\|_1 \quad (\text{dual norm})$$

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BPM with ℓ_∞ Balls

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BPM run

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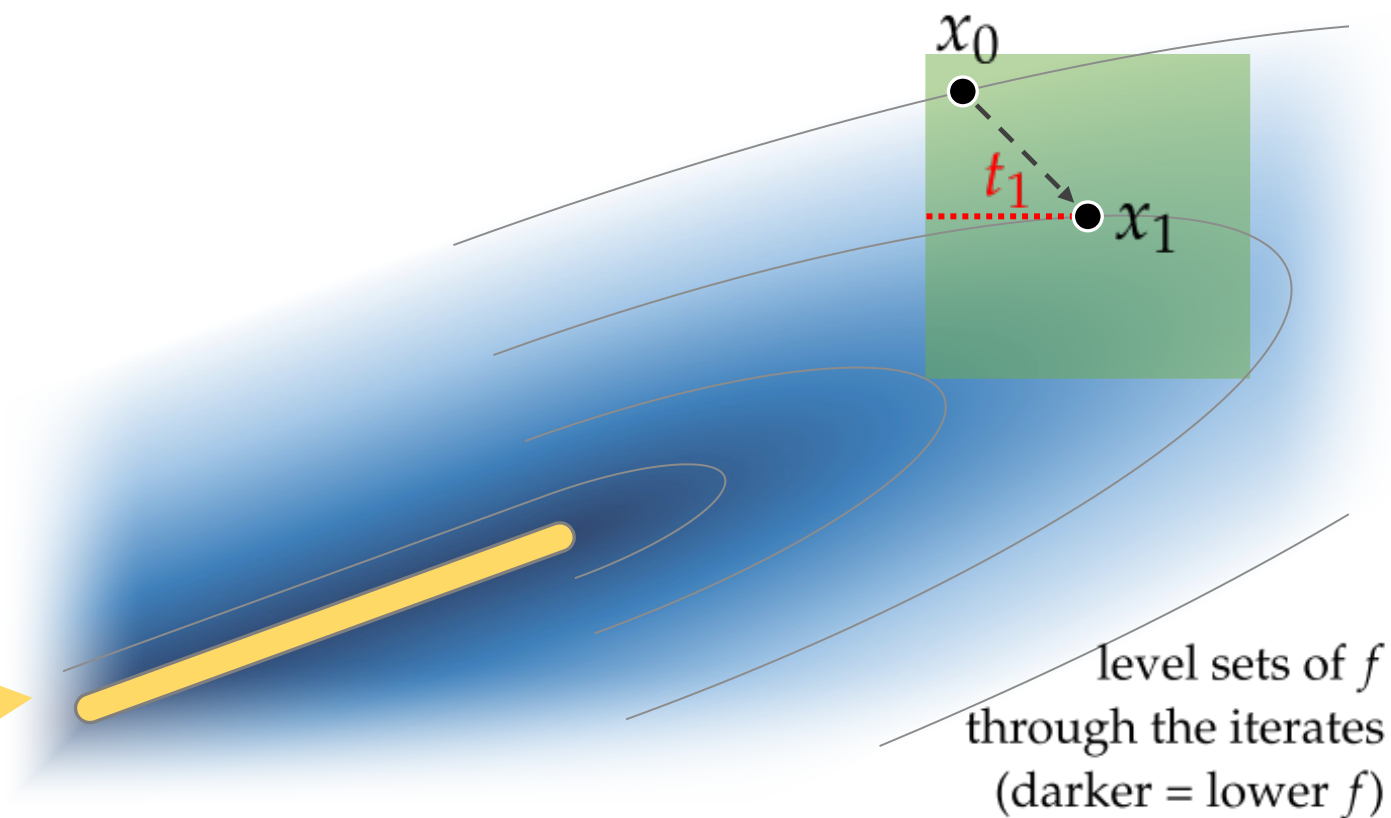
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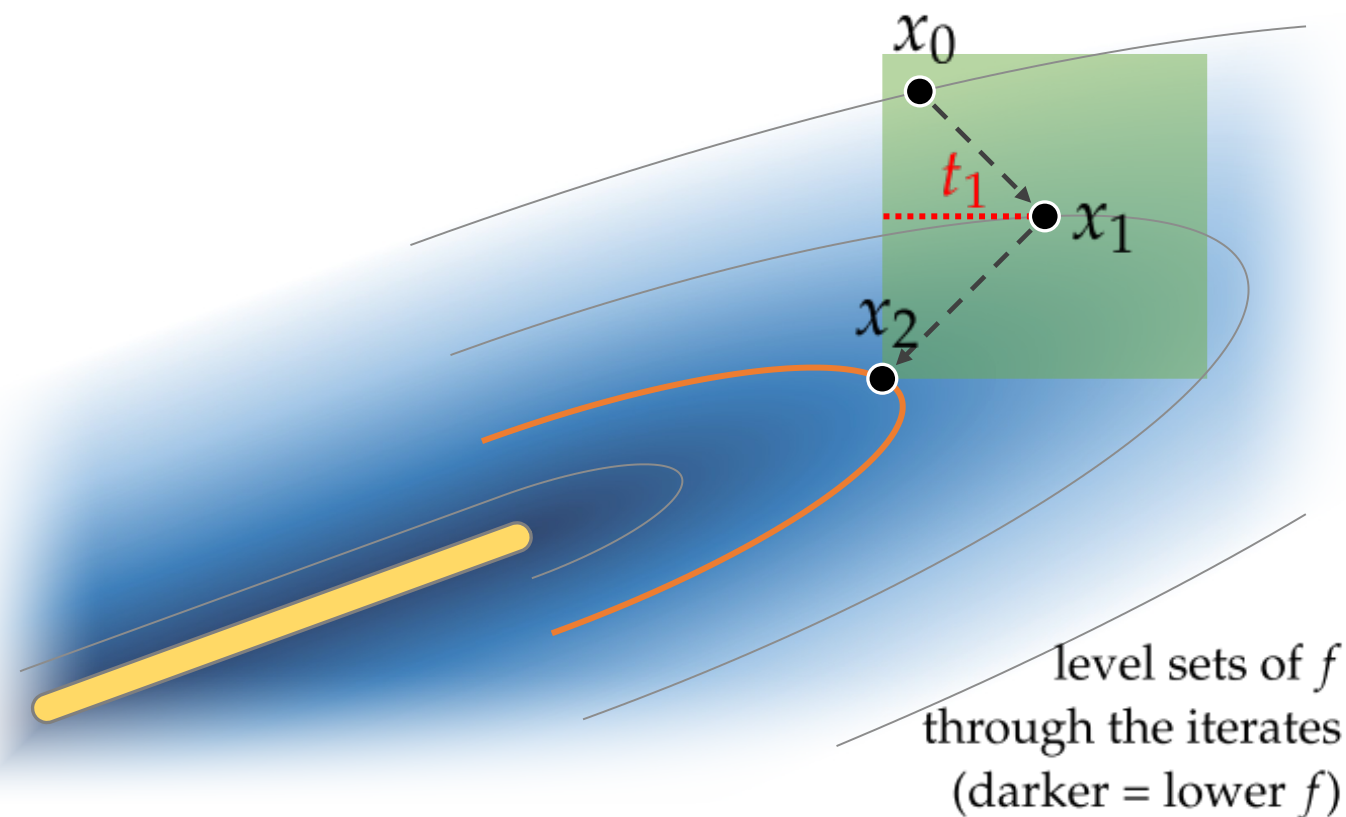
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$t_1 = 1.3$	$d_2 = 2.92$	$f_2 = 5.35$	$g_2 = 5.91$

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BPM with ℓ_∞ Balls

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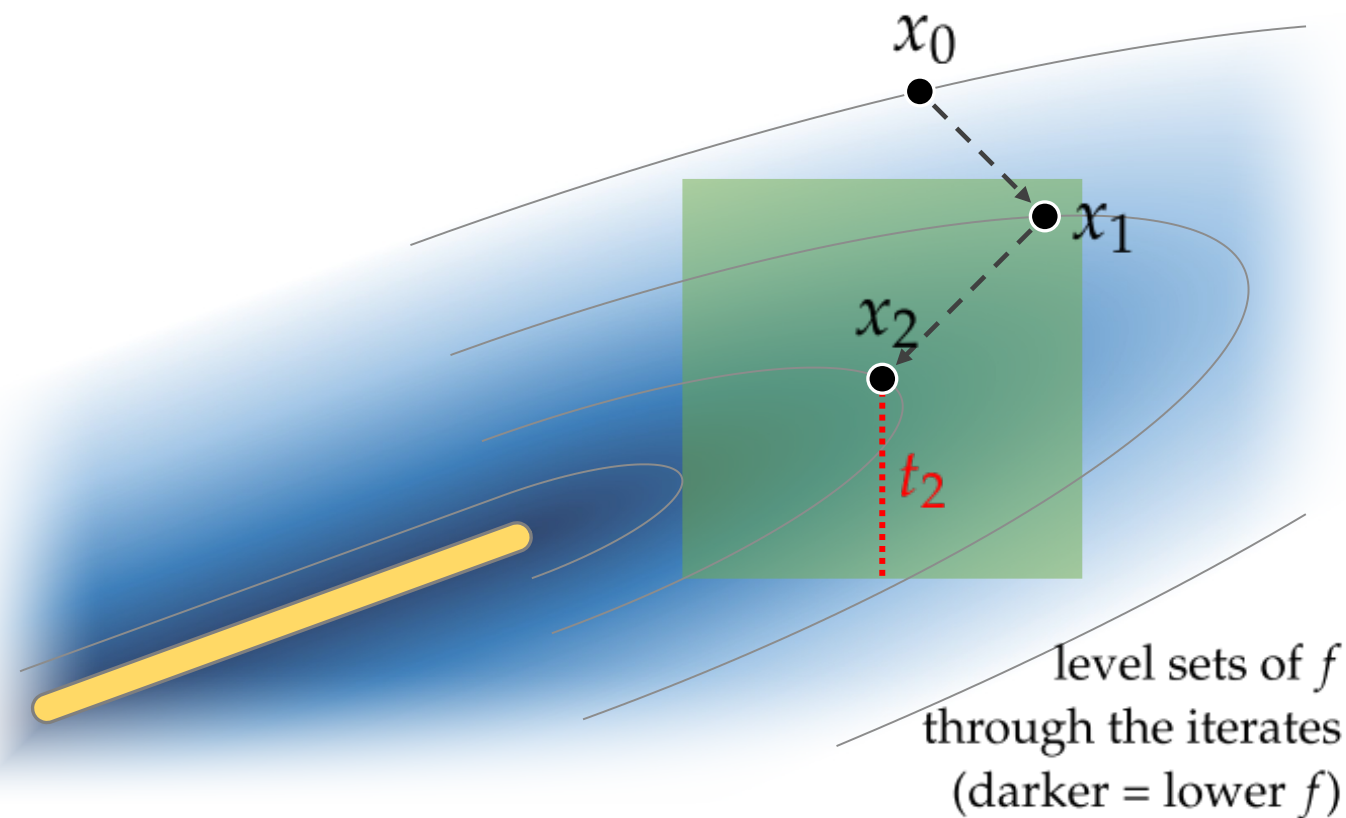
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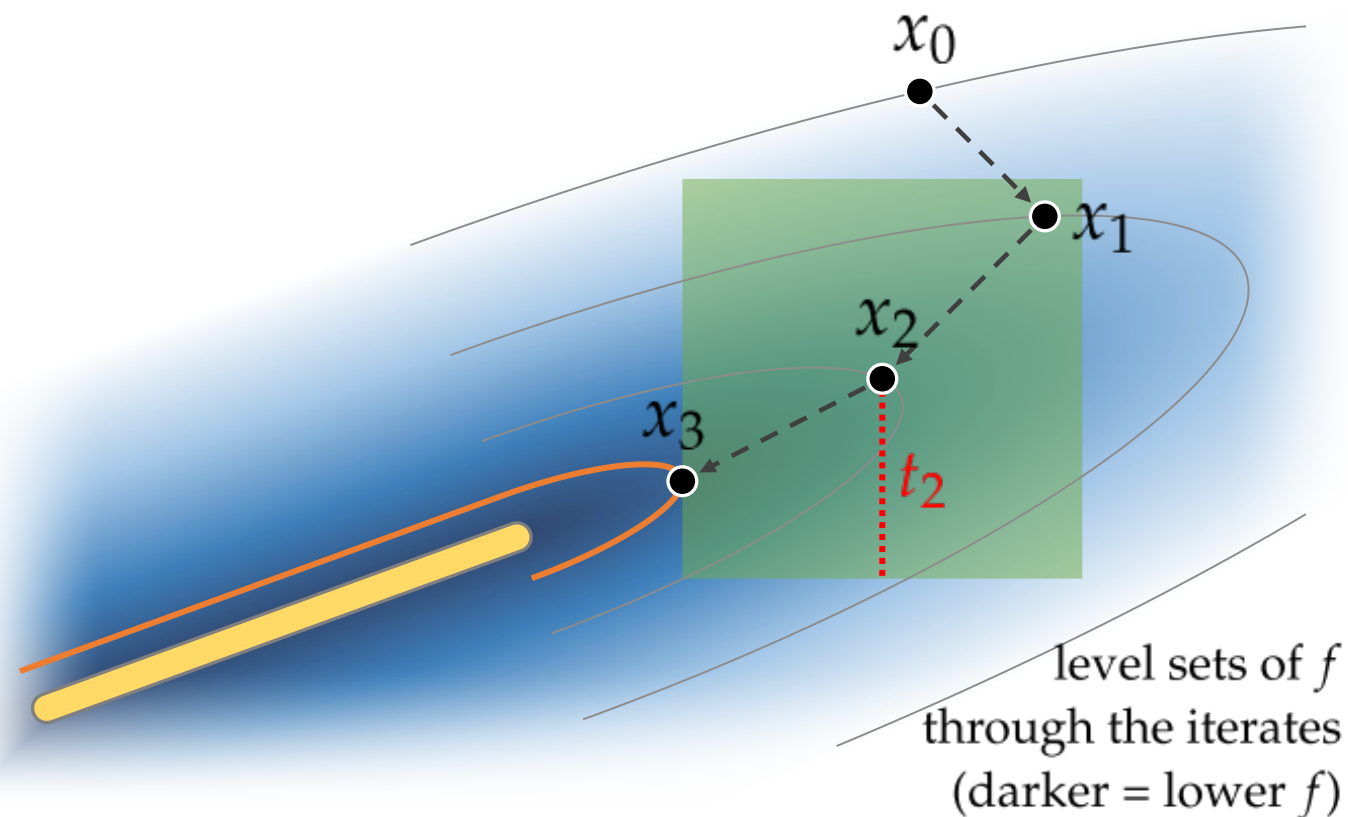
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BPM with ℓ_∞ Balls

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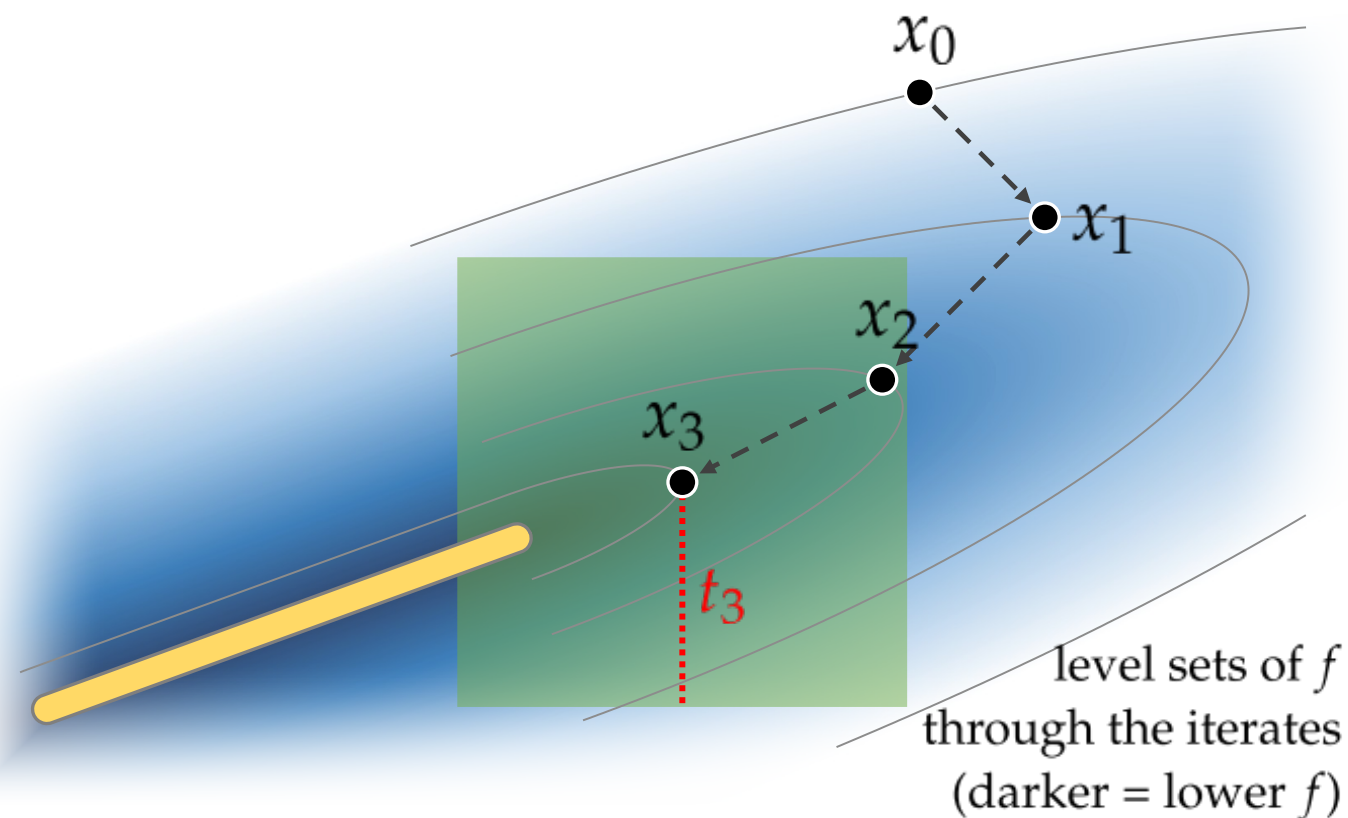
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BPM with ℓ_∞ Balls

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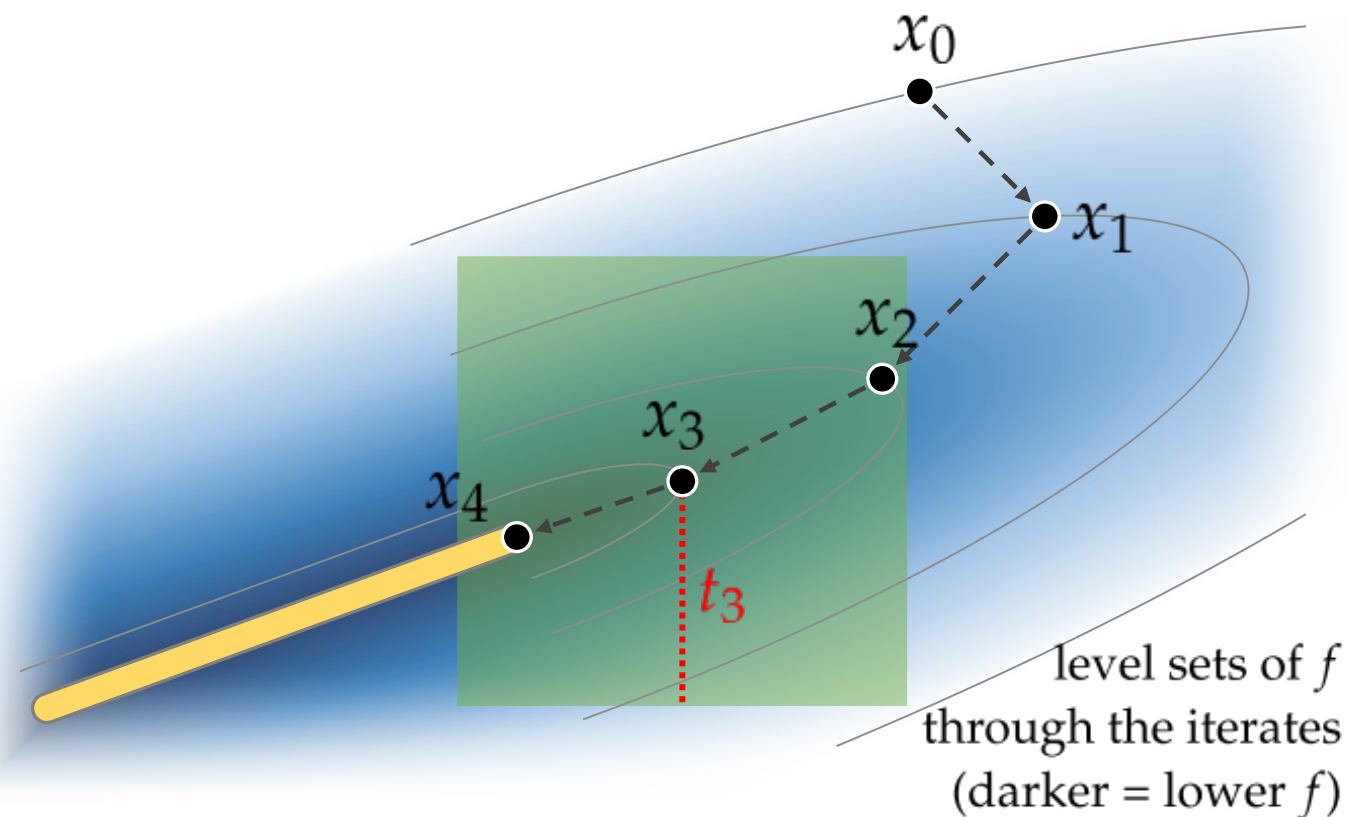
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$t_1 = 1.3$	$d_2 = 2.92$	$f_2 = 5.35$	$g_2 = 5.91$
$t_2 = 1.6$	$d_3 = 1.32$	$f_3 = 0.98$	$g_3 = 1.49$
$t_3 = 1.8$	$d_4 = 0.00$	$f_4 = 0.00$	$g_4 = 0.00$

$$\operatorname{Argmin} f = [-2a, 2a]$$

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BPM with ℓ_∞ Balls

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BPM run

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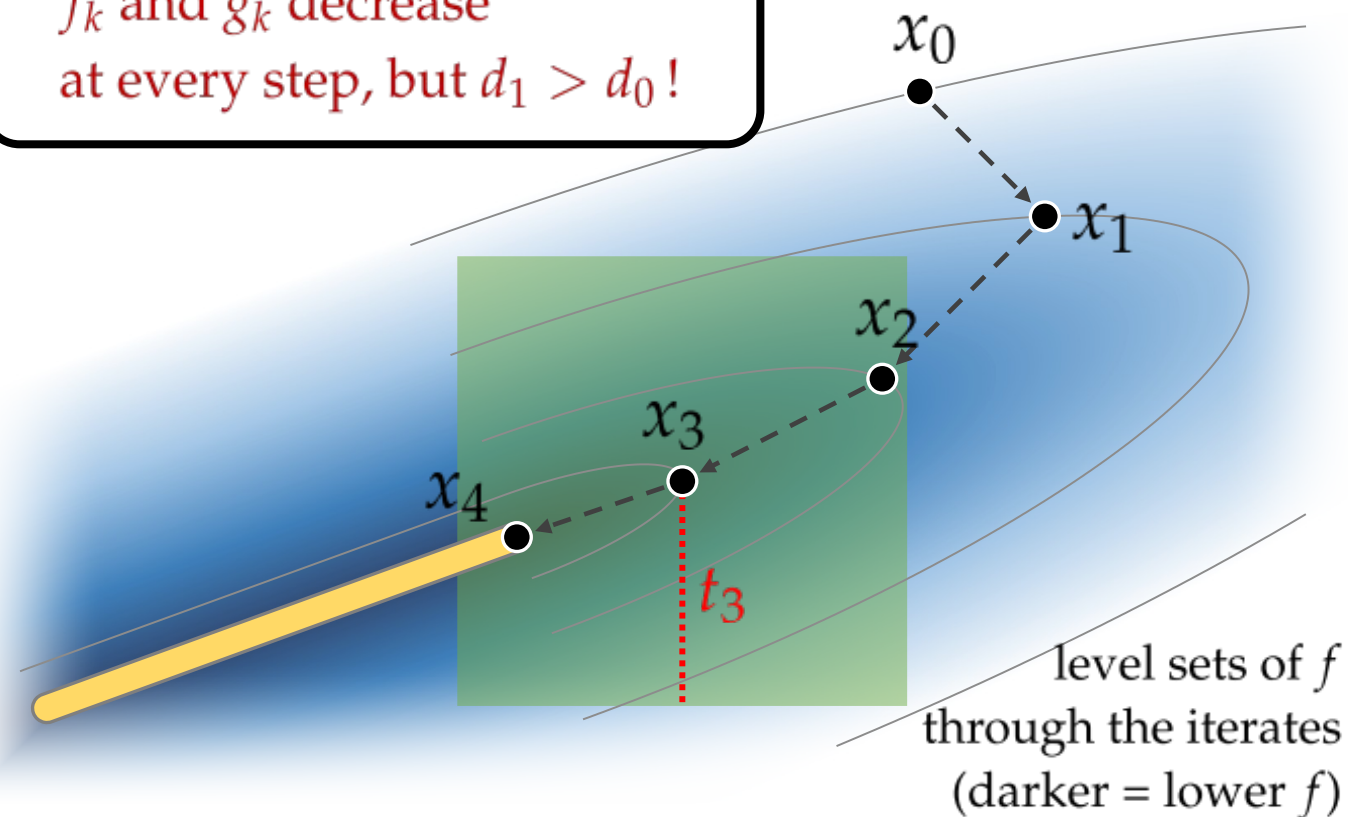
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	$d_0 = 3.57$	$f_0 = 49.53$	$g_0 = 43.60$
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$t_2 = 1.6$	$d_3 = 1.32$	$f_3 = 0.98$	$g_3 = 1.49$
$t_3 = 1.8$	$d_4 = 0.00$	$f_4 = 0.00$	$g_4 = 0.00$

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_\star := \min f = 0$$

f_k and g_k decrease
at every step, but $d_1 > d_0$!



Linear Programming via BPM with ℓ_∞ Balls

$$x_{k+1} \in \operatorname{Argmin} \{ \langle c, y \rangle : y \in P, \|y - x_k\|_\infty \leq t \}$$

Example: $f(x) = \langle c, x \rangle + \delta_P(x)$, $P = \{x \in \mathbb{R}^2 : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$, $m = 12$, $c = (1, 2)$, $t = 1.4$



P.R.

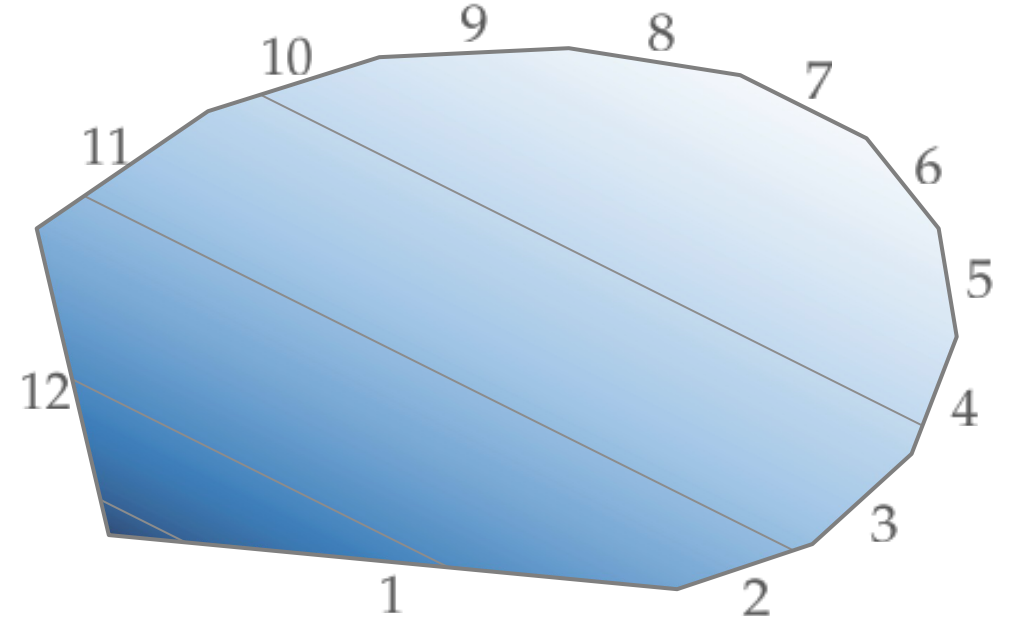
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Manuscript, 2026

Linear Programming via BPM with ℓ_∞ Balls

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level sets of f
(darker = lower f)



P.R.

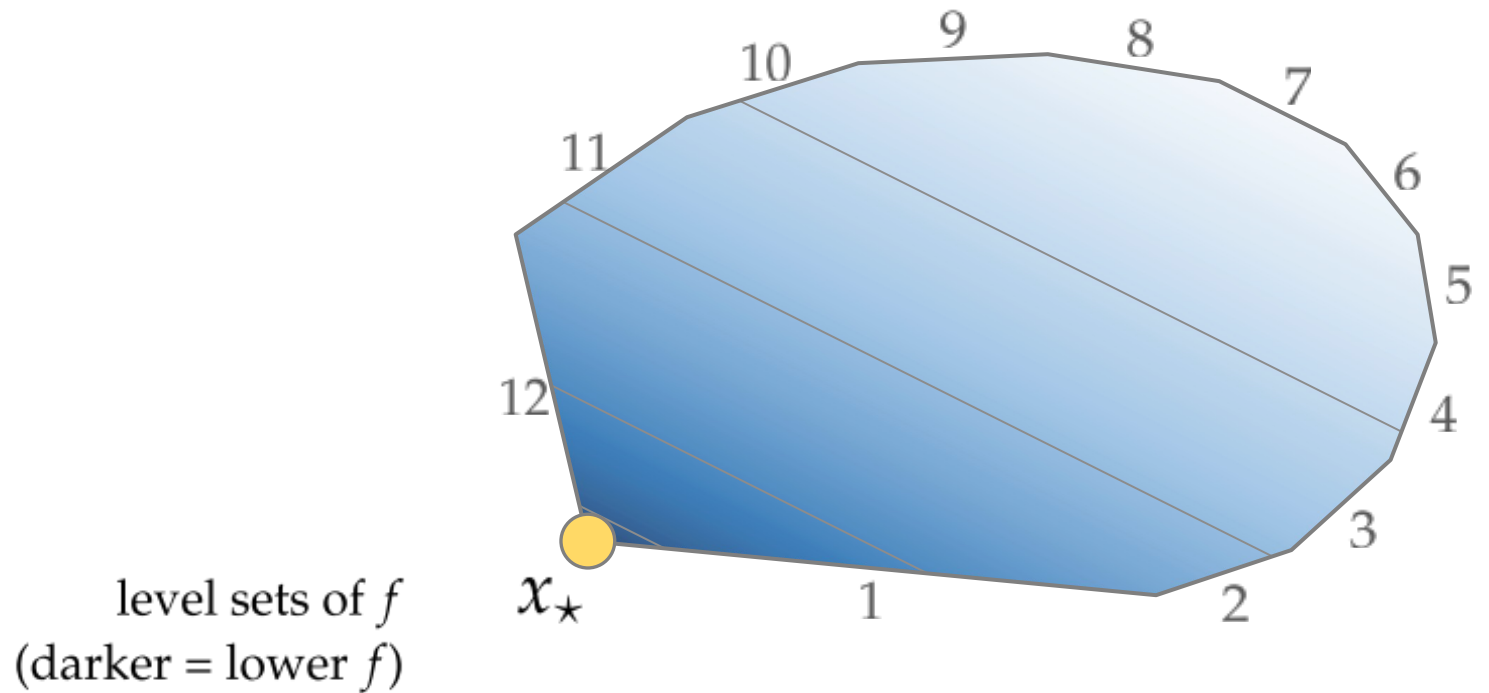
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Example: $f(x) = \langle c, x \rangle + \delta_P(x)$, $P = \{x \in \mathbb{R}^2 : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$, $m = 12$, $c = (1, 2)$, $t = 1.4$

BPM run

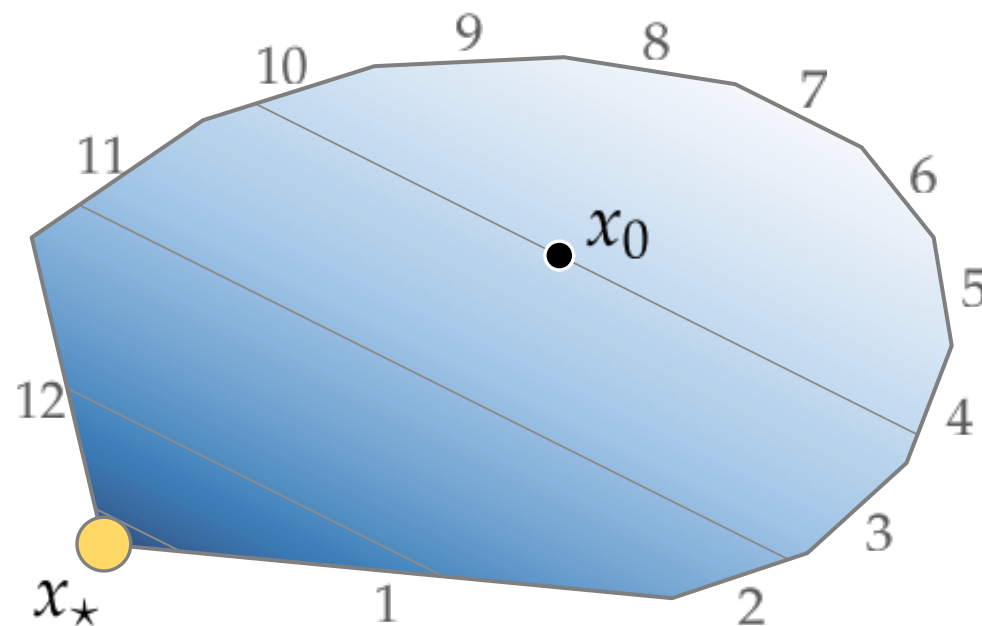
$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

$$d_0 = 5.05 \quad f_0 = 11.45$$

level sets of f
(darker = lower f)



P.R.

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BPM run

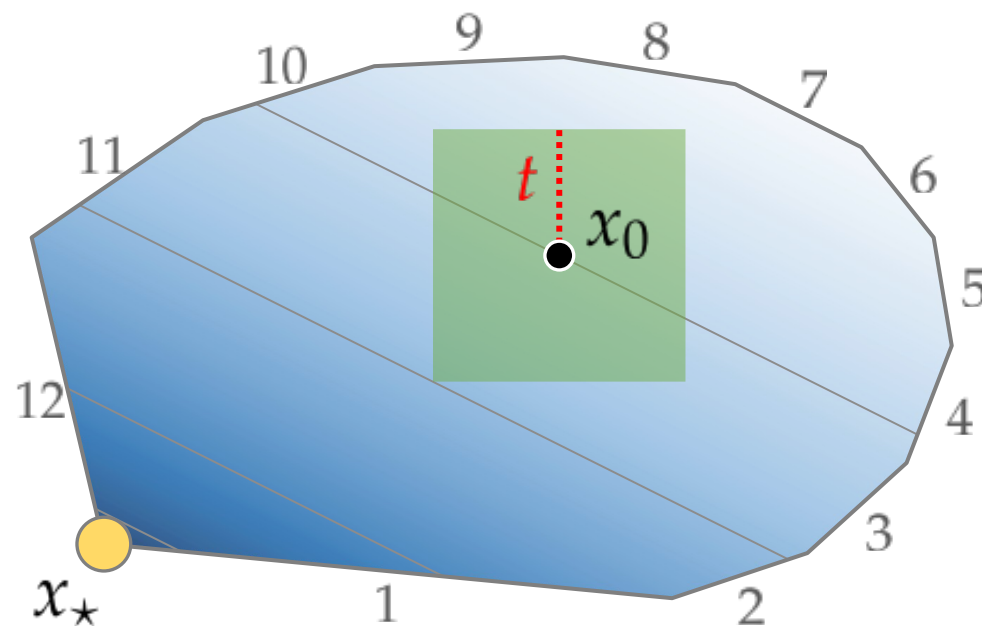
$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

level sets of f
(darker = lower f)



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BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

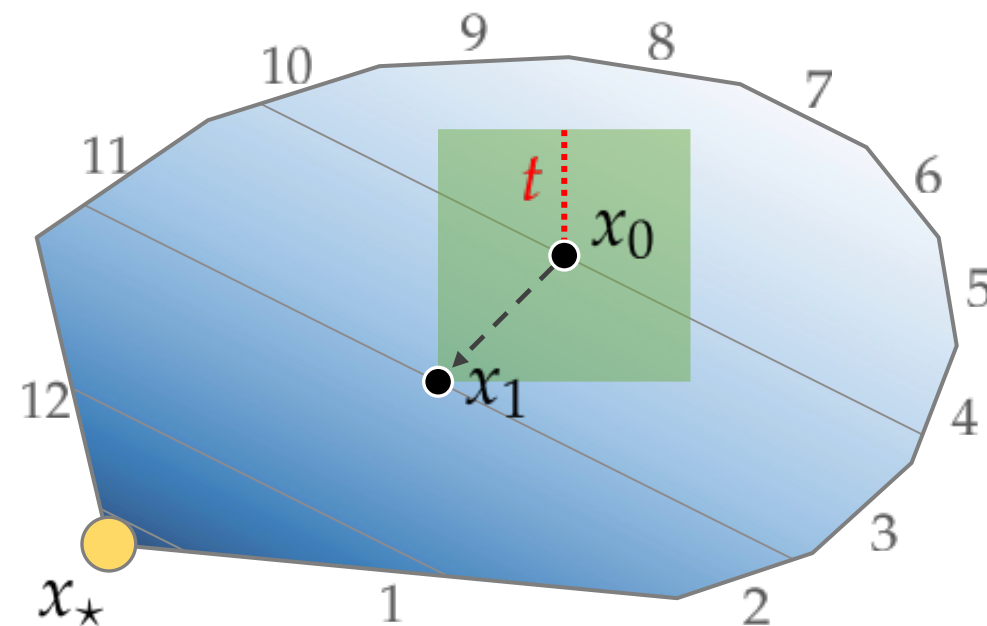
$$f_k = \langle c, x_k \rangle - f_\star$$

$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

$$d_1 = 3.65 \quad f_1 = 7.25$$

level sets of f
(darker = lower f)



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Example: $f(x) = \langle c, x \rangle + \delta_P(x)$, $P = \{x \in \mathbb{R}^2 : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$, $m = 12$, $c = (1, 2)$, $t = 1.4$

BPM run

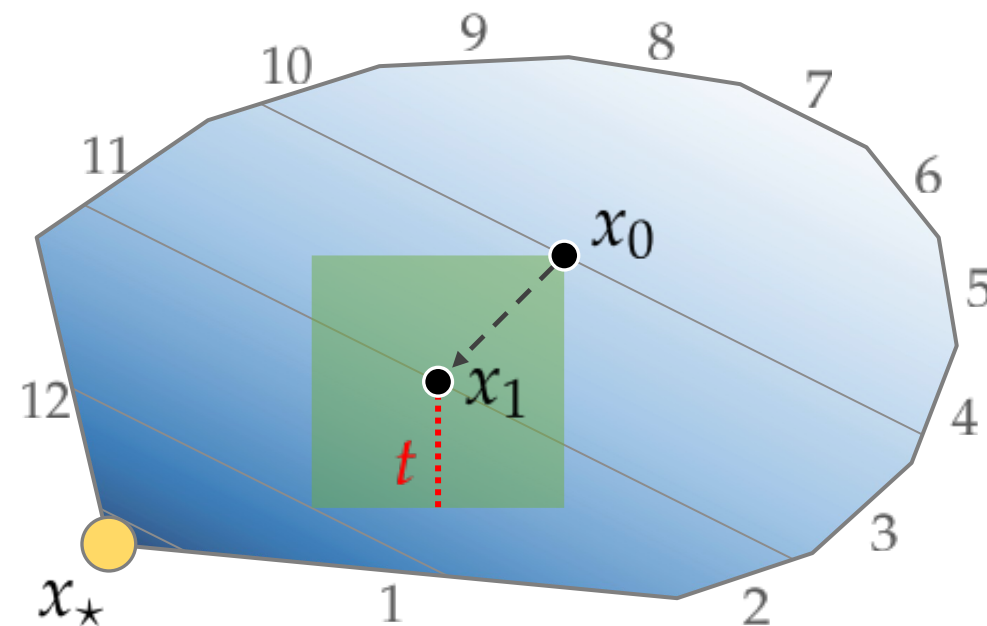
$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

$$d_1 = 3.65 \quad f_1 = 7.25 \quad \mathcal{I}_1 = \emptyset$$



level sets of f
(darker = lower f)



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BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

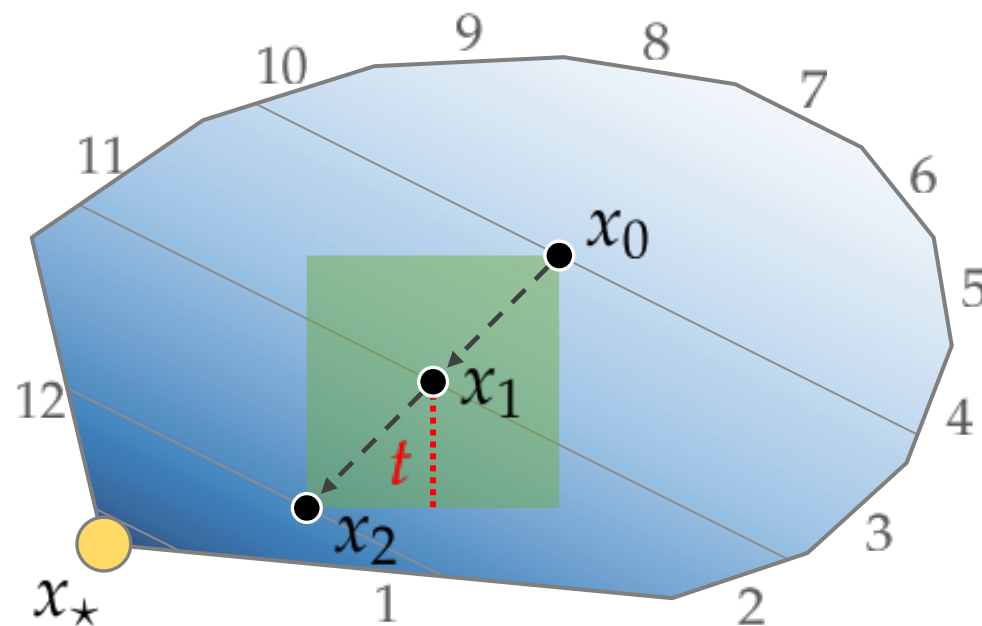
$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

$$d_1 = 3.65 \quad f_1 = 7.25 \quad \mathcal{I}_1 = \emptyset$$

$$d_2 = 2.25 \quad f_2 = 3.05$$

level sets of f
(darker = lower f)



P.R.

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Linear Programming via BPM with ℓ_∞ Balls

$$\begin{aligned} x_{k+1} &\in \operatorname{Argmin} \{ \langle c, y \rangle : y \in P, \|y - x_k\|_\infty \leq t \} \\ &= \operatorname{Argmin} \{ \langle c, y \rangle : \langle a_i, y \rangle \leq b_i \ \forall i \in \mathcal{I}_k, \|y - x_k\|_\infty \leq t \} \end{aligned}$$

Example: $f(x) = \langle c, x \rangle + \delta_P(x)$, $P = \{x \in \mathbb{R}^2 : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$, $m = 12$, $c = (1, 2)$, $t = 1.4$

BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

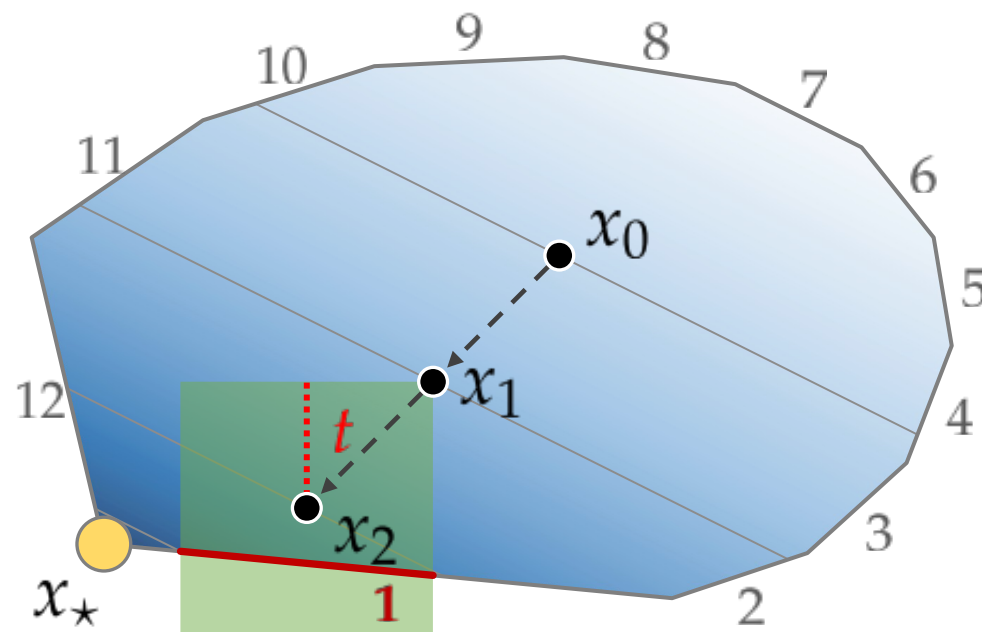
$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

$$d_1 = 3.65 \quad f_1 = 7.25 \quad \mathcal{I}_1 = \emptyset$$

$$d_2 = 2.25 \quad f_2 = 3.05 \quad \mathcal{I}_2 = \{1\}$$

level sets of f
(darker = lower f)



P.R.

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Programming: A Screened Broximal Approach

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Linear Programming via BPM with ℓ_∞ Balls

$$x_{k+1} \in \operatorname{Argmin} \{ \langle c, y \rangle : y \in P, \|y - x_k\|_\infty \leq t \}$$

$$= \operatorname{Argmin} \{ \langle c, y \rangle : \langle a_i, y \rangle \leq b_i \ \forall i \in \mathcal{I}_k, \|y - x_k\|_\infty \leq t \}$$

Example: $f(x) = \langle c, x \rangle + \delta_P(x)$, $P = \{x \in \mathbb{R}^2 : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$, $m = 12$, $c = (1, 2)$, $t = 1.4$

BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

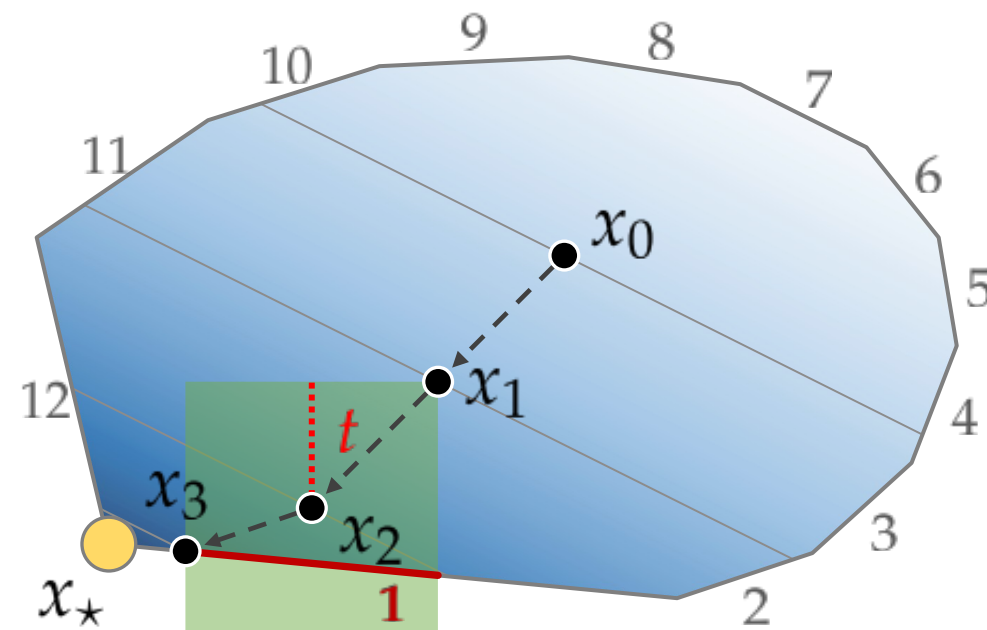
$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

$$d_1 = 3.65 \quad f_1 = 7.25 \quad \mathcal{I}_1 = \emptyset$$

$$d_2 = 2.25 \quad f_2 = 3.05 \quad \mathcal{I}_2 = \{1\}$$

$$d_3 = 0.85 \quad f_3 = 0.69$$

level sets of f
(darker = lower f)



P.R.

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Linear Programming via BPM with ℓ_∞ Balls

$$x_{k+1} \in \operatorname{Argmin} \{ \langle c, y \rangle : y \in P, \|y - x_k\|_\infty \leq t \}$$

$$= \operatorname{Argmin} \{ \langle c, y \rangle : \langle a_i, y \rangle \leq b_i \ \forall i \in \mathcal{I}_k, \|y - x_k\|_\infty \leq t \}$$

Example: $f(x) = \langle c, x \rangle + \delta_P(x)$, $P = \{x \in \mathbb{R}^2 : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$, $m = 12$, $c = (1, 2)$, $t = 1.4$

BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

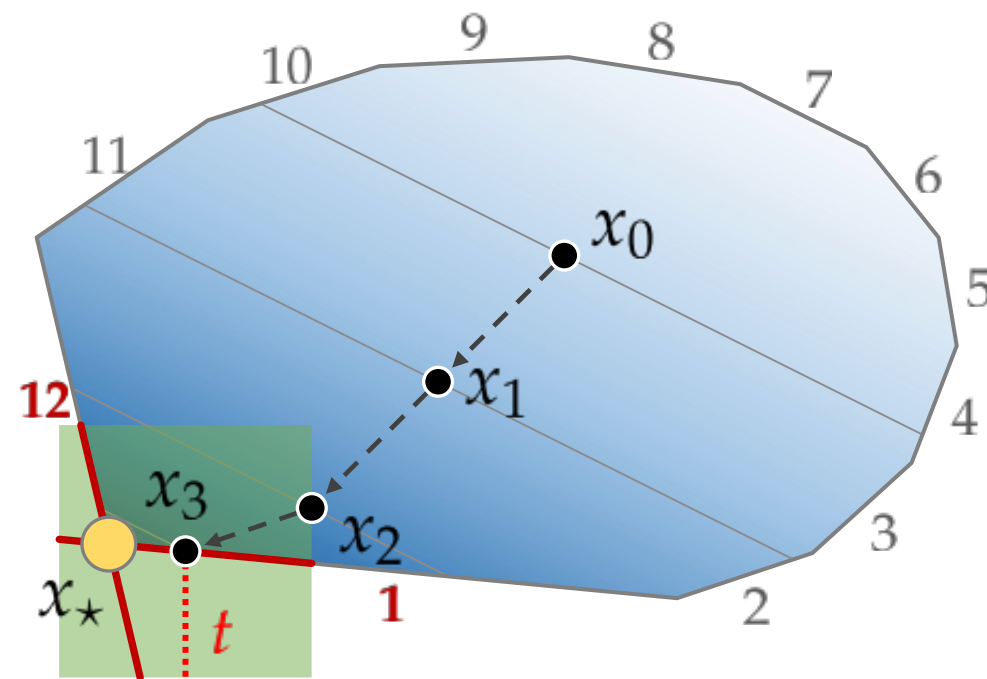
$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

$$d_1 = 3.65 \quad f_1 = 7.25 \quad \mathcal{I}_1 = \emptyset$$

$$d_2 = 2.25 \quad f_2 = 3.05 \quad \mathcal{I}_2 = \{1\}$$

$$d_3 = 0.85 \quad f_3 = 0.69 \quad \mathcal{I}_3 = \{1, 12\}$$

level sets of f
(darker = lower f)



P.R.

Rethinking Gradient Descent for Linear
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Linear Programming via BPM with ℓ_∞ Balls

$$x_{k+1} \in \operatorname{Argmin} \{ \langle c, y \rangle : y \in P, \|y - x_k\|_\infty \leq t \}$$

$$= \operatorname{Argmin} \{ \langle c, y \rangle : \langle a_i, y \rangle \leq b_i \ \forall i \in \mathcal{I}_k, \|y - x_k\|_\infty \leq t \}$$

Example: $f(x) = \langle c, x \rangle + \delta_P(x)$, $P = \{x \in \mathbb{R}^2 : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$, $m = 12$, $c = (1, 2)$, $t = 1.4$

BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

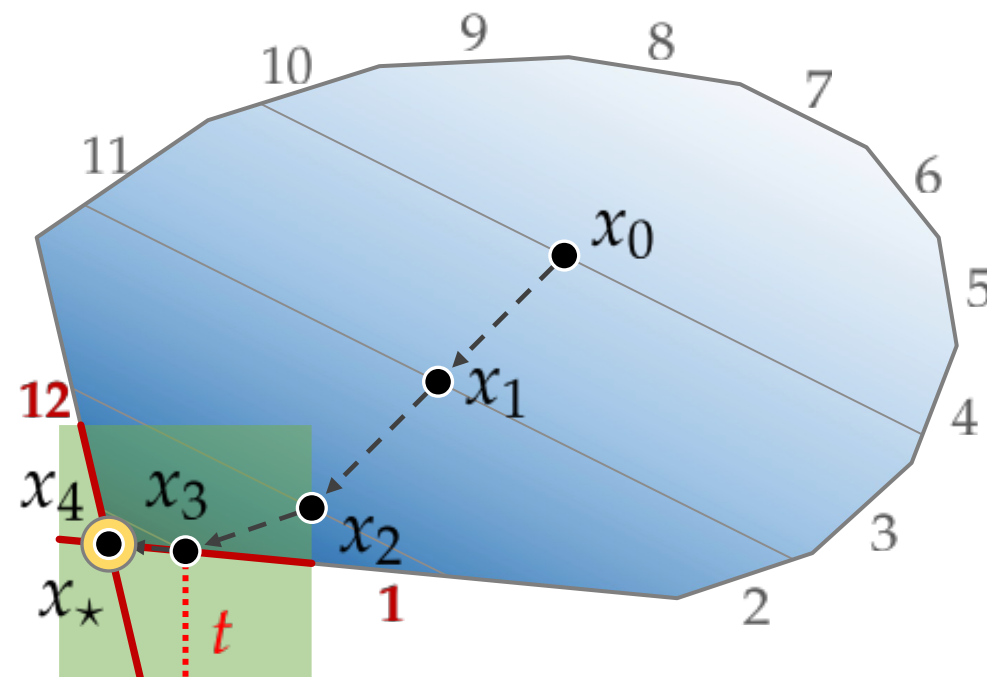
$$d_1 = 3.65 \quad f_1 = 7.25 \quad \mathcal{I}_1 = \emptyset$$

$$d_2 = 2.25 \quad f_2 = 3.05 \quad \mathcal{I}_2 = \{1\}$$

$$d_3 = 0.85 \quad f_3 = 0.69 \quad \mathcal{I}_3 = \{1, 12\}$$

$$d_4 = 0.00 \quad f_4 = 0.00 \quad x_4 = x_\star$$

level sets of f
(darker = lower f)



P.R.

**Rethinking Gradient Descent for Linear
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Linear Programming via BPM with ℓ_∞ Balls

$$x_{k+1} \in \operatorname{Argmin} \{ \langle c, y \rangle : y \in P, \|y - x_k\|_\infty \leq t \}$$

$$= \operatorname{Argmin} \{ \langle c, y \rangle : \langle a_i, y \rangle \leq b_i \ \forall i \in \mathcal{I}_k, \|y - x_k\|_\infty \leq t \}$$

Example: $f(x) = \langle c, x \rangle + \delta_P(x)$, $P = \{x \in \mathbb{R}^2 : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$, $m = 12$, $c = (1, 2)$, $t = 1.4$

BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

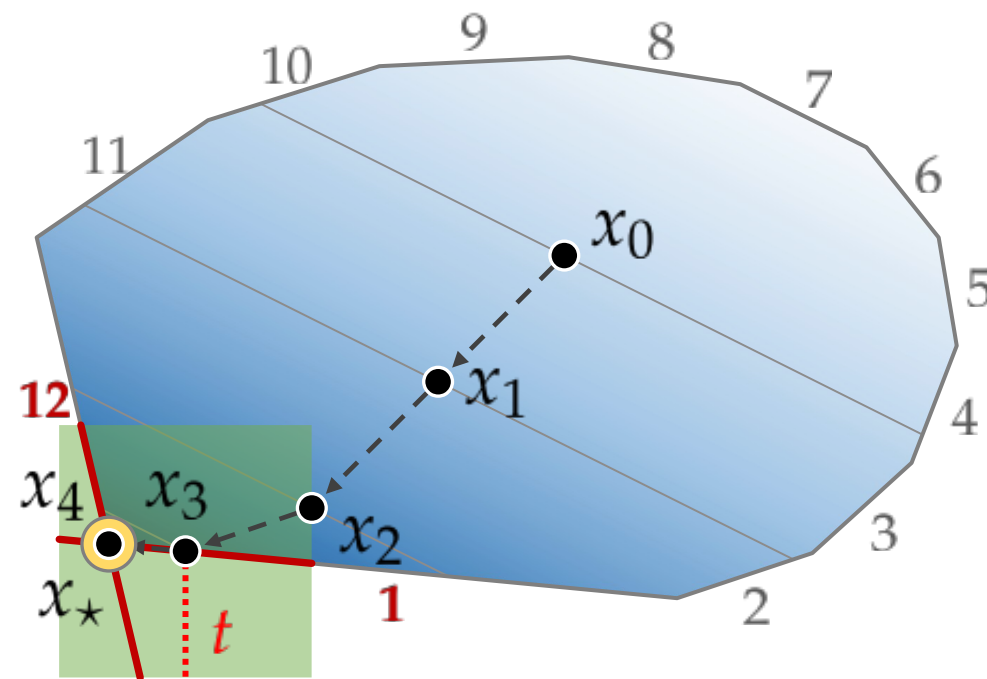
$$d_1 = 3.65 \quad f_1 = 7.25 \quad \mathcal{I}_1 = \emptyset$$

$$d_2 = 2.25 \quad f_2 = 3.05 \quad \mathcal{I}_2 = \{1\}$$

$$d_3 = 0.85 \quad f_3 = 0.69 \quad \mathcal{I}_3 = \{1, 12\}$$

$$d_4 = 0.00 \quad f_4 = 0.00 \quad x_4 = x_\star$$

level sets of f
(darker = lower f)



P.R.

Rethinking Gradient Descent for Linear Programming: A Screened Broximal Approach
Manuscript, 2026

LP $\longrightarrow$ 4 small LPs, one per BPM step;
each uses only $|\mathcal{I}_k| \leq 2$ of the $m = 12$ constraints

Part 2

Broximal Point Method: Convergence Theory



Kaja Grunkowska, Hanmin Li, Aadi Rane, P.R.

The ball-proximal (= "broximal") point method: a new algorithm, convergence theory, and applications

arXiv:2502.02002, 2025



P.R., Hanmin Li

A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026

Problem Formulation & Assumptions

$$\min_{x \in \mathbb{R}^d} f(x)$$

$$x_{k+1} \in \operatorname{Argmin} \{ f(y) : \|y - x_k\|_2 \leq t_k \}$$

The Ball-Proximal ("BProximal") Point Method: a New Algorithm, Convergence Theory, and Applications

Kaj Gerdikova¹, Hamza Li¹, Adi Rao^{1,2}, Peter Richtárik¹

Abstract

Non-convex and non-concave global optimization poses significant challenges across various applications, where finding global minima/maxima often remains elusive. We propose the *Ball-Proximal Point Method*, *BProximal*, *Ball-Proximal Point Method*, or *BPPM* for short – a novel algorithm inspired by the classical Proximal Point Method (PPM) (Rockafellar, 1976), which, as we show, sheds new light on several fundamental optimization principles and phenomena, including non-convex and non-concave optimization, acceleration, overrelaxation, adaptive stepsize selection, and more rigorous methods. As the core of BPPM is the ball-proximal "Proximal" operator, which iterates the classical proximal operator by replacing the quadratic distance penalty by a ball constraint. Surprisingly, and in sharp contrast with the sublinear rate of PPM in the nonconvex/concave regime, we prove that BPPM converges linearly and in a finite number of steps in the same regime. Furthermore, by introducing the concept of ball-convergence, we prove that BPPM iterates the same global convergence guarantees under weaker assumptions, making it a powerful tool for a broader class of practically non-convex optimization problems. Our findings also place the role of a proximal method beyond the development of practically effective algorithms and algorithms, e.g., proximal distance, adaptive stepsize, acceleration (Abbott & Boyd, 2020), and "PPM" as a key concept (Zhang et al., 2023), we believe that BPPM should be understood as the same manner as the proximal method requires for further developments.

1. Introduction

The minimization of non-convex functions is a fundamental challenge across many fields, including machine learning, optimization, applied mathematics, signal processing, and operations research. Finding such problems is integral to most machine learning algorithms aiming at both training and inference, where non-convex objectives or constraints are often necessary to capture complex prediction tasks.

1.1. Global nonconvex optimization

In this paper, we propose a new *non-convex* algorithm (see Section 1.1) capable of finding global minimizers for a specific (new) class of non-convex functions. In particular, we introduce an algorithmic framework designed to solve the (generally) non-convex optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is bounded below and which means that the set $\operatorname{dom} f := \{x \in \mathbb{R}^d : f(x) < +\infty\}$ is nonempty, closed, and has at least one minimizer. We let A be the set of all minimizers of f , and $f^* := \min_x f(x)$.

1.1.1. The ball-proximal operator

A key inspiration for our method stems from the well-known Proximal Point Method (PPM) (Rockafellar, 1976), which iteratively adds a quadratic penalty term to the objective and solves. For some classes of non-convex/minimal problems at each step. Building on this idea, we introduce the ball-proximal "Proximal" operator

Definition 1.1 (Ball-Proximal Operator). The ball-proximal ("Proximal") operator with radius $r > 0$ associated with a function $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is given by

$$\operatorname{ballprox}_f(x) := \operatorname{argmin}_{y \in B(x, r)} f(y), \quad (2)$$

where $B(x, r) := \{y \in \mathbb{R}^d : \|y - x\|_2 \leq r\}$ is the standard Euclidean norm.

According to this definition, for a given point $x \in \mathbb{R}^d$, $\operatorname{ballprox}_f(x)$ returns the minimizers of f within the ball of radius r centered at x .

A Sharper Theory of Ball-Proximal Optimization: Convergence and Radius Selection

Peter Richtárik¹, Hamza Li¹
King Abdullah University of Science and Technology
Thuwal 23955-6906, Saudi Arabia
28 September 2026

Abstract

We study the exact Euclidean ball-proximal point method for proper, closed, convex functions, where each iteration minimizes the objective over a ball centered at the current point. Assuming the objective f is the distance of squared distance (quadratic) from an algorithm value, minimizers, and the proximal distance to a solution. For convex and $r > 0$ and initial distance d_0 , the distance and the algorithm after k iterations is at most d_0 times multiplied by $\exp(-k/r^2)$.

We also investigate the convergence of the algorithm for non-convex functions, where the algorithm value converges to the infimum and the distance to the solution set. If the ball is convex, the algorithm converges to a globally optimal point. Ball-convergence gives finite stepsize bounds for bounded functions and a constant radius comparison bound $d_0 \leq 2/r^2$, where d_0 is the initial distance. In particular, we provide a sharp bound that the algorithm converges to a solution set.

We also identify a minimum proximal distance bound factor and radius selection rule based on the algorithm value, stepsize, and radius selection. Together, these results strengthen the foundations and convergence guarantees of the method without requiring smoothness or strong convexity.

Keywords: ball-proximal point method, convex optimization, finite termination, self-contracted sequences, radius selection, convex optimization.
Mathematics Subject Classification (2020): Primary 90C30; Secondary 90C06, 49J55.

1 Introduction

We study the convex optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (3)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, closed, and convex. Unless stated otherwise, we assume unless that the infimum is attained, i.e., $f^* < +\infty$. We focus on the exact ball-proximal iteration considered by Censor et al. (2020, Appendix A) and subsequently developed as the ball-proximal (proximal) point framework by Gerdikova et al. (2023). Starting from x^0 , the ball-proximal point method (BPPM) minimizes f over a Euclidean ball centered at the current point

$$x^{k+1} \in \operatorname{ballprox}_f(x^k) := \operatorname{argmin}_{y \in B(x^k, r_k)} f(y), \quad r_k > 0. \quad (\text{BPPM})$$

Problem Formulation & Assumptions

$$\min_{x \in \mathbb{R}^d} f(x)$$

$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_2 \leq t_k\}$$

$$f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\} \text{ is proper, closed \& convex}$$

Abstract

Non-smooth and non-convex global optimization poses significant challenges across various applications, where standard gradient-based methods often struggle. We propose the *Rail Primal/Primal Method*, *Rail Primal Method*, or *Rail Primal Method* (RPM) for short, a novel algorithm.

1. Introduction

The minimization of non-convex functions is a fundamental challenge across many fields, including machine learning, optimization, applied mathematics, signal processing and operations research. Solving such problems is integral to most machine learning algorithms arising in both training and inference, where non-convex objectives or constraints are often necessary to capture complex prediction tasks.

1.1. Global nonconvex optimization

In this paper, we propose a new meta-algorithm (see Section 1.3) capable of finding global minimizers for a specific (new) class of non-convex functions. In particular, we introduce an algorithmic framework designed to solve the (recently) non-convex optimization problems

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is assumed to be proper (which means that the set $\text{dom} f := \{x \in \mathbb{R}^d : f(x) < +\infty\}$ is nonempty), closed, and have at least one minimizer. We let X_f be the set of all minimizers of f , and $f_* := \min_x f(x)$.

1.2. The half-proximal operator

Definition 1.1 (Ball-Primal Operator). The ball-primal ("briental") operator with radius $r > 0$ associated with a function $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is given by

$$\text{bri}_r f(x) := \arg \min f(z), \quad (2)$$

According to this definition, for a given input point $x \in \mathbb{R}^d$, $\text{best}_f^t(x)$ returns the minimizer(s) of f within the ball of radius t centered at x .

*The work of Hadi Ramezani was conducted during a VISRP internship at KAUST.

Alstrom

We study the exact Euclidean ball-proximal point method for proper, closed, convex functions, where each iteration minimizes the objective over a ball centered at the current point. Retaining the objective gap is the decrease of squared distance yields sharper bounds on objective values, stationarity, and the symmetric Bregman distance to a minimizer. For constant radius $\varepsilon > 0$ and initial distance $D_0 > 0$ to the solution set, the objective gap after K iterations is at most its initial value multiplied by $\exp(-2K\varepsilon^2/D_0^2)$.

We characterize convergence for arbitrary positive radial sequences. If their sum diverges the method reaches a minimizer in finitely many iterations whenever one exists; otherwise the objective values converge to the infimum and the iterates escape every bounded set. The radii are summable, the iterates converge to a possibly nonoptimal point. Self-constriction gives finite trajectory length for bounded iterates and a constant-radius termination bound $O_d(1 + D_0/c)$, whose implicit constant depends only on the dimension. A polyhedral hull shows that the dimension-independent quadratic bound remains asymptotically sharp.

we also study a randomized successive geometric decay factor and analyze adaptive rules based on subgradients or objective gaps, ϵ -optimal reformulation, and relaxed updates. Together, these results strengthen the foundations and convergence guarantees of the methods without assuming smoothness or strong convexity.

Keywords: ball-proximal point method; convex optimization; finite termination; self-consequences; radius selection; oracle complexity.

Mathematics Subject Classification (2020): Primary 90C25; Secondary 90C60, 65K05.

1 Introduction

We study the convex optimization probl

$$\max_{x \in X} f(x)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, closed, and convex. Unless stated otherwise, we neither that the infimum is attained nor that f is smooth. We focus on the exact ball iteration considered by [Carmon et al. \(2020, Appendix A\)](#) and subsequently develop the ball-primal (biprimal) point framework by [Grustkowska et al. \(2023\)](#). Starting off [Carmon et al. \(2020\)](#) (biprimal) point method (BBP) minimizes f over a sequence of

Figure 1

Problem Formulation & Assumptions

$$\min_{x \in \mathbb{R}^d} f(x)$$

$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_2 \leq t_k\}$$

$$f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\} \text{ is proper, closed \& convex}$$

$$x_\star \in \operatorname{Argmin} f \neq \emptyset, \quad f_\star := \min f$$

arXiv:2502.0302v1 [math.OA] 4 Feb 2025

The Ball-Proximal ("Brossimal") Point Method:
a New Algorithm, Convergence Theory, and Applications

Kaj Gerdikova¹ · Hanxin Li¹ · Adi Rao^{1,2} · Peter Richtárik¹

Abstract

Non-convex and non-smooth global optimization poses significant challenges across various applications, where finding global minima is often intractable. We propose the *Ball-Proximal Point Method*, *Proximal Point Method*, or *Ball-Point Method* (BPM) for short – a novel algorithm inspired by the classical Proximal Point Method (PPM) [Rockafellar, 1976], which, as we show, sheds new light on several fundamental optimization principles and phenomena, including non-convex and non-smooth optimization, acceleration, overrelaxation, adaptive stepsize selection, and trust-region methods. At the core of (BPM) lies the ball-proximal "Proximal" operator, which iterates the classical proximal operator by replacing the quadratic distance penalty by a ball constraint. Surprisingly, and in sharp contrast with the sublinear rate of PPM in the nonconvex convex regime, we prove that BPM converges linearly and in a finite number of steps in the same regime. Furthermore, by introducing the concept of *ball-convexity*, we prove that BPM attains the same global convergence guarantees under weaker assumptions, making it a powerful tool for a broader class of practically non-convex optimization problems. For the PPM, this role is played by a conceptual method inspiring the development of practically efficient algorithms and algorithms, e.g., parallel descent, adaptive step sizes, acceleration (Nesterov, 2004; Sca, 2000), and "PP" as known (Ouyang et al., 2023), we believe that BPM should be understood as the same manner as the proximal and subgradients for further developments.

1. Introduction

The minimization of non-convex functions is a fundamental challenge across many fields, including machine learning, optimization, applied mathematics, signal processing, and operations research. Finding such problems is integral to most machine learning algorithms aiming to both training and inference, where non-convex objectives or constraints are often necessary to capture complex patterns in data.

1.1. Global non-convex optimization

In this paper, we propose a new, *simple algorithm* (see Section 1.1) capable of finding global minimizers for a specific (new) class of non-convex functions. In particular, we introduce an algorithm framework designed to solve the (generally) non-convex optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is bounded below and which means that the set $\operatorname{dom} f := \{x \in \mathbb{R}^d : f(x) < +\infty\}$ is nonempty, closed, and has a finite non-emptiness. We let X be the set of all minimizers of f and $f_\star := \min_x f(x)$.

1.1.1. The ball-proximal operator

Also inspired by our method stems from the well-known Proximal Point Method (PPM) [Rockafellar, 1976], which iteratively adds a quadratic penalty term to the objective function. For non-convex and non-smooth optimization problems at each step. Building on this idea, we introduce the ball-proximal "Proximal" operator

Definition 1.1 (Ball-Proximal Operator). The ball-proximal ("Proximal") operator with radius $r > 0$ associated with a function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is given by

$$\operatorname{ball-prox}_r f(x) := \argmin_{y \in \mathbb{R}^d} \{f(y) : \|y - x\|_2 \leq r\} \text{ and } \frac{1}{2} \|x - y\|_2^2 \text{ in the standard Euclidean norm.}$$

According to this definition, for a given point $x \in \mathbb{R}^d$, $\operatorname{ball-prox}_r f(x)$ returns the minimizers of f within the ball of radius r centered at x .

A Sharper Theory of Ball-Proximal Optimization:
Convergence and Radius Selection

Peter Richtárik¹ · Hanxin Li¹
King Abdullah University of Science and Technology
Tharad 2005-6906, Saudi Arabia
28 September 2026

Abstract

We study the exact Euclidean ball-proximal point method for proper, closed, convex functions, where each iteration minimizes the objective over a ball centered at the current point. Assuming the objective f is the maximum of a signed distance (plus a large bound on the objective value, ultimately, and the constant Bregman distance to a solution). For convex and $r > 0$ and initial distance Δ_0 , the distance after the algorithm after k iterations is at most its initial value multiplied by $\exp(-k \Delta_0 / \Delta_0)$.

We also investigate convergence for arbitrary points and non-convex functions. The method makes a transition to globally near-optimal solutions in convex optimization, the objective values converge to the infimum and the feasible region converges to the set of all minimizers. We also investigate the convergence of the method for non-convex functions. The method makes a transition to globally near-optimal solutions in convex optimization, the objective values converge to the infimum and the feasible region converges to the set of all minimizers. We also investigate the convergence of the method for non-convex functions. The method makes a transition to globally near-optimal solutions in convex optimization, the objective values converge to the infimum and the feasible region converges to the set of all minimizers.

We also identify a minimum number of iterations that depend on the dimension, the radius, and the initial distance. We also identify a minimum number of iterations that depend on the dimension, the radius, and the initial distance. We also identify a minimum number of iterations that depend on the dimension, the radius, and the initial distance.

Keywords: ball-proximal point method, convex optimization, finite termination, self-contracted sequences, online algorithms, convex optimization.

Mathematics Subject Classification (2010): Primary 90C30; Secondary 90C06, 49J55.

1 Introduction

We study the convex optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, closed, and convex. Unless stated otherwise, we assume further that the following is satisfied: f is μ -strongly convex. We focus on the exact ball-proximal iteration considered by Csiszar et al. (2020) (Appendix A) and subsequently developed as the ball-proximal (biproximal) point framework by Gerdikova et al. (2025). Starting from x^0 and r , the ball-proximal point method (BPM) minimizes f over a Euclidean ball centered at the current point

$$x^{k+1} \in \argmin_{y \in \mathbb{R}^d} \{f(y) : \|y - x^k\|_2 \leq r\} \quad \text{for } k \geq 0. \quad (\text{BPM})$$

1

Problem Formulation & Assumptions

arXiv:2502.0202v1 [math.OA] 4 Feb 2025

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a New Algorithm, Convergence Theory, and Applications

Kaj Gerschlager¹, Hanmin Li¹, Adi Ros^{1,2}, Peter Richtárik¹

Abstract

Non-convex and non-smooth global optimization poses significant challenges across various applications, where finding global minima is often intractable. We propose the *Ball-Proximal Point Method*, *Proximal Point Method*, or *Ball-Point Method* (BPM) for this – a novel algorithm inspired by the classical Proximal Point Method (PPM) [Rockafellar, 1976], which, as we show, sheds new light on several fundamental optimization principles and phenomena, including non-convex and non-smooth optimization, acceleration, smoothing, adaptive stepsize selection, and trust-region methods. As the core of BPM is the ball-proximal "Proximal" operator, which iterates the classical proximal operator by replacing the quadratic distance penalty by a ball constraint. Surprisingly, and in sharp contrast with the sublinear rate of PPM in the non-convex convex regime, we prove that BPM converges linearly and in a finite number of steps in the same regime. Furthermore, by introducing the concept of *ball-convexity*, we prove that BPM attains the same global convergence guarantees under weaker assumptions, making it a powerful tool for a broader class of practically non-convex optimization problems. Our key idea gives the birth of a conceptual method inspiring the development of practically efficient algorithms and algorithmic directions, e.g., parallel descent, adaptive step sizes, acceleration (Ros & Yin, 2020), and "PP" as kernel (Zhang et al., 2023), we believe that BPM should be understood as the same manner as the proximal and subgradients for further developments.

1. Introduction

The minimization of non-convex functions is a fundamental challenge across many fields, including machine learning, optimization, applied mathematics, signal processing, and operations research. Finding such problems is integral to most machine learning algorithms aiming to both training and inference, where non-convex objectives or constraints are often necessary to capture complex patterns in data.

In this paper, we propose a new, *simple algorithm* (see Section 1.1) capable of finding global minimizers for a specific (new) class of non-convex functions. In particular, we introduce an algorithmic framework designed to solve the (generally) non-convex optimization problem

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1.1. The ball-proximal operator

Also inspired by our method stems from the well-known Proximal Point Method (PPM) [Rockafellar, 1976], which iteratively adds a quadratic penalty term to the objective and solves. For some closed and convex bounded subproblems at each step. Building on this idea, we introduce the ball-proximal "Proximal" operator

Definition 1.1 (Ball-Proximal Operator). The ball-proximal ("Proximal") operator with radius $r > 0$ associated with a function $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is given by

$$\text{ball-prox}_r f(x) := \arg \min_{y \in \mathbb{B}(x, r)} f(y), \quad (2)$$

where $\mathbb{B}(x, r) := \{y \in \mathbb{R}^d : \|y - x\|_2 \leq r\}$ is the standard Euclidean ball.

According to this definition, for a given point $x \in \mathbb{R}^d$, $\text{ball-prox}_r f(x)$ returns the minimizers of f within the ball of radius r centered at x .

$$\min_{x \in \mathbb{R}^d} f(x)$$
$$x_{k+1} \in \text{Argmin} \{ f(y) : \|y - x_k\|_2 \leq t_k \}$$

$f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, closed & convex

$$x_\star \in \text{Argmin} f \neq \emptyset, \quad f_\star := \min f$$

$$f(x_0) \neq +\infty$$

A Sharper Theory of Ball-Proximal Optimization:
Convergence and Radius Selection

Peter Richtárik¹, Hanmin Li¹
King Abdullah University of Science and Technology
Tharad 2005-6906, Saudi Arabia
28 September 2026

Abstract

We study the exact Euclidean ball-proximal point method for proper, closed, convex functions, where each iteration minimizes the objective over a ball centered at the current point. Assuming the objective f is the maximum of squared distances (quadratic bounds on objective values, statements), and the constraint (Euclidean distance to a solution). For constant radius $r > 0$ and initial distance d_0 , the iteration ends, the algorithm pays after K iterations it is at most its initial value multiplied by $\exp(-\text{BPM}[d_0])$.

We discover an unexpected property of iterative point-to-set mappings. If the initial radius r is chosen to be the maximum of squared distances (quadratic bounds on objective values, statements), and the constraint (Euclidean distance to a solution), the algorithm pays after K iterations it is at most its initial value multiplied by $\exp(-\text{BPM}[d_0])$.

We also identify a minimum number of iterations (lower bound) and a maximum number of iterations (upper bound) for the algorithm to converge. We also identify a minimum number of iterations (lower bound) and a maximum number of iterations (upper bound) for the algorithm to converge. We also identify a minimum number of iterations (lower bound) and a maximum number of iterations (upper bound) for the algorithm to converge.

Keywords: ball-proximal point method, convex optimization, finite termination, self-contracted mappings, online algorithms, convex optimization.

1 Introduction

We study the convex optimization problem

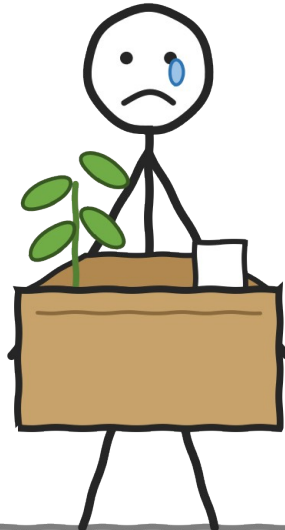
$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, closed, and convex. Unless stated otherwise, we assume unless that the following is satisfied: we let $f_\star := \min_x f(x)$. We focus on the exact ball-proximal iteration considered by Csiszar et al. (2020) (Appendix A) and subsequently developed as the ball-proximal (biproximal) point framework by Gerschlager et al. (2023). Starting from x^0 , the ball-proximal point method (BPM) minimizes f over a Euclidean ball centered at the current point

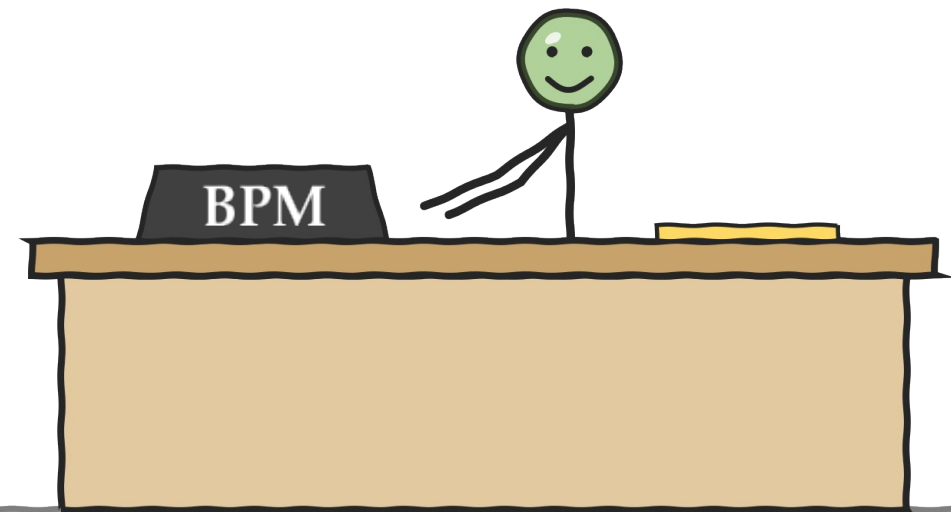
$$x^{k+1} \in \arg \min_{y \in \mathbb{B}(x^k, r)} f(y), \quad \text{for } k \geq 0. \quad (\text{BPM})$$

The Condition Number

You can't just leave me out.
I've been in every convergence bound
for fifty years.



We've decided to go
in a different direction.



Fixed-Radius Theory: Distance



Kaja Grunkowska, Hanmin Li, Aadi Rane, P.R.

**The ball-proximal (= "broximal") point method: a new algorithm,
convergence theory, and applications**

arXiv:2502.02002, 2025

Fixed-Radius Theory: Distance

$$d_k^2 \leq \max \left\{ d_0^2 - k t^2, 0 \right\}$$



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$$g_k = \text{dist}_2(0, \partial f(x_k))$$



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Fixed-Radius Theory: Distance

$$d_k^2 \leq \max \left\{ d_0^2 - k \textcolor{red}{t}^2, 0 \right\}$$

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$$f_k = f(x_k) - f_\star$$

$$g_k = \text{dist}_2(0, \partial f(x_k))$$

$$k \geq \frac{d_0^2}{\textcolor{red}{t}^2} \implies d_k = 0$$



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$$k \geq \frac{d_0^2}{t^2} \implies d_k = 0$$

No dependence
on a condition
number!



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Fixed-Radius Theory: Function



P.R., Hanmin Li

A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026

Fixed-Radius Theory: Function

$$f_k \leq f_0 \cdot e^{-2k^2 t^2 / d_0^2}$$



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$$k \geq \frac{d_0}{\sqrt{2} t} \sqrt{\ln \frac{f_0}{\varepsilon}} \implies f_k \leq \varepsilon$$



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Fixed-Radius Theory: Gradient



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Fixed-Radius Theory: Gradient

$$g_{k+1} \leq g_k$$



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Fixed-Radius Theory: Gradient

$$g_{k+1} \leq g_k$$

$$g_k \leq \frac{f_0}{t} \cdot e^{-2(k-1)^2 t^2 / d_0^2}$$



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$$k \geq 1 + \frac{d_0}{\sqrt{2} t} \sqrt{\ln \frac{f_0}{t \varepsilon}} \implies g_k \leq \varepsilon$$



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A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026

Arbitrary-Radius Theory: Convergence

Theorem. f proper, closed, convex, $x_0 \in \text{dom } f$, any radii $t_k > 0$. Exactly one case applies:



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A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026 (Theorem 4.1)

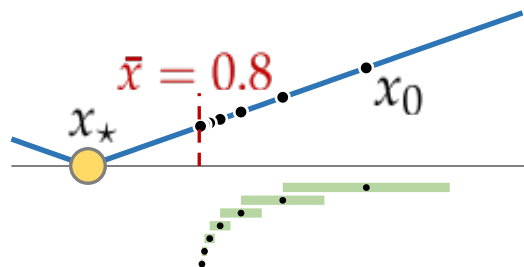
Arbitrary-Radius Theory: Convergence

Theorem. f proper, closed, convex, $x_0 \in \text{dom } f$, any radii $t_k > 0$. Exactly one case applies:

(i) $\sum_k t_k < \infty$:

$x_k \rightarrow \bar{x}$, $f(x_k) \downarrow f(\bar{x})$

$\bar{x}$ need *not* be a minimizer



$$f(x) = |x|, \quad x_0 = 2$$

$$t_k = 0.6 \cdot 2^{-k} \Rightarrow x_k \rightarrow 0.8$$



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A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026 (Theorem 4.1)

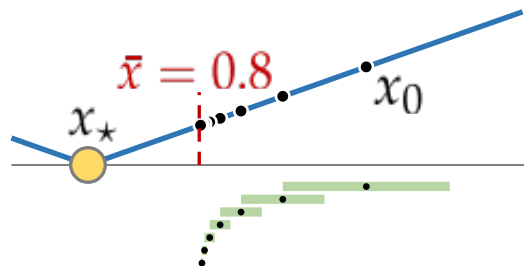
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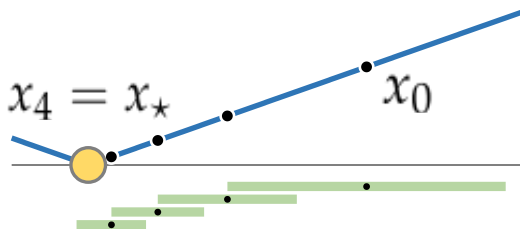


$$f(x) = |x|, \quad x_0 = 2$$
$$t_k = 0.6 \cdot 2^{-k} \Rightarrow x_k \rightarrow 0.8$$

(ii) $\sum_k t_k = \infty$, $\text{Argmin } f \neq \emptyset$:

$x_K \in \text{Argmin } f$ for some $K < \infty$

finite termination



$$f(x) = |x|, \quad x_0 = 2$$
$$t_k = \frac{1}{k+1} \Rightarrow x_4 = 0$$



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A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026 (Theorem 4.1)

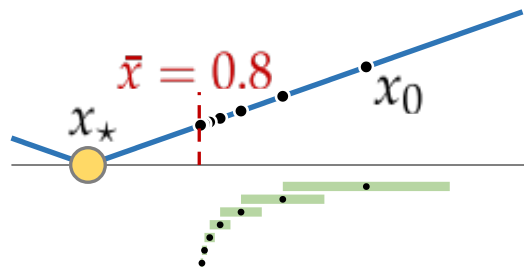
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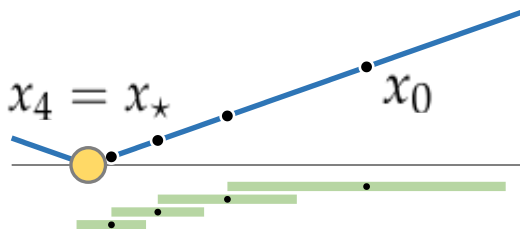
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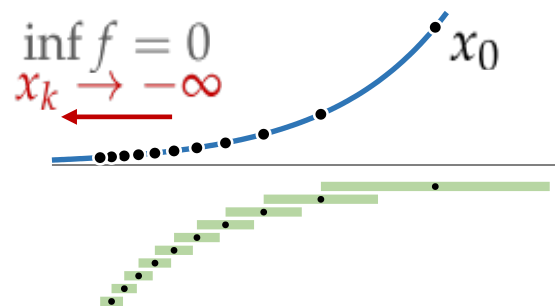
$$f(x) = |x|, \quad x_0 = 2$$

$$t_k = \frac{1}{k+1} \Rightarrow x_4 = 0$$

(iii) $\sum_k t_k = \infty$, $\text{Argmin } f = \emptyset$:

$f(x_k) \downarrow \inf f$, $\|x_k\|_2 \rightarrow \infty$

also when $\inf f = -\infty$



$$f(x) = e^x, \quad x_0 = 1$$

$$t_k = \frac{1}{k+1} \Rightarrow x_k \rightarrow -\infty$$



P.R., Hanmin Li

A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026 (Theorem 4.1)

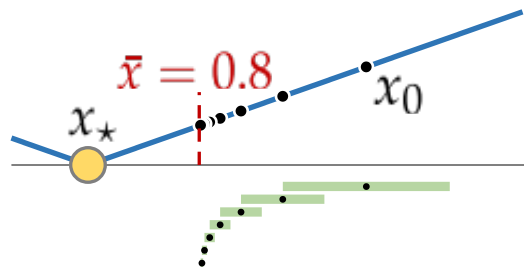
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$x_k \rightarrow \bar{x}$, $f(x_k) \downarrow f(\bar{x})$

$\bar{x}$ need *not* be a minimizer



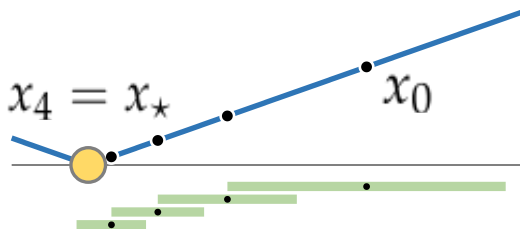
$$f(x) = |x|, \quad x_0 = 2$$

$$t_k = 0.6 \cdot 2^{-k} \Rightarrow x_k \rightarrow 0.8$$

(ii) $\sum_k t_k = \infty$, $\text{Argmin } f \neq \emptyset$:

$x_K \in \text{Argmin } f$ for some $K < \infty$

finite termination



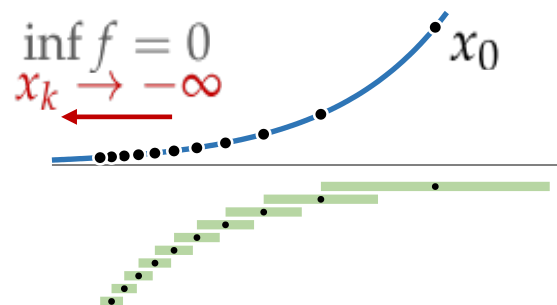
$$f(x) = |x|, \quad x_0 = 2$$

$$t_k = \frac{1}{k+1} \Rightarrow x_4 = 0$$

(iii) $\sum_k t_k = \infty$, $\text{Argmin } f = \emptyset$:

$f(x_k) \downarrow \inf f$, $\|x_k\|_2 \rightarrow \infty$

also when $\inf f = -\infty$



$$f(x) = e^x, \quad x_0 = 1$$

$$t_k = \frac{1}{k+1} \Rightarrow x_k \rightarrow -\infty$$

In particular, $\sum_k t_k = \infty \implies f(x_k) \downarrow \inf f$; radii may shrink, e.g. $t_k = 1/(k+1)$.

Conversely, $\sum_k t_k = \infty$ is *necessary* for this, for every proper closed convex f and x_0 (see (i)).



P.R., Hanmin Li

A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026 (Theorem 4.1)

Part 3

Proximal Point Method: Nonconvex Regime



Kaja Grunkowska, Hanmin Li, Xun Qian, P.R.
Proximal alignment for global non-convex optimization
arXiv:2604.13483, 2026

2D Example 1: GD Gets Stuck

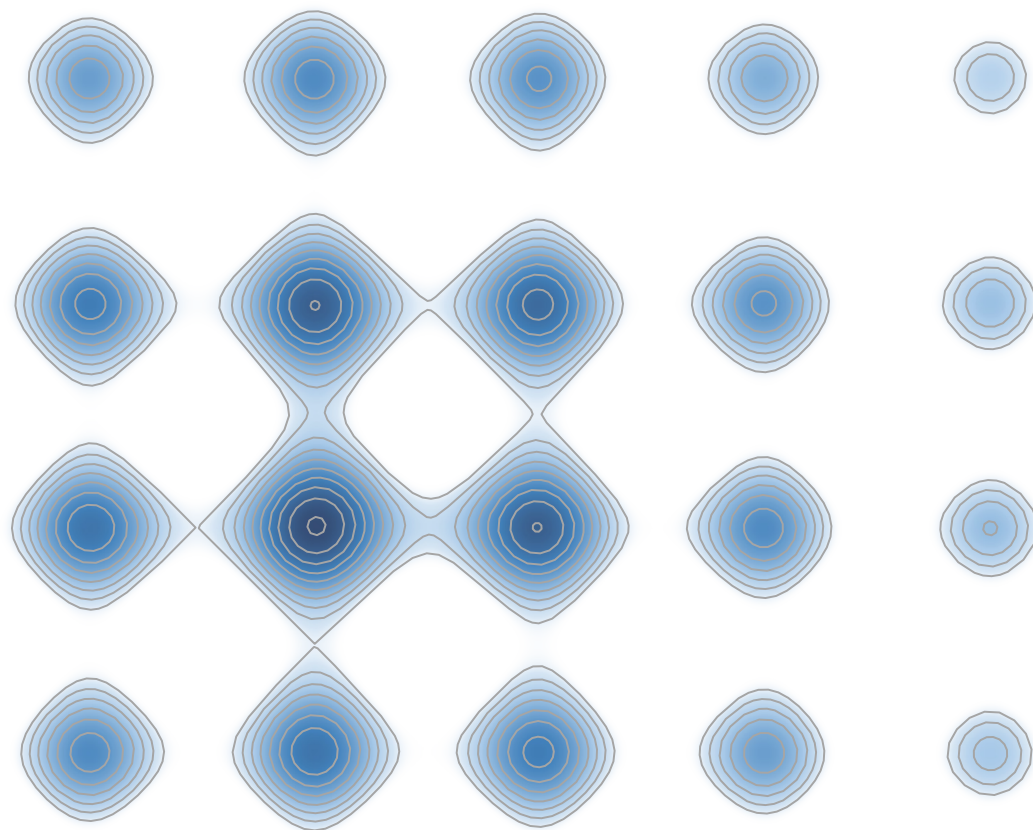
$$f(x) = \|x\|_2 + 10 \sin \zeta_1 + 10 \sin \zeta_2, \quad x = (\zeta_1, \zeta_2) \in \mathbb{R}^2$$

2D version of the 1D example $|x| + 10 \sin x$ (Grunkowska et al., 2026)

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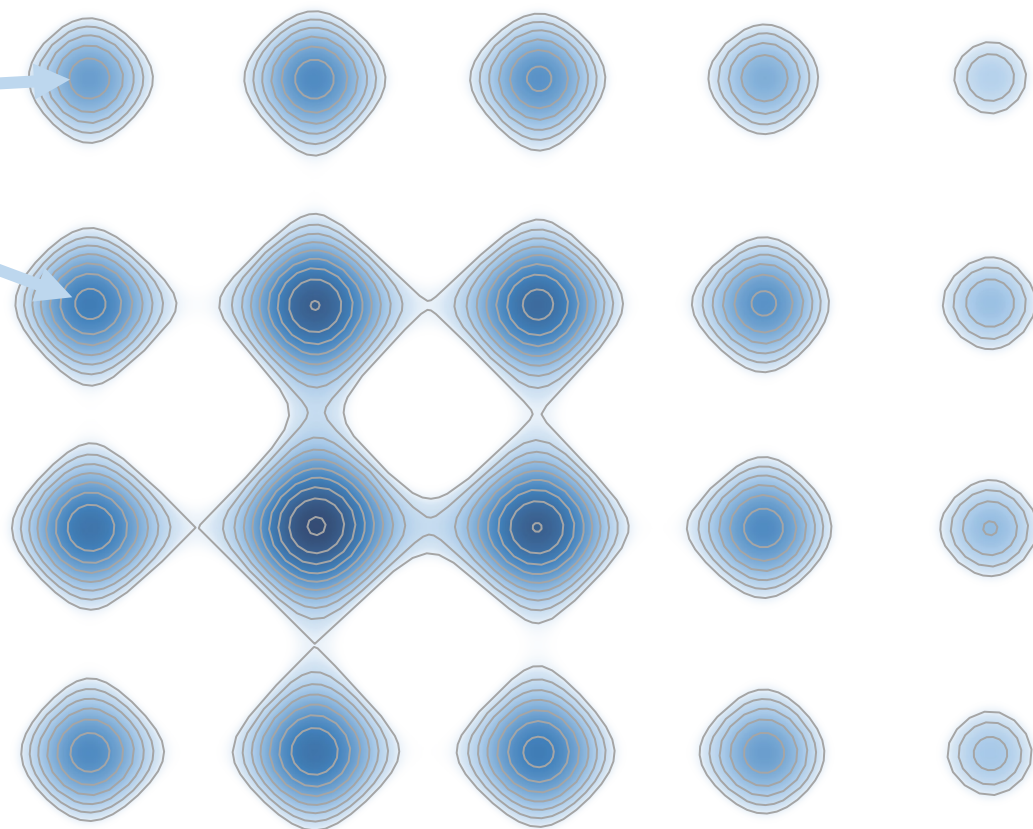
level sets of f (darker = lower f)

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2D version of the 1D example $|x| + 10 \sin x$ (Grunkowska et al., 2026)

non-optimal
local minima



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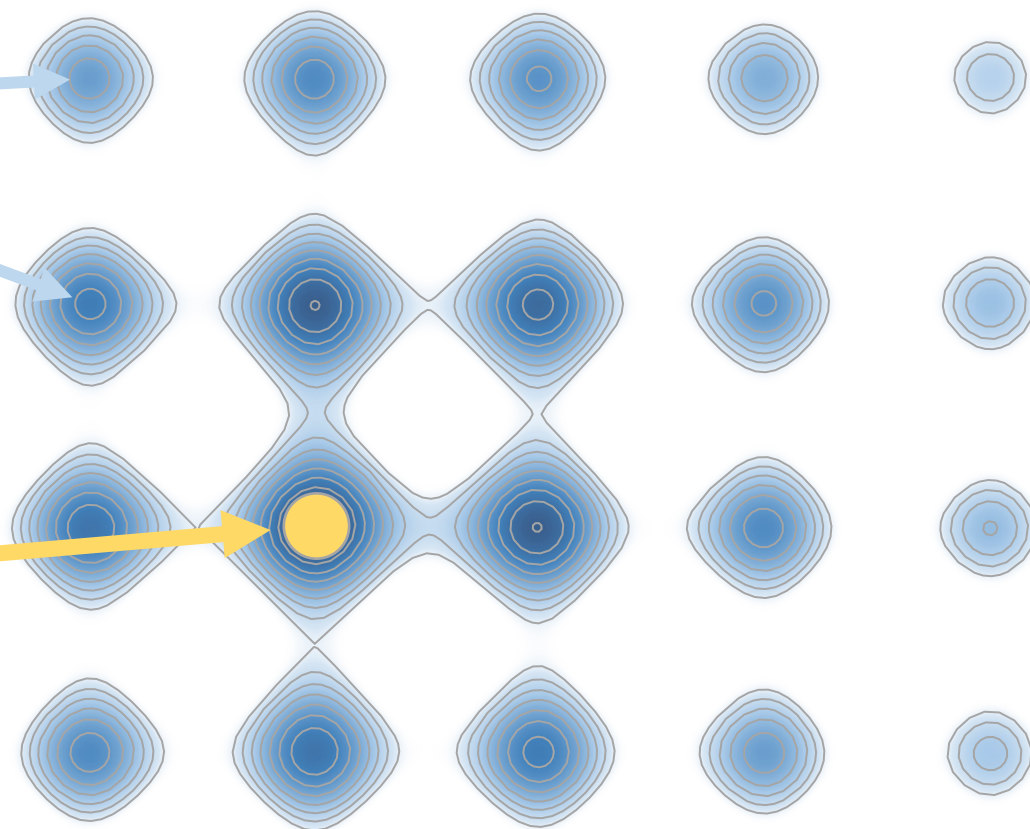
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non-optimal
local minima

Argmin f

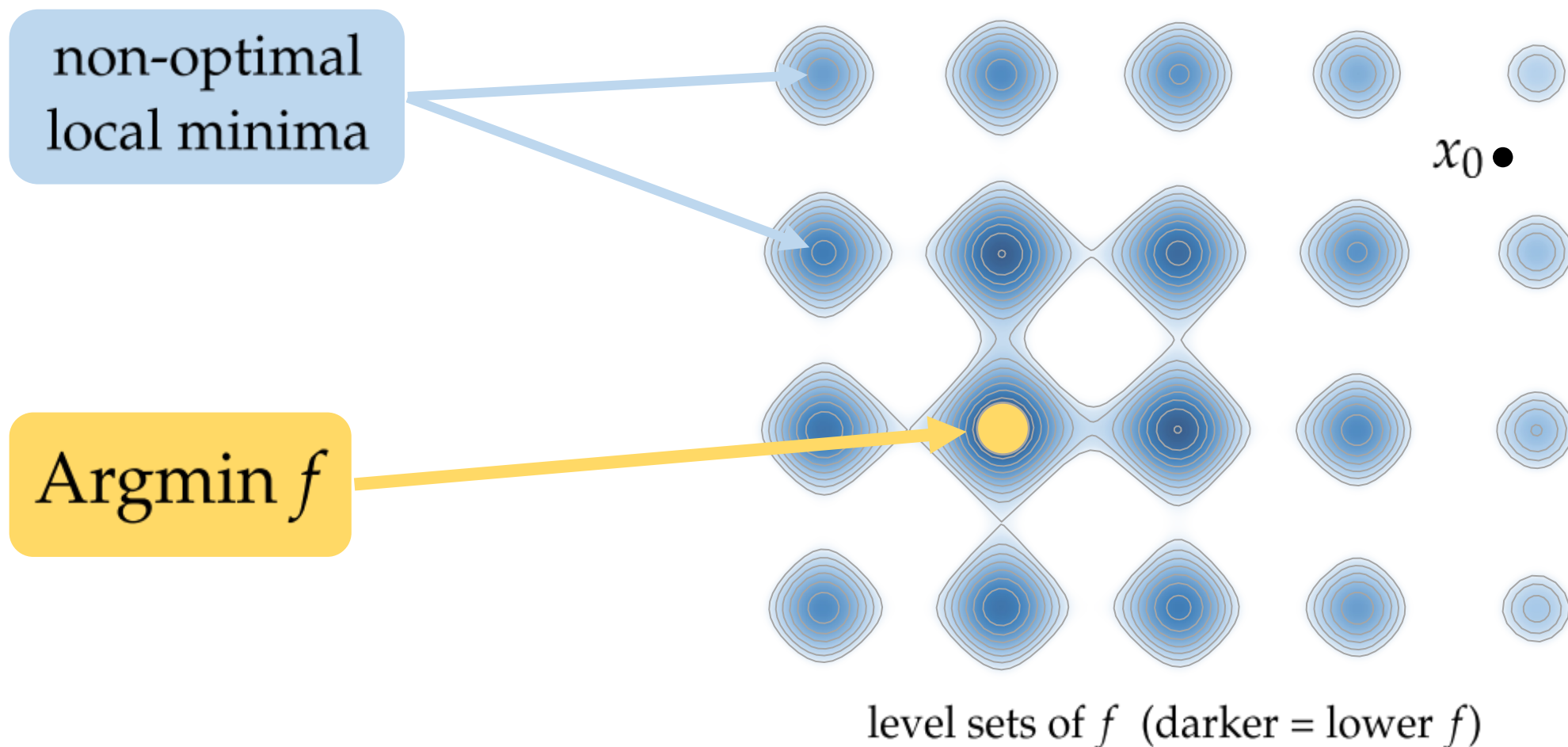


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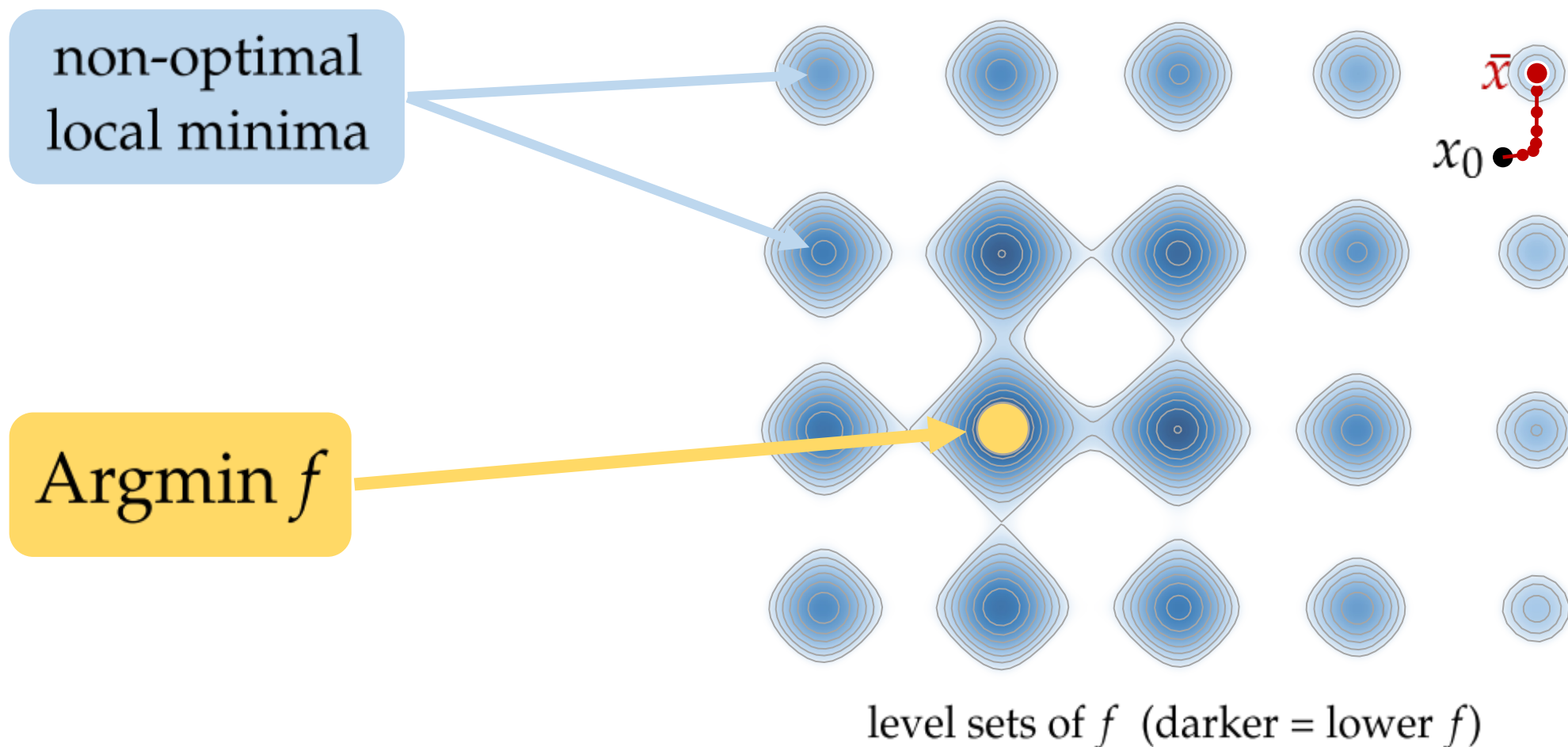
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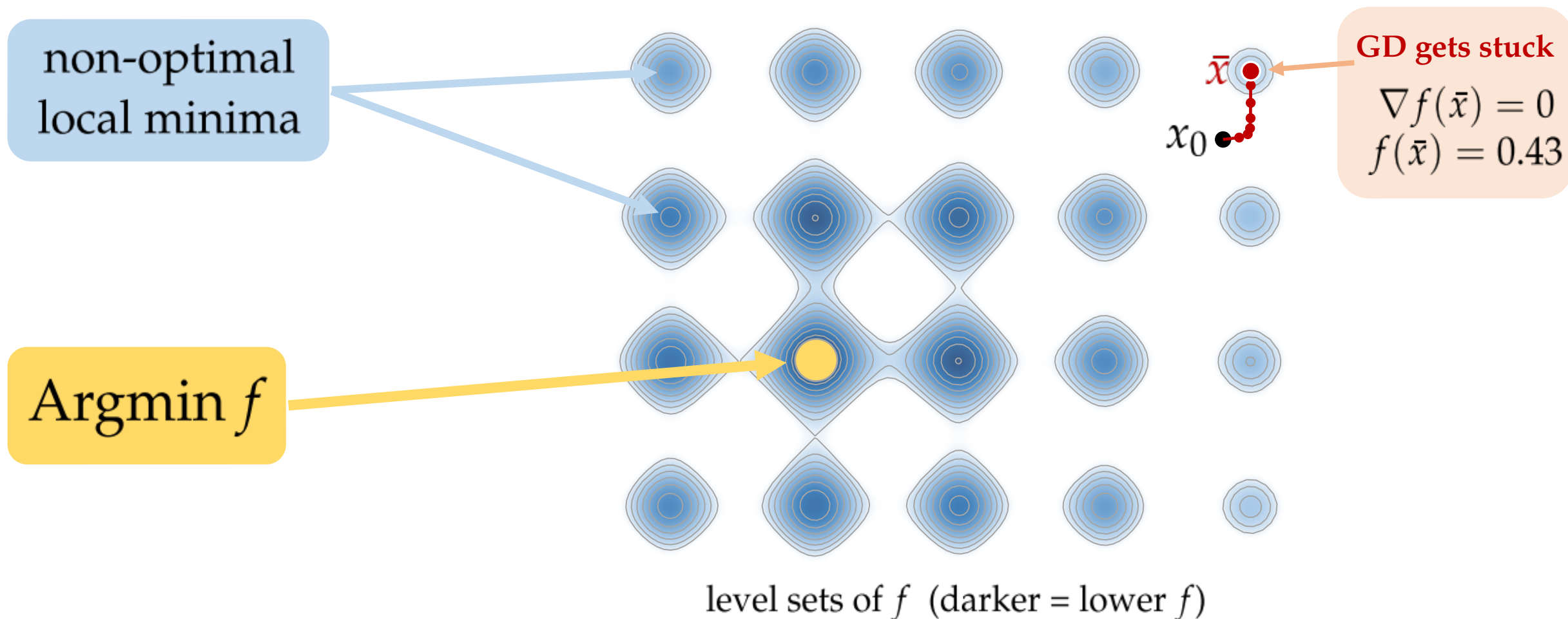
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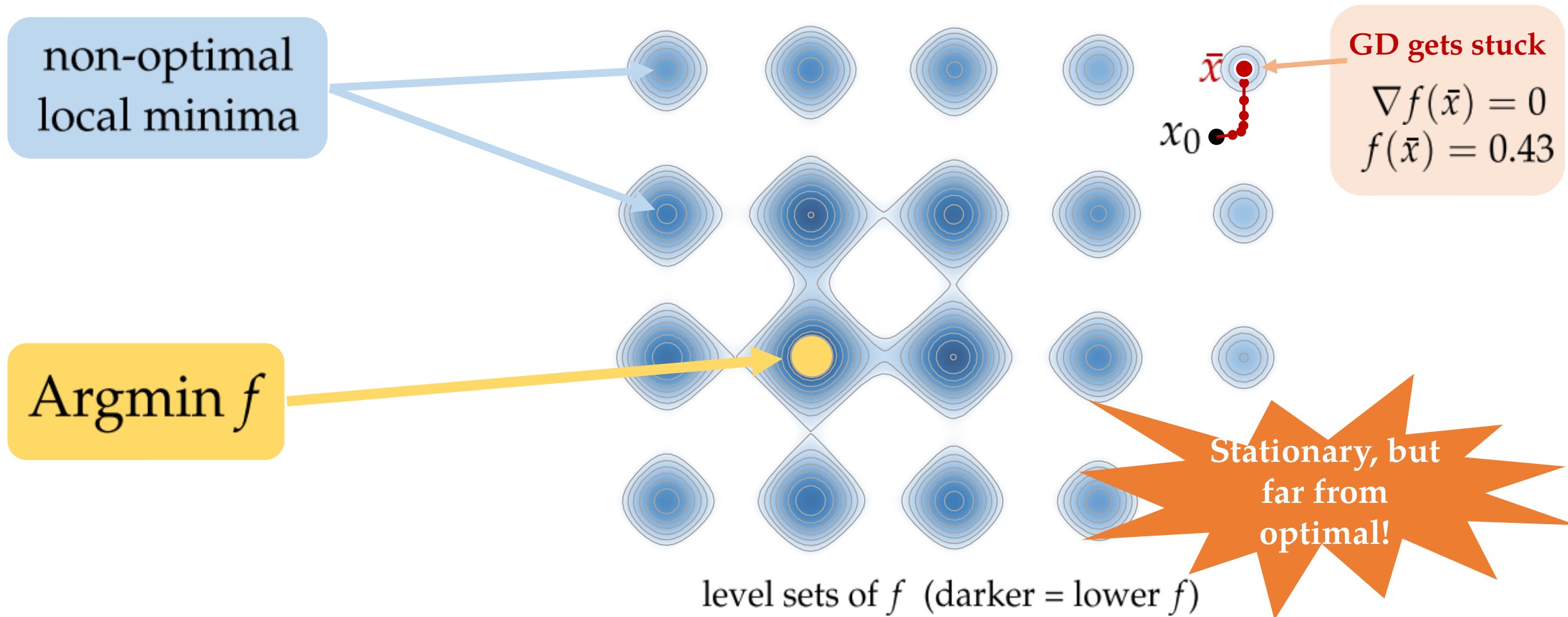
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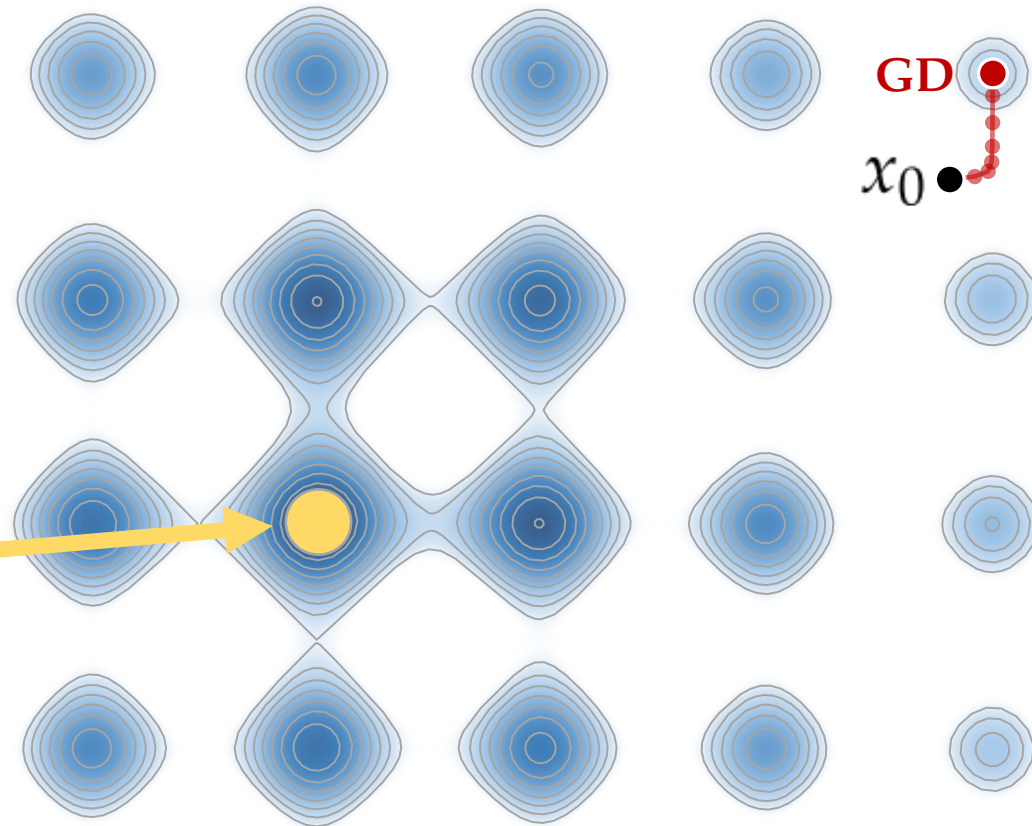


2D Example 1: BPM Jumps Over Local Minima

$$x_{k+1} \in \text{brox}_f^t(x_k) = \text{Argmin} \{f(y) : \|y - x_k\|_2 \leq t\}, \quad t = 2\pi$$

BPM function values
 $f(x_0) = 24.90$

Argmin f



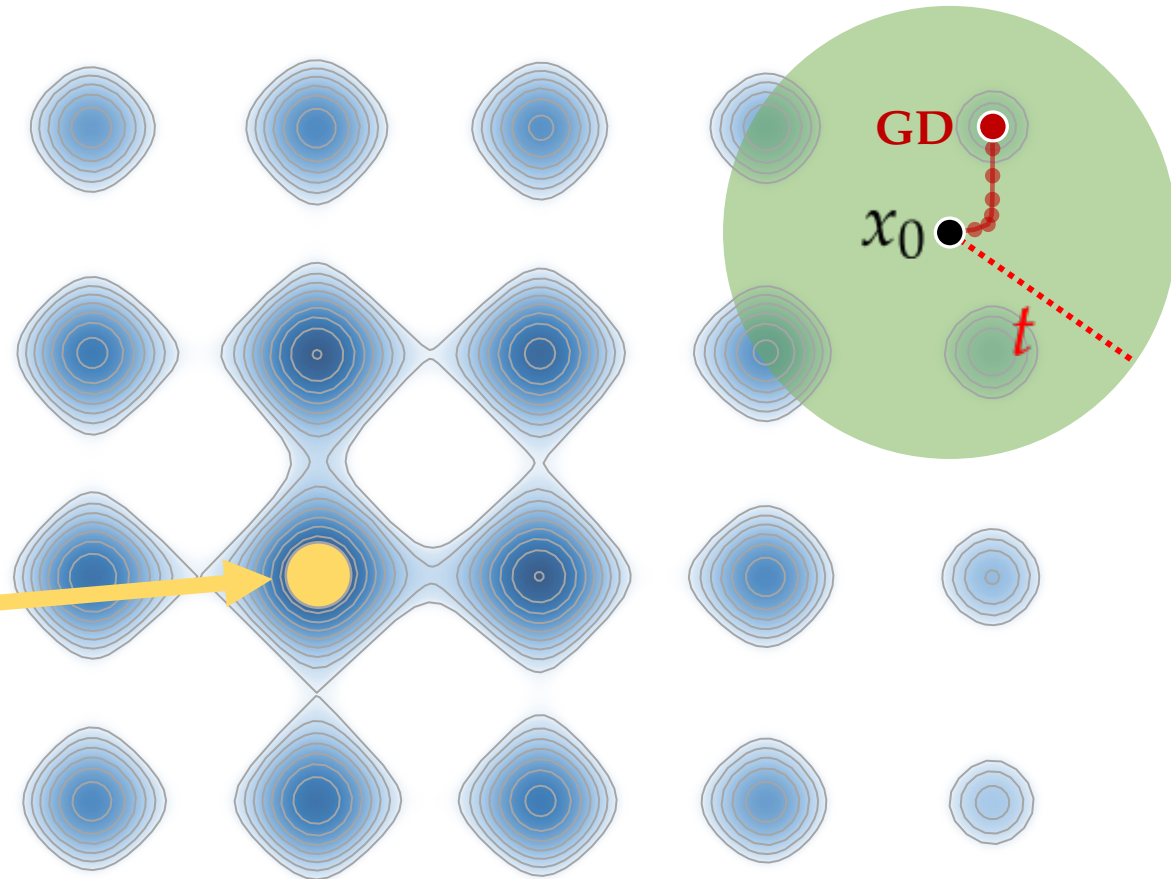
level sets of f (darker = lower f)

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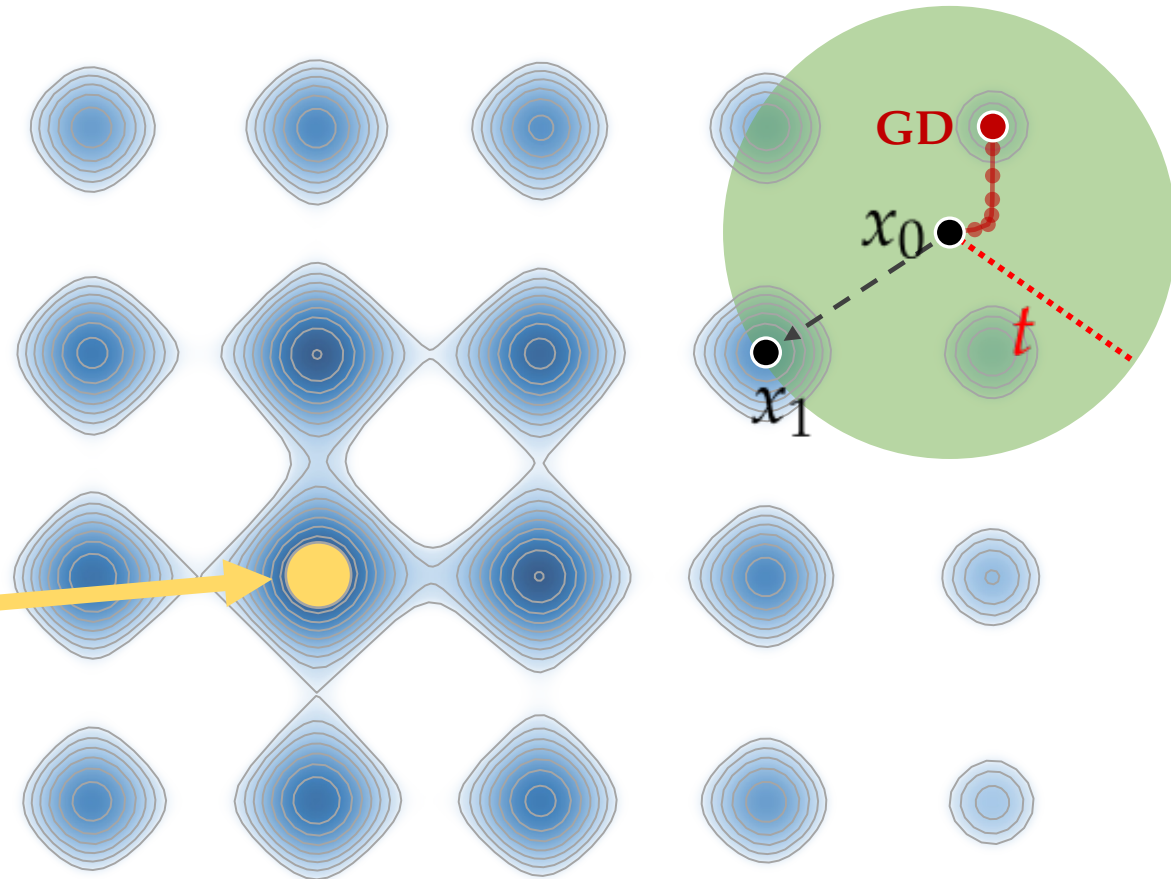
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BPM function values

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$$f(x_1) = -8.09$$

Argmin f



level sets of f (darker = lower f)

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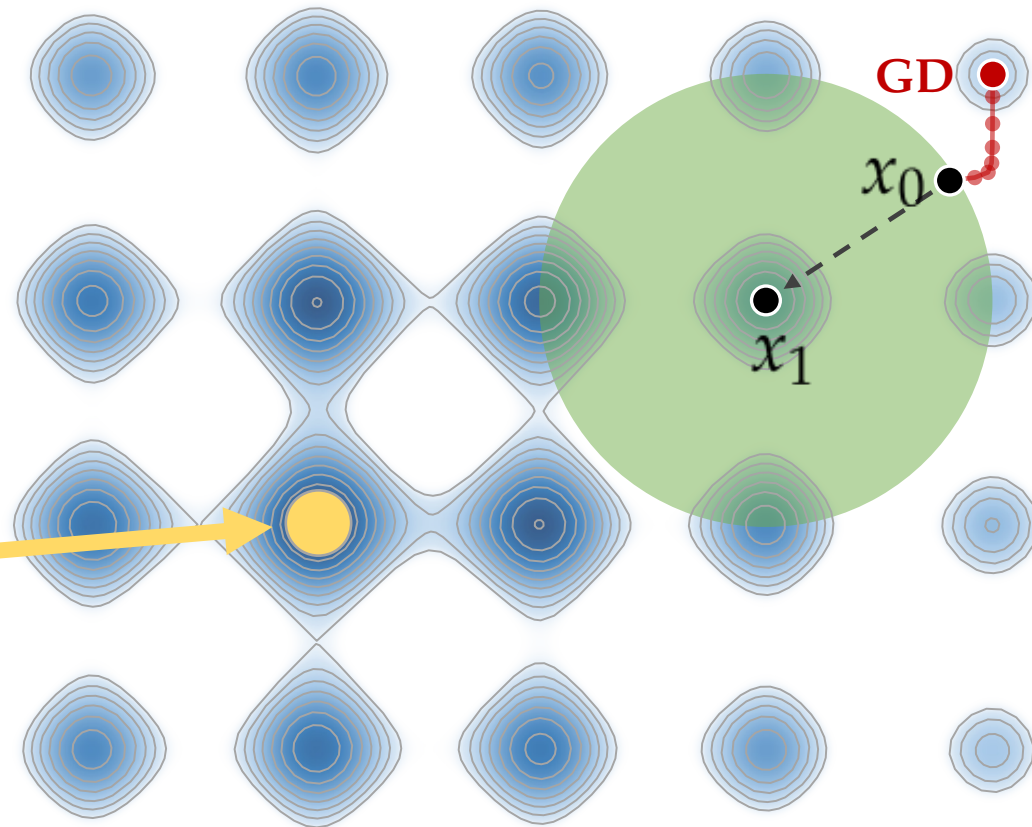
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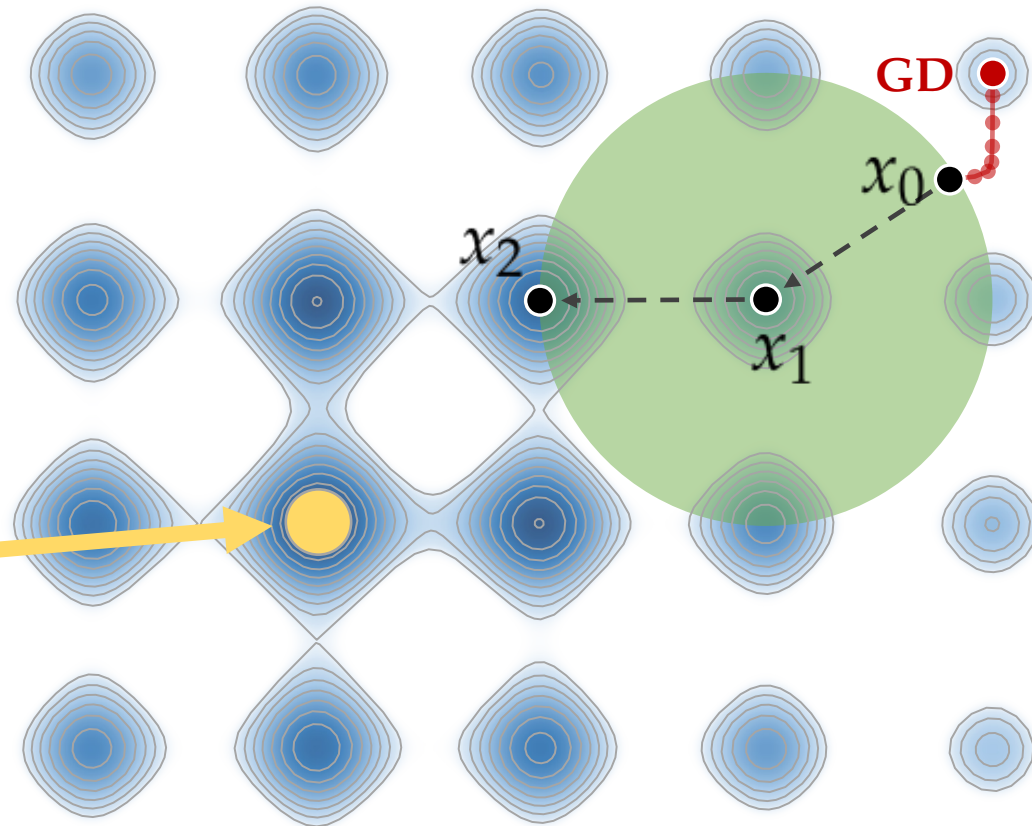
BPM function values

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$$f(x_2) = -13.39$$

Argmin f



level sets of f (darker = lower f)

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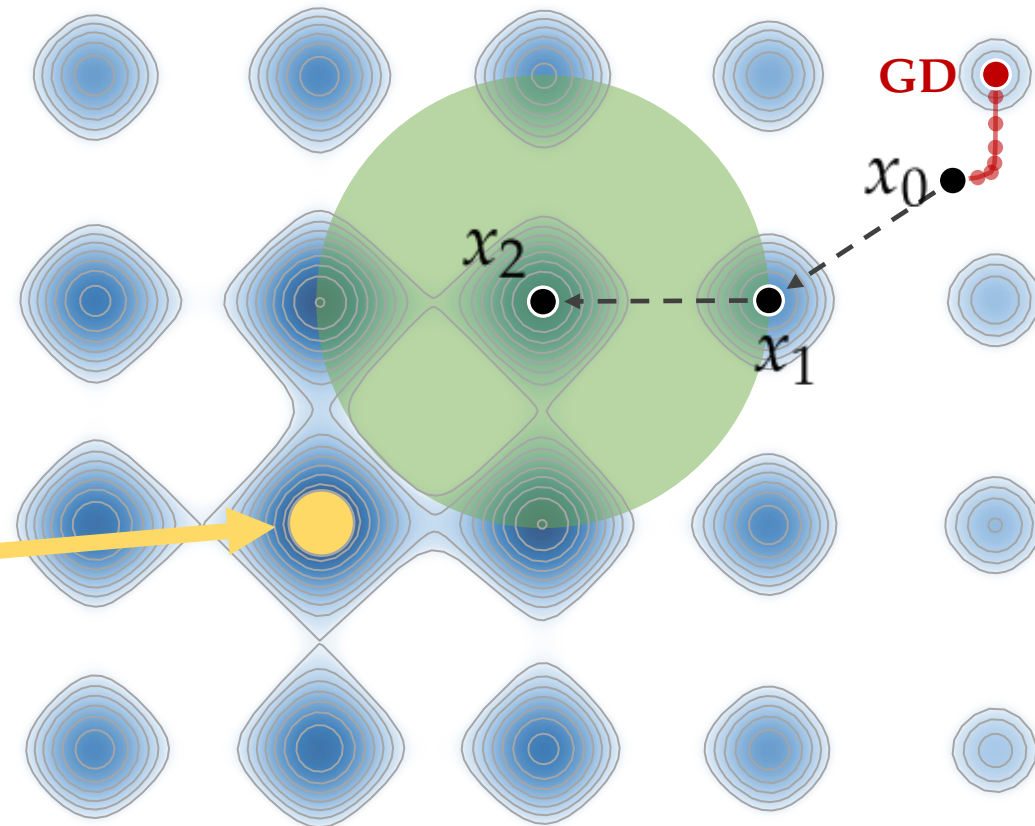
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level sets of f (darker = lower f)

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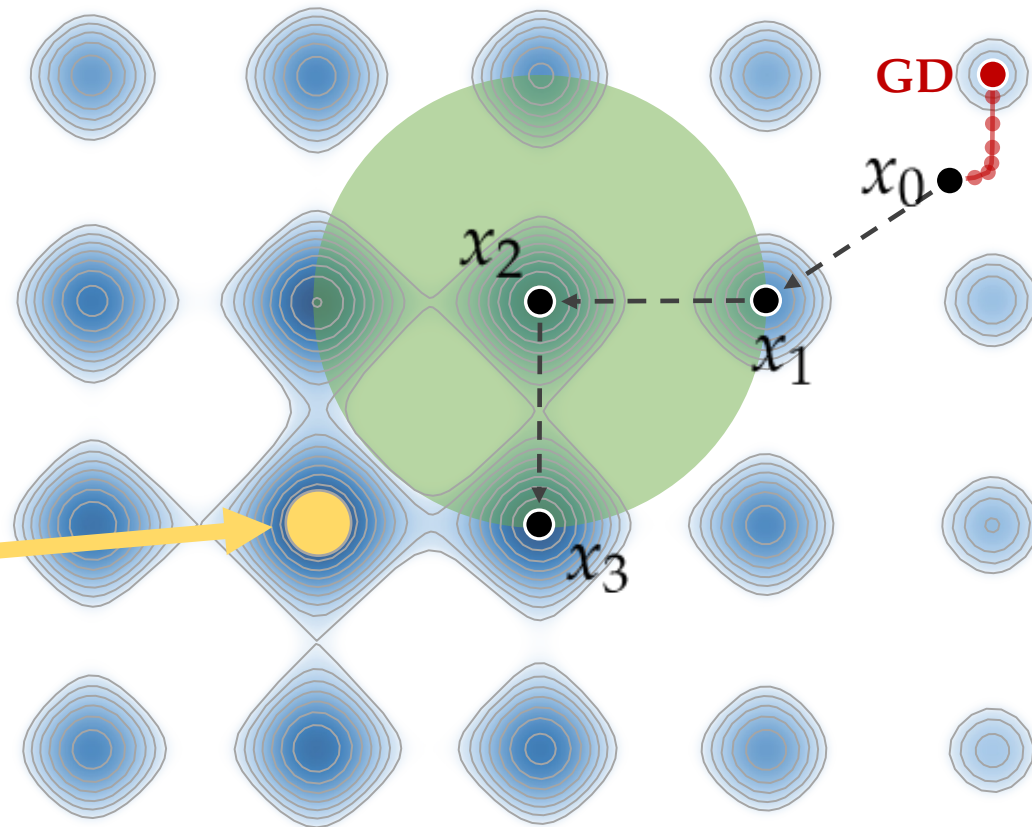
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Argmin f



level sets of f (darker = lower f)

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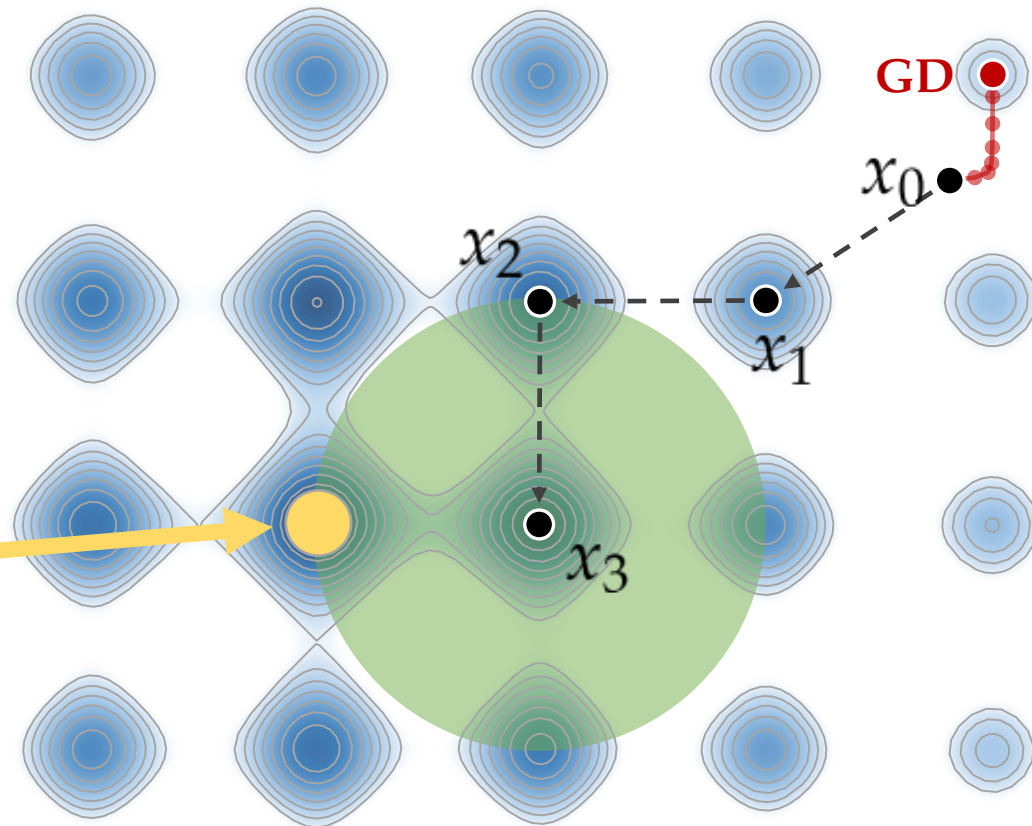
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level sets of f (darker = lower f)

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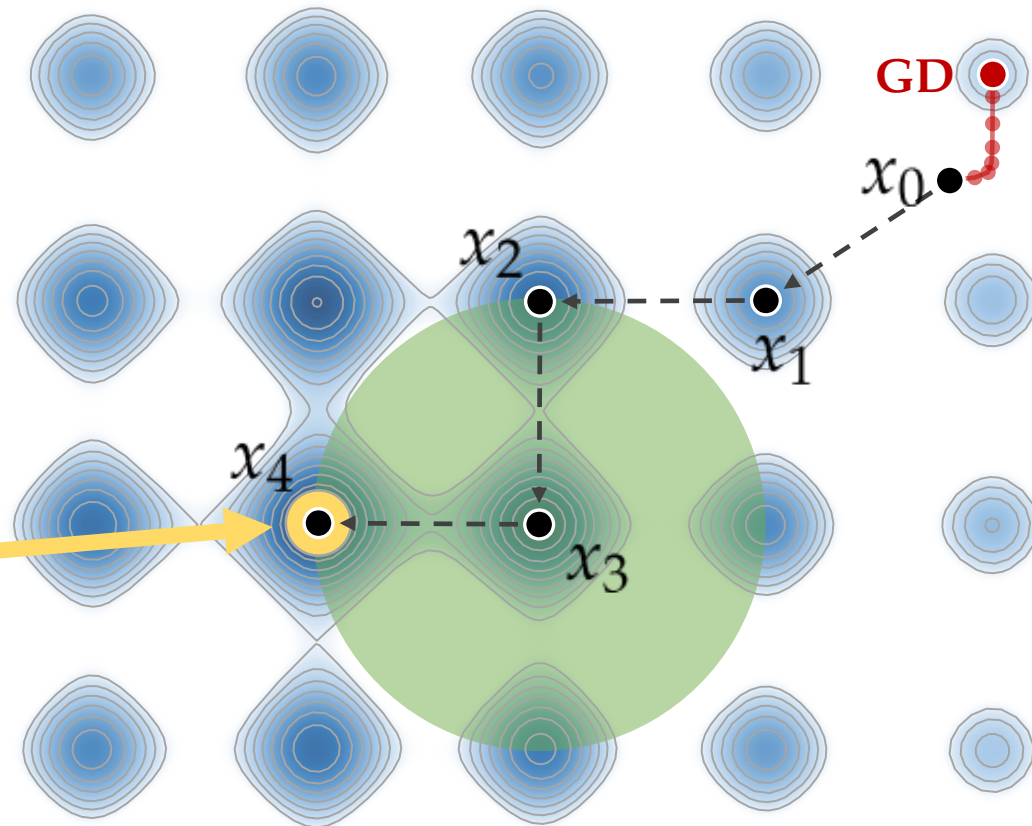
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$$f(x_4) = -17.83 = f_\star$$

Argmin f



level sets of f (darker = lower f)

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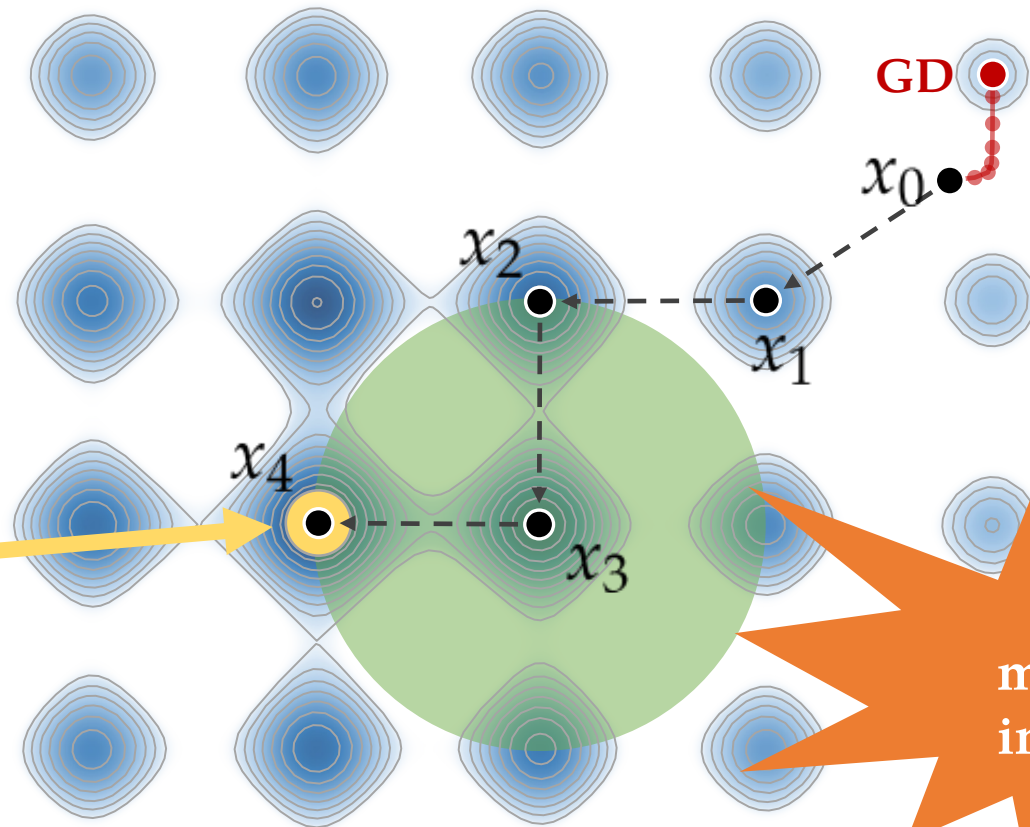
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Argmin f



level sets of f (darker = lower f)

Global
minimum
in 4 steps!

2D Example 2: GD Gets Stuck

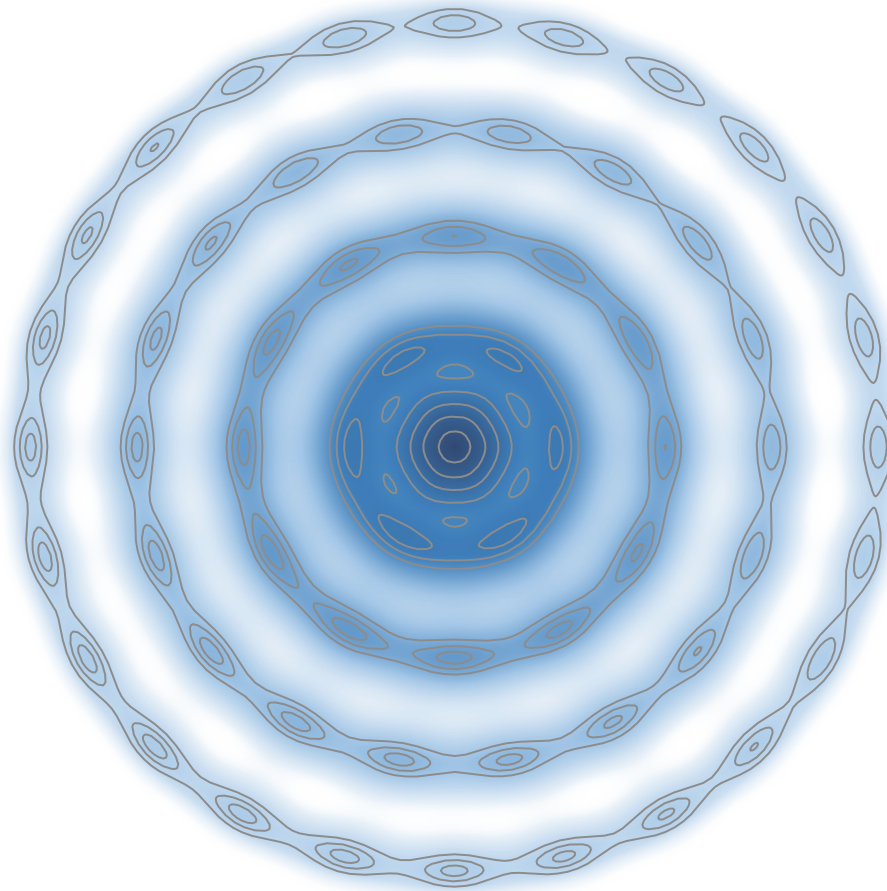
$$f(x) = \varphi(\|x\|_2) + \eta \text{ripple}(x), \quad \varphi(r) = r + 20 w(r) \sin^2(r/2)$$

radial analogue of the 1D example $|x| + 10 \sin x$; $w(r) = 1 - e^{-r^2/64}$; $0 \leq \text{ripple} \leq 1$, $\text{ripple}(0) = 0$; $\eta = 5$

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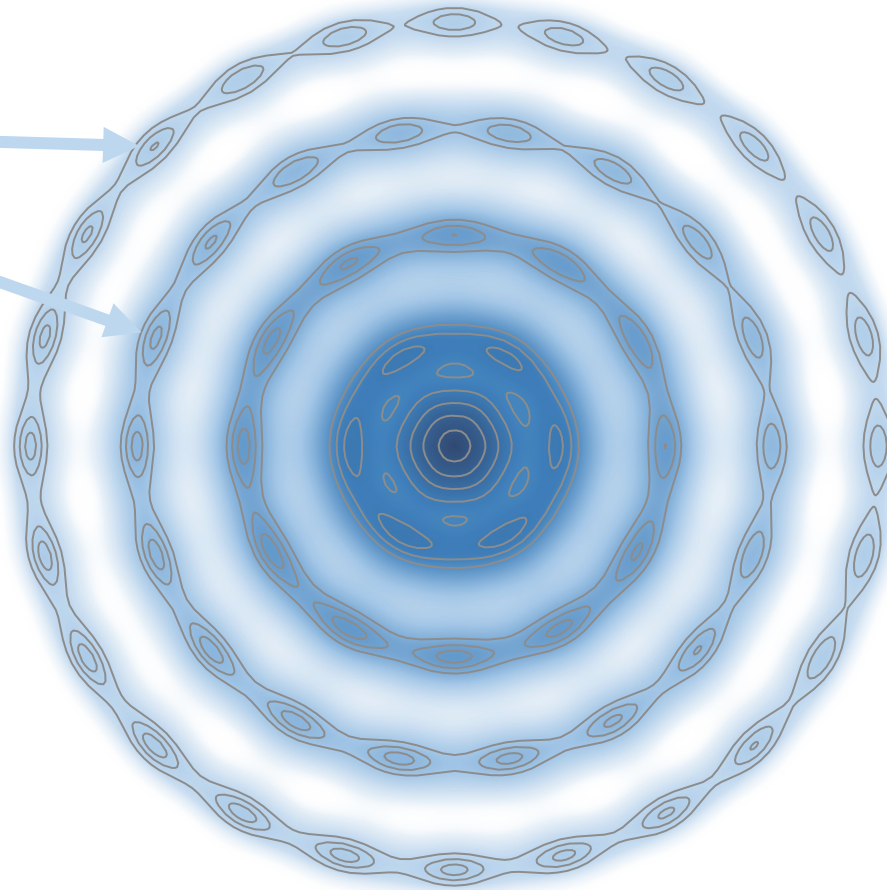
level sets of f
(darker = lower f)

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60 non-optimal
local minima



level sets of f
(darker = lower f)

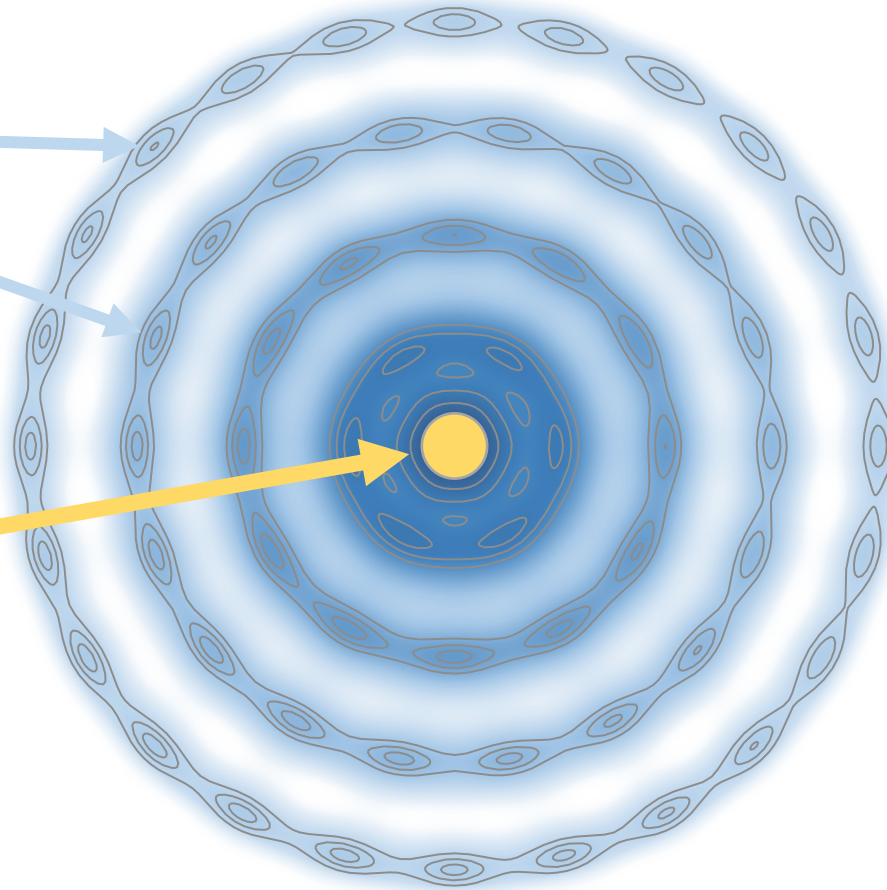
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60 non-optimal
local minima

Argmin f



level sets of f
(darker = lower f)

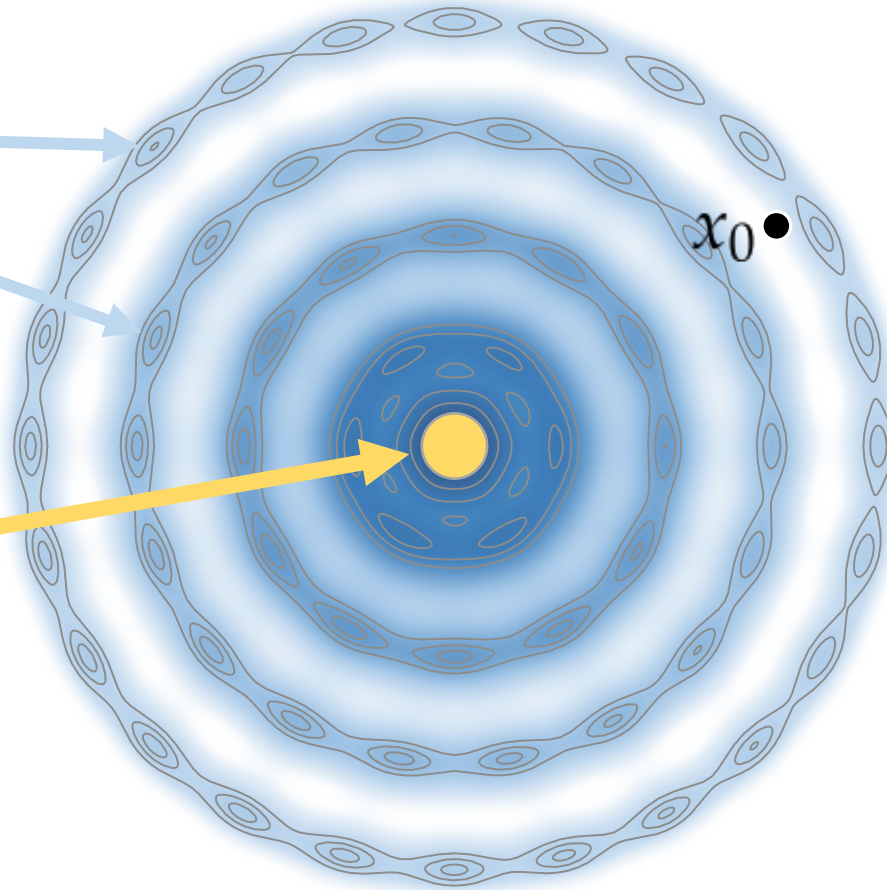
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level sets of f
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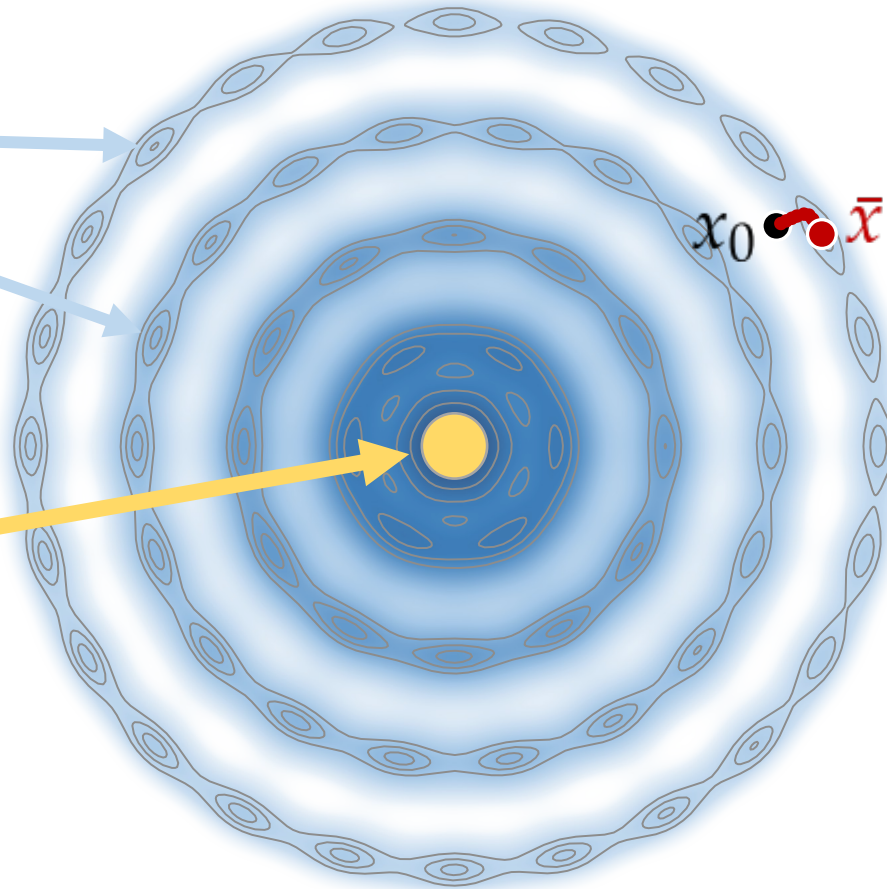
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60 non-optimal
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level sets of f
(darker = lower f)

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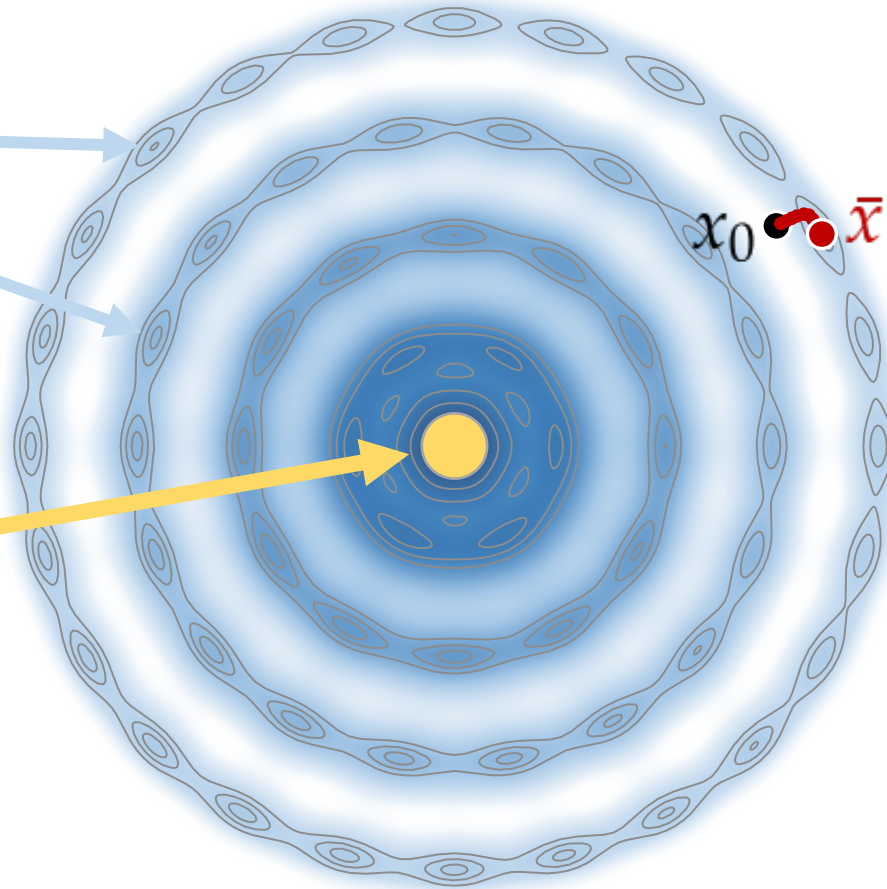
60 non-optimal
local minima

Argmin f

GD gets stuck

$$\begin{aligned} \nabla f(\bar{x}) &= 0 \\ f(\bar{x}) &= 26.30 \end{aligned}$$

level sets of f
(darker = lower f)



2D Example 2: GD Gets Stuck

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60 non-optimal
local minima

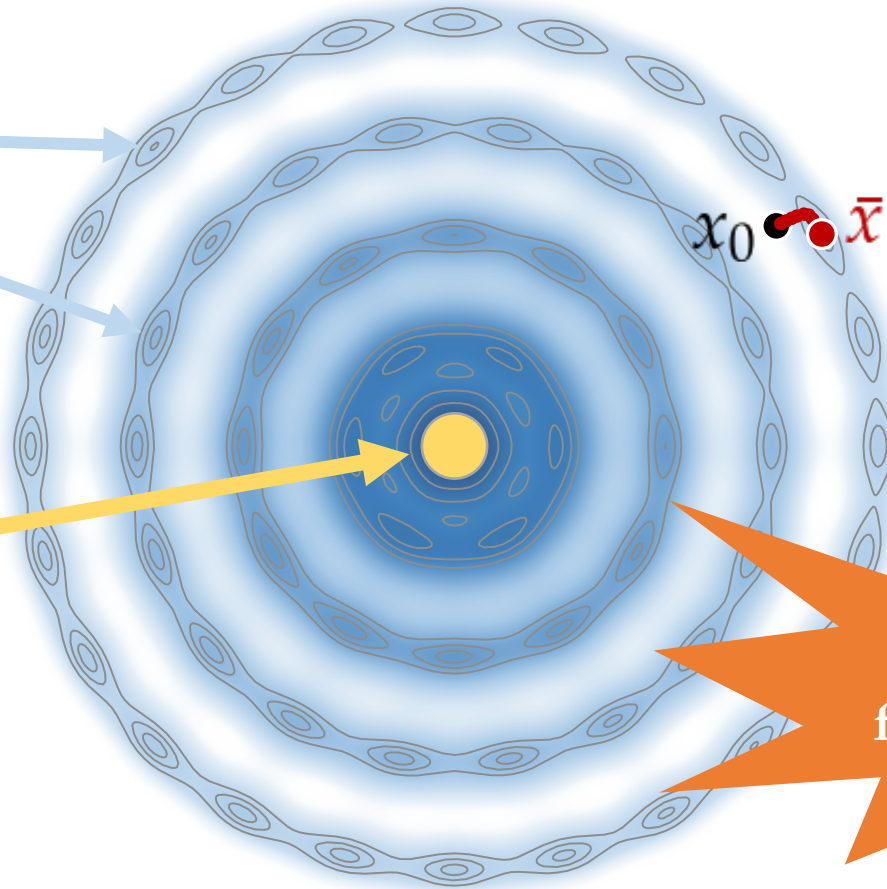
GD gets stuck

$$\begin{aligned} \nabla f(\bar{x}) &= 0 \\ f(\bar{x}) &= 26.30 \end{aligned}$$

level sets of f
(darker = lower f)

Argmin f

Stationary, but
far from optimal!

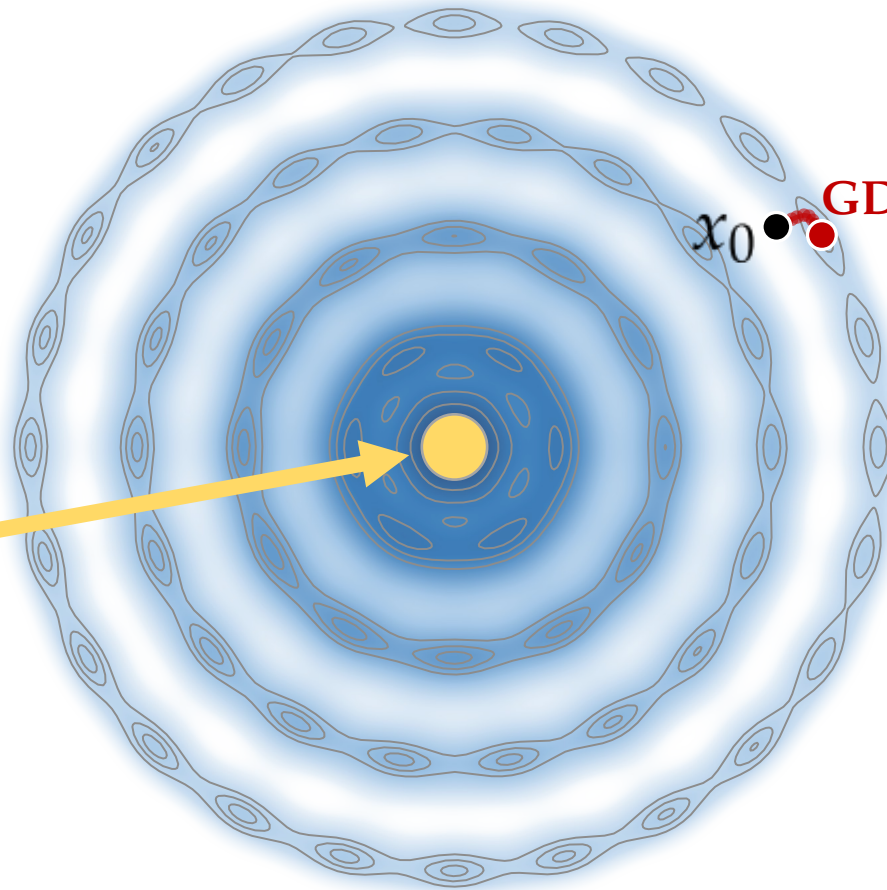


2D Example 2: BPM Jumps Over Local Minima

$$x_{k+1} \in \text{brox}_f^t(x_k) = \text{Argmin} \{f(y) : \|y - x_k\|_2 \leq t\}, \quad t = 2.5\pi$$

BPM function values
 $f(x_0) = 41.64$

Argmin f



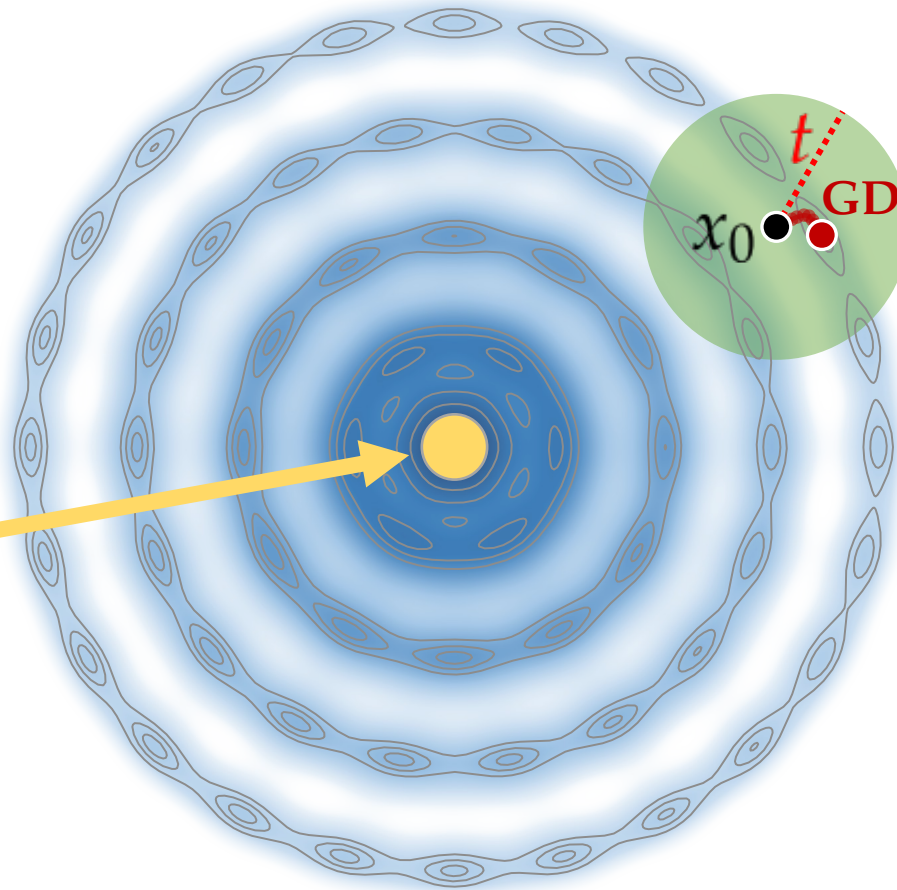
level sets of f
(darker = lower f)

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Argmin f



level sets of f
(darker = lower f)

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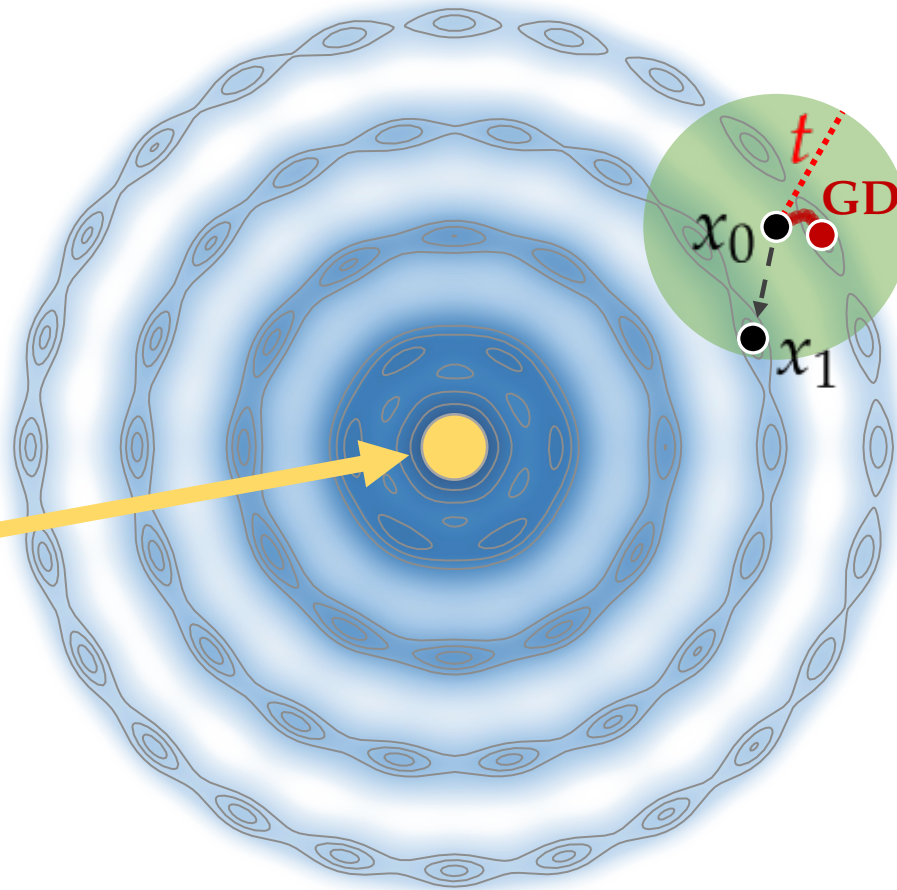
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BPM function values

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$$f(x_1) = 19.89$$

Argmin f



level sets of f
(darker = lower f)

2D Example 2: BPM Jumps Over Local Minima

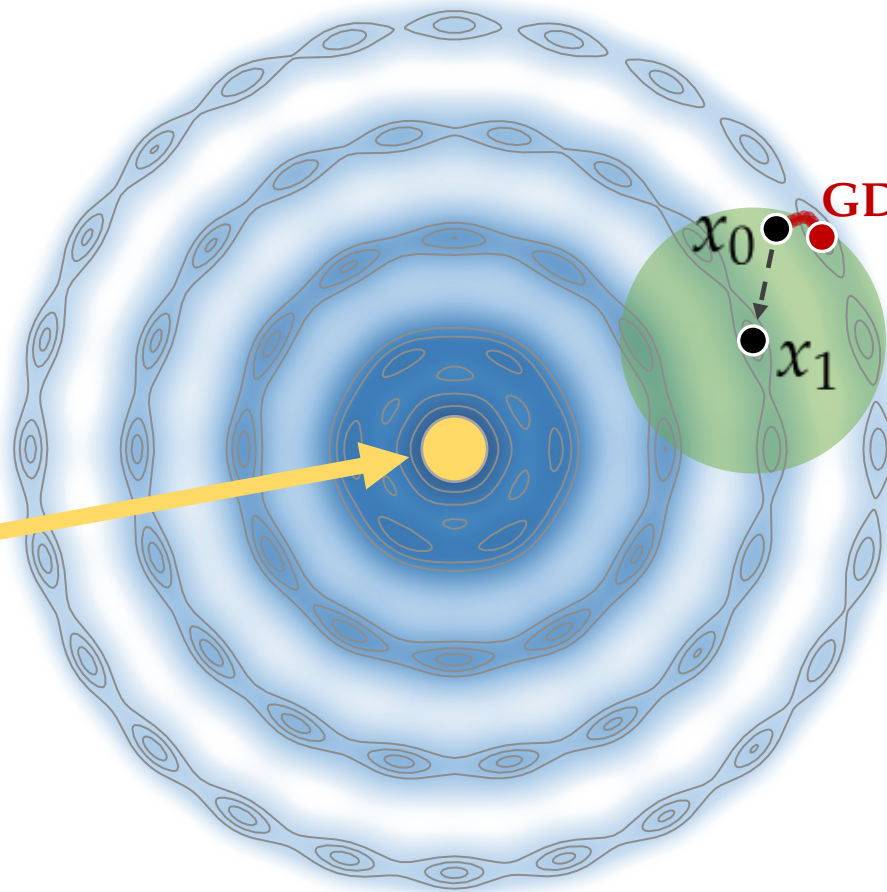
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level sets of f
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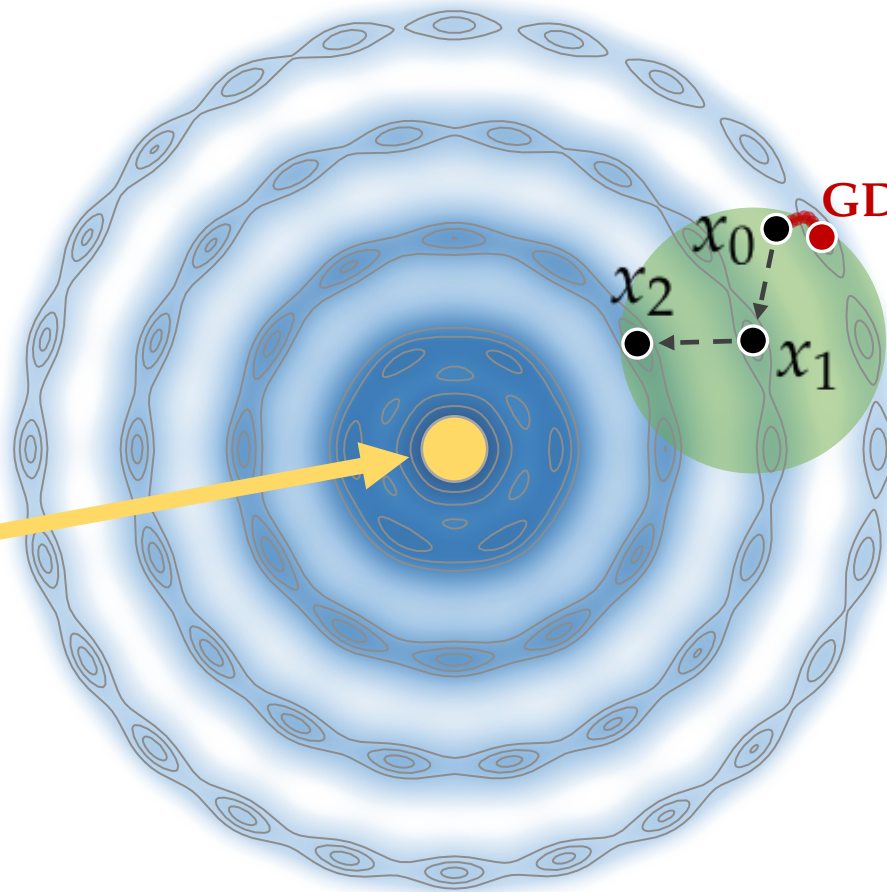
BPM function values

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Argmin f



level sets of f
(darker = lower f)

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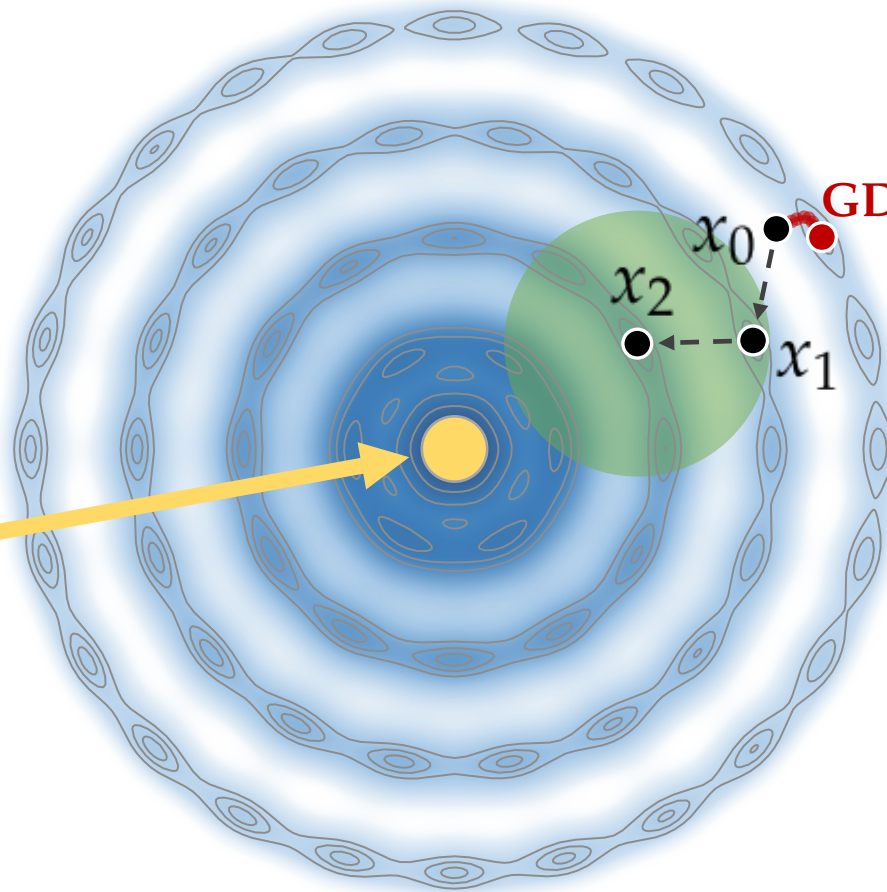
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level sets of f
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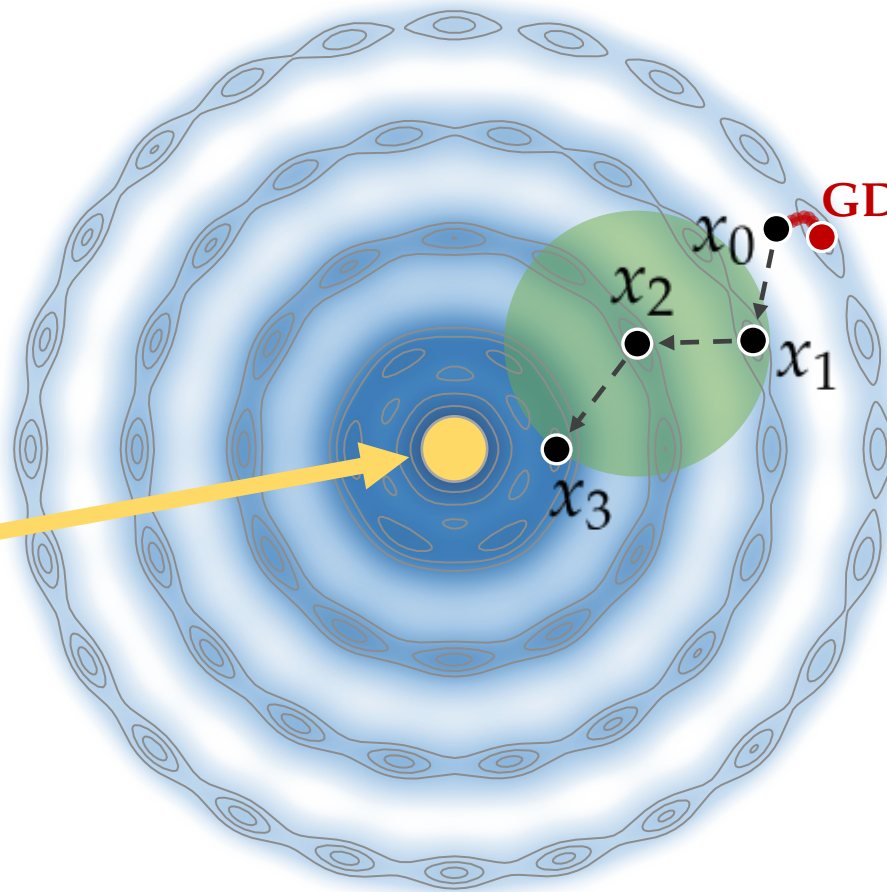
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$$f(x_3) = 6.41$$

Argmin f



level sets of f
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BPM function values

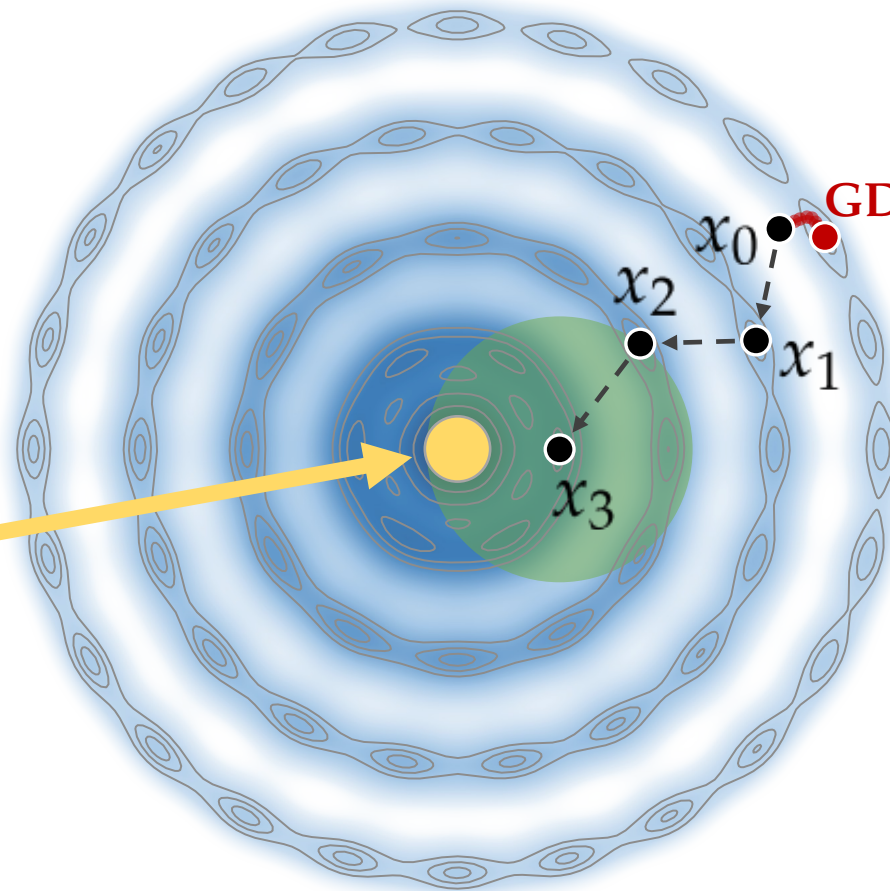
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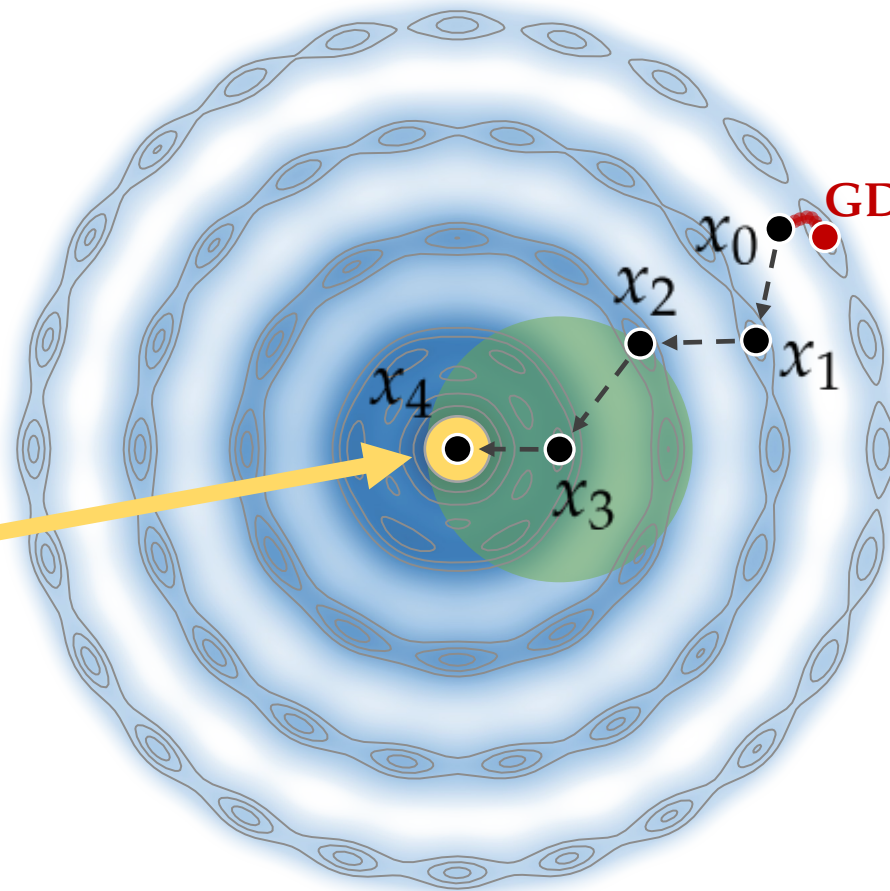
$$f(x_1) = 19.89$$

$$f(x_2) = 13.32$$

$$f(x_3) = 6.41$$

$$f(x_4) = 0.00 = f_\star$$

Argmin f



level sets of f
(darker = lower f)

2D Example 2: BPM Jumps Over Local Minima

$$x_{k+1} \in \text{brox}_f^t(x_k) = \text{Argmin} \{f(y) : \|y - x_k\|_2 \leq t\}, \quad t = 2.5\pi$$

BPM function values

$$f(x_0) = 41.64$$

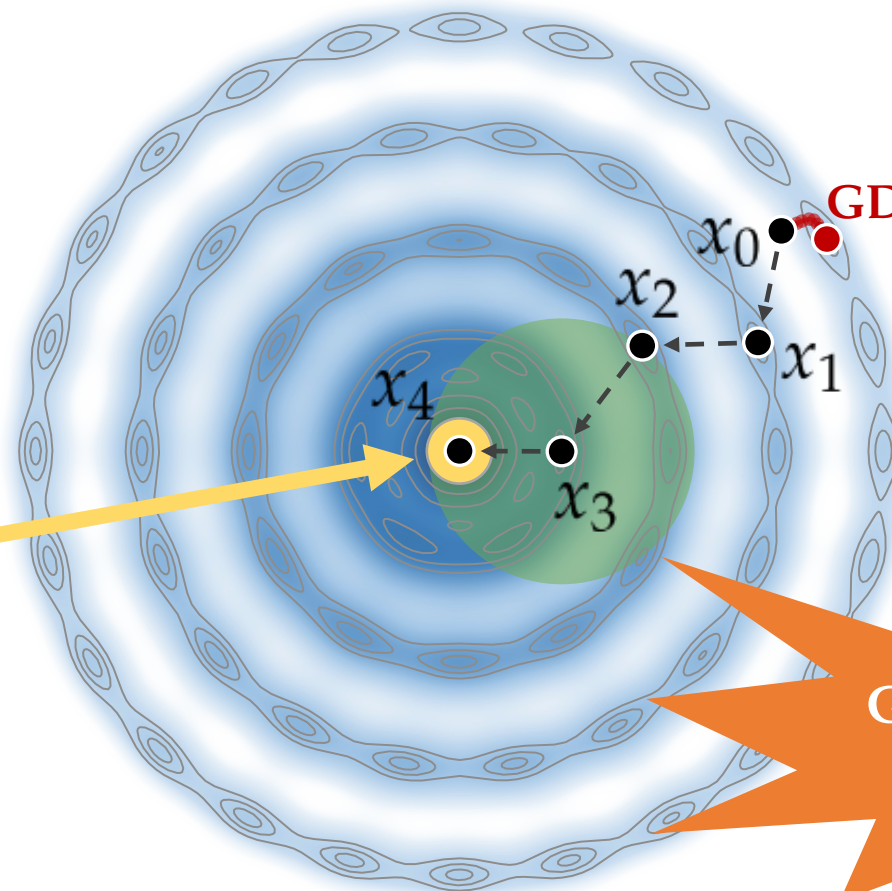
$$f(x_1) = 19.89$$

$$f(x_2) = 13.32$$

$$f(x_3) = 6.41$$

$$f(x_4) = 0.00 = f_\star$$

Argmin f



level sets of f
(darker = lower f)

**Global minimum
in 4 steps!**

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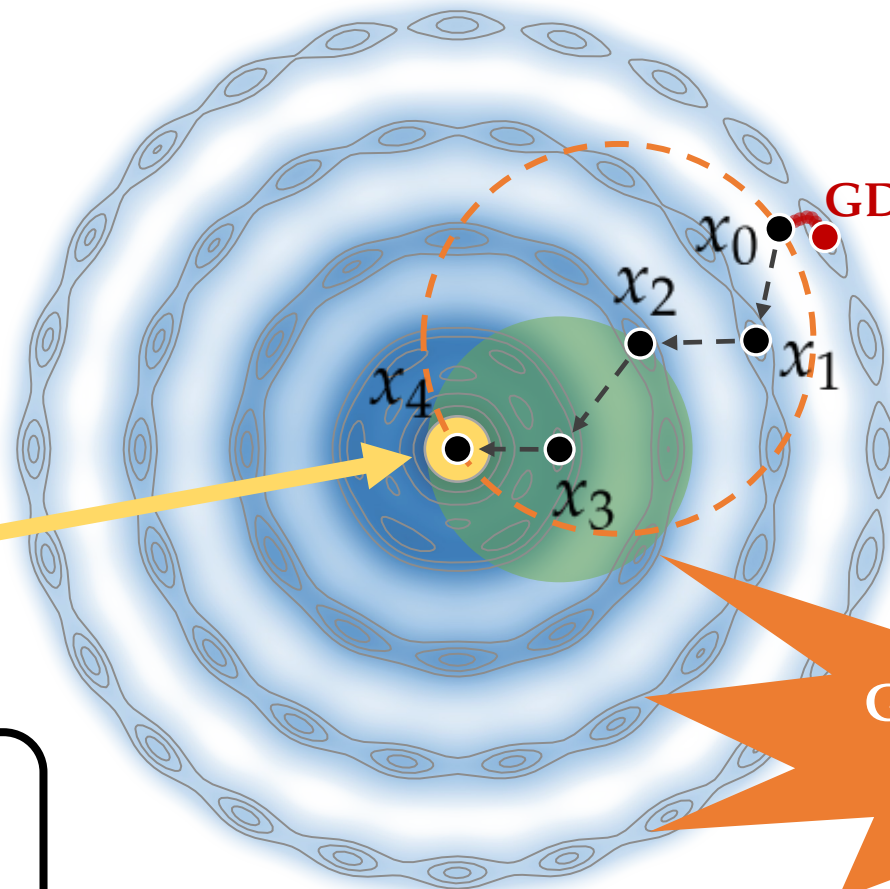
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Argmin f

$f \in \mathcal{F}_I^{\text{BA}}(t) \Rightarrow$ Theorem 4.1 applies

Brox-aligned: each step lands in the ball
with diameter $[x_k, x_\star]$ (dashed: $k = 0$)



level sets of f
(darker = lower f)

**Global minimum
in 4 steps!**

Brox-Aligned Nonconvex Functions

Broximal Alignment for Global Non-Convex Optimization

Table 1. Comparison of convex and non-convex function classes. **Non-Convex** = does not require f to be convex; **Non-Differentiable** = does not require f to be differentiable; **Stationary pts** = allows existence of stationary points that are not global minima; **Non-unique min** = allows existence of multiple global minima; **Disconnected** $\mathcal{X}_f$: the set of minimizers need not be connected.

Function class	Non-Convex	Non-Differentiable	Stationary pts	Non-unique min	Disconnected $\mathcal{X}_f$
Strongly convex	✗	✓	✗	✗	✗
Convex	✗	✓	✗	✓	✗
Pseudoconvex (Mangasarian, 1975)	✓	✗	✗	✓	✗
Quasiconvex (Arrow & Enthoven, 1961)	✓	✓	✓	✓	✗
Star-convex (Nesterov & Polyak, 2006)	✓	✗	✗	✓	✗
Quasar convex (Hardt et al., 2018)	✓	✗	✗	✓	✗
Aiming (Liu et al., 2023)	✓	✗	✗	✓	✗
Polyak-Łojasiewicz (Polyak, 1963; Łojasiewicz, 1963)	✓	✗	✗	✗	✗
Quadratic growth (Bonnans & Ioffe, 1995)	✓	✓	✓	✓	✓
$\mathcal{F}_{\mathbf{x}}^{\text{BA}}(t)$ (ours)	✓	✓	✓	✓	✓

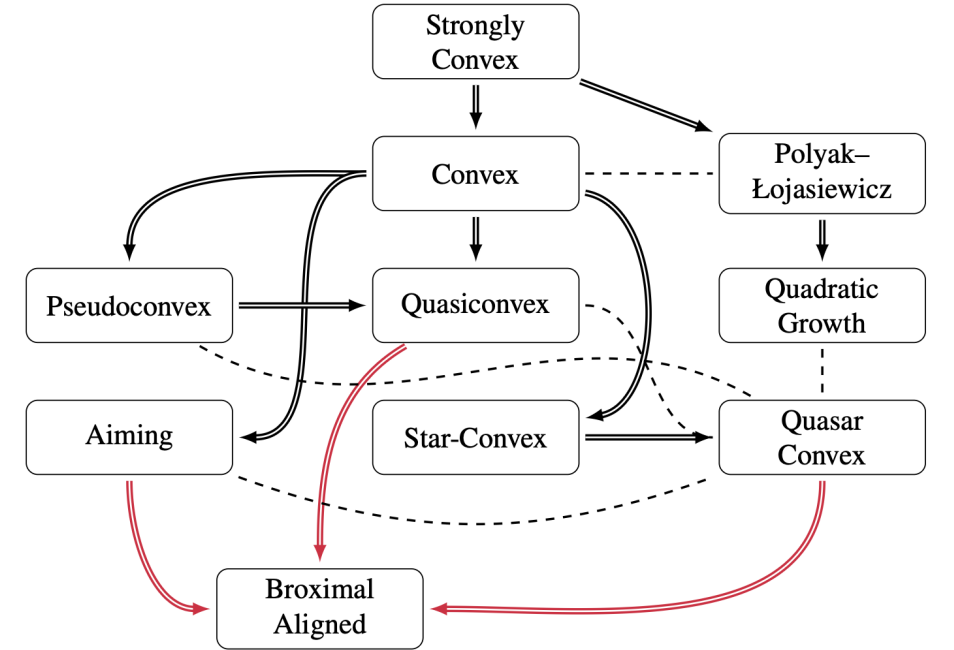


Figure 3. **Hierarchy of assumptions.** Solid arrows encode one-way implications, dashed lines denote mutual non-implications, and **red** arrows indicate conditions guaranteeing Broximal Alignment for any $t > 0$. For simplicity, we assume f is L -smooth (i.e., $\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \forall x, y \in \mathbb{R}^d$) with domain $\text{dom } f = \mathbb{R}^d$. See the appendix and Karimi et al. (2016); Hinder et al. (2019); Khanh & Lara (2025) for proofs.

Brox-Aligned Nonconvex Functions

Broximal Alignment for Global Non-Convex Optimization

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Quasar convex (Hardt et al., 2018)	✓	✗	✗	✓	✗
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$\mathcal{F}_{\mathbf{x}}^{\text{BA}}(t)$ (ours)	✓	✓	✓	✓	✓

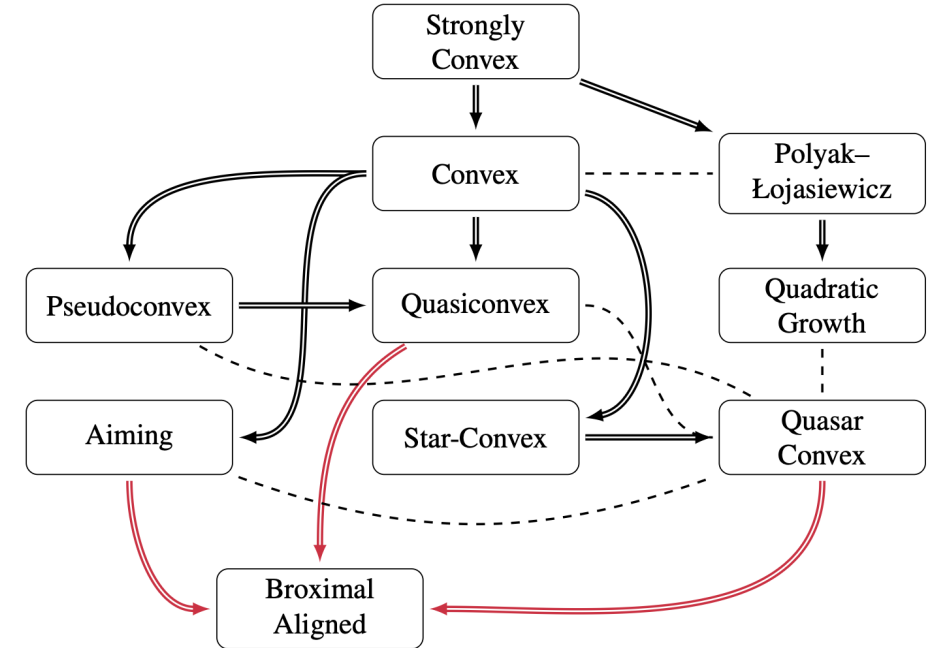


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Kaja Grunkowska, Hanmin Li, Xun Qian, P.R.
Broximal alignment for global non-convex optimization
arXiv:2604.13483, 2026

Part 4
Proximal Point Method:
Practical Approximations

GD = Linearized BPM

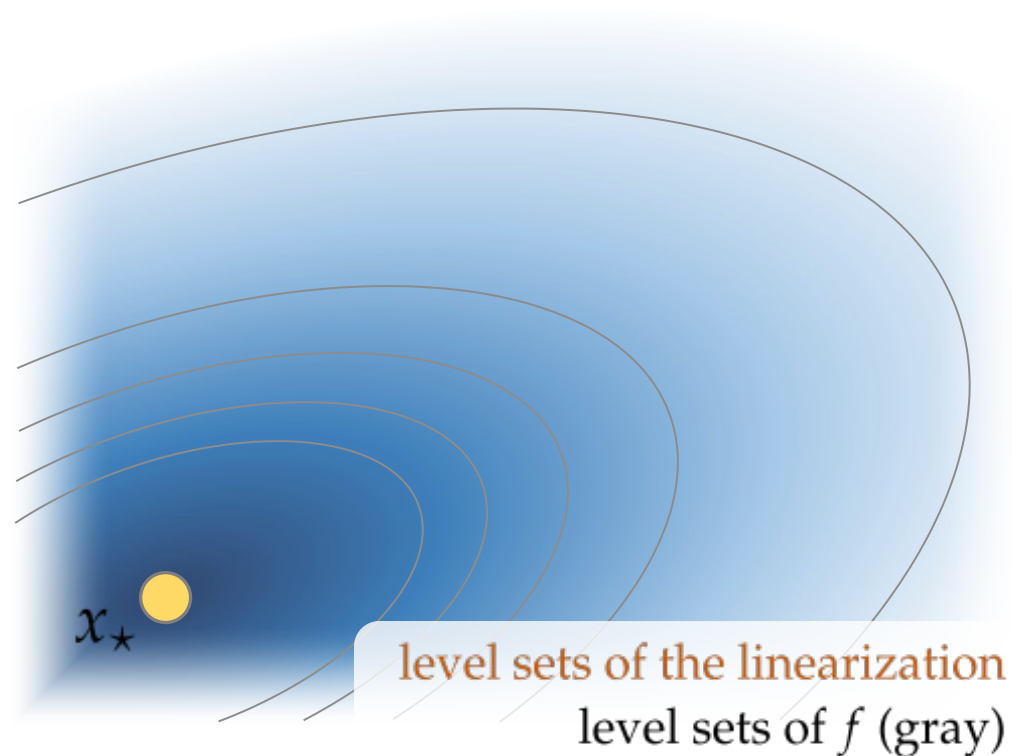
$$x_{k+1} = \operatorname{argmin} \left\{ \underbrace{f(x_k) + \langle \nabla f(x_k), y - x_k \rangle}_{\text{linearization of } f \text{ at } x_k} : \|y - x_k\|_2 \leq t_k \right\} = x_k - \gamma \nabla f(x_k) \quad \text{if } t_k = \gamma \|\nabla f(x_k)\|_2$$

$$\text{Example: } f(x) = \frac{1}{2} \langle a, x \rangle^2 + 2 \langle b, x \rangle^2, \quad a = (\cos 20^\circ, \sin 20^\circ), \quad b = (-\sin 20^\circ, \cos 20^\circ)$$

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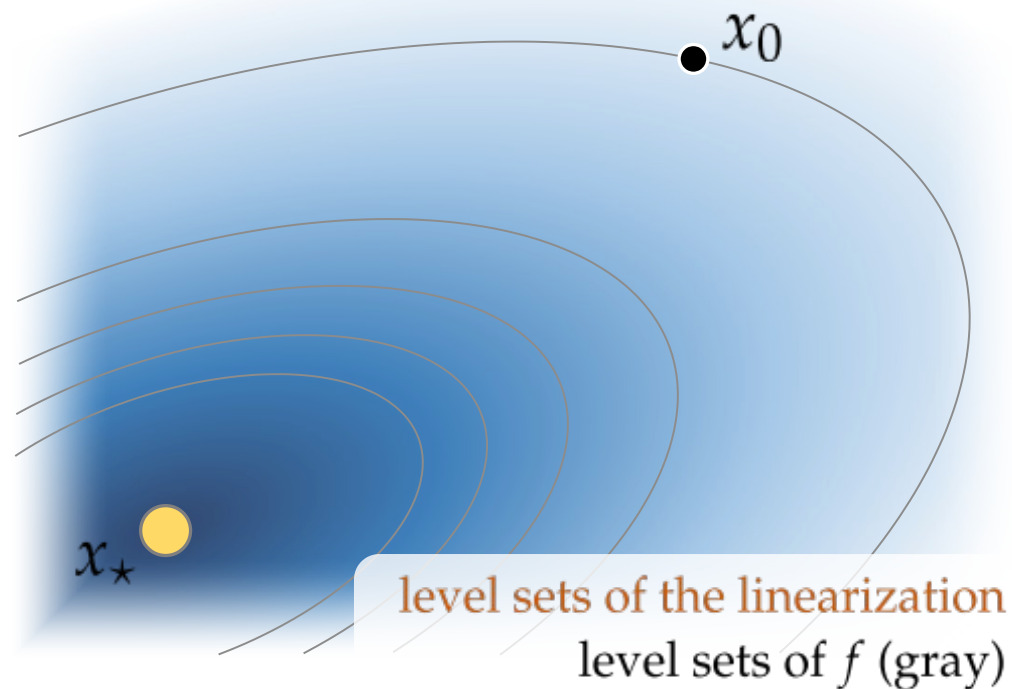


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Linearize the objective inside the ball



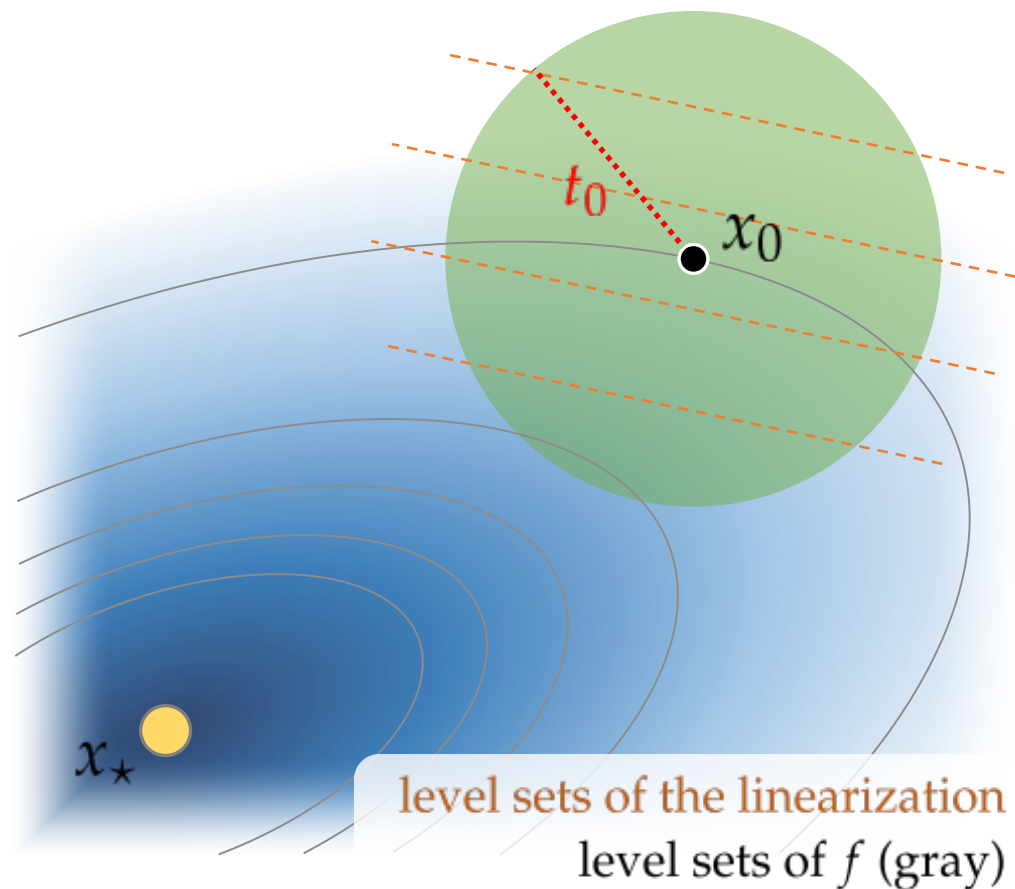
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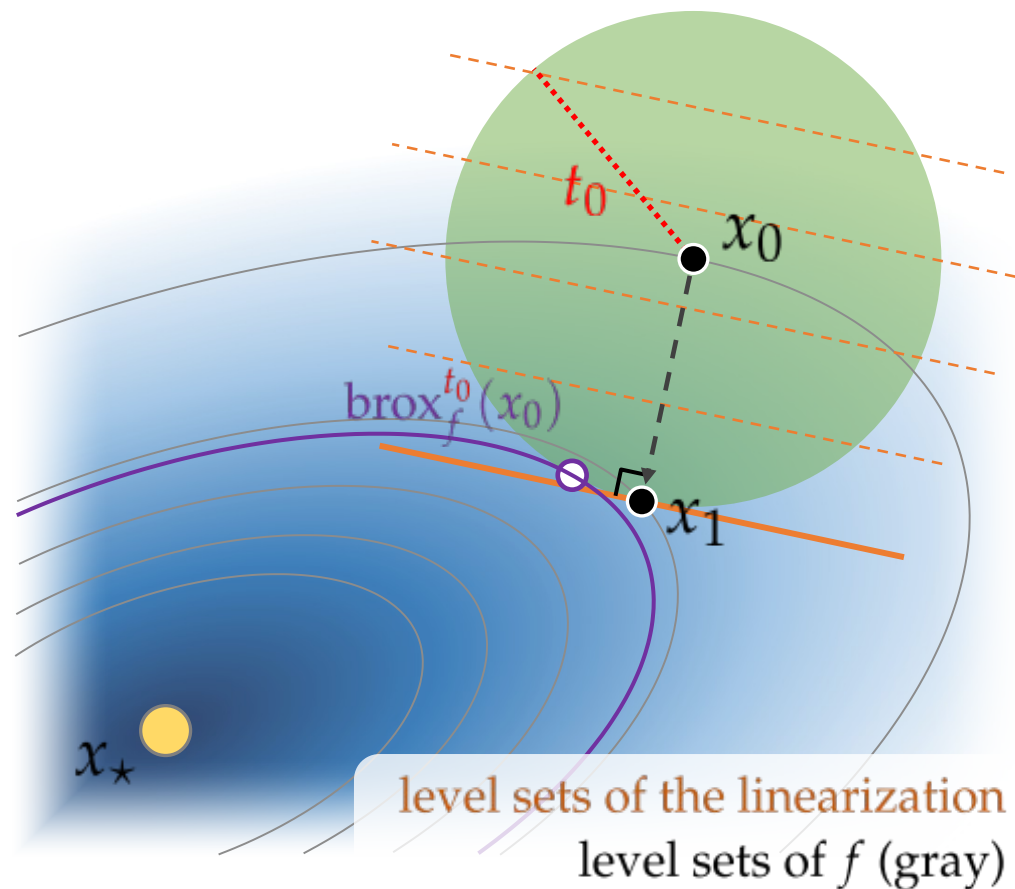
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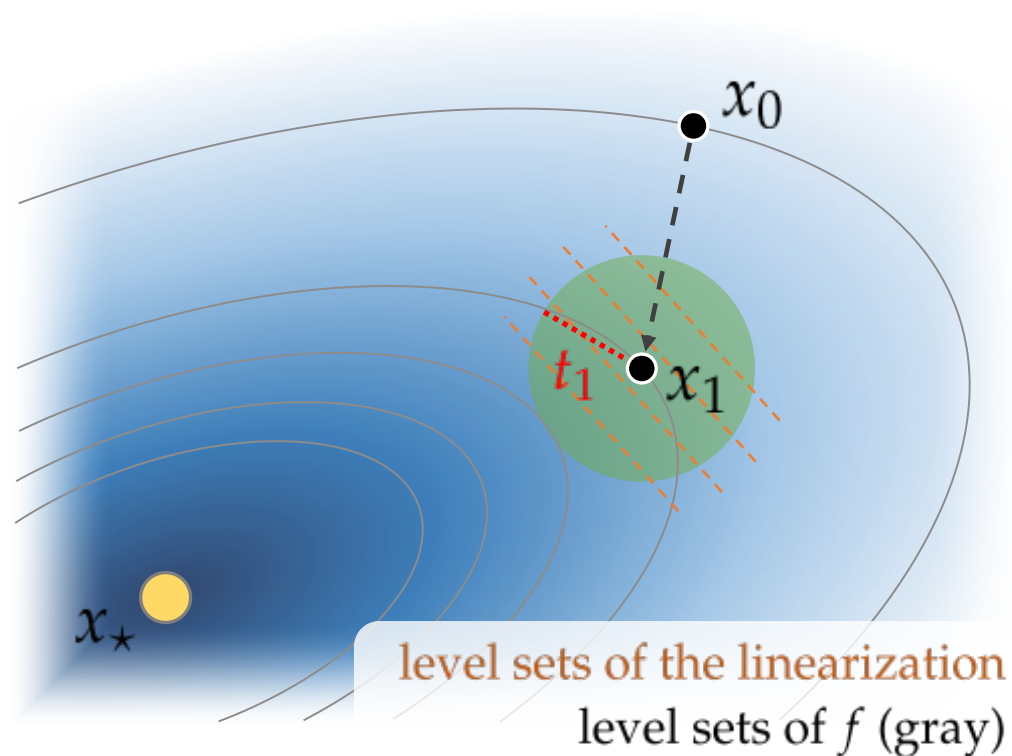
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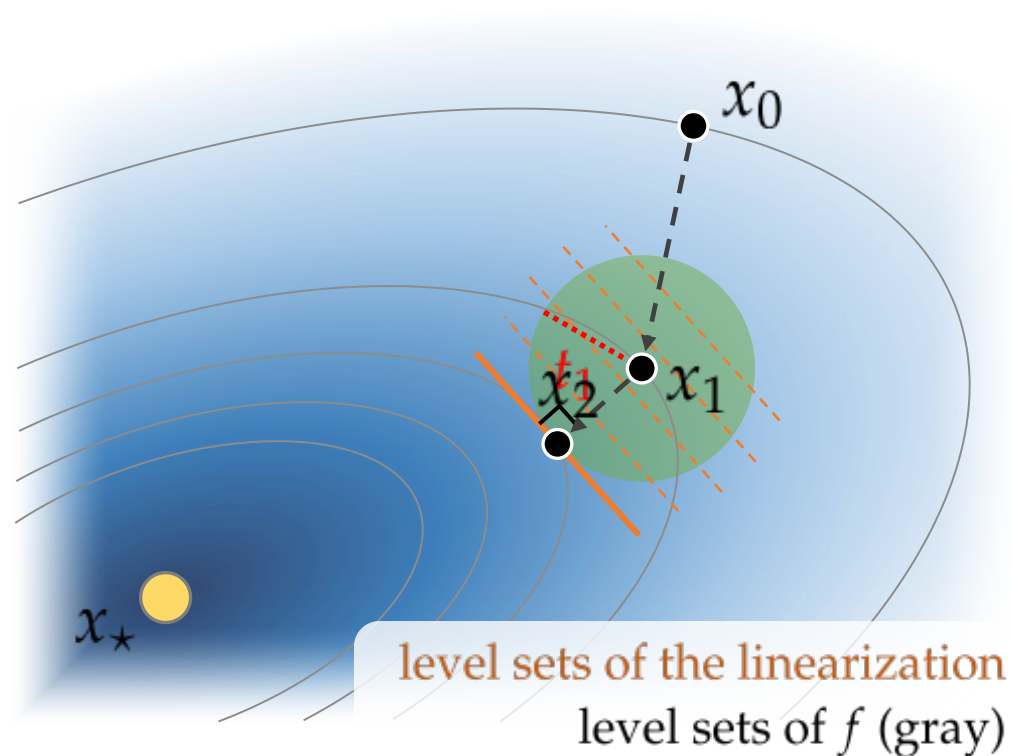
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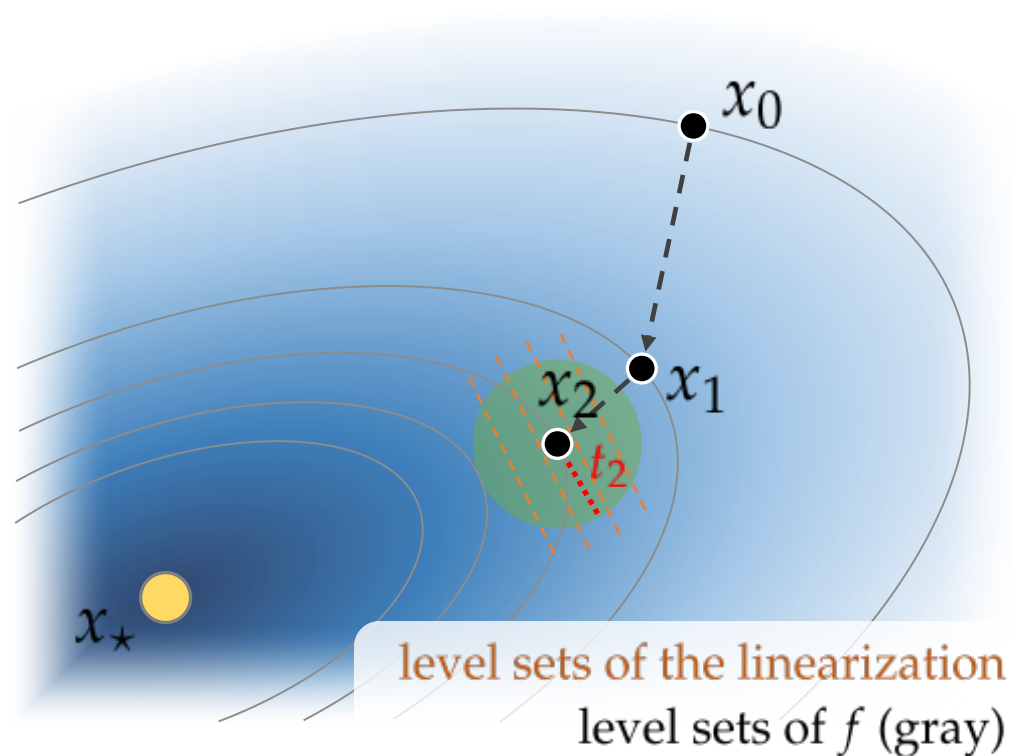
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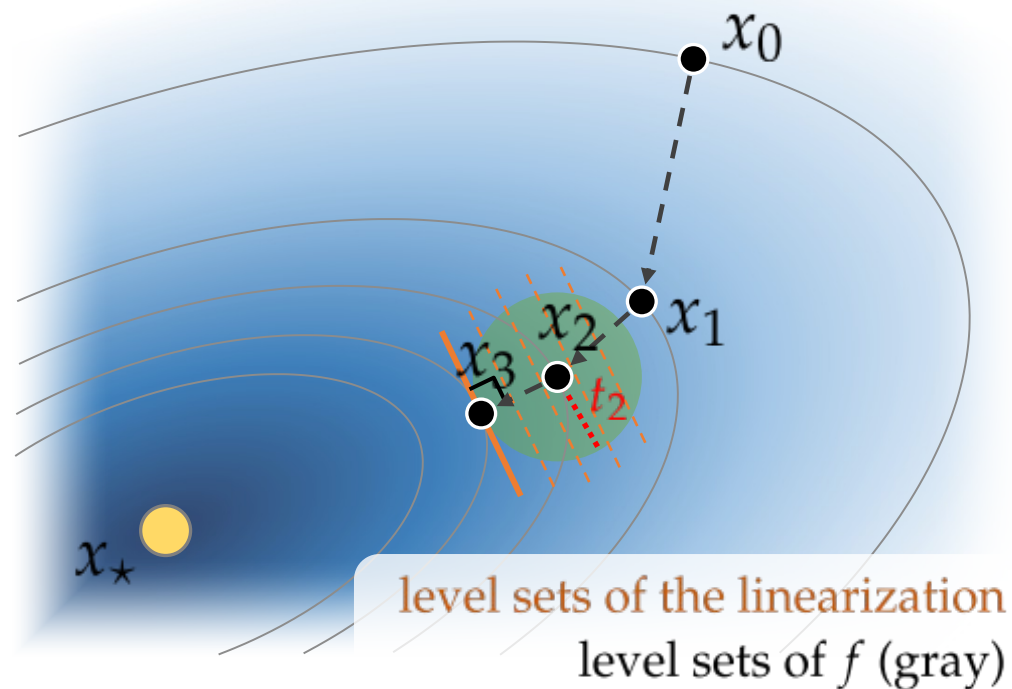
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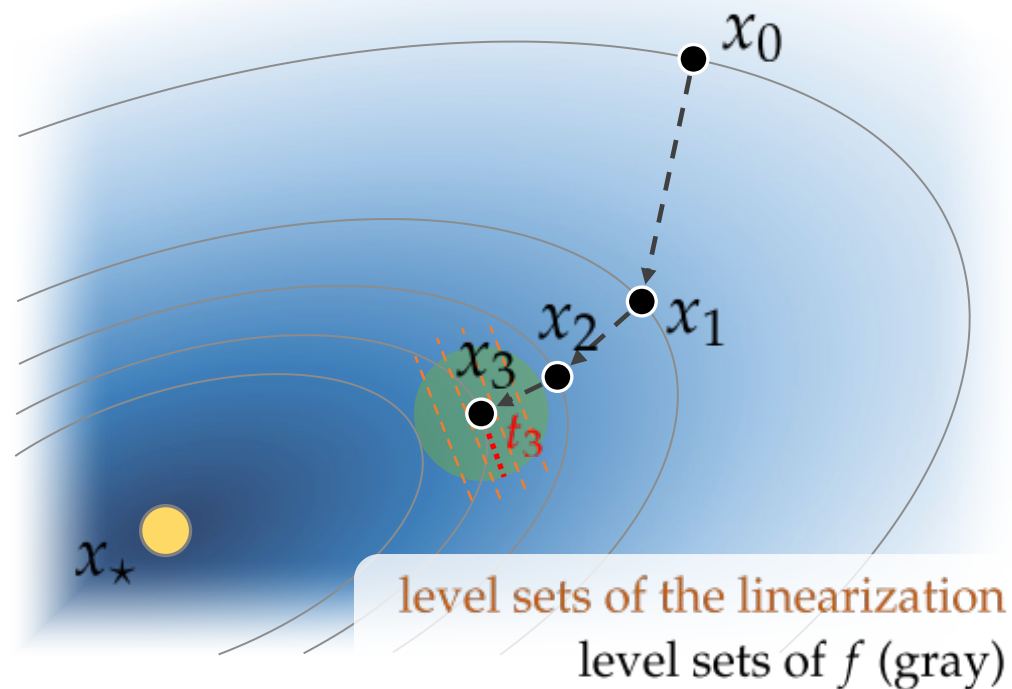
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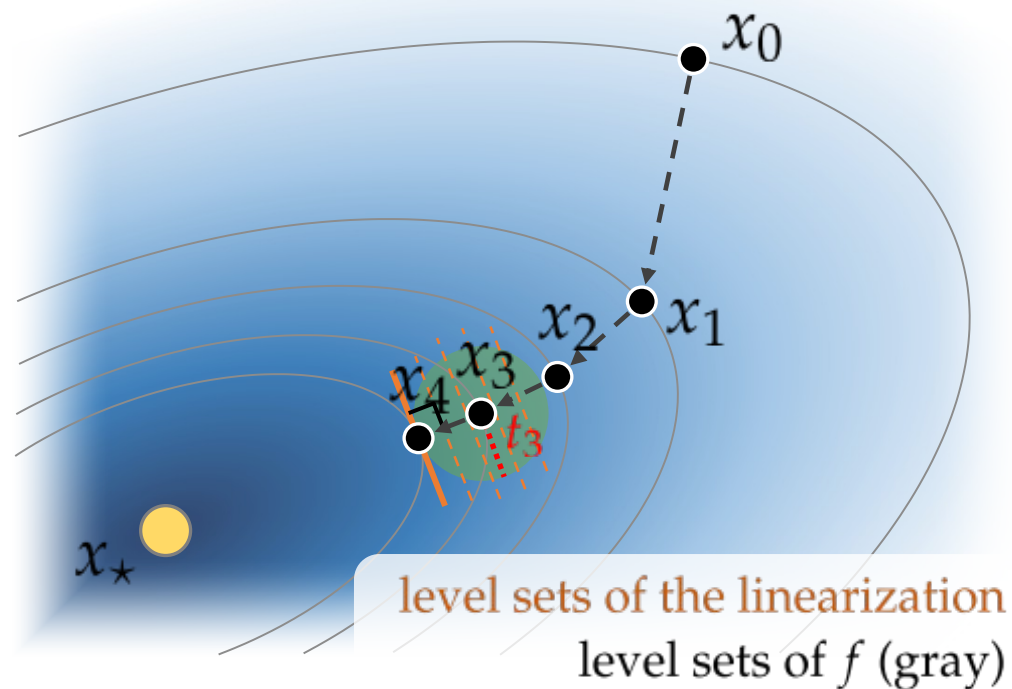
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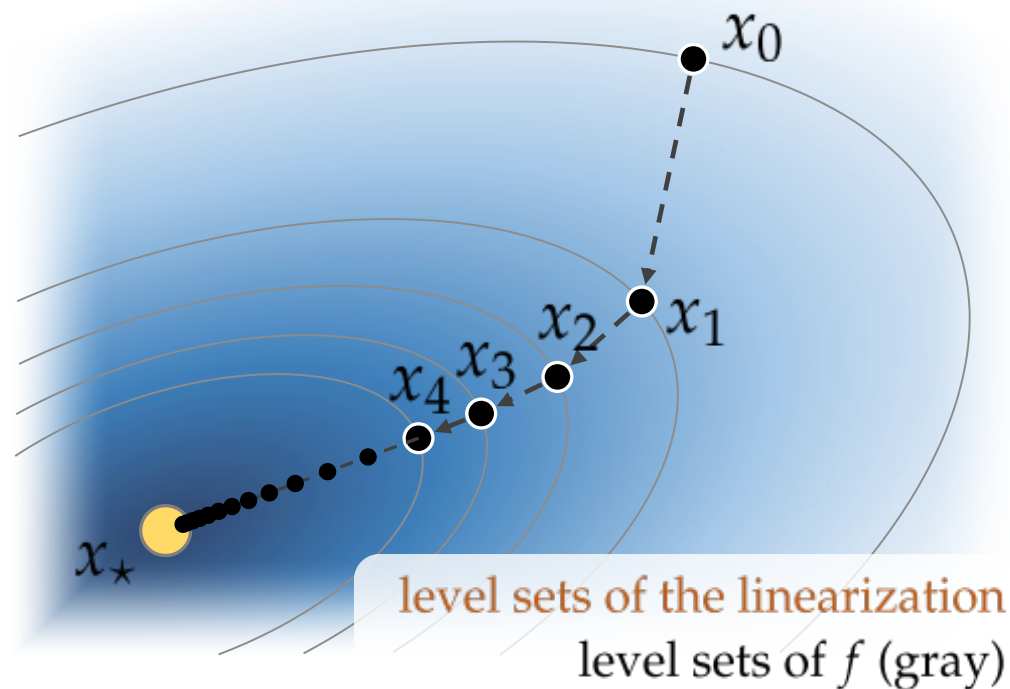
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GD = BPM on the linearization of f ,
with radius $t_k = \gamma \|\nabla f(x_k)\|_2$



SGD = Stochastically Linearized BPM

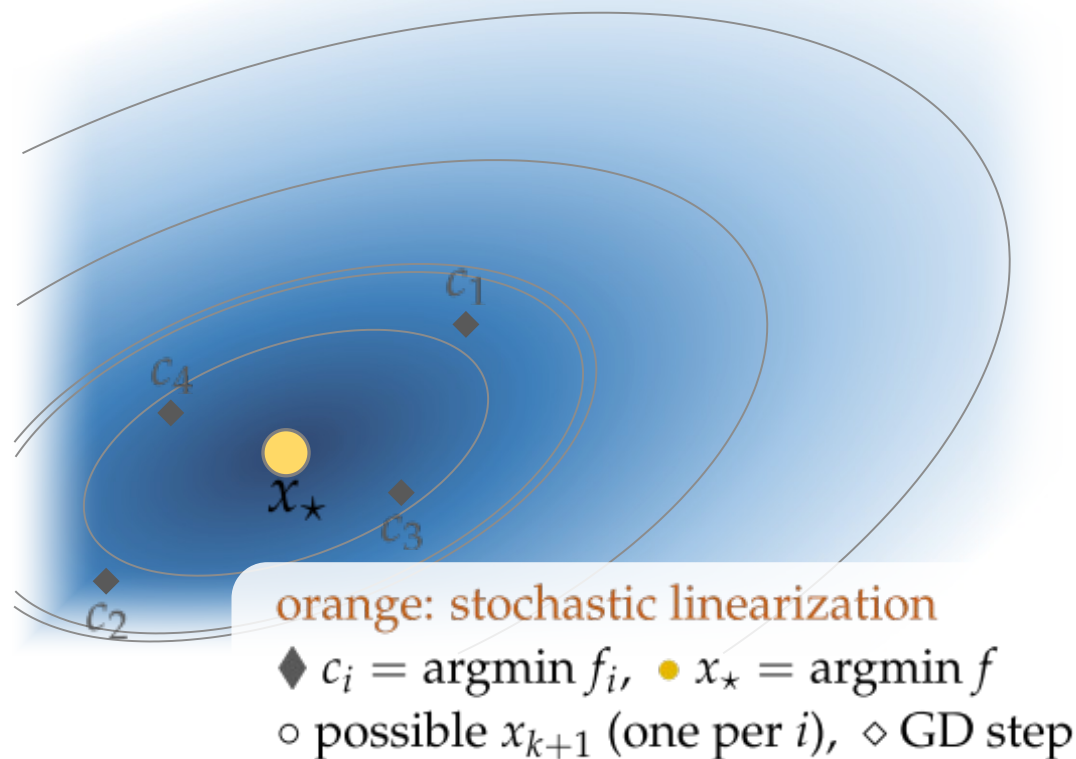
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Example: $f = \frac{1}{4} \sum_{i=1}^4 f_i$, $f_i(x) = \frac{1}{2} \langle a, x - c_i \rangle^2 + 2 \langle b, x - c_i \rangle^2$ (a, b as on the GD slide)

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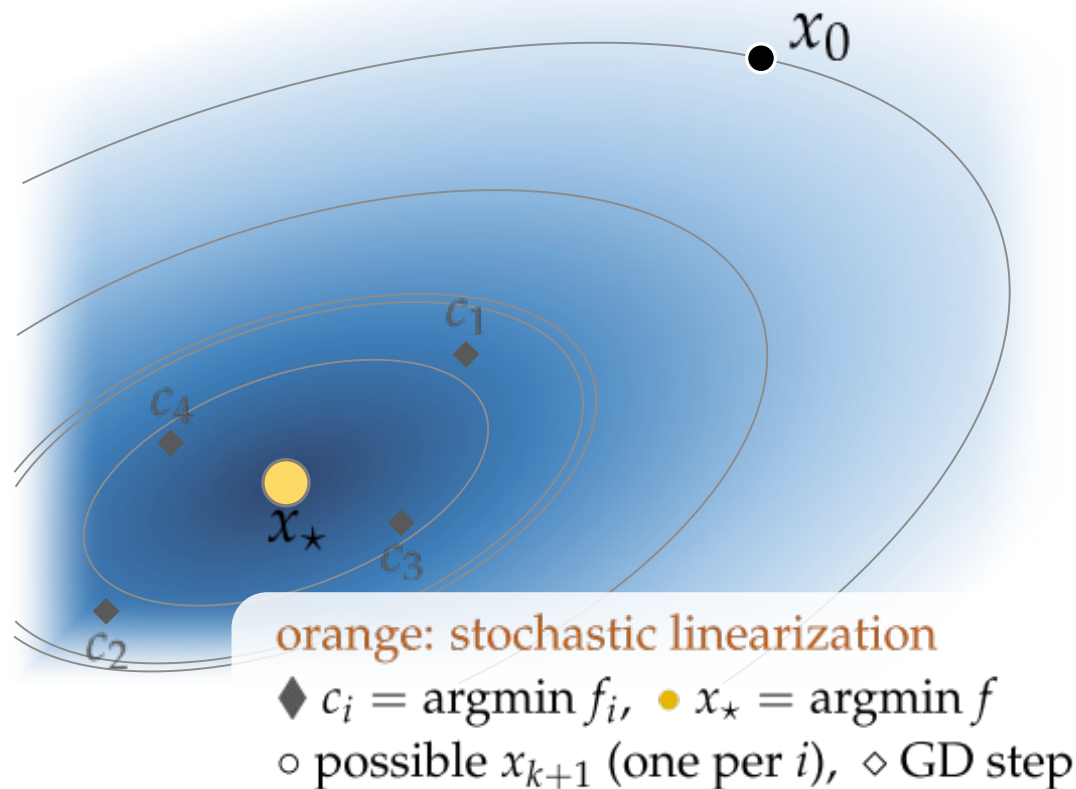
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Linearize a random summand inside the ball

$$\ell_k(y) := f_{i_k}(x_k) + \langle \nabla f_{i_k}(x_k), y - x_k \rangle$$

$$\mathbb{E}_{i_k}[\ell_k(y)] = f(x_k) + \langle \nabla f(x_k), y - x_k \rangle \quad (\text{unbiased})$$



SGD = Stochastically Linearized BPM

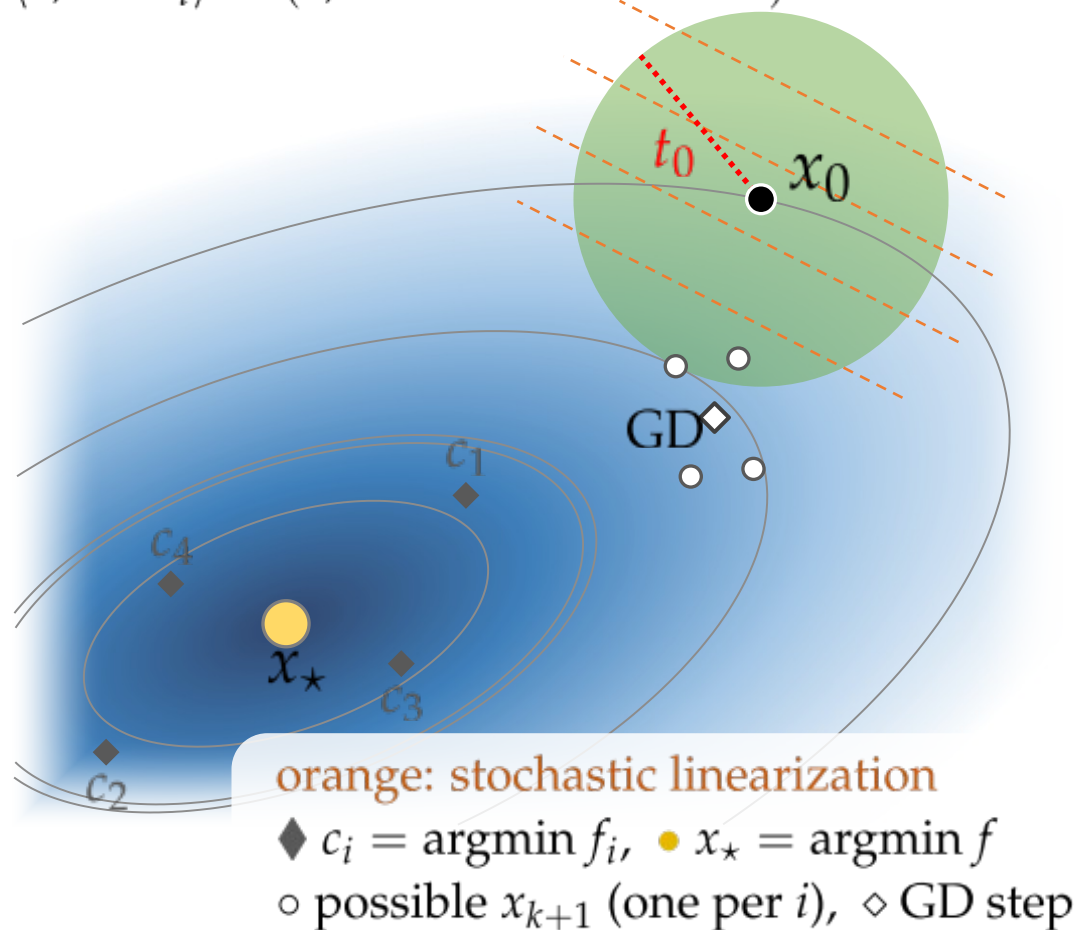
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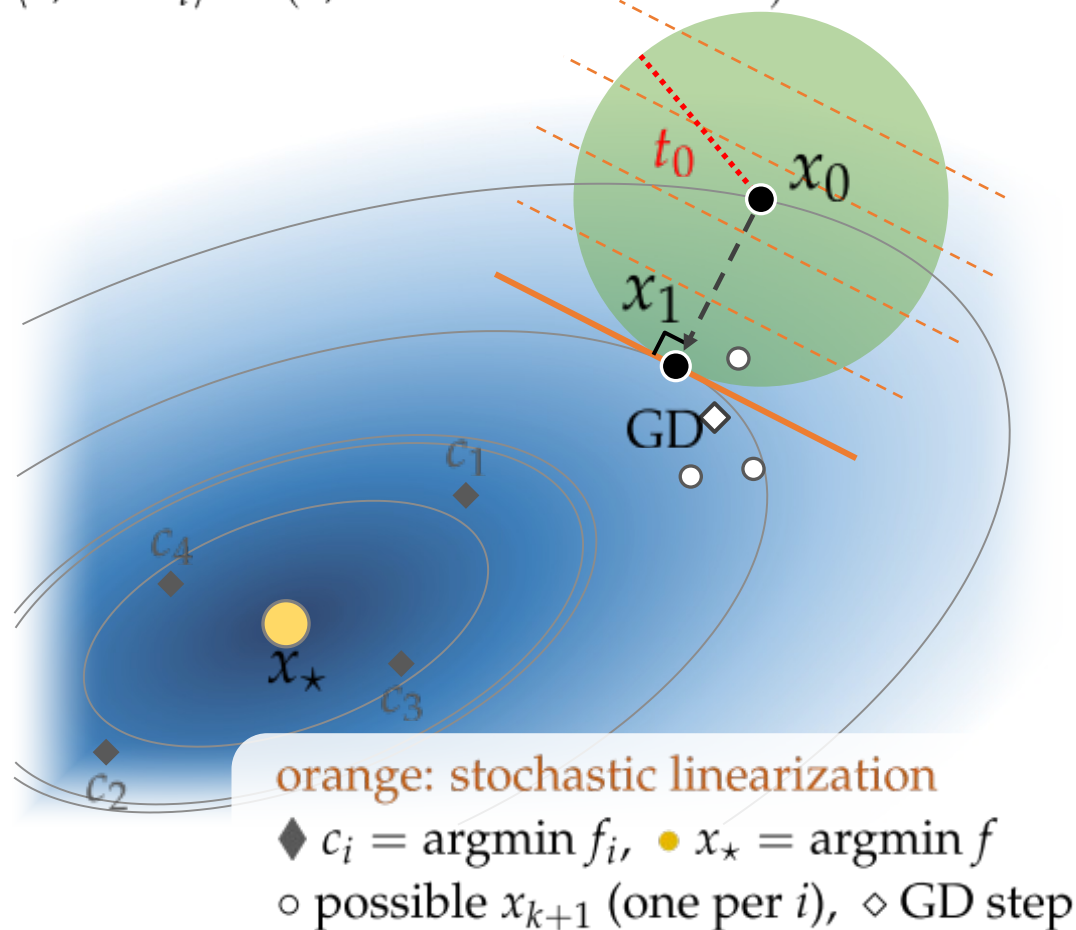
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Linearize a random summand inside the ball

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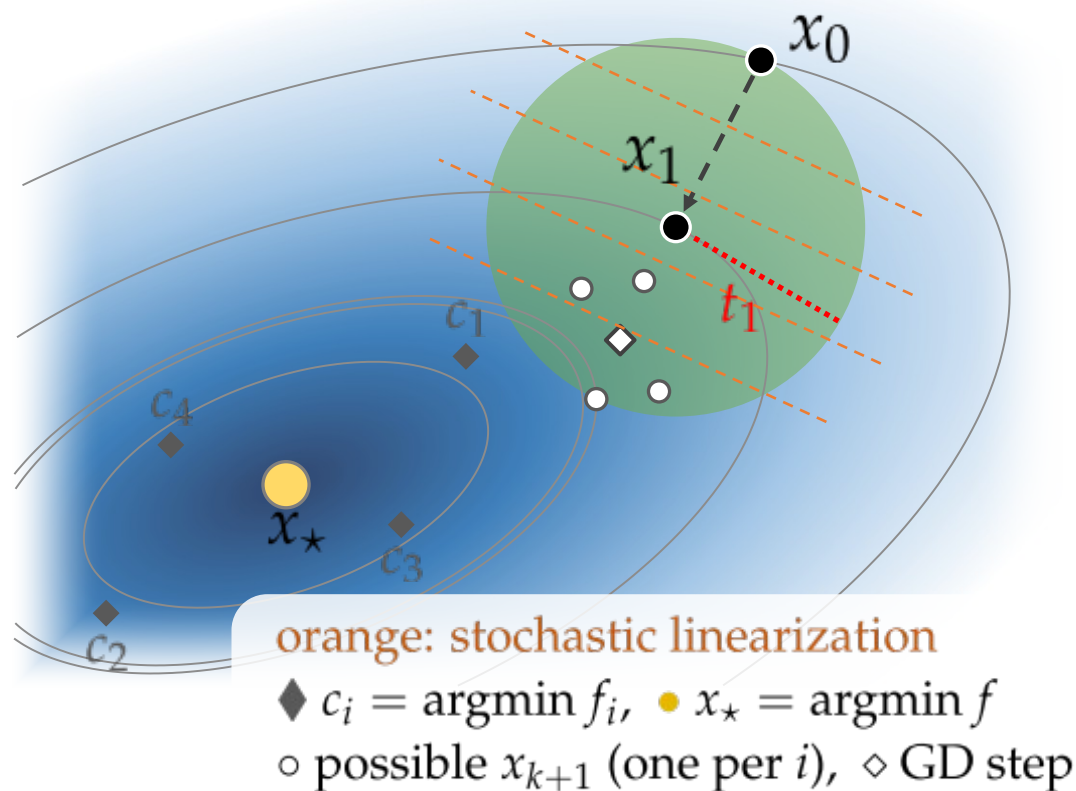
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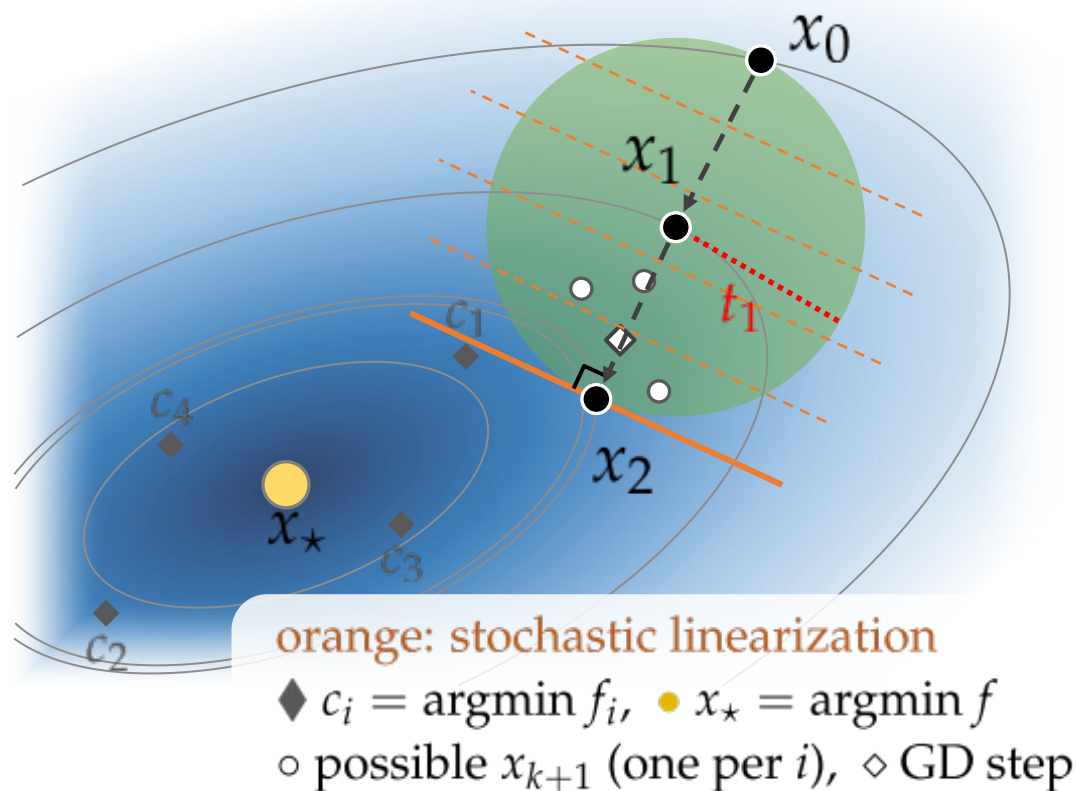
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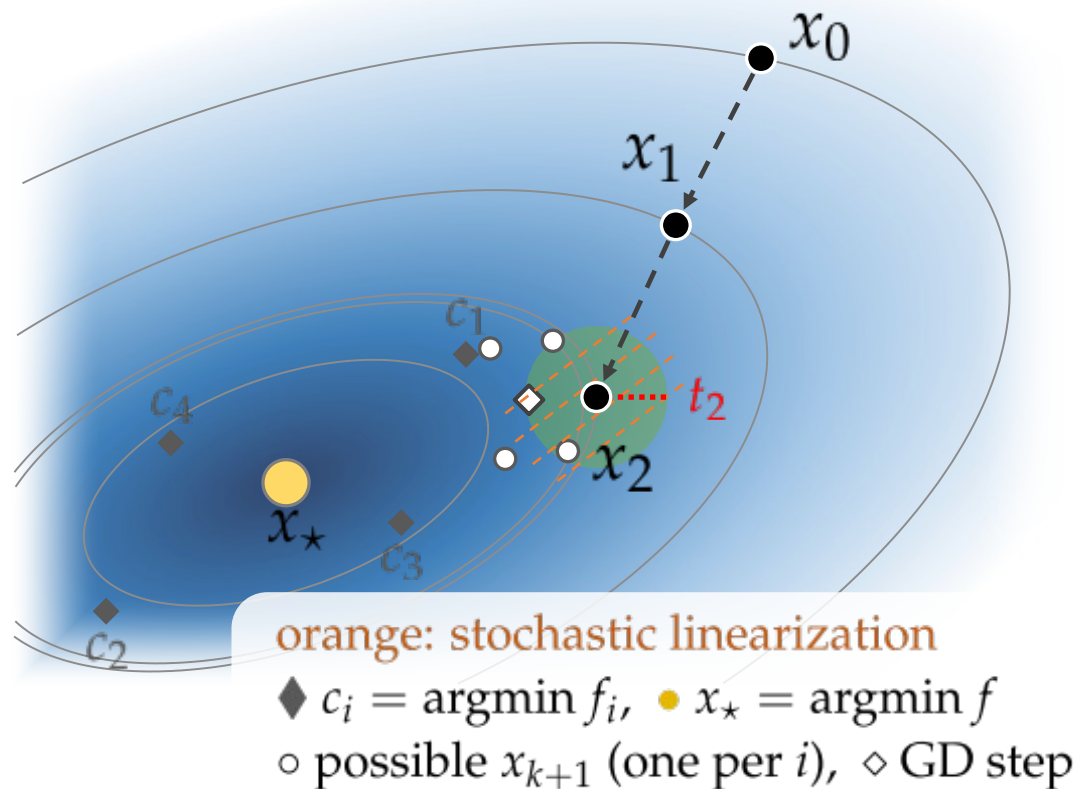
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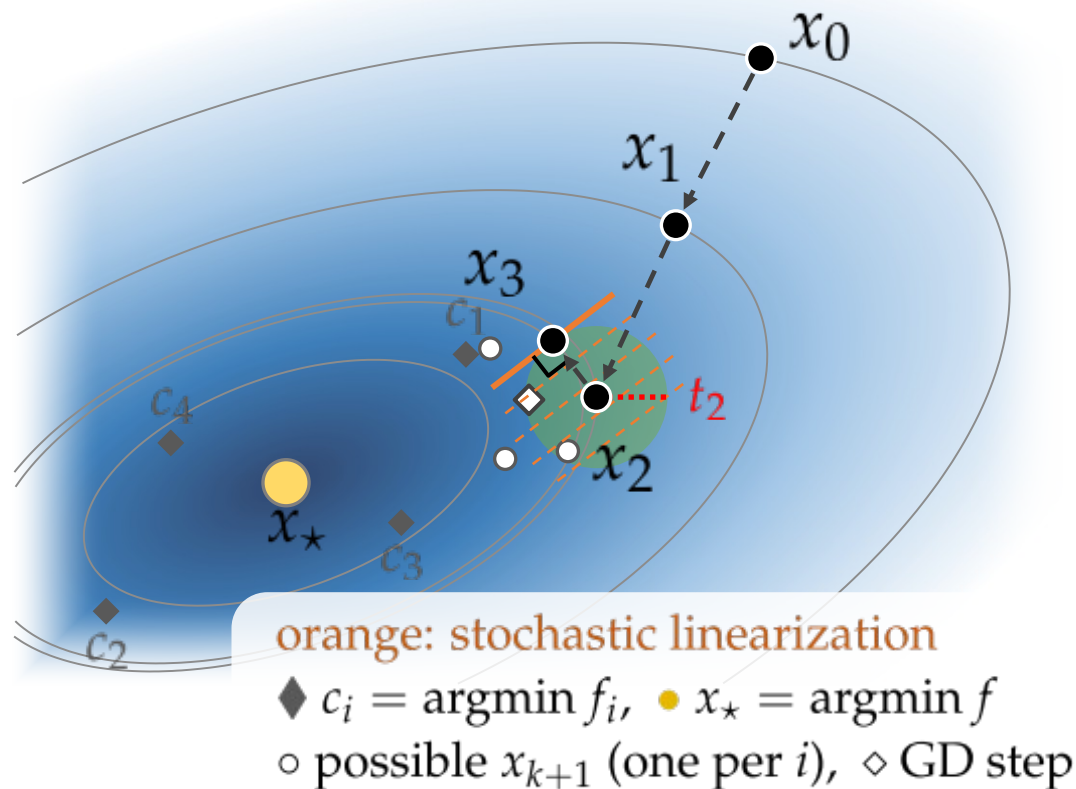
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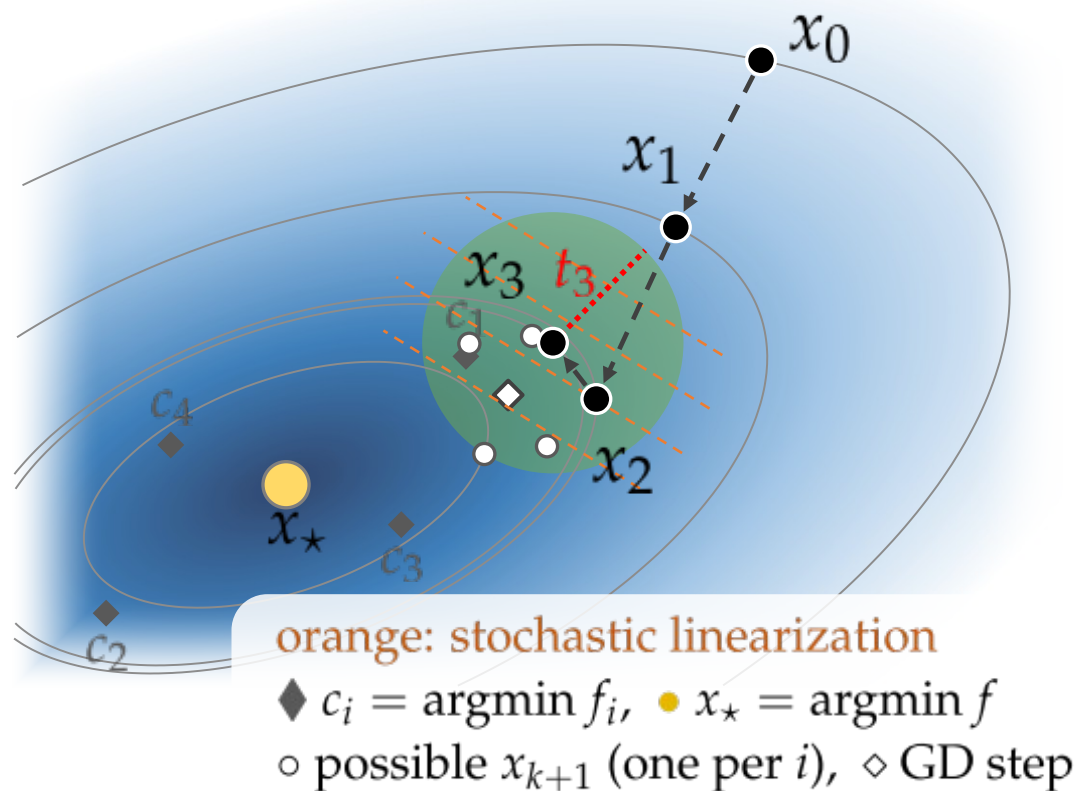
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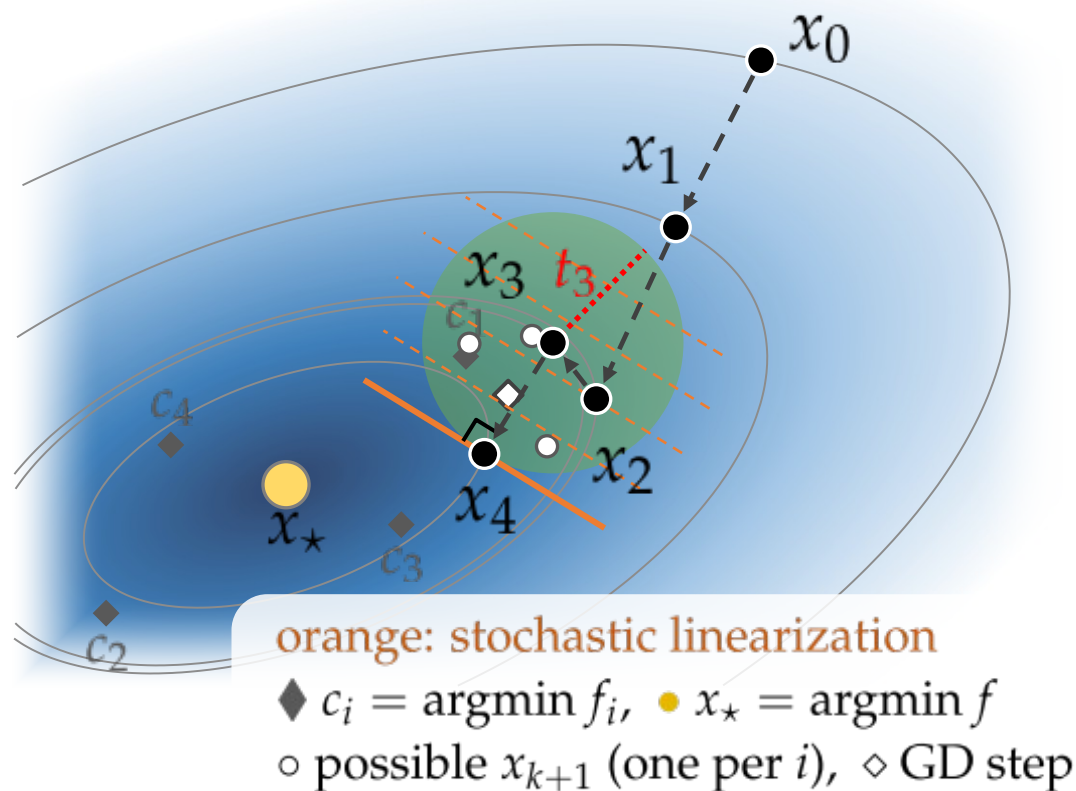
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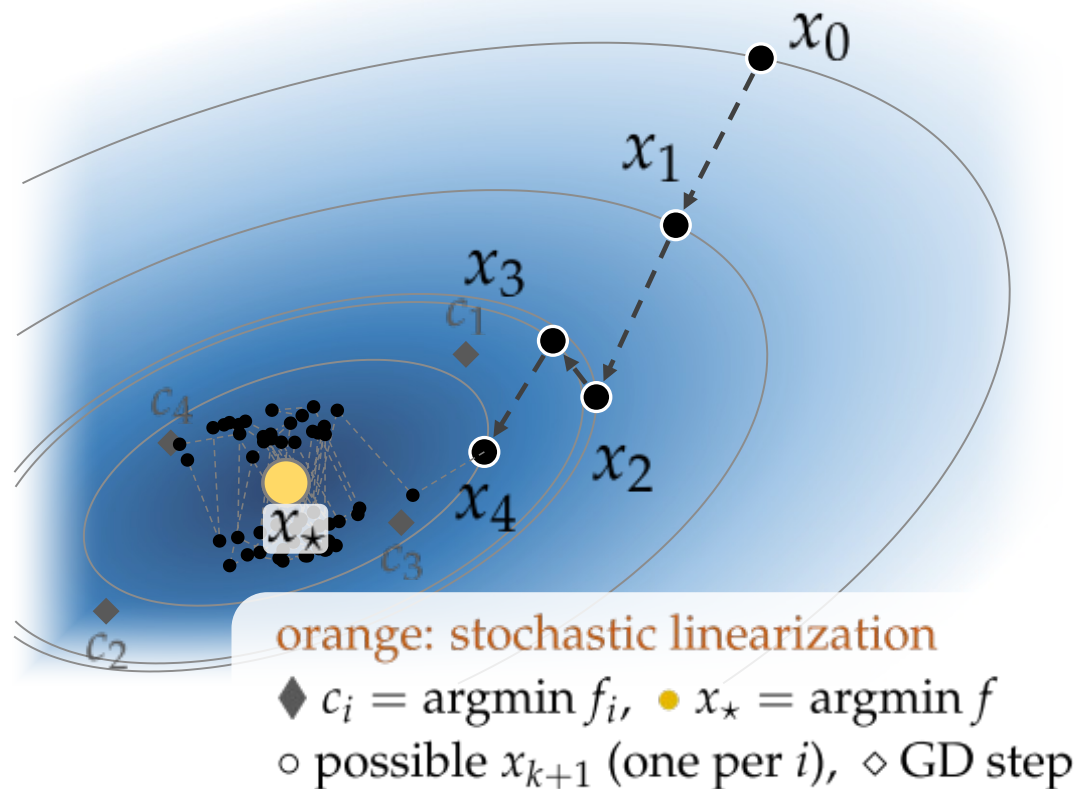
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SGD = BPM on a random, unbiased linearization of f .
 The radius does not vanish at x_* : SGD hovers around x_* .



SGD with Spectral-Norm Balls

$$X_{k+1} \in \operatorname{Argmin} \left\{ \underbrace{f_{i_k}(X_k) + \langle G_k, Y - X_k \rangle}_{\text{stochastic linearization of } f \text{ at } X_k} : \|Y - X_k\|_{2 \rightarrow 2} \leq t_k \right\} \ni X_k - t_k U_k V_k^\top$$

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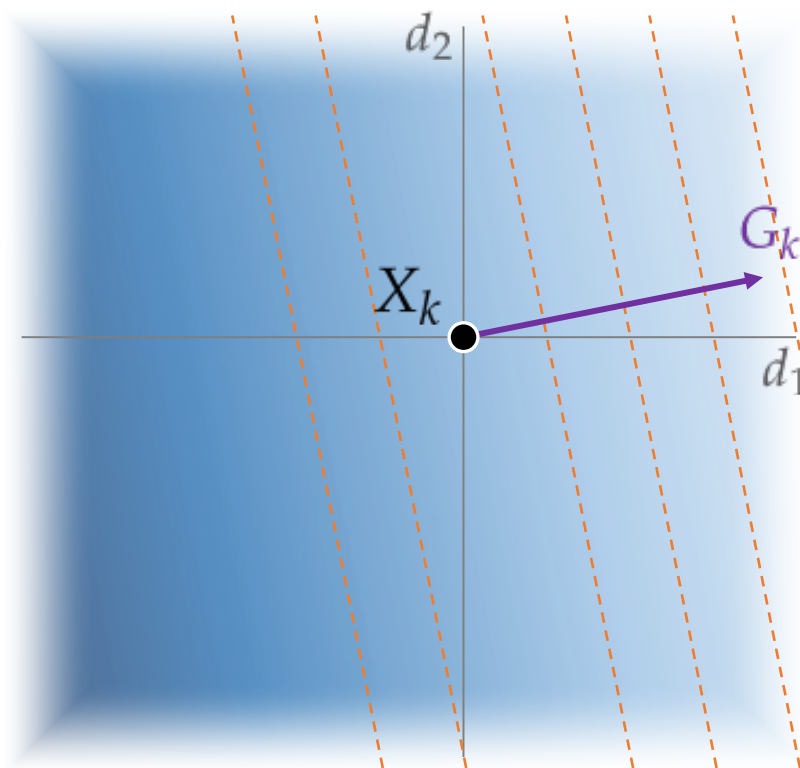
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Linearize, then minimize over a spectral ball

$$\|D\|_F = \|\sigma(D)\|_2, \quad \|D\|_{2 \rightarrow 2} = \|\sigma(D)\|_\infty$$

$$\|D\|_* = \|\sigma(D)\|_1 \text{ (nuclear norm)}$$



green: $\|D\|_{2 \rightarrow 2} \leq t_k$
dashed: $\|D\|_F \leq t_k$
orange: $\langle G_k, \cdot \rangle$
 $\sigma(G_k) = (3, 0.6)$



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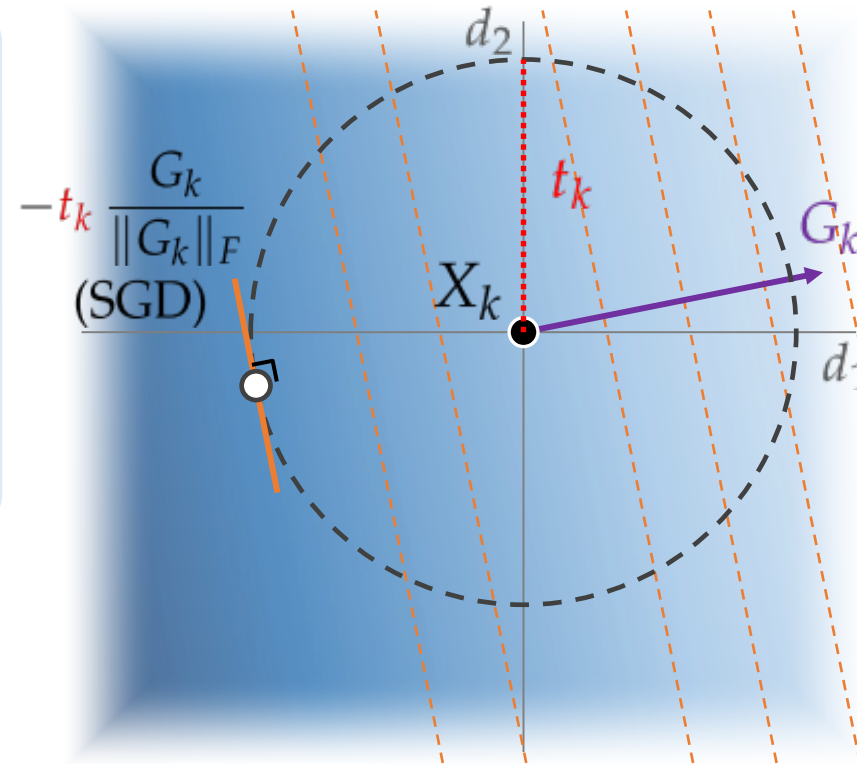
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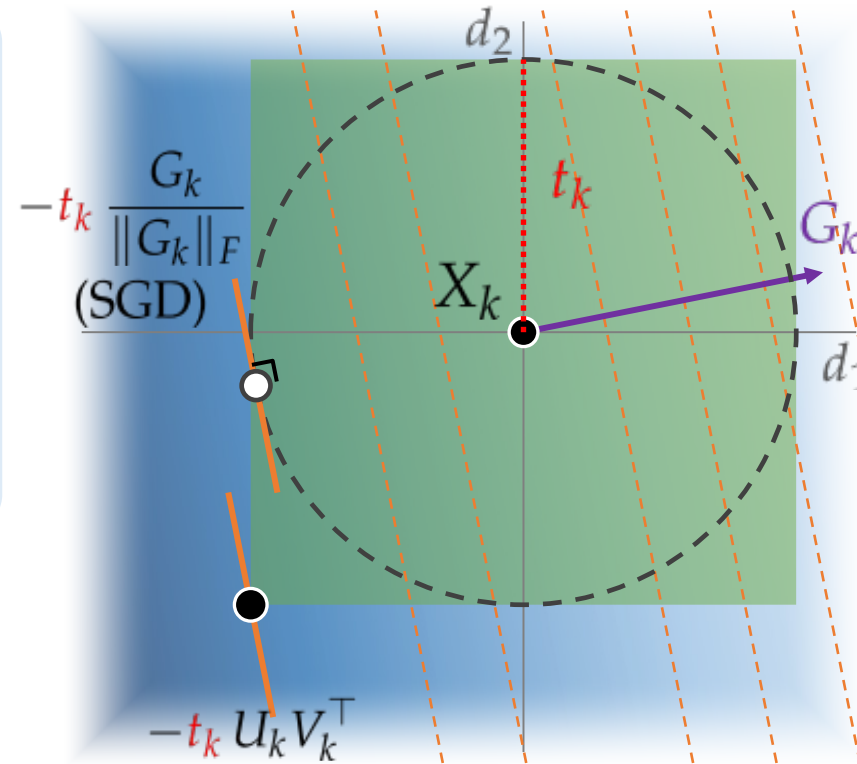
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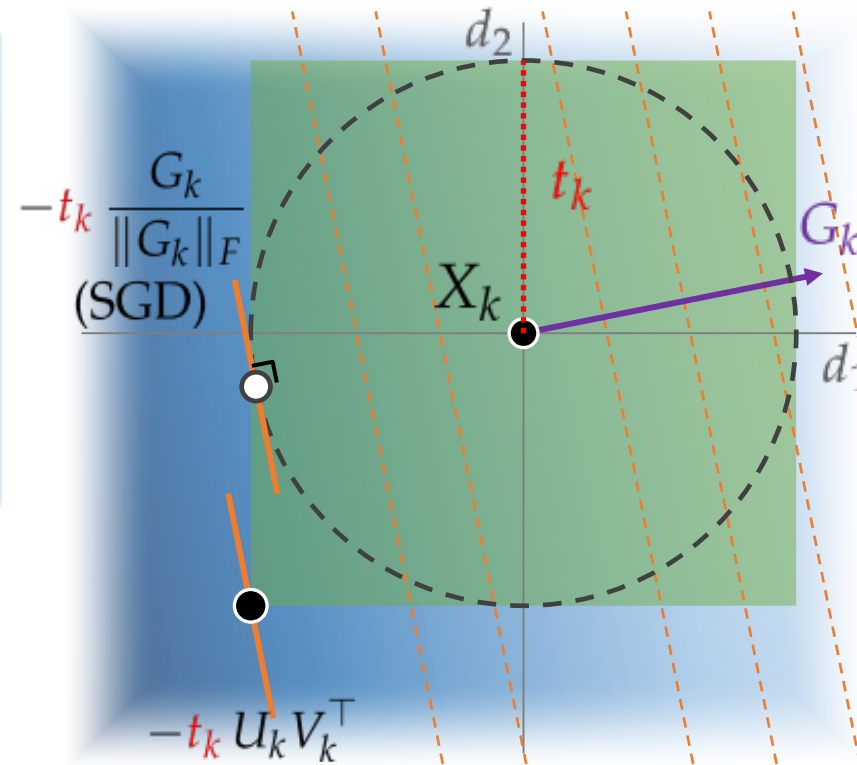
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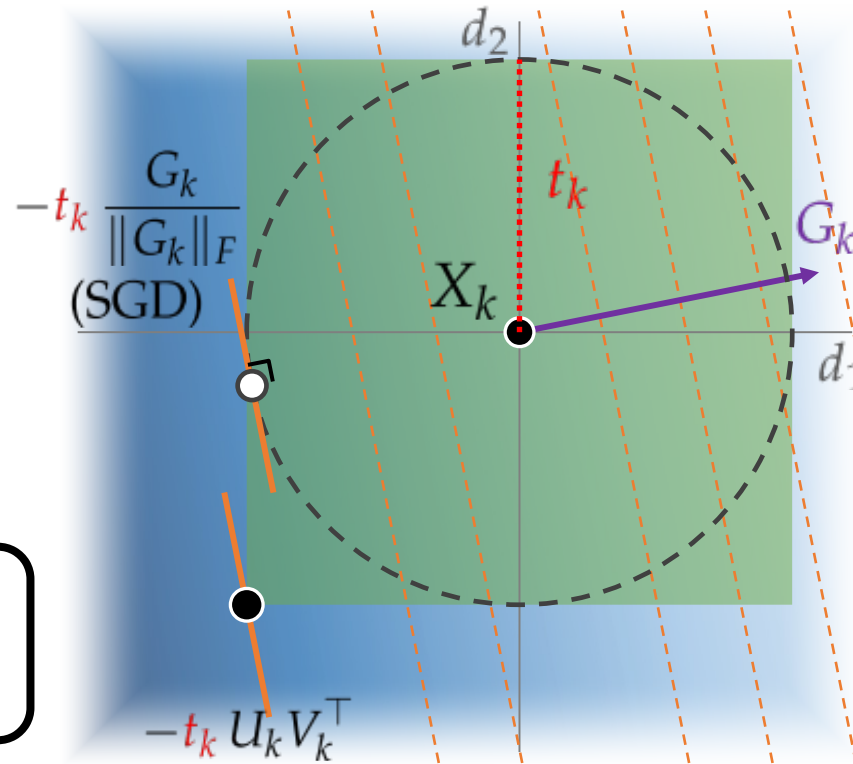
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Same recipe as SGD, only the ball changes:
the step is the orthogonalized gradient $-t_k U_k V_k^\top$.



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Muon: Adding Momentum and Newton–Schulz

$$M_k = \beta M_{k-1} + (1 - \beta) G_k, \quad X_{k+1} = X_k - \eta \text{NS}_5(M_k) \approx X_k - \eta U_k V_k^\top \quad (\text{Jordan et al., 2024})$$

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Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, Jeremy Bernstein

Muon: An optimizer for hidden layers in neural networks

kellerjordan.github.io/posts/muon, 2024

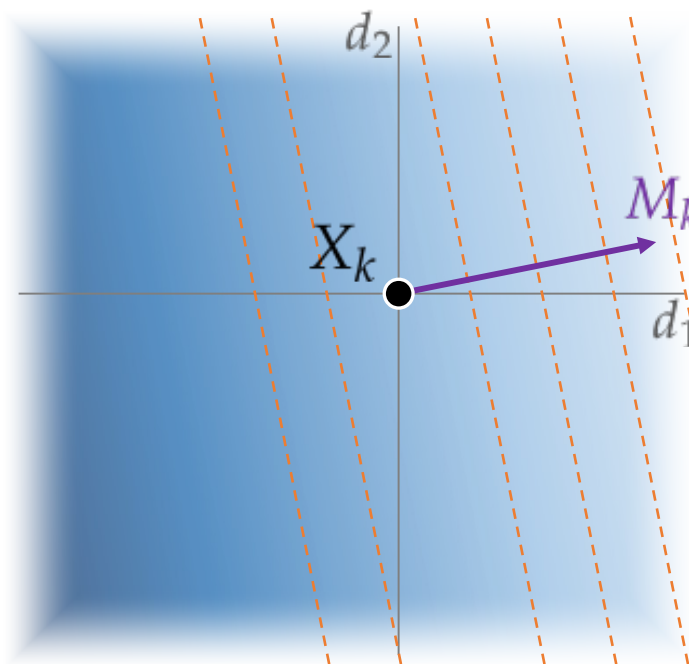
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Spectral SGD (previous slide) + three changes

1. momentum: linearize with M_k , not G_k



green: $\|D\|_{2 \rightarrow 2} \leq \eta$
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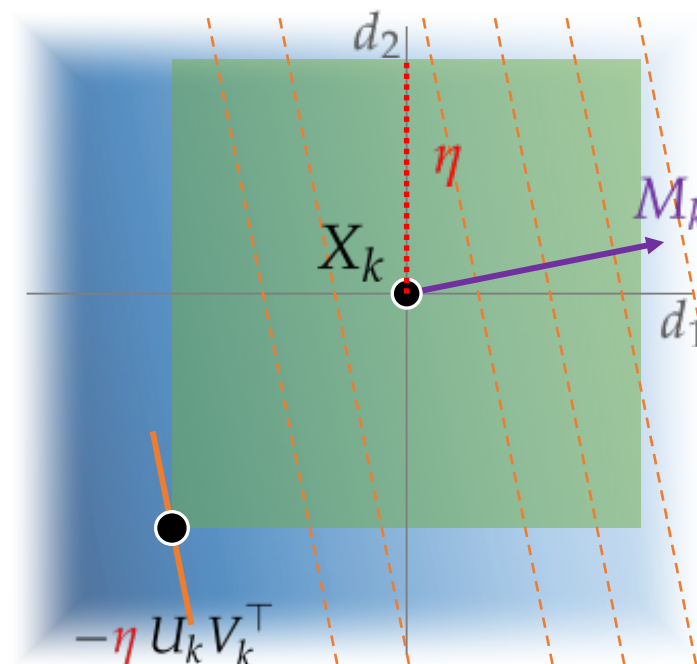
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Spectral SGD (previous slide) + three changes

1. momentum: linearize with M_k , not G_k
2. fixed radius: $t_k \equiv \eta$ (fixed-radius theory)



green: $\|D\|_{2 \rightarrow 2} \leq \eta$
dashed: $\|D\|_F \leq \eta$
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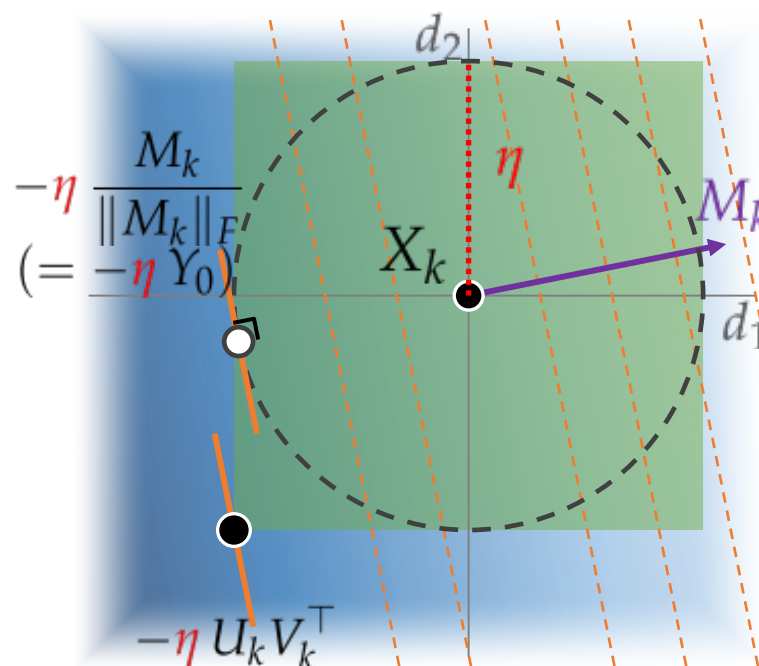
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$$M_k = U_k \Sigma_k V_k^\top \text{ (reduced SVD) : } X_k - \eta U_k V_k^\top \in \text{Argmin} \{ \langle M_k, Y - X_k \rangle : \|Y - X_k\|_{2 \rightarrow 2} \leq \eta \}$$

Spectral SGD (previous slide) + three changes

1. momentum: linearize with M_k , not G_k
2. fixed radius: $t_k \equiv \eta$ (fixed-radius theory)
3. inexact ball step: $U_k V_k^\top \approx \text{NS}_5(M_k)$



green: $\|D\|_{2 \rightarrow 2} \leq \eta$
dashed: $\|D\|_F \leq \eta$
orange: $\langle M_k, \cdot \rangle$
purple: $-\eta Y_j$
 $\sigma(M_k) = (3, 0.6)$



Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, Jeremy Bernstein

Muon: An optimizer for hidden layers in neural networks

kellerjordan.github.io/posts/muon, 2024

Muon: Adding Momentum and Newton–Schulz

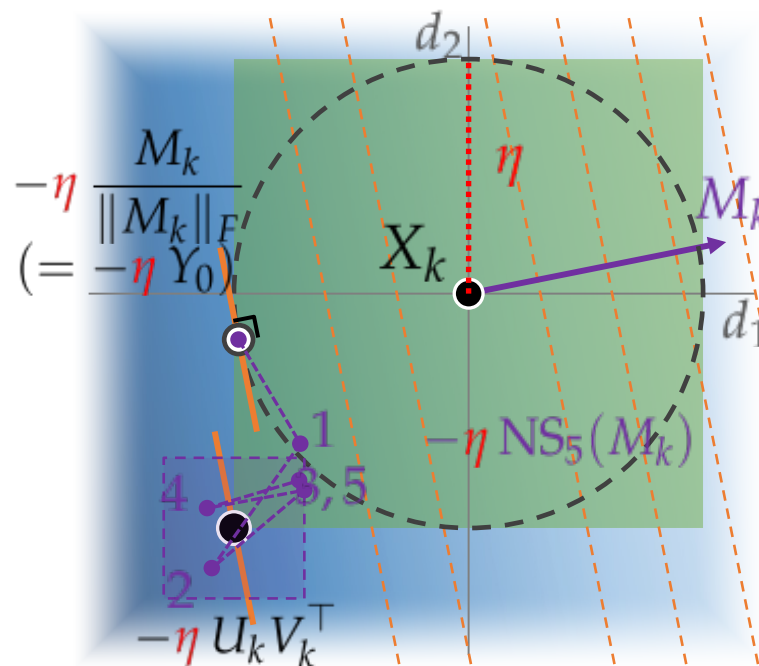
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NS₅: 5 Newton–Schulz steps, $Y_0 \rightarrow Y_5 \approx U_k V_k^\top$



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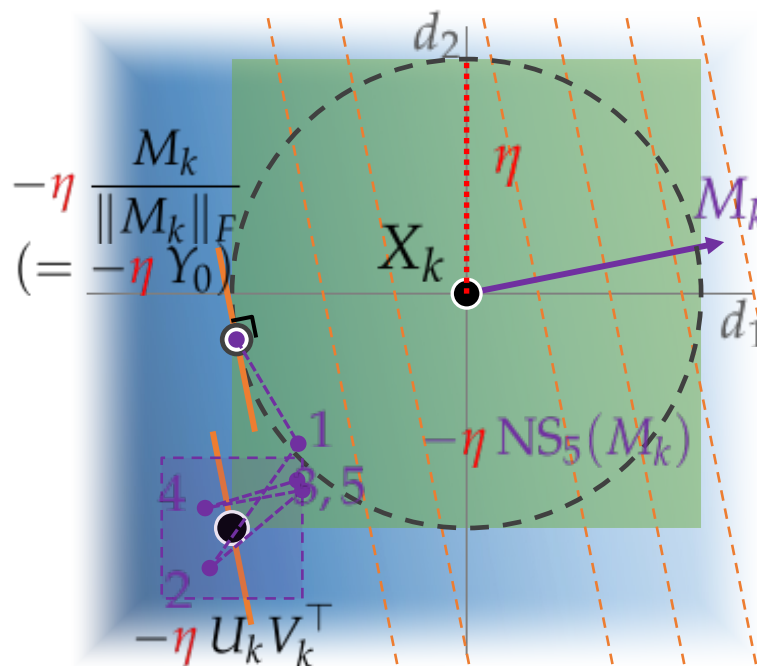
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NS₅: 5 Newton–Schulz steps, $Y_0 \rightarrow Y_5 \approx U_k V_k^\top$

Muon $\approx$ BPM: *momentum* linearization,
spectral ball of *fixed* radius η ,
ball step computed *inexactly*.



green: $\|D\|_{2 \rightarrow 2} \leq \eta$
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BroxGD: Linearized BPM with Constraints

$$x_{k+1} \in \operatorname{Argmin} \left\{ \underbrace{f(x_k) + \langle \nabla f(x_k), y - x_k \rangle}_{\text{linearization of } f \text{ at } x_k} : \underbrace{y \in \mathcal{X}}_{\text{kept exactly}}, \|y - x_k\|_2 \leq t_k \right\}$$

Problem: $\min_{x \in \mathcal{X}} f(x)$, $\mathcal{X}$ closed, convex $\iff \min_x f(x) + \iota_{\mathcal{X}}(x)$: BPM with only f linearized



P.R., Kaja Grunkowska, Hanmin Li

Broximal gradient descent: a projection-free sister of projected gradient descent

arXiv:2605.08850, 2026



P.R., Ammar Mahran

Local LMO is Secretly a Projection Method!

arXiv:2609.33383, 2026

BroxGD: Linearized BPM with Constraints

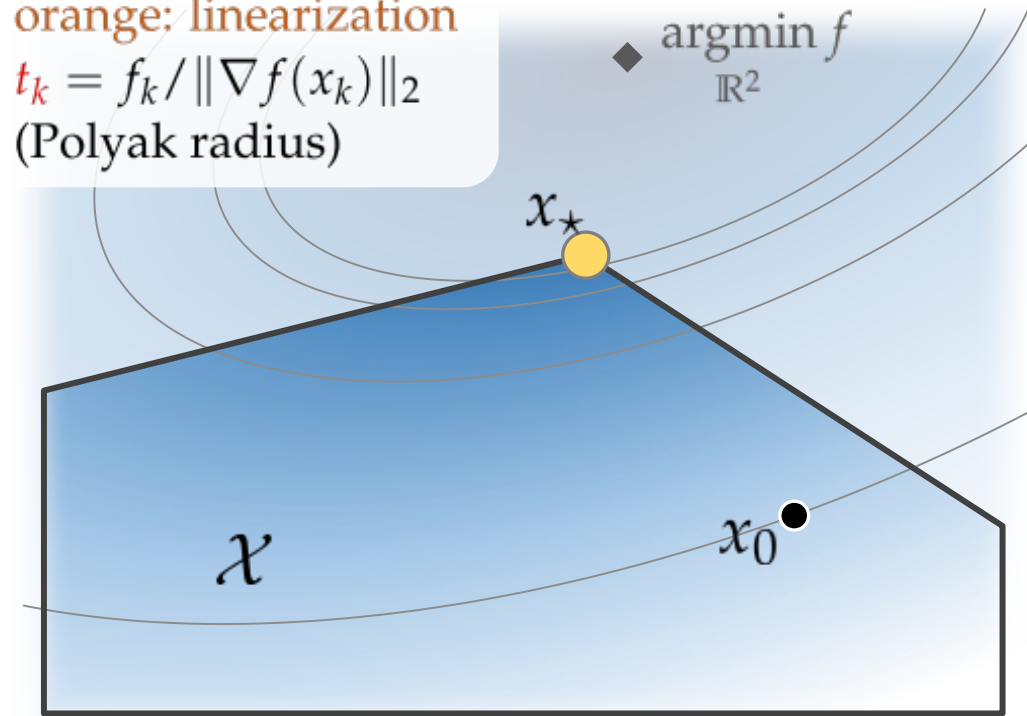
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Between projected GD and Frank–Wolfe
local LMO on $\mathcal{X} \cap \mathcal{B}(x_k, t_k)$: no projection

orange: linearization

$t_k = f_k / \|\nabla f(x_k)\|_2$
(Polyak radius)



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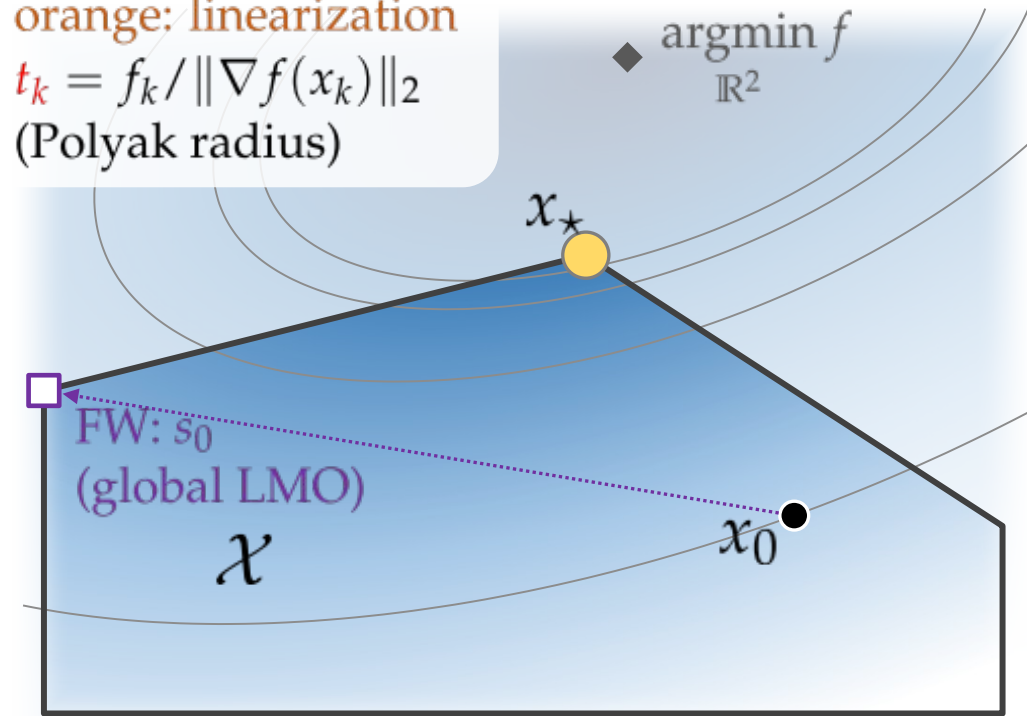
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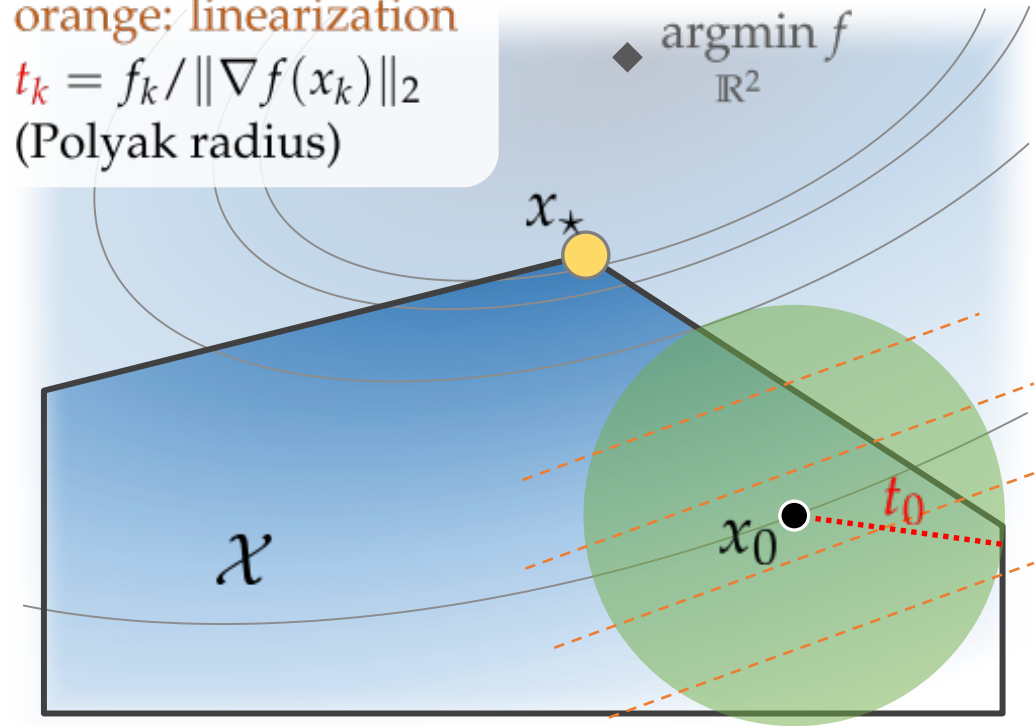
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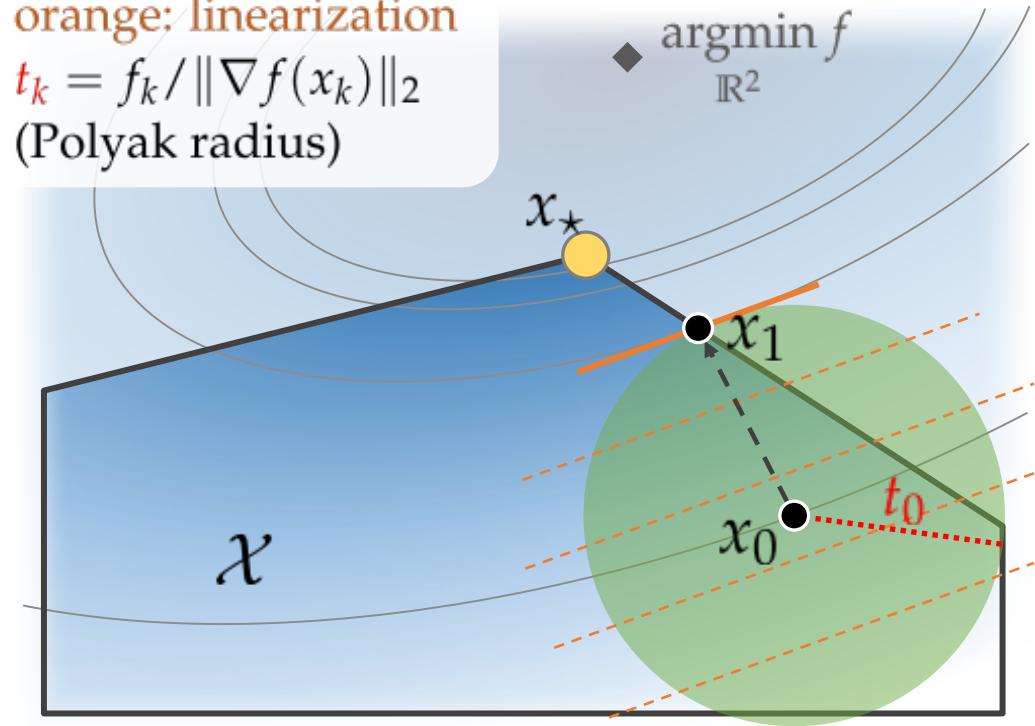
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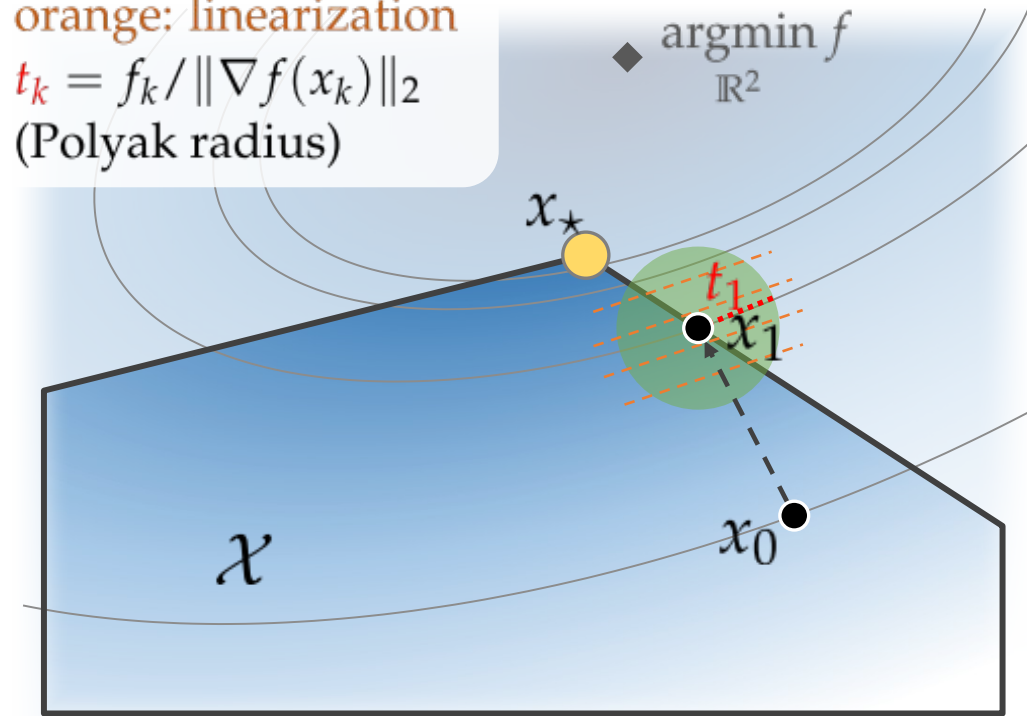
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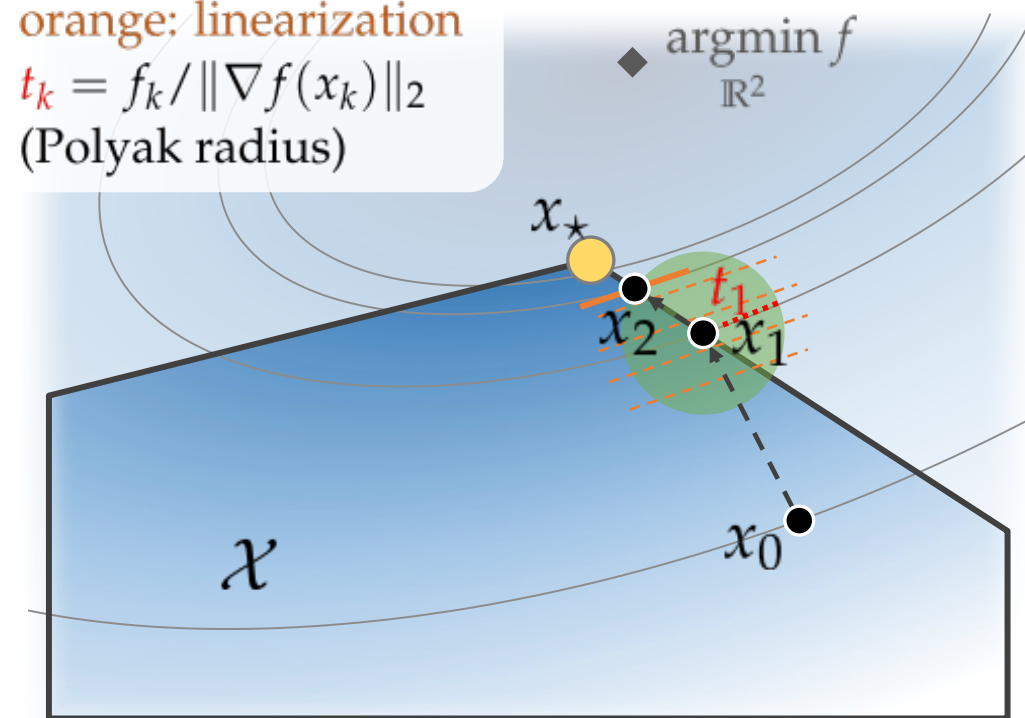
ProjGD: soft ball, needs Proj_χ

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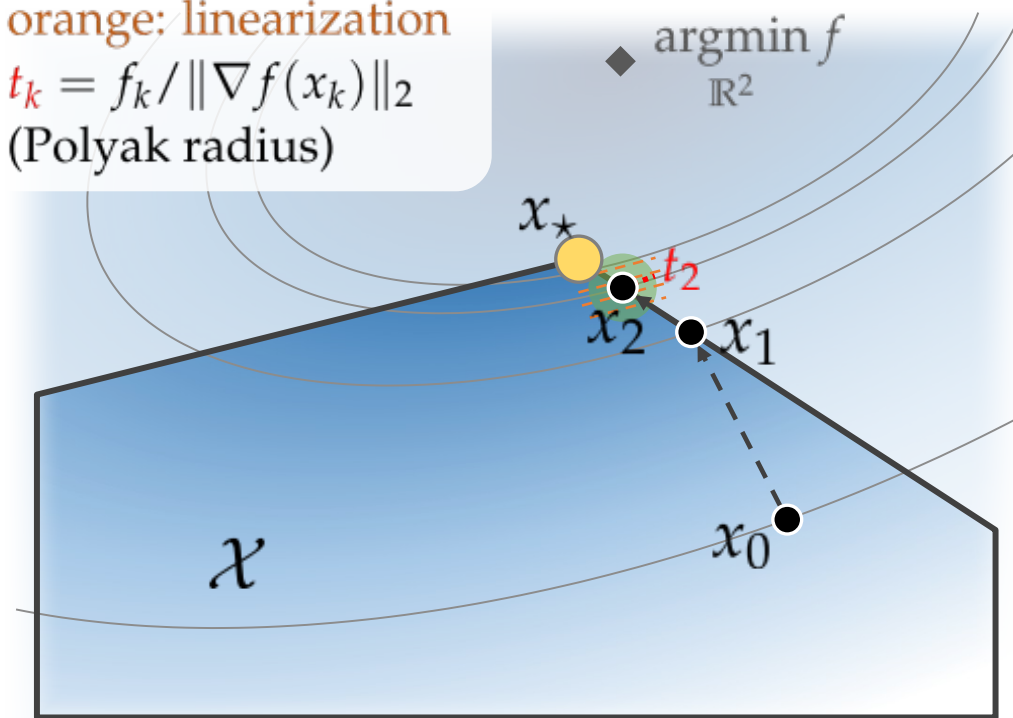
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Between projected GD and Frank–Wolfe

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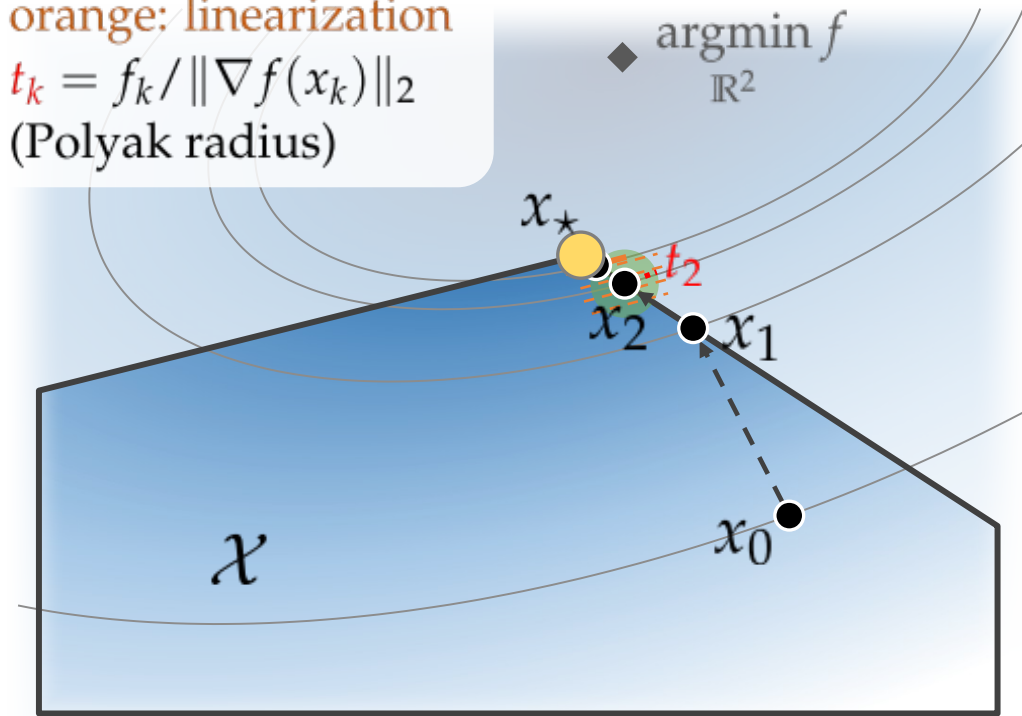
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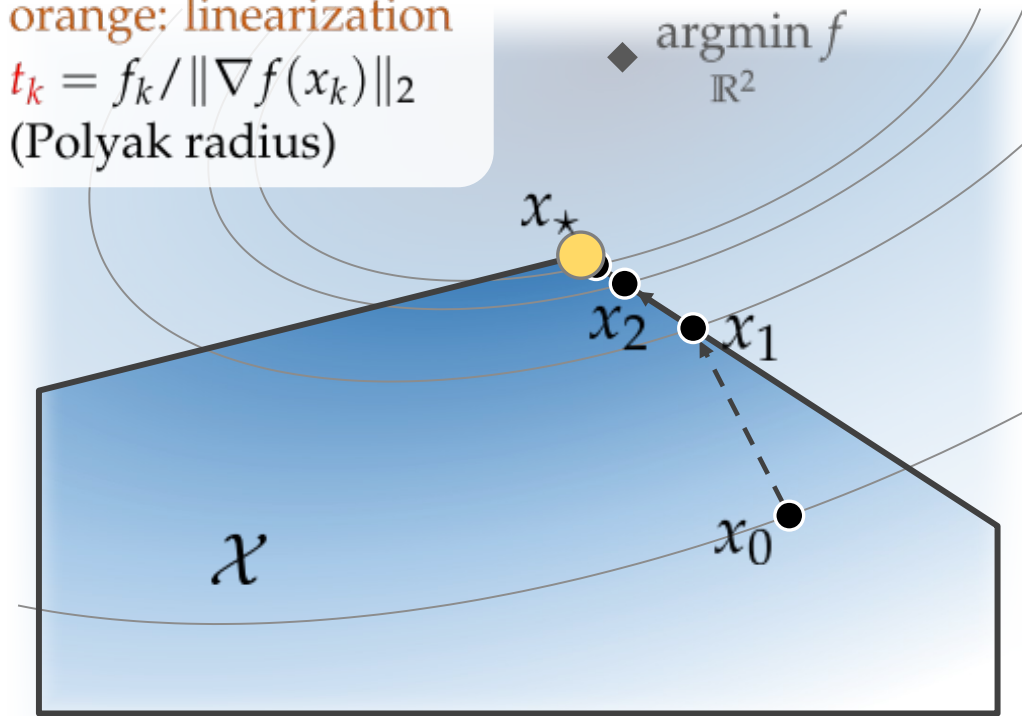
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BroxGD = BPM on $f + \iota_{\mathcal{X}}$ with only f linearized.

Smooth convex, $t_k = R/\sqrt{K}$: $f(\hat{x}_K) - f_{\star} \leq LR^2/(2K)$.

orange: linearization

$t_k = f_k / \|\nabla f(x_k)\|_2$
(Polyak radius)



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arXiv:2609.33383, 2026

Part 5

Proximal Point Method: Four Formulations



P.R., Hanmin Li

A sharper theory of ball-proximal optimization: convergence and radius selection

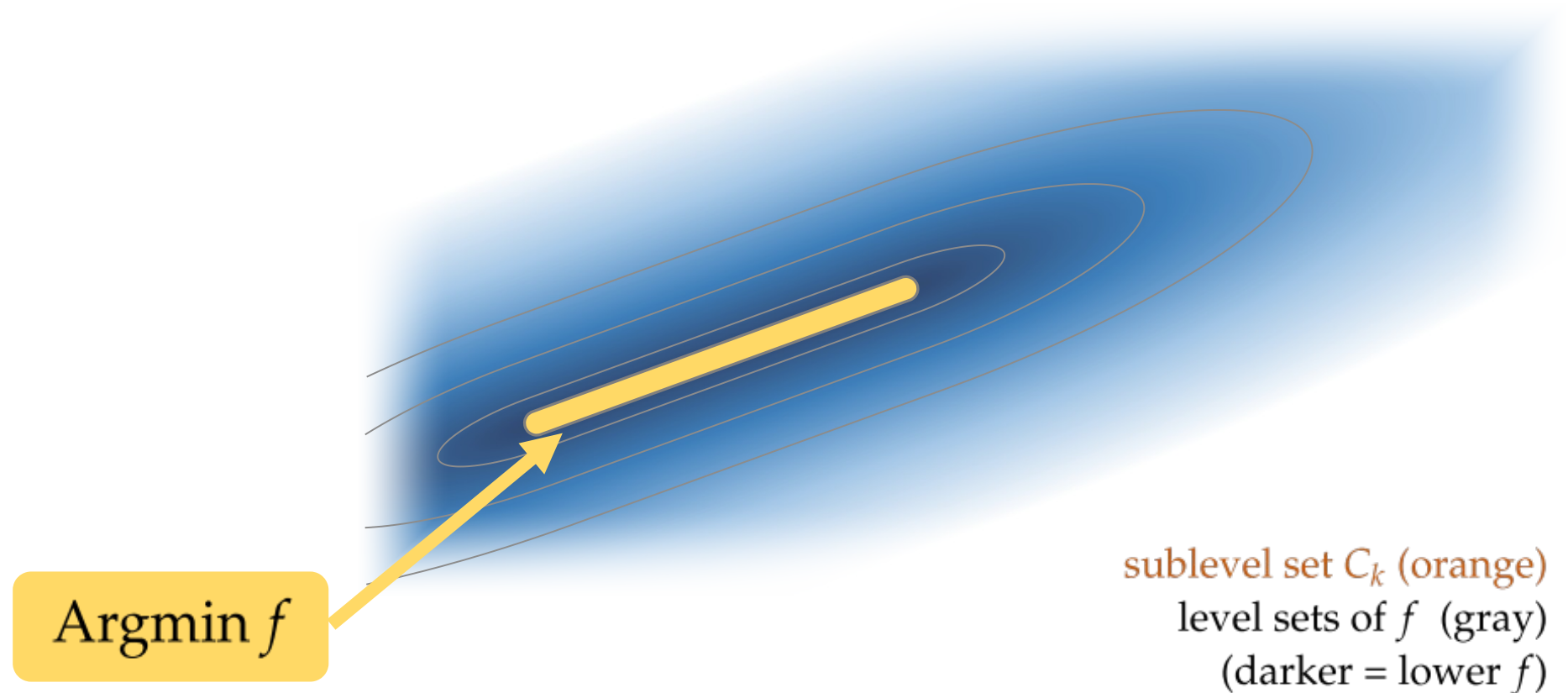
arXiv:2609.35147, 2026

BPM = Projection onto Sublevel Sets

$$x_{k+1} \in \text{brox}_f^t(x_k) \implies x_{k+1} = \text{Proj}_{C_k}(x_k), \quad C_k := \{y : f(y) \leq f(x_{k+1})\}$$

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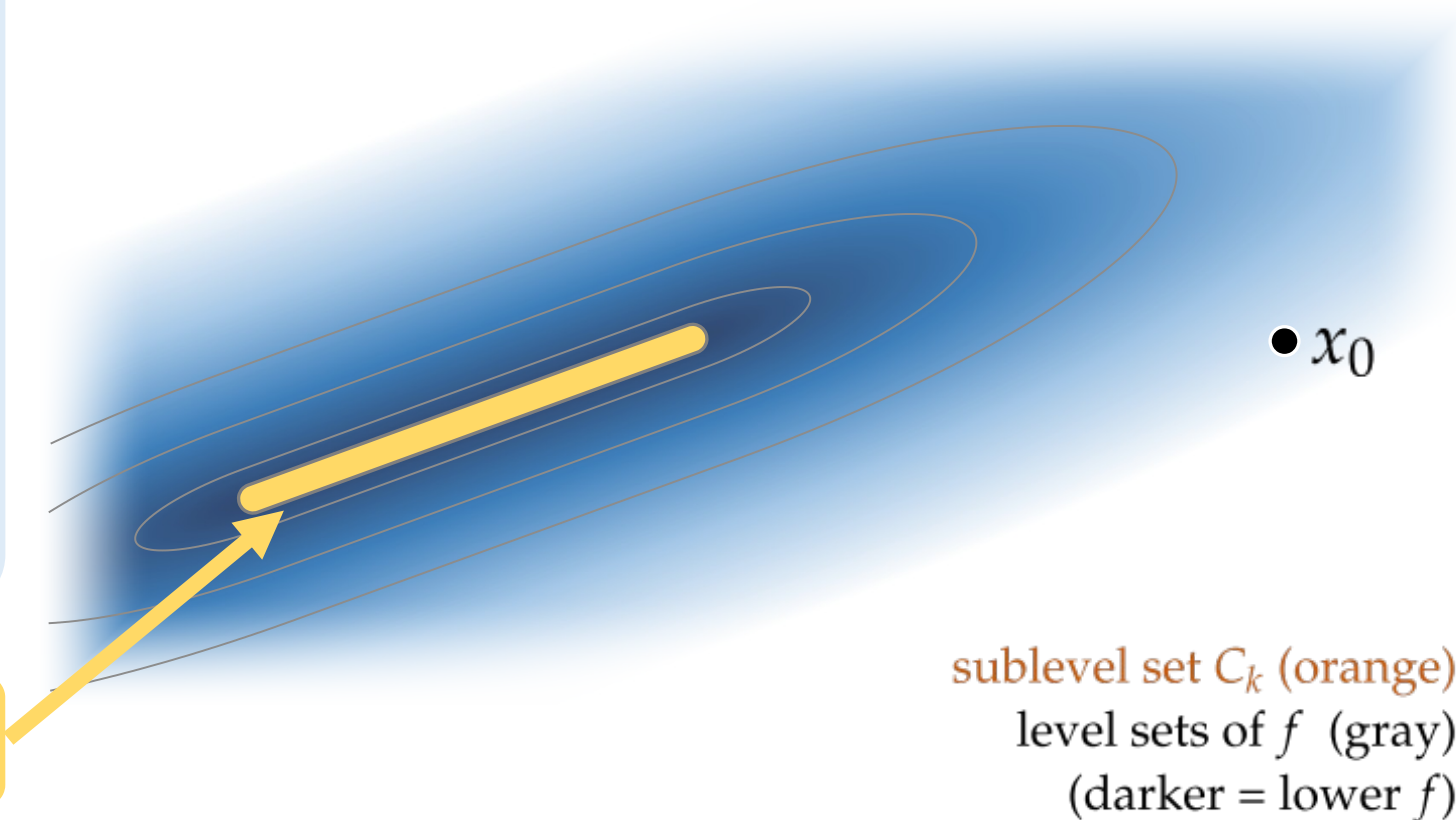
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Sublevel projection

$$\min_{\|y-x_k\|_2 \leq t} f(y) = \alpha_k$$

$$\iff \min_{f(y) \leq \alpha_k} \|y - x_k\|_2 = t$$

Argmin f



BPM = Projection onto Sublevel Sets

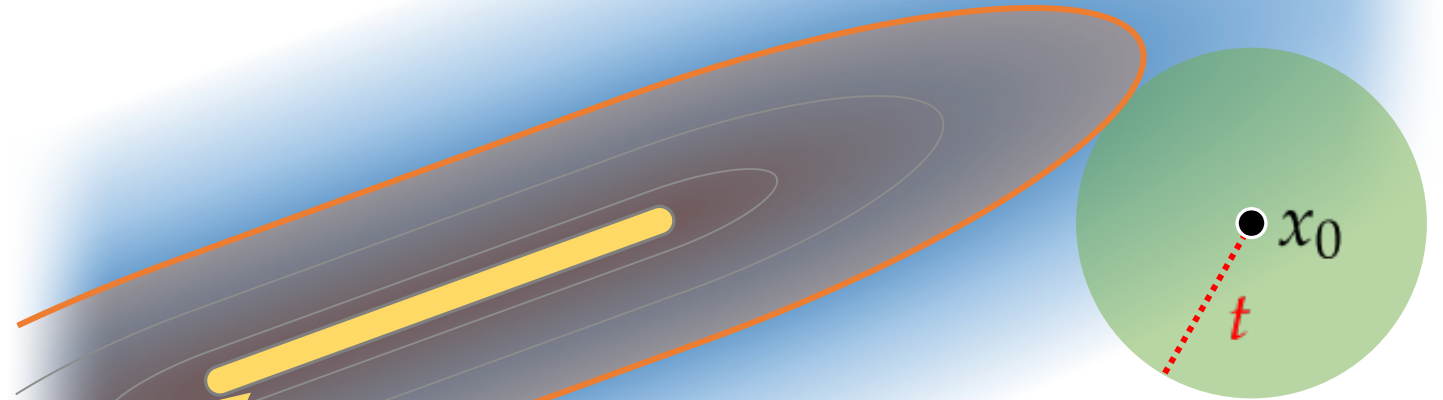
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sublevel set C_k (orange)
level sets of f (gray)
(darker = lower f)

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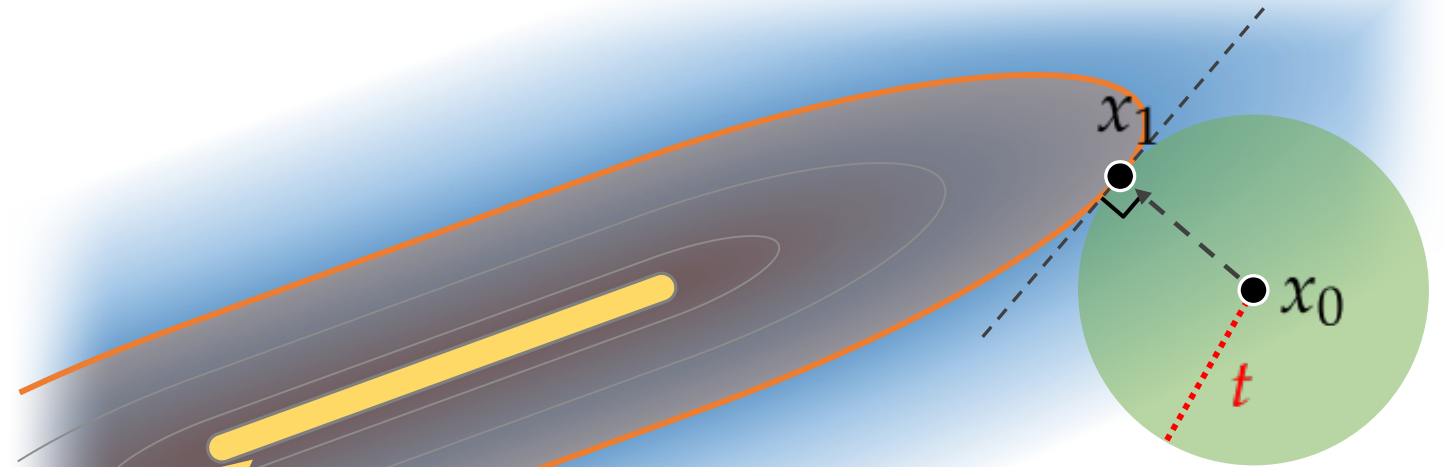
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$$C_0 \supseteq C_1 \supseteq \dots \supseteq \text{Argmin } f$$

$$\implies d_{k+1}^2 \leq d_k^2 - t^2$$

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sublevel set C_k (orange)
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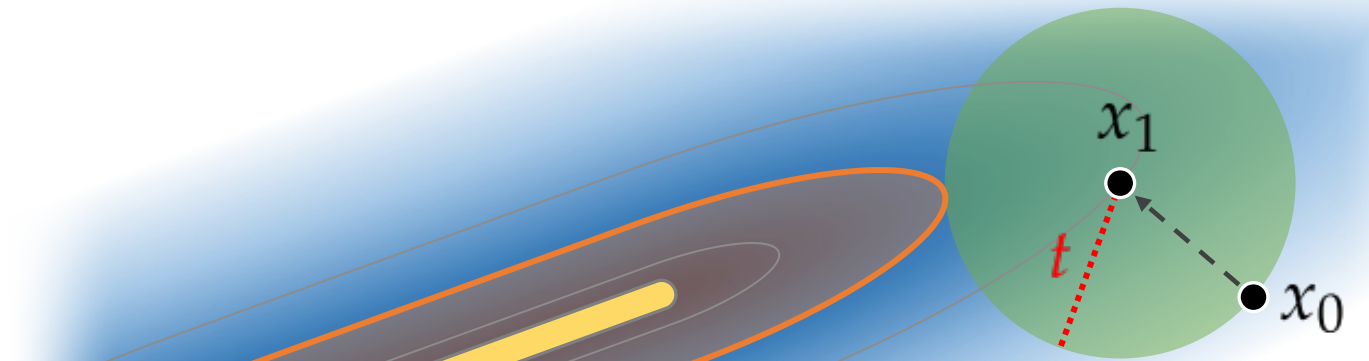
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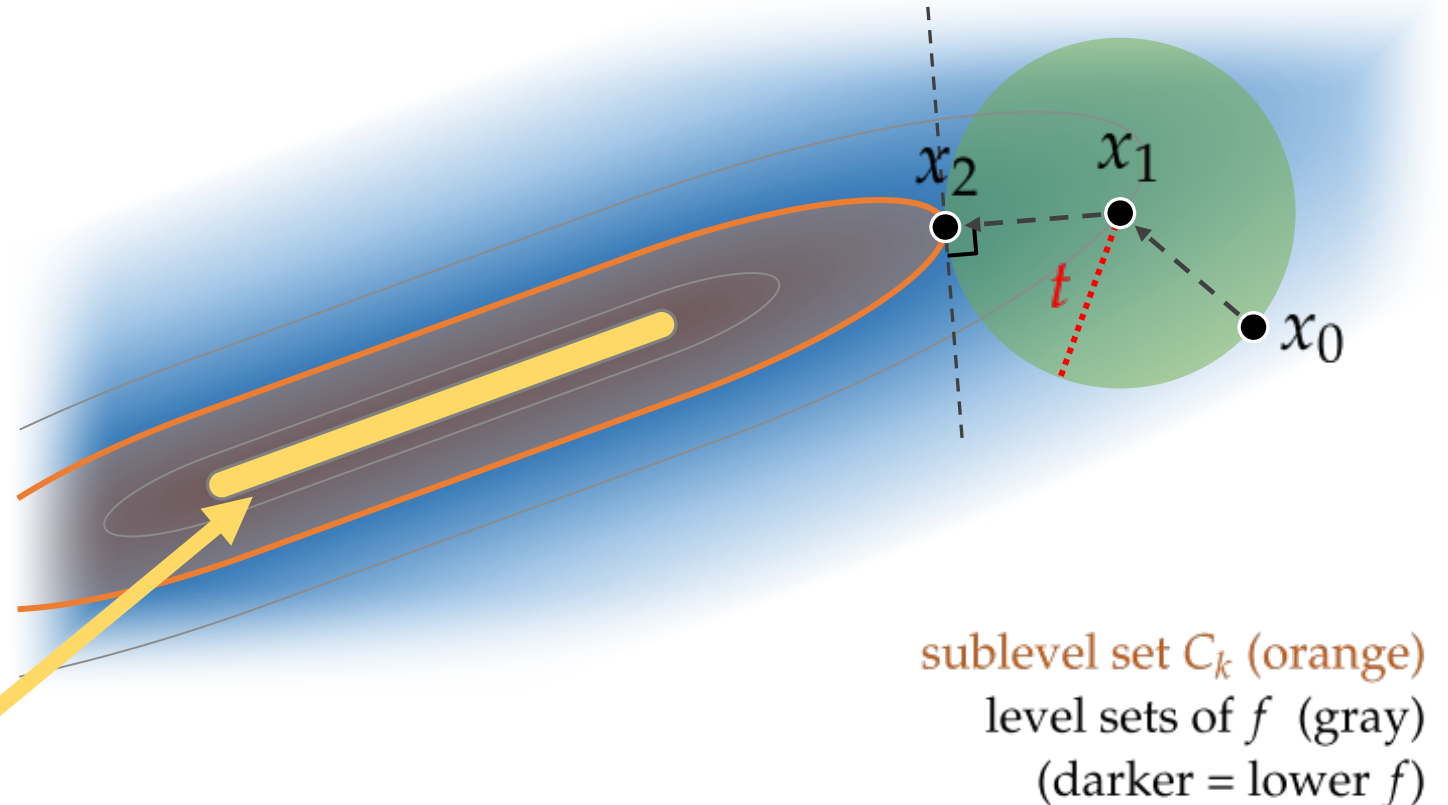
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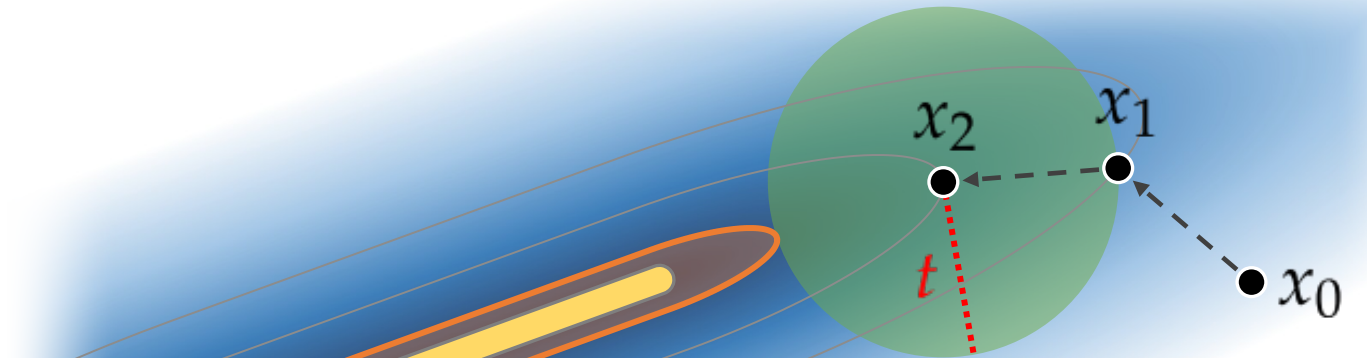
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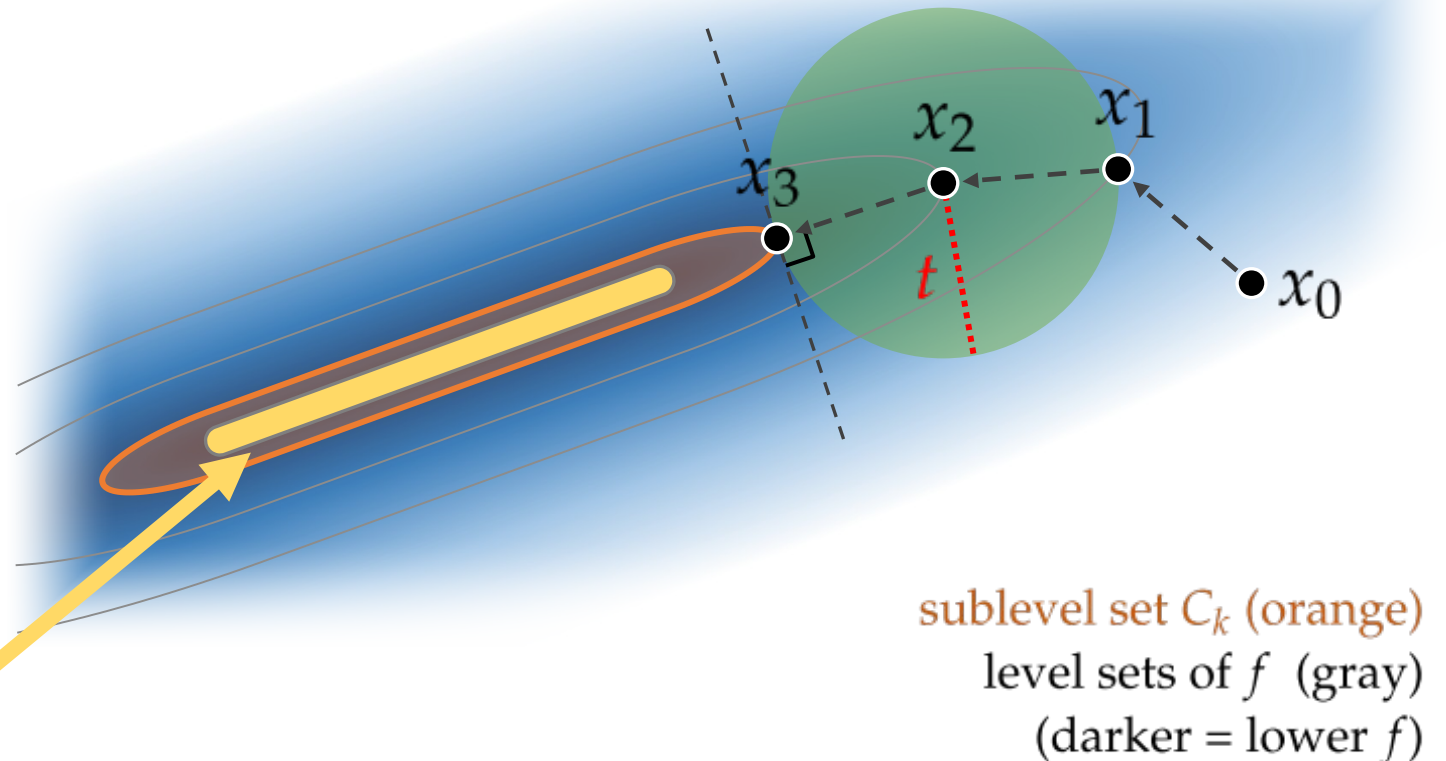
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$$C_0 \supseteq C_1 \supseteq \dots \supseteq \text{Argmin } f$$

$$\implies d_{k+1}^2 \leq d_k^2 - t^2$$

Argmin f



BPM = Projection onto Sublevel Sets

$$x_{k+1} \in \text{brox}_f^t(x_k) \implies x_{k+1} = \text{Proj}_{C_k}(x_k), \quad C_k := \{y : f(y) \leq f(x_{k+1})\}$$

Sublevel projection

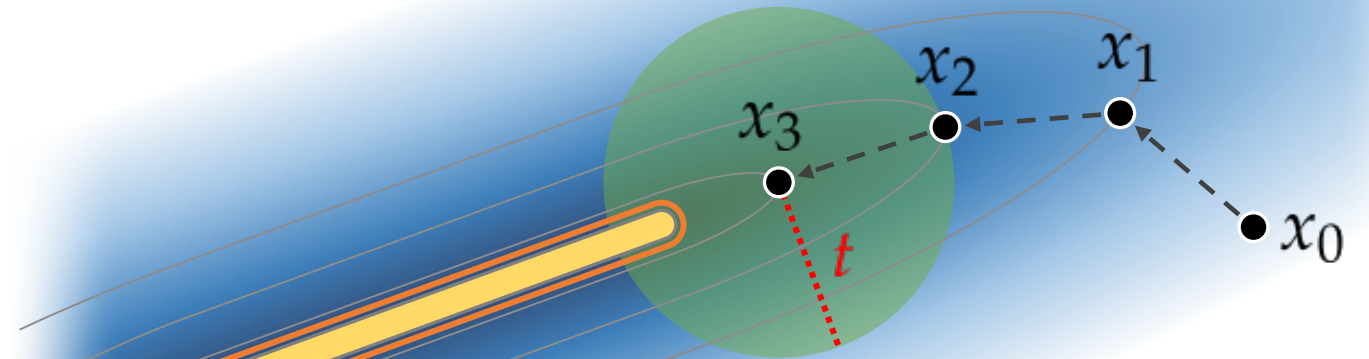
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Argmin f



sublevel set C_k (orange)
level sets of f (gray)
(darker = lower f)

BPM = Projection onto Sublevel Sets

$$x_{k+1} \in \text{brox}_f^t(x_k) \implies x_{k+1} = \text{Proj}_{C_k}(x_k), \quad C_k := \{y : f(y) \leq f(x_{k+1})\}$$

Sublevel projection

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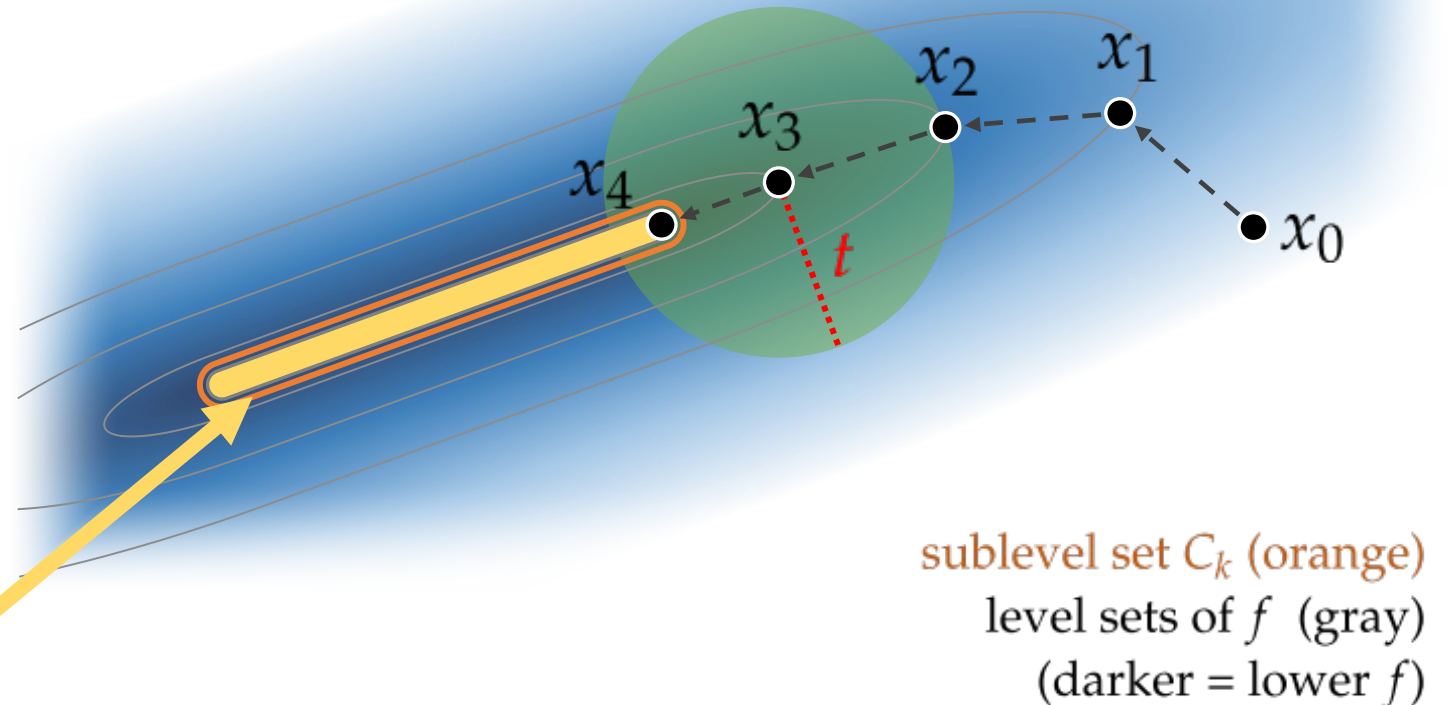
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$$\alpha_3 = 0 = f_* : x_4 = \text{Proj}_{\text{Argmin } f}(x_3)$$

Argmin f

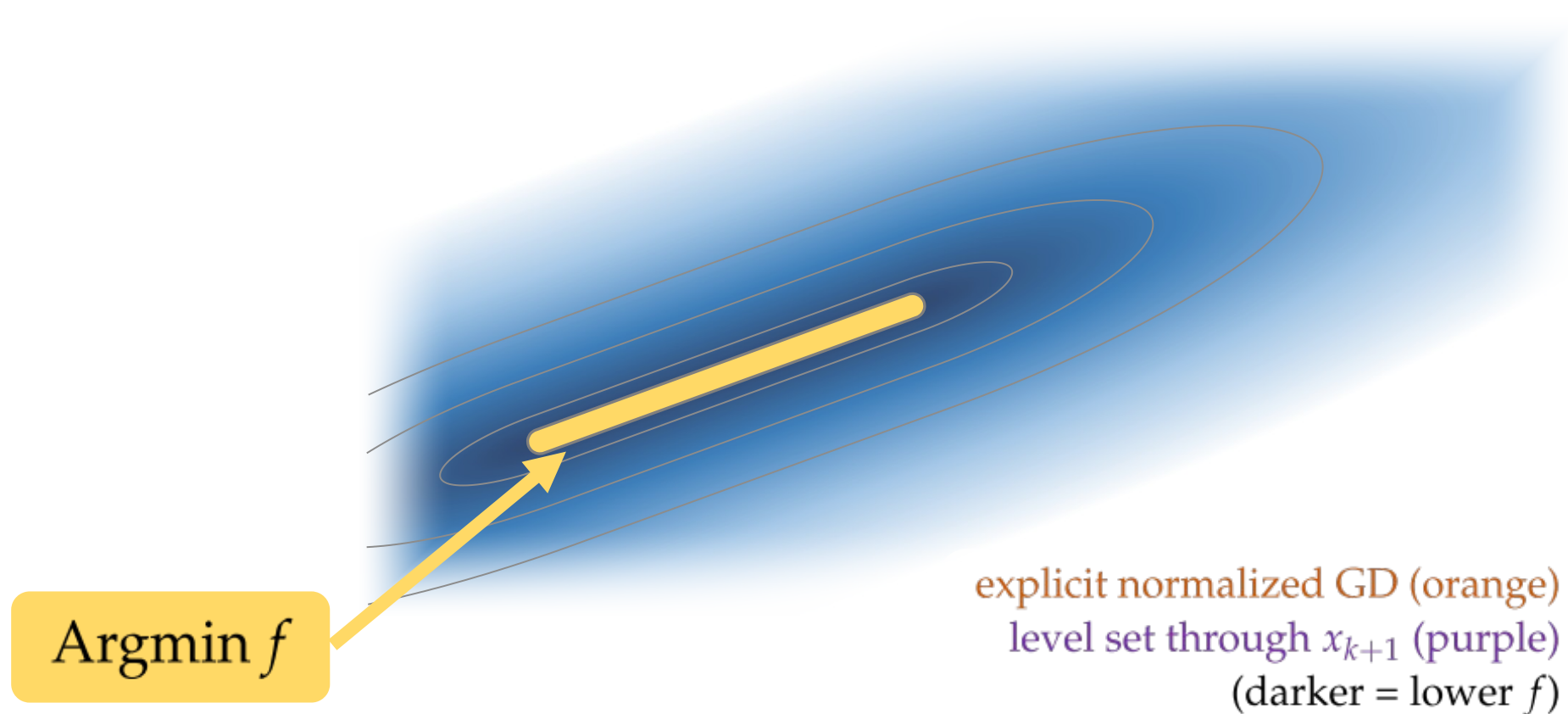


BPM = Implicit Normalized GD

$$x_{k+1} \in \text{brox}_f^t(x_k) \implies x_{k+1} = x_k - t \frac{\nabla f(x_{k+1})}{\|\nabla f(x_{k+1})\|_2}$$

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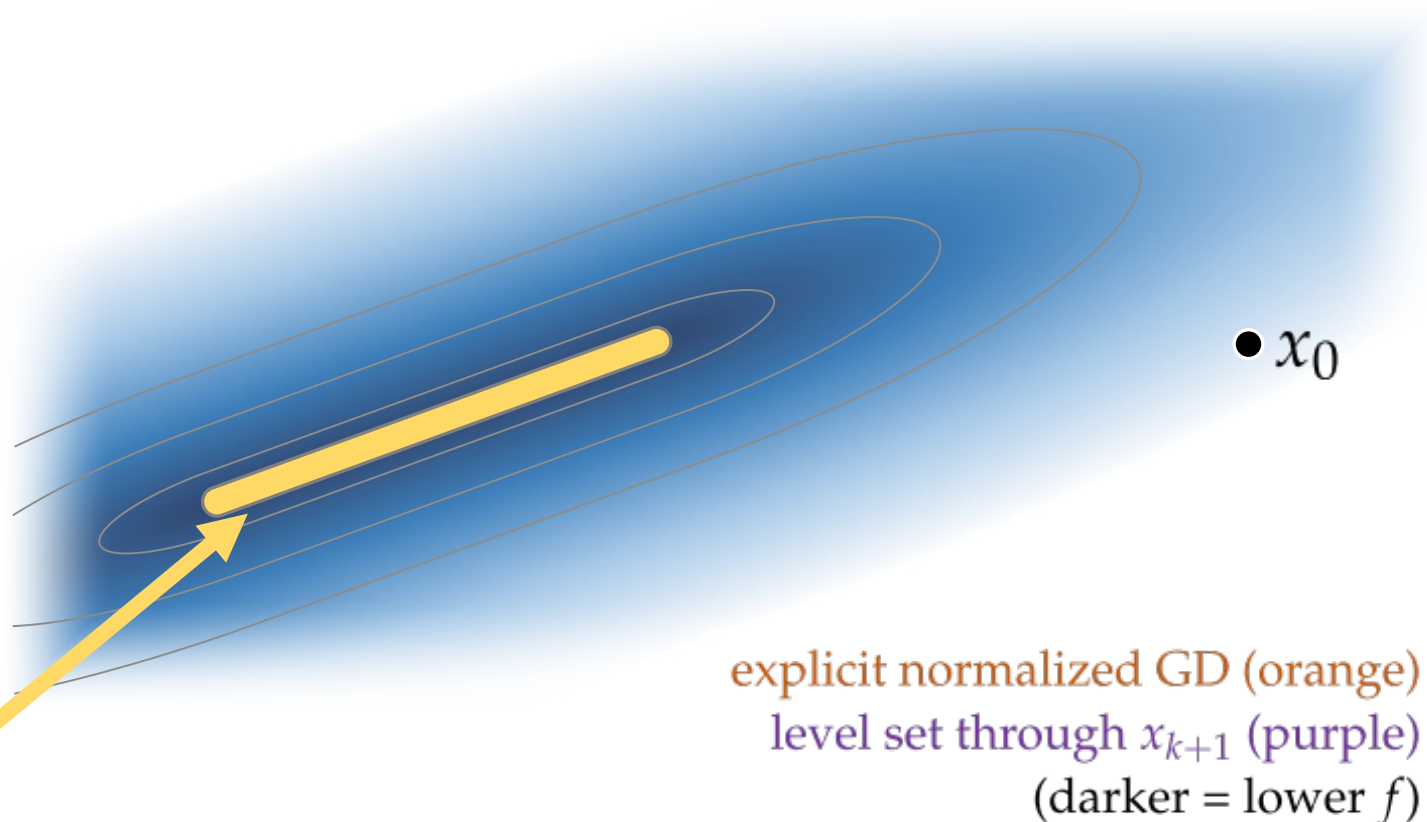
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Implicit (BPM) vs explicit

Argmin f

• x_0

explicit normalized GD (orange)
level set through x_{k+1} (purple)
(darker = lower f)



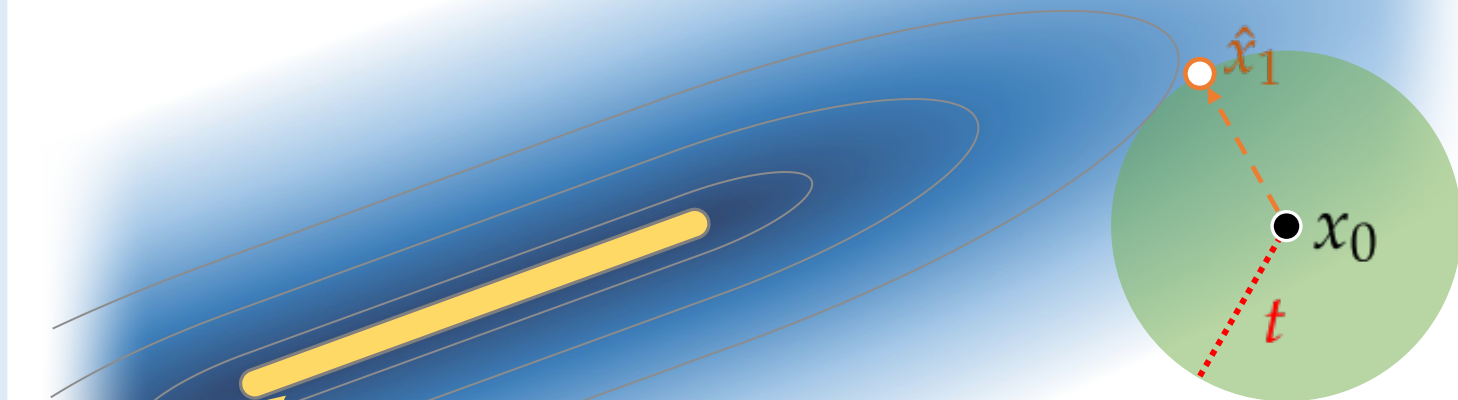
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$$\hat{x}_{k+1} = x_k - t \frac{\nabla f(x_k)}{\|\nabla f(x_k)\|_2} \quad (\text{explicit})$$

Argmin f



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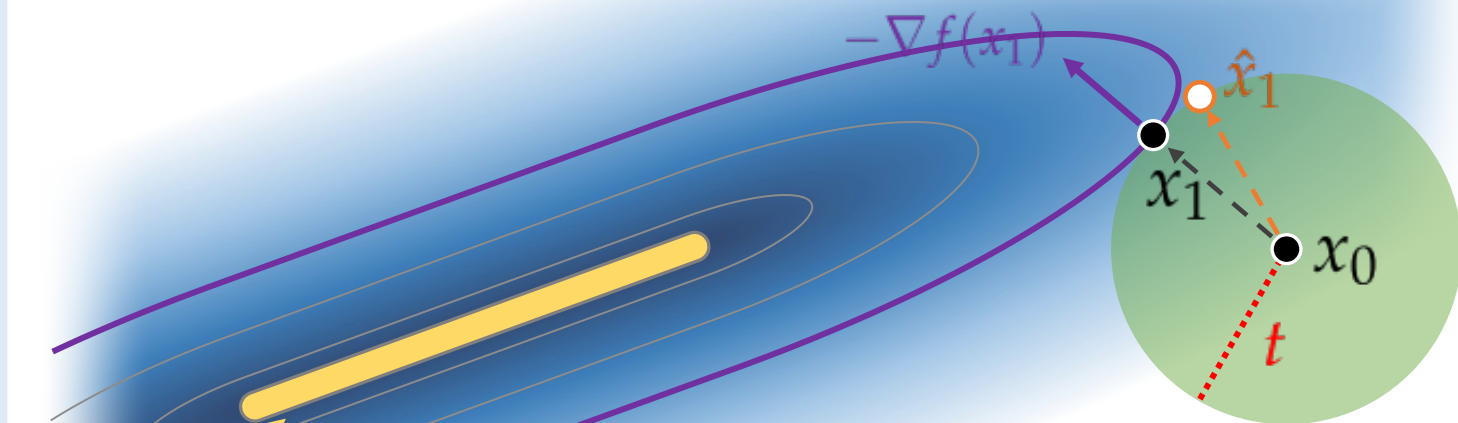
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Argmin f



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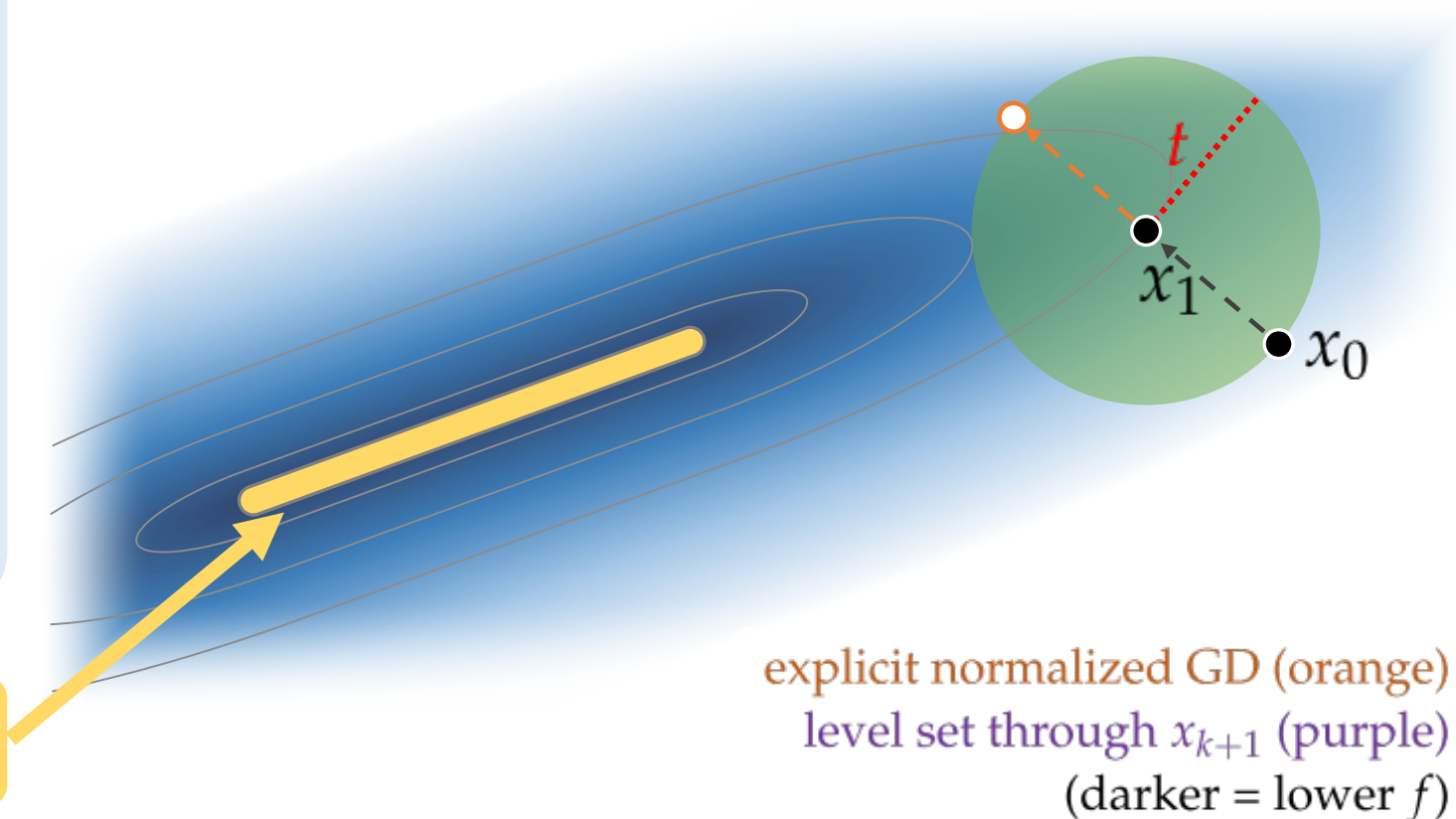
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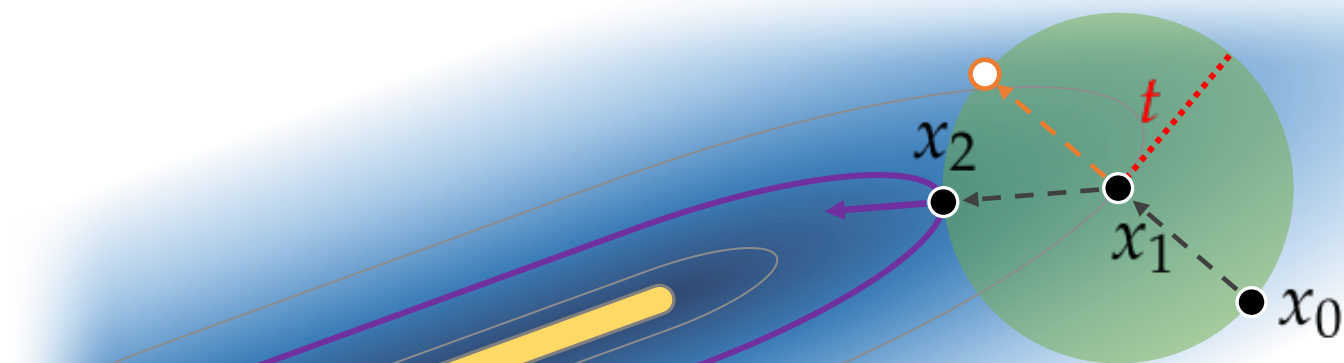
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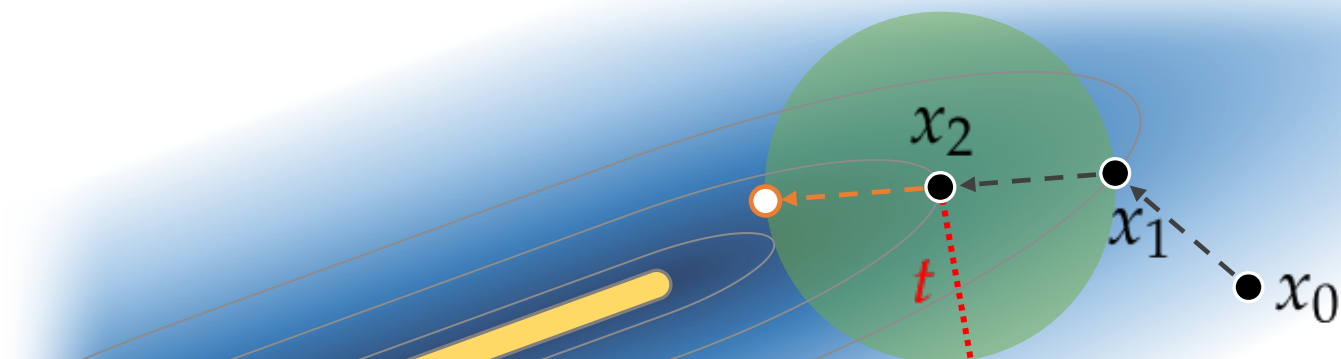
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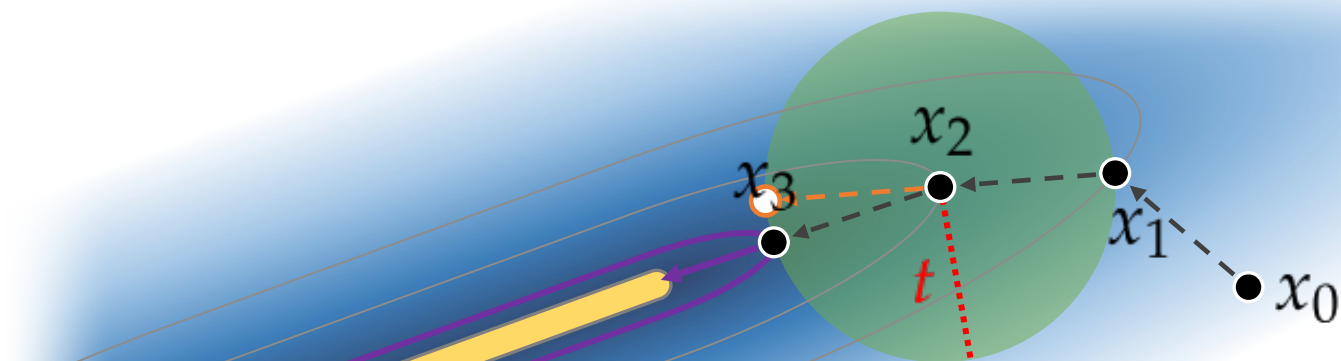
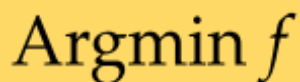
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level set through x_{k+1} (purple)
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BPM = Implicit Normalized GD

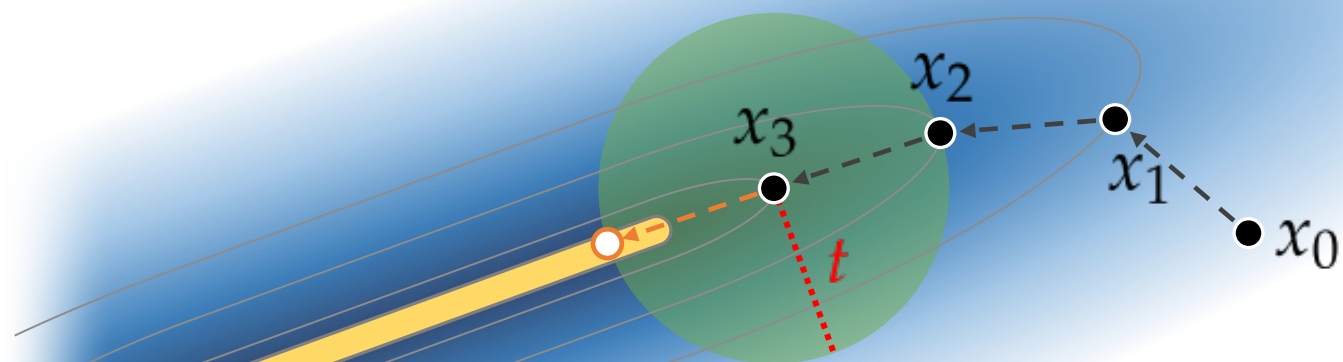
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Argmin f



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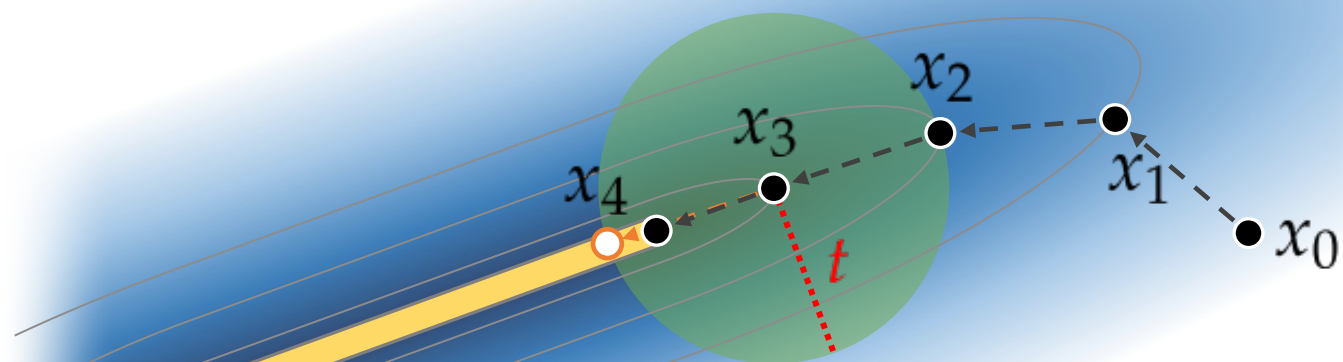
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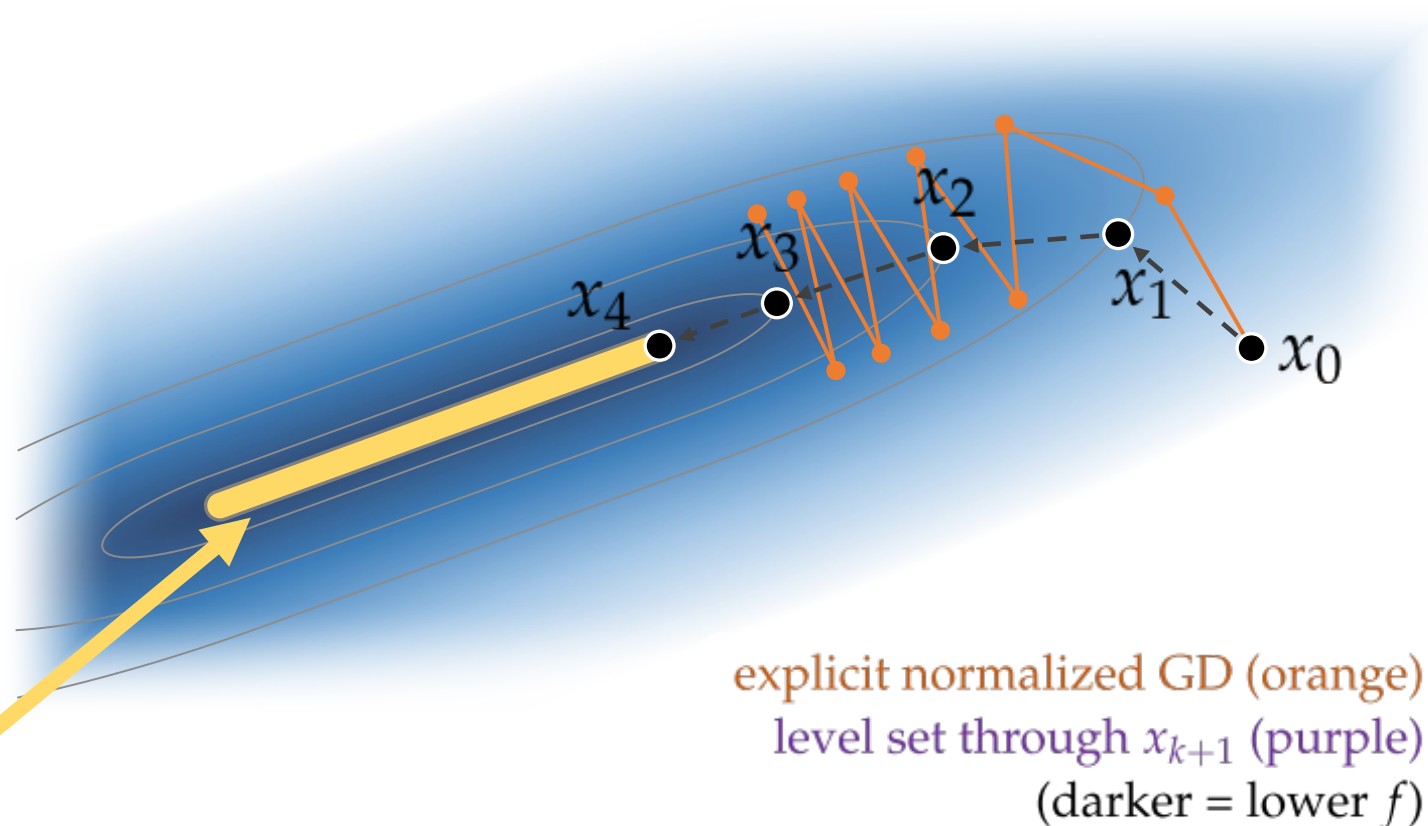
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BPM: $f(x_4) = f_\star$

normalized GD: $f(\hat{x}_{10}) = 5.48$

Argmin f

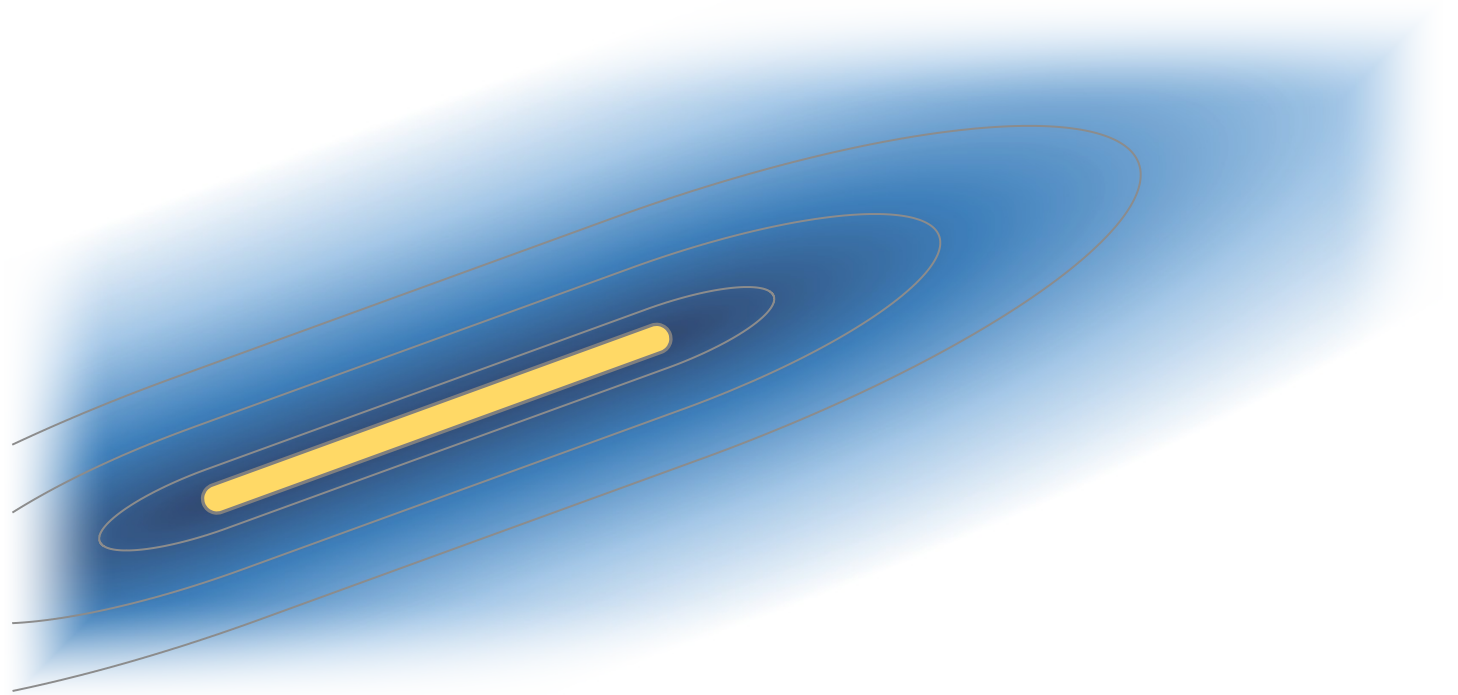


BPM = Normalized GD on the Ball Envelope

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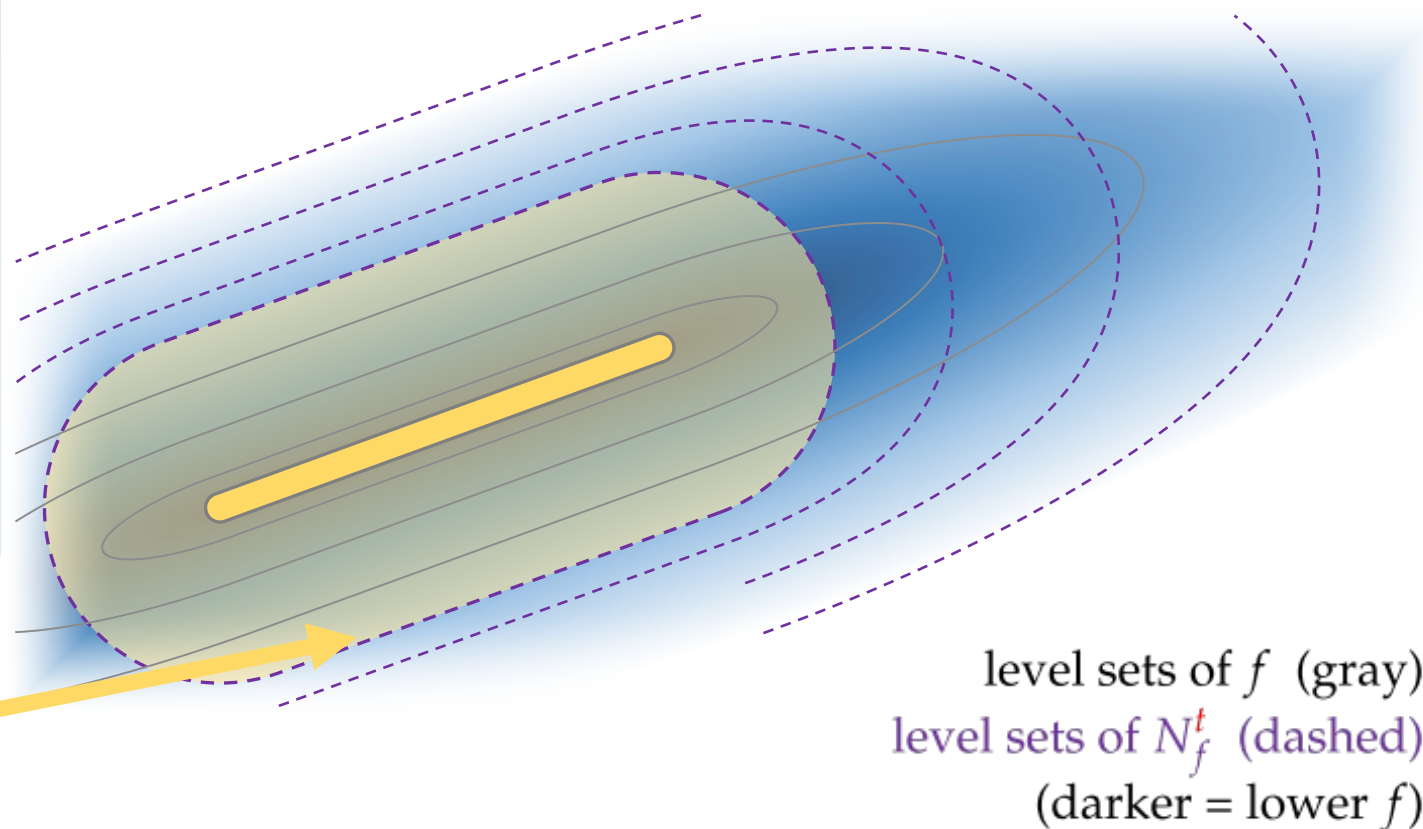
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Ball envelope

$$N_f^t(x) := \min_{\|y-x\|_2 \leq t} f(y)$$

$$\{N_f^t \leq c\} = \{f \leq c\} + \mathcal{B}(0, t)$$



$$\text{Argmin } N_f^t = \text{Argmin } f + \mathcal{B}(0, t)$$

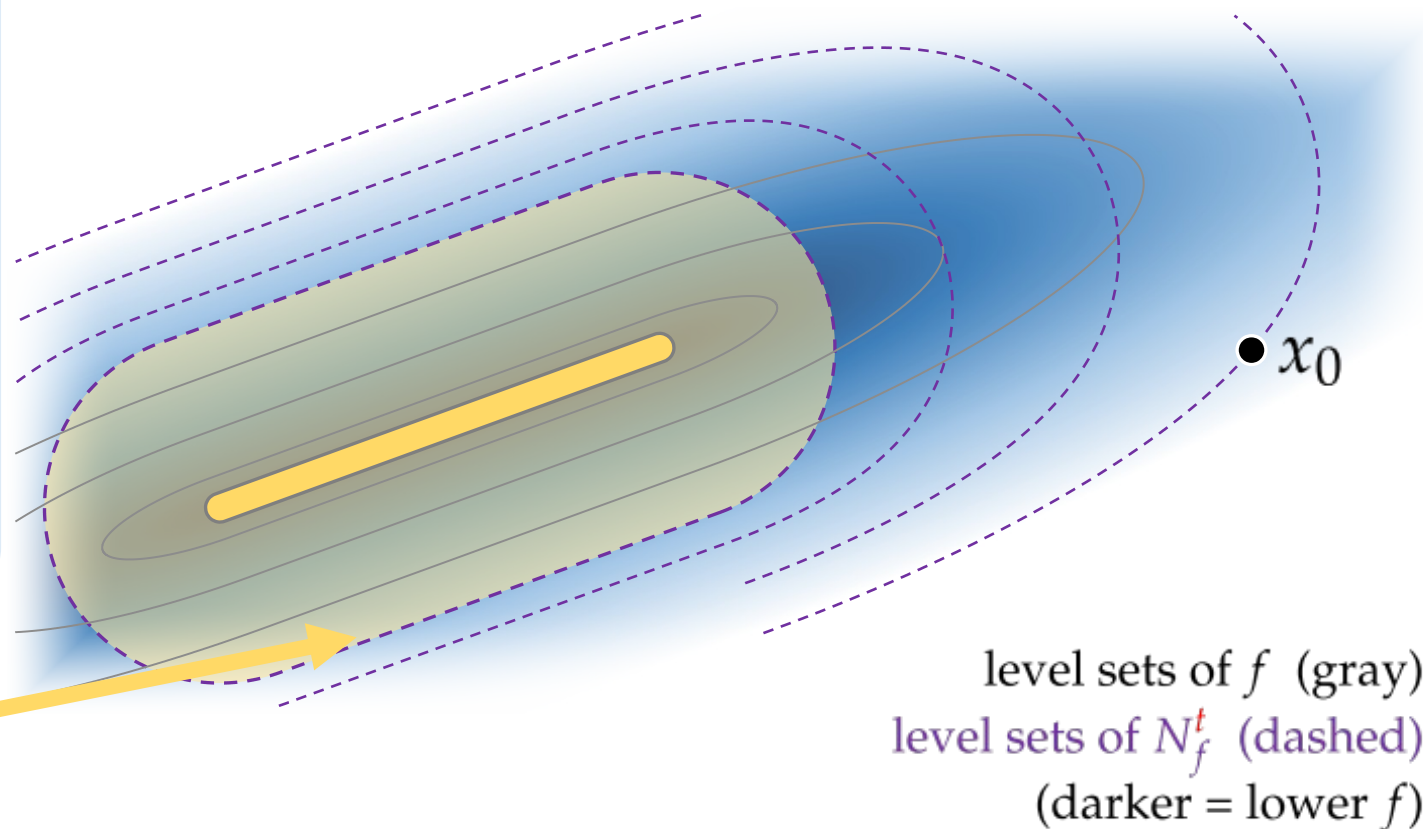
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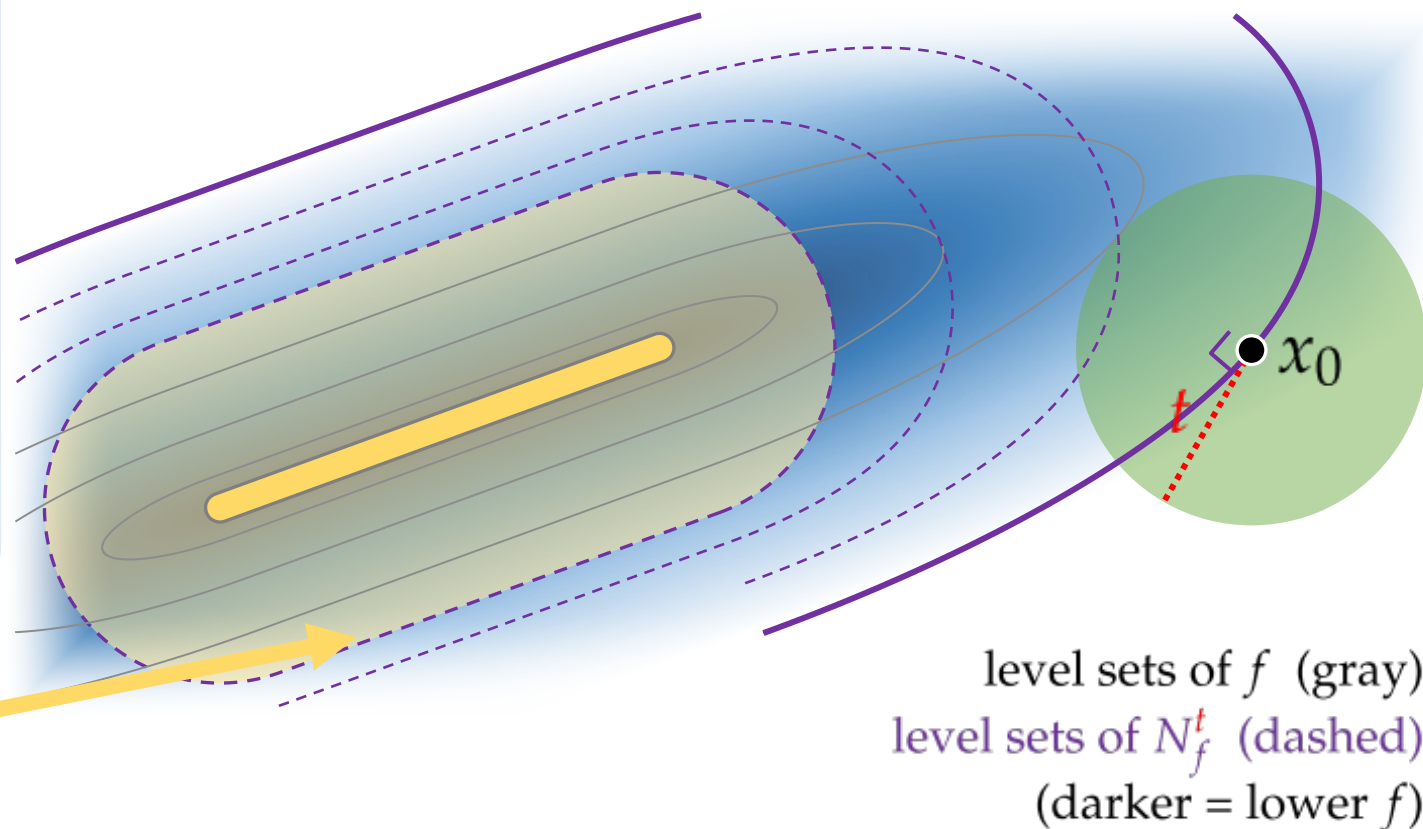
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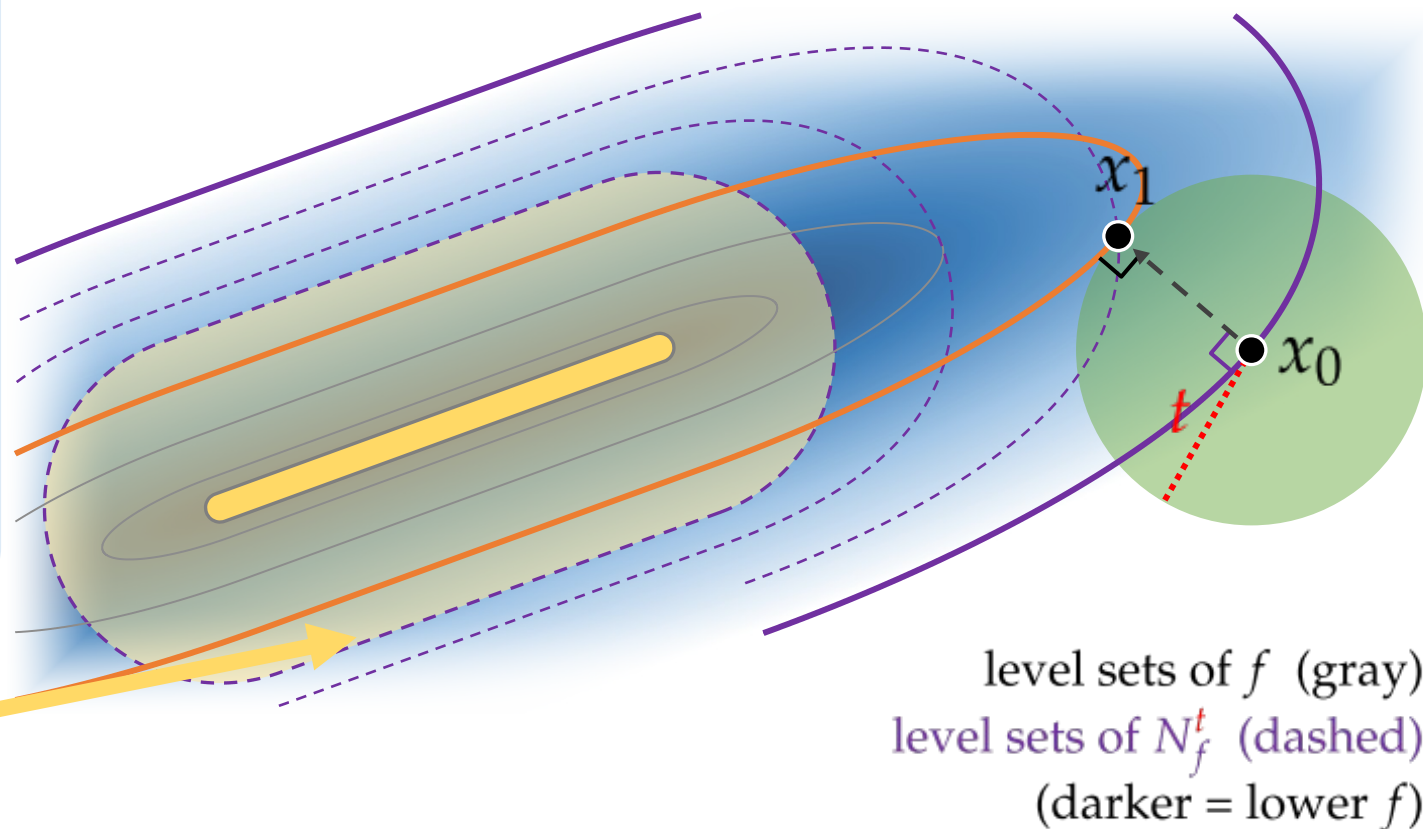
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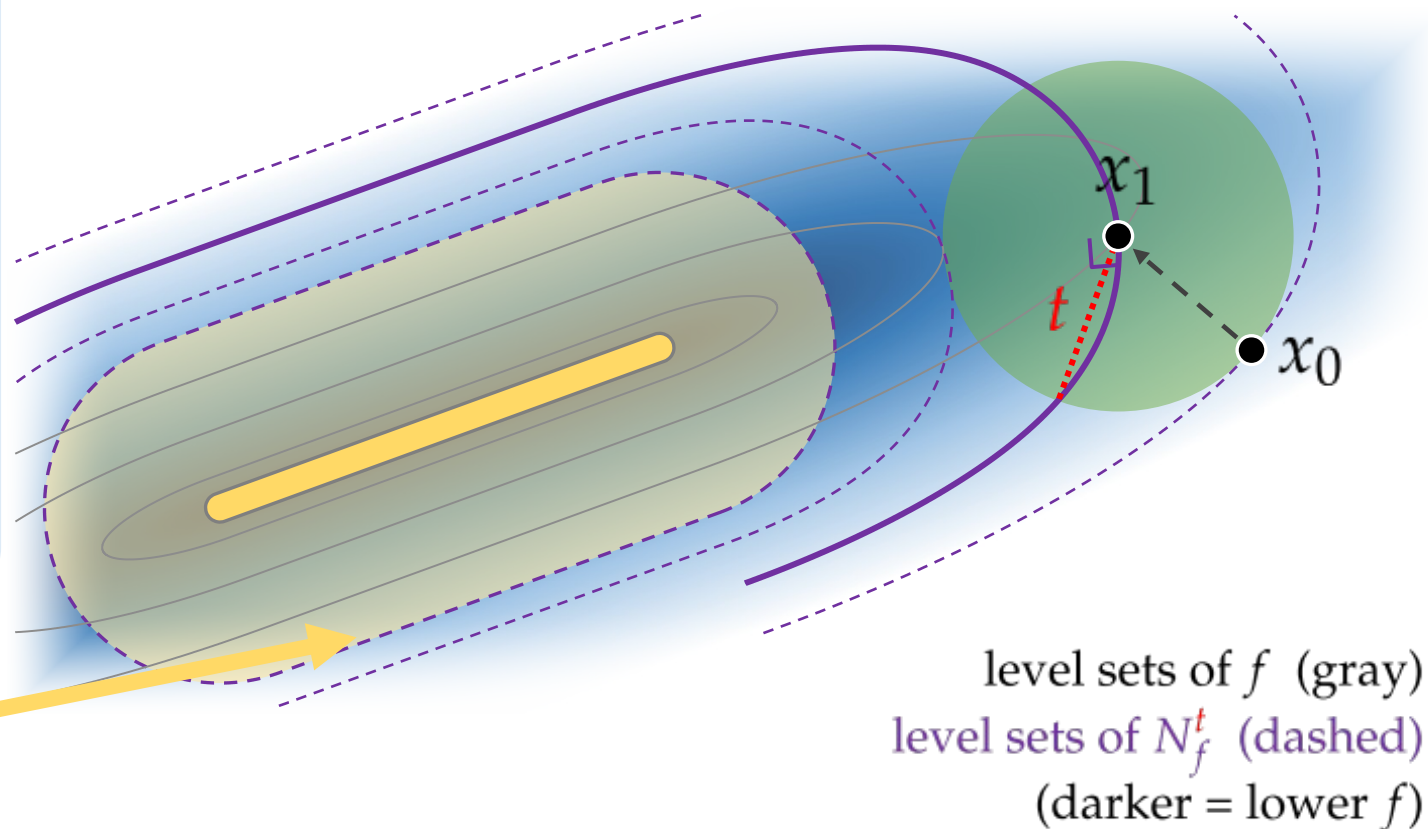
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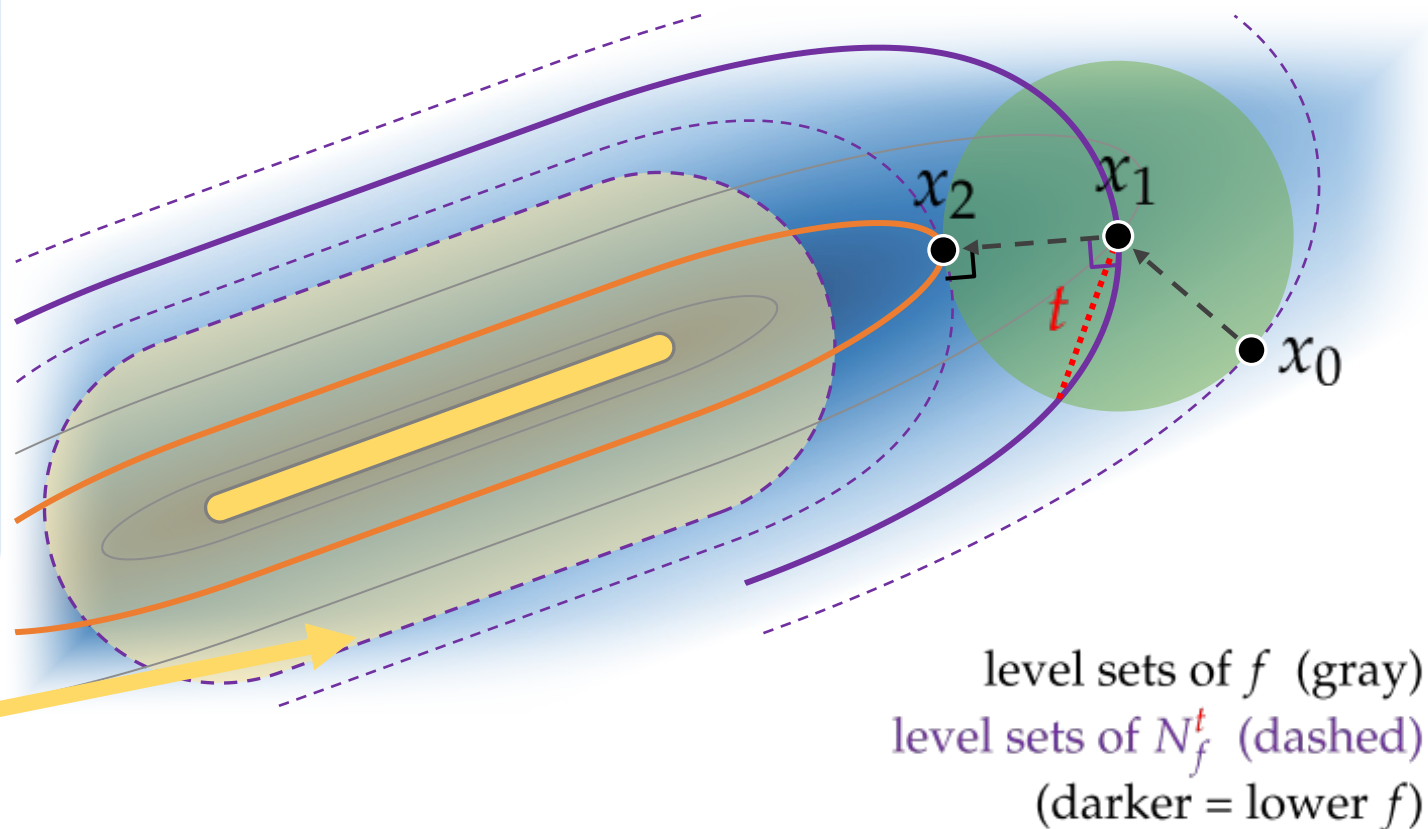
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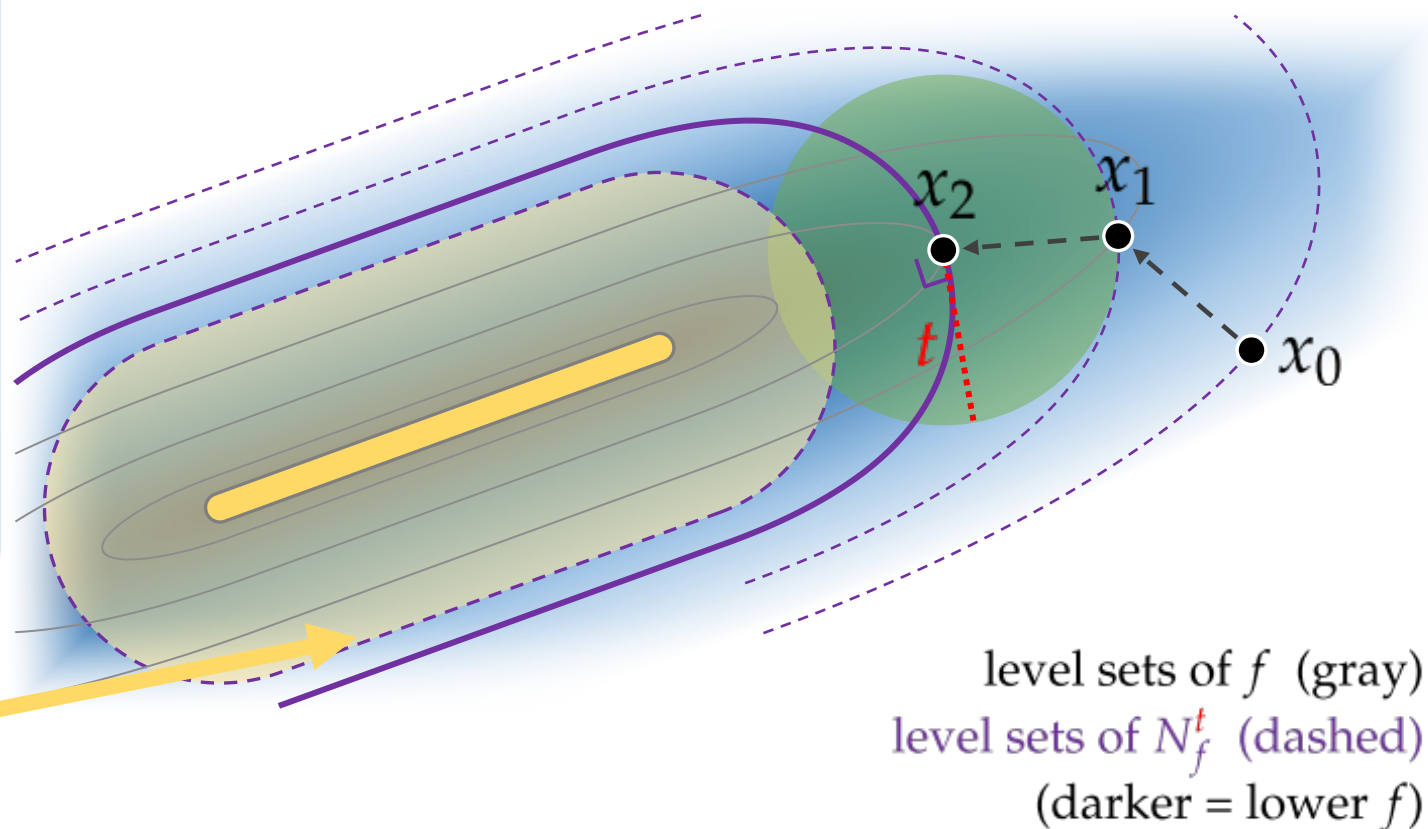
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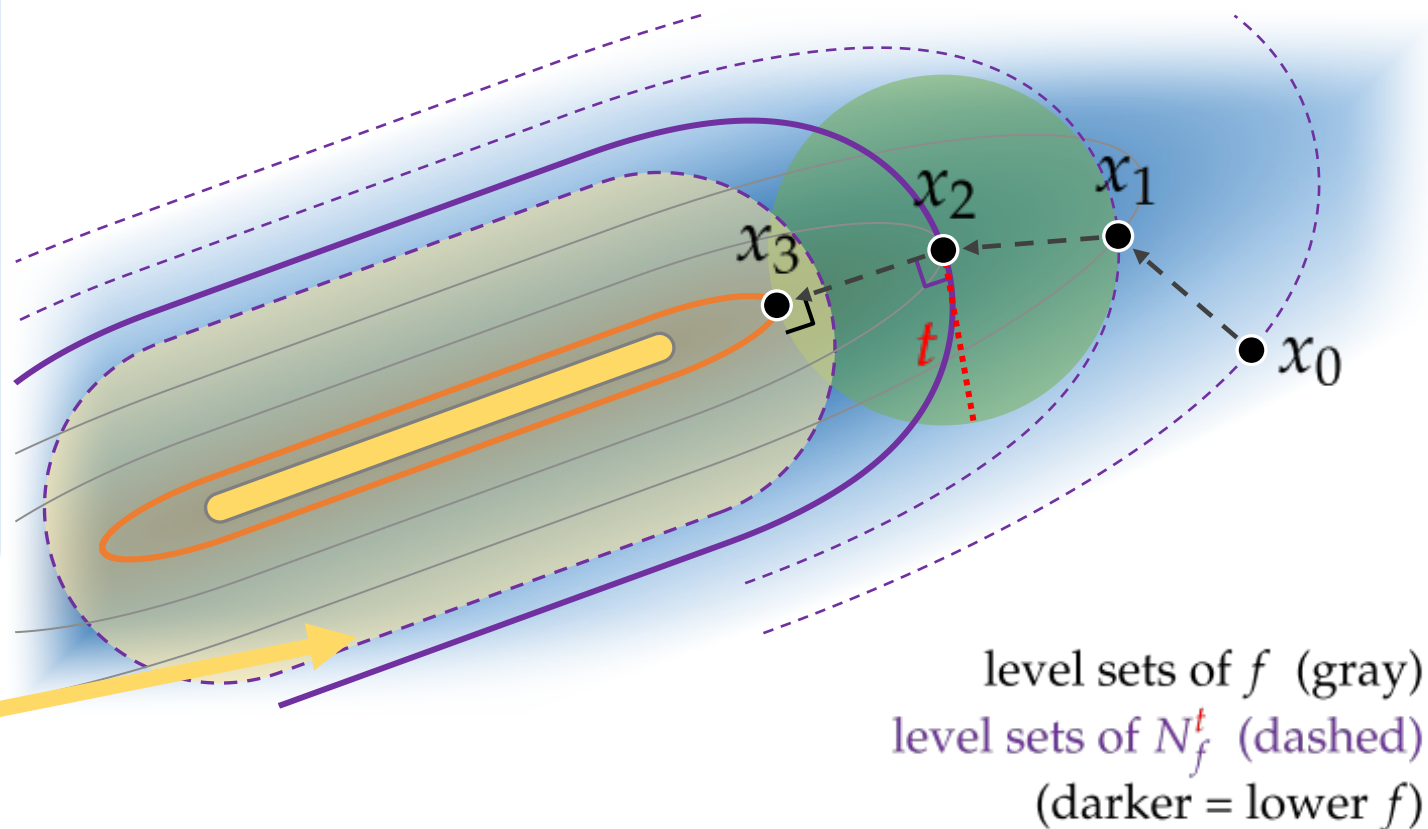
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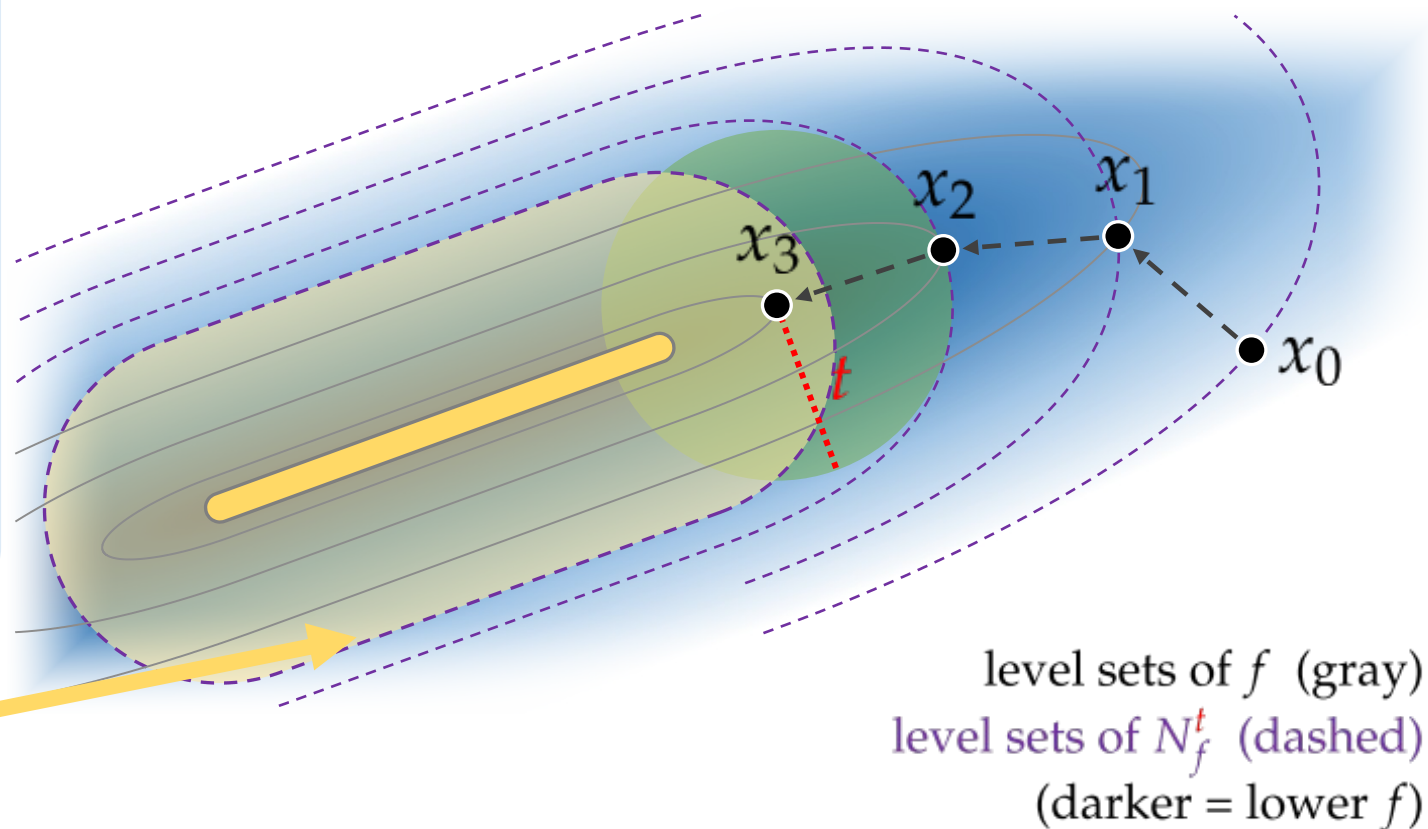
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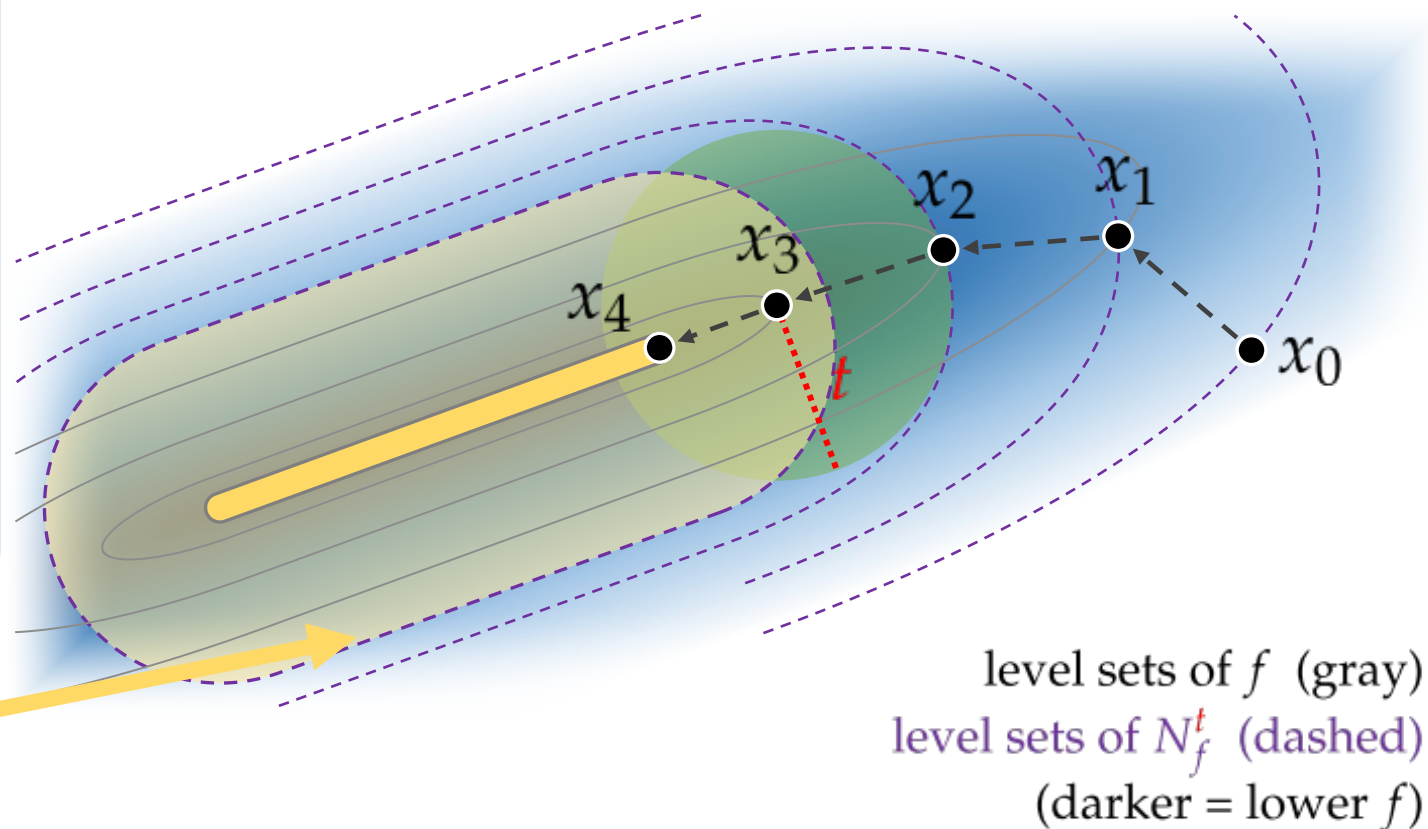
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$$N_f^t(x_3) = 0 \implies x_4 \in \text{Argmin } f$$

$$\text{Argmin } N_f^t = \text{Argmin } f + \mathcal{B}(0, t)$$



BPM = Implicit Regularized Newton

$$x_{k+1} = x_k - \left(\bar{H}_k + \underbrace{\lambda_k I}_{\text{regularization}} \right)^{-1} \nabla f(x_k)$$

Example: $f(x) = \frac{1}{3}(|\langle a, x \rangle|^3 + 8|\langle b, x \rangle|^3)$, $t = 0.6$



P.R.

**The Design and Analysis of Newton-Type Algorithms as
Approximate Broximal Point Methods**

Manuscript, 2026

BPM = Implicit Regularized Newton

averaged Hessian

$$\bar{H}_k := \int_0^1 \nabla^2 f(x_k + \tau(x_{k+1} - x_k)) d\tau$$

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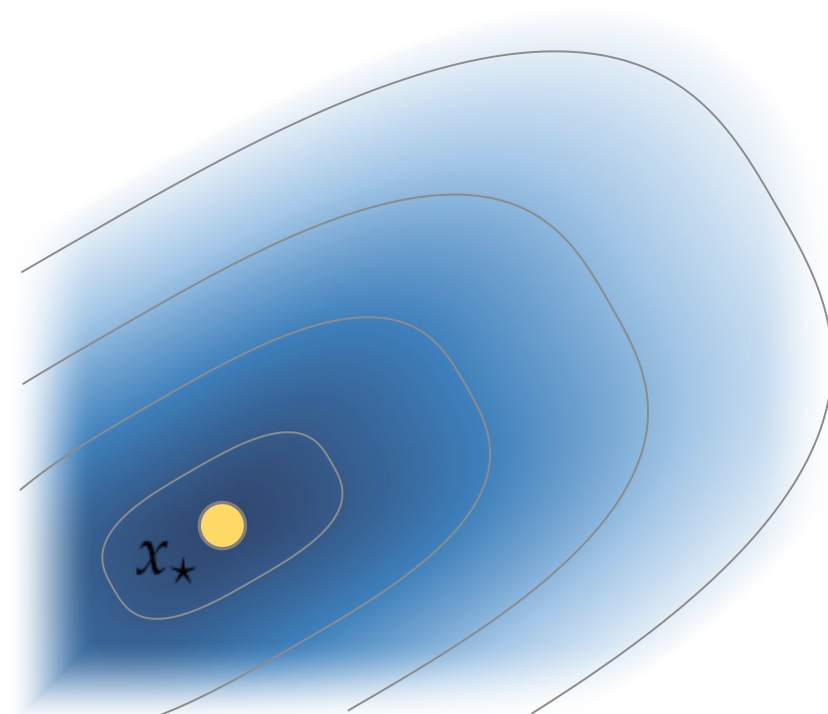
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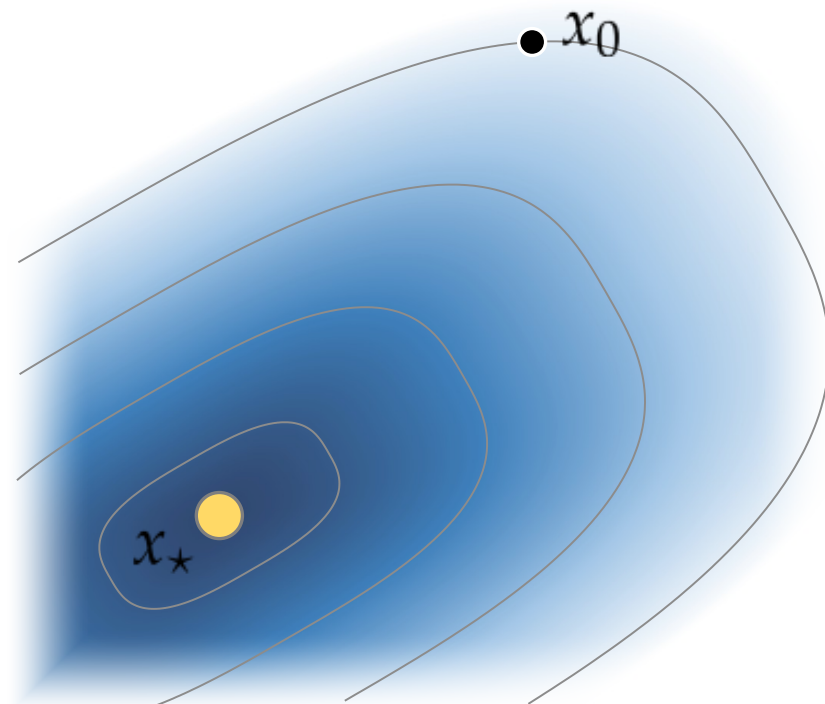
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Example: $f(x) = \frac{1}{3}(|\langle a, x \rangle|^3 + 8|\langle b, x \rangle|^3)$, $t = 0.6$

Why? (convex, twice-differentiable objective)

optimality: $\nabla f(x_{k+1}) + \lambda_k s_k = 0$, $\lambda_k = \|\nabla f(x_{k+1})\|_2 / t$



P.R.

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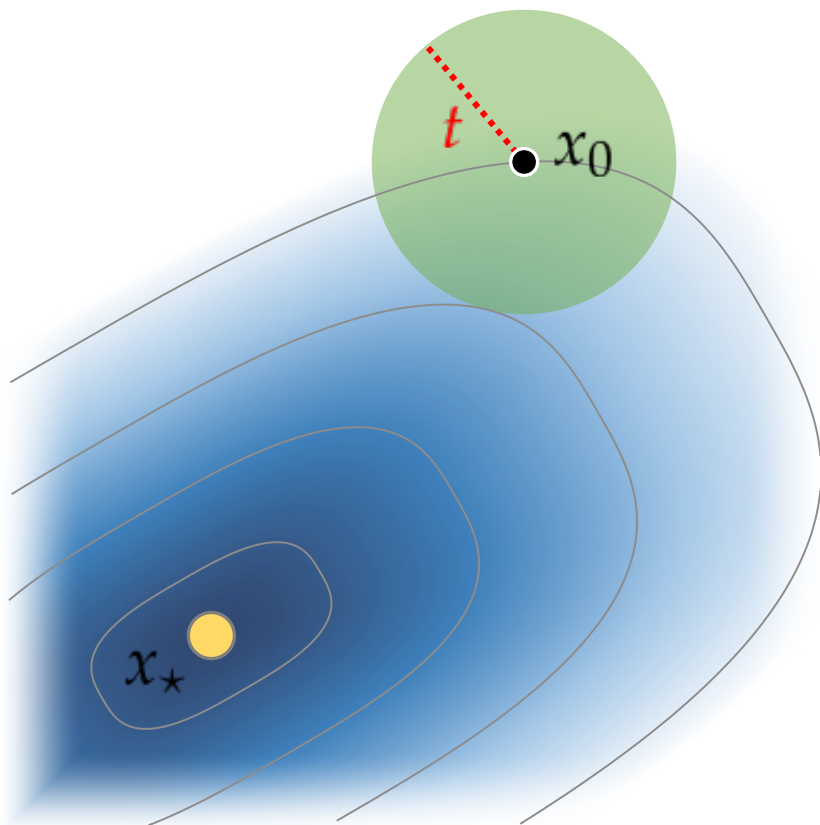
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Approximate Broximal Point Methods

Manuscript, 2026

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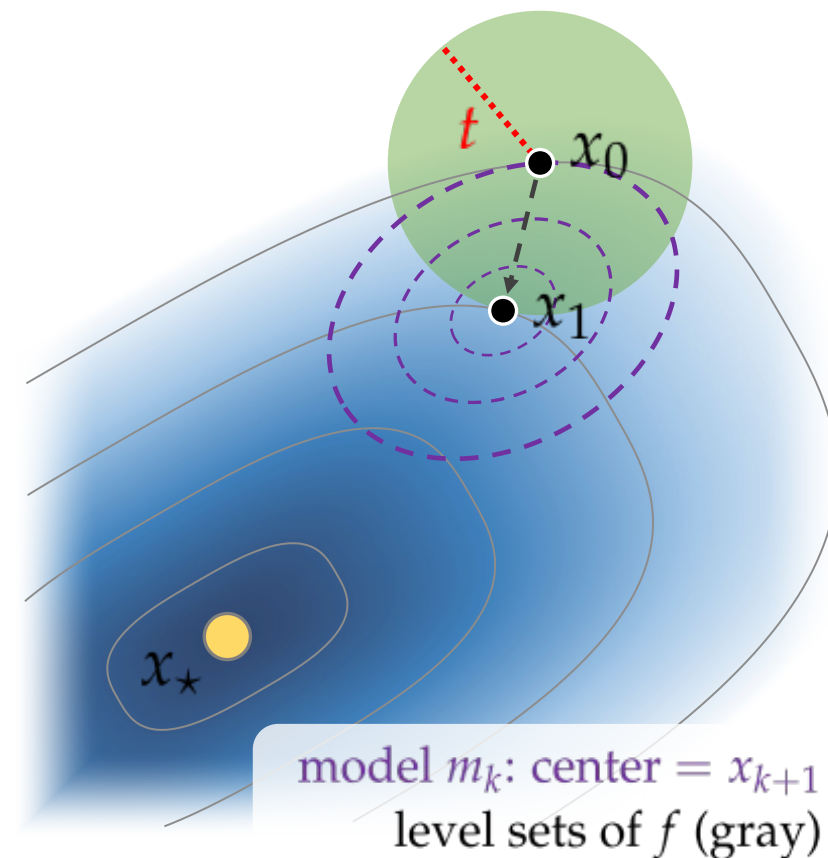
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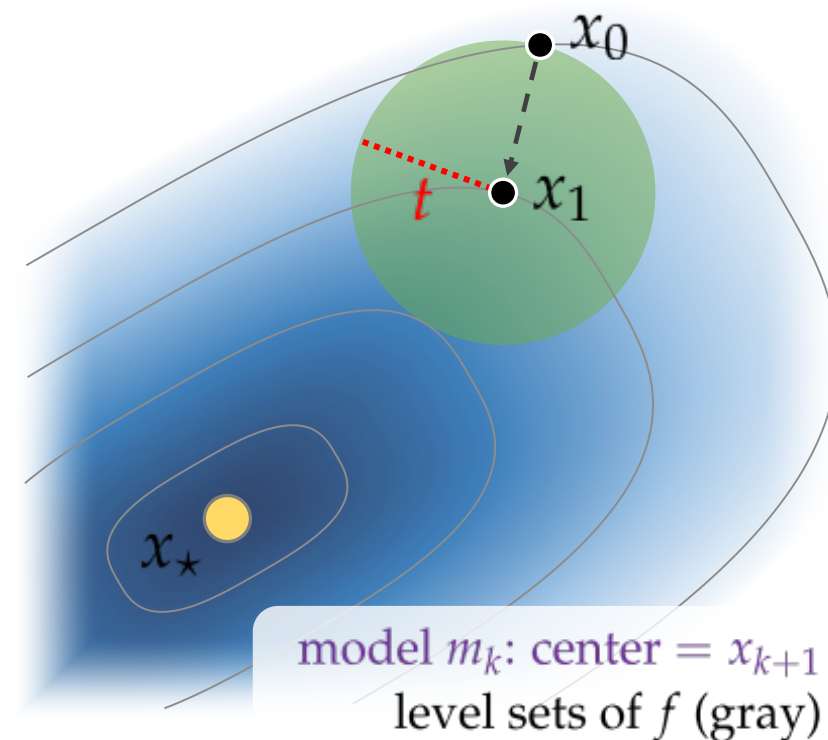
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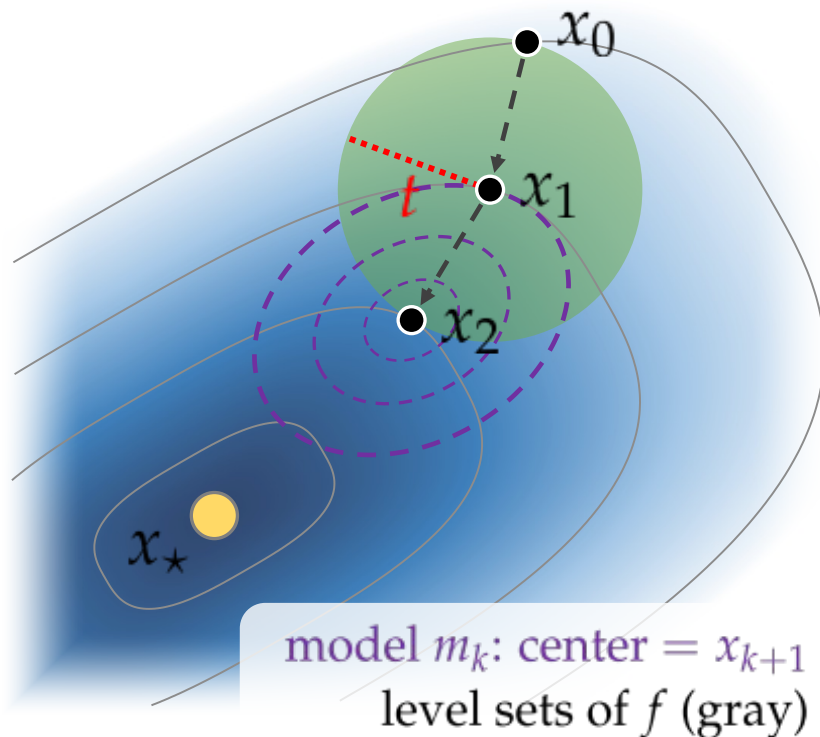
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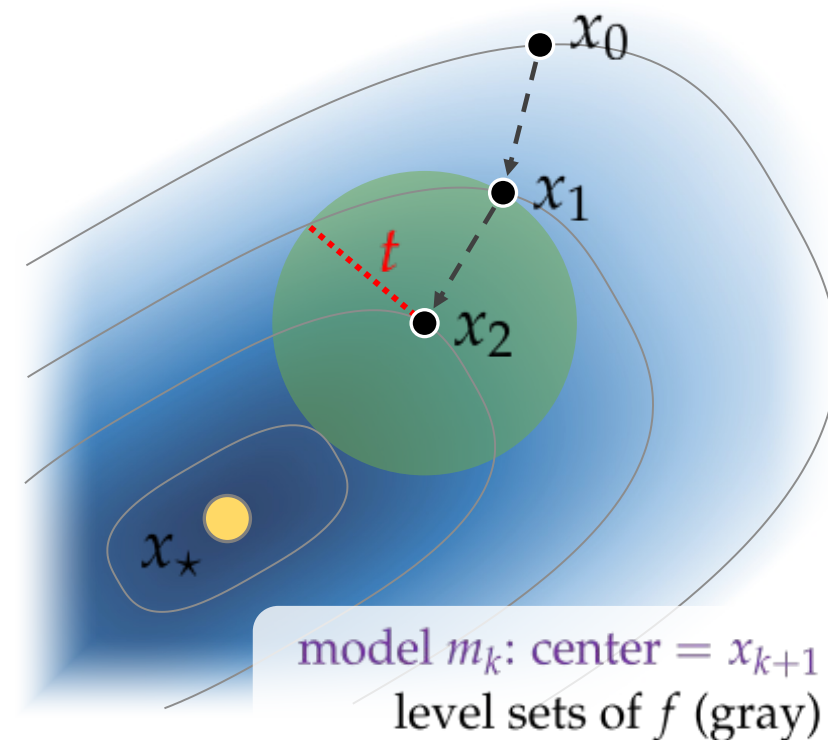
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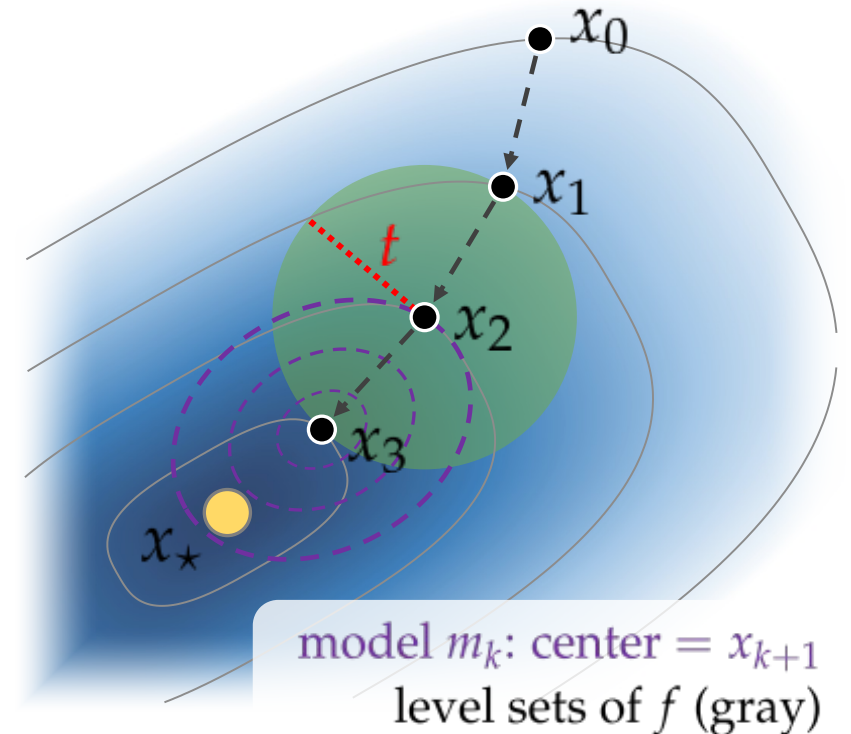
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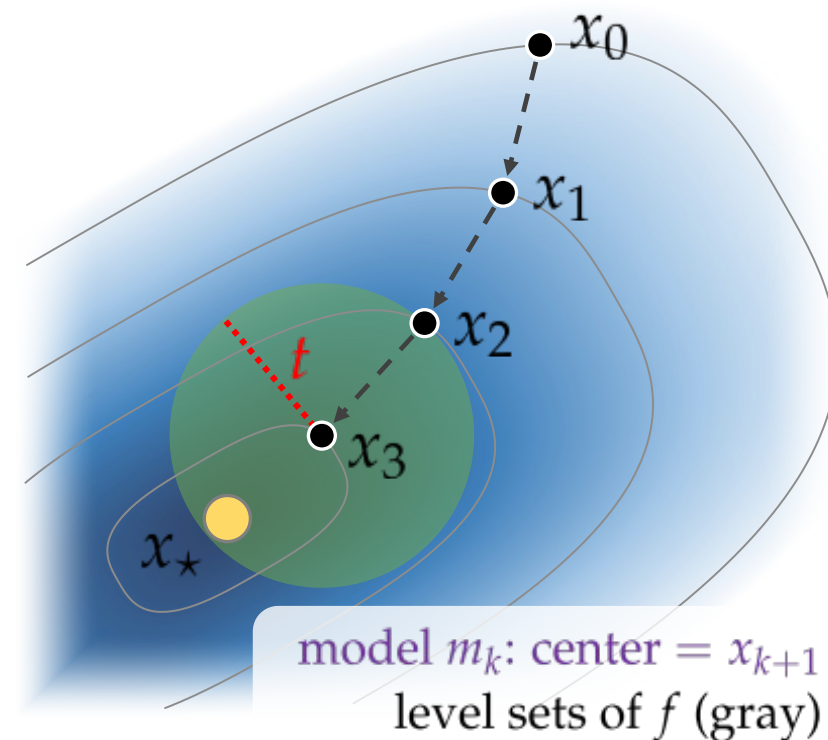
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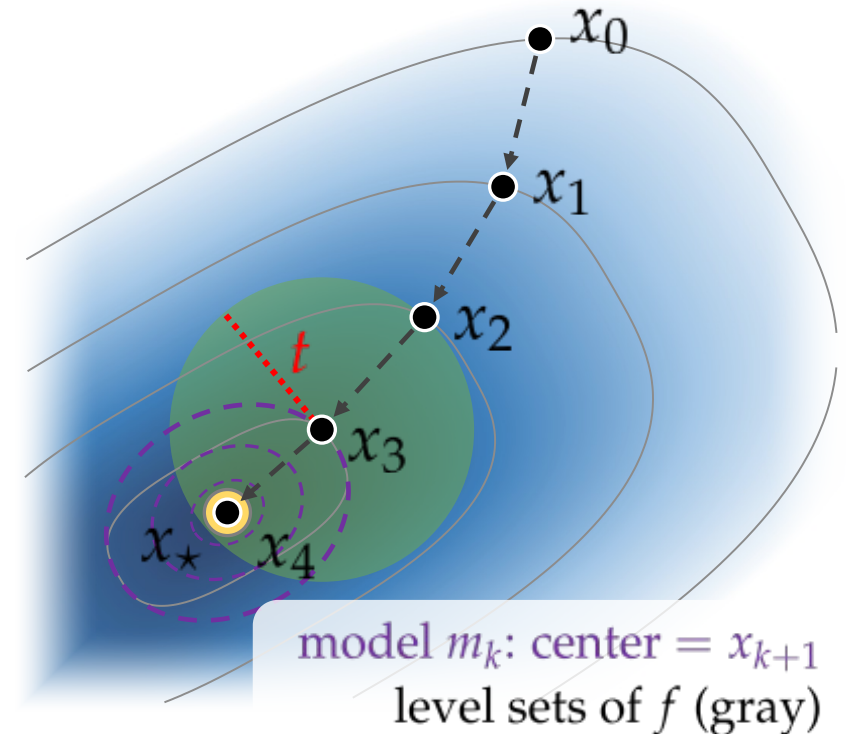
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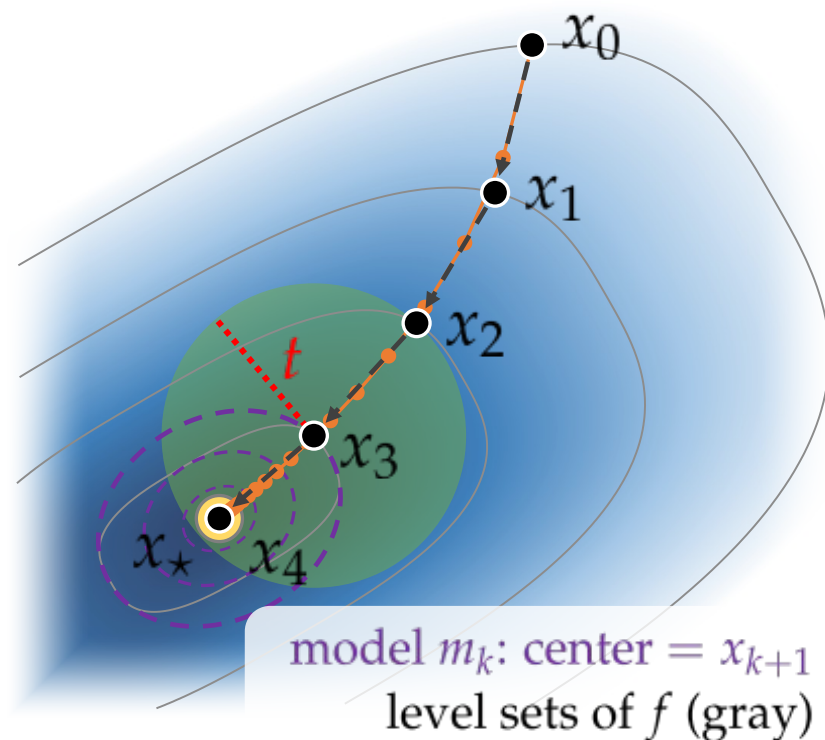
Computable: replace $\bar{H}_k \rightarrow \nabla^2 f(x_k)$ and $\lambda_k \rightarrow \sqrt{\sigma \|\nabla f(x_k)\|_2}$
 $\Rightarrow$ regularized Newton (Mishchenko, 2023): orange, $\sigma = 8$



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Conclusions

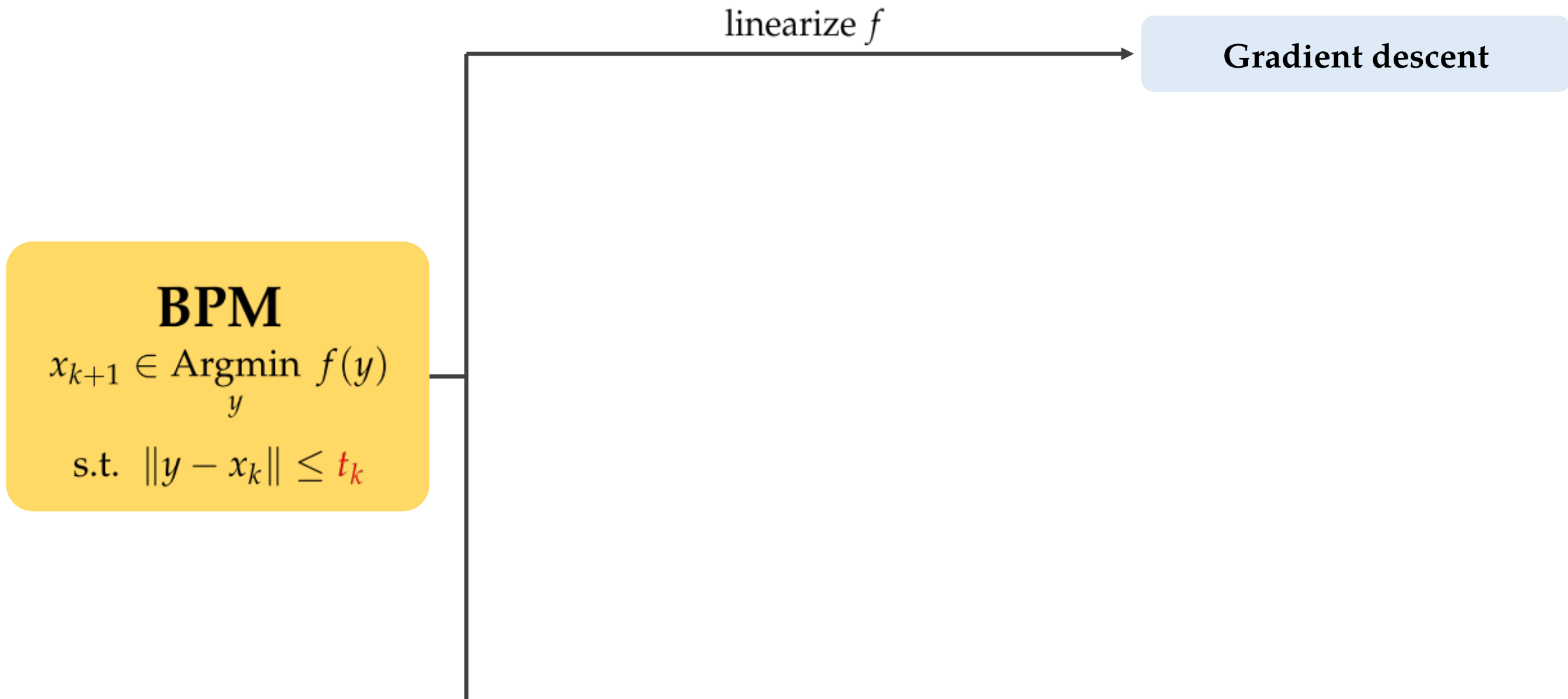
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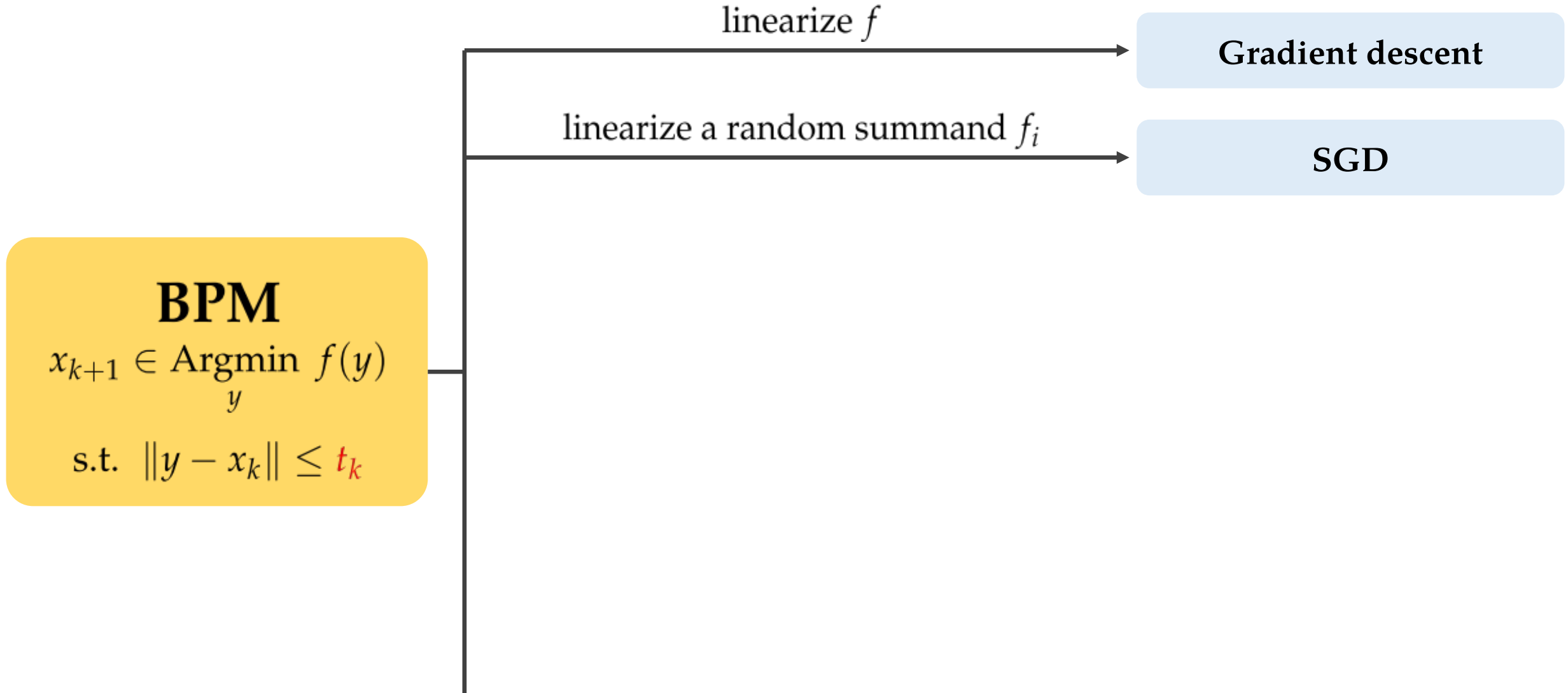
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ball $\rightarrow$ penalty $\|y - x_k\|_2^2 / (2\gamma)$

Proximal point method

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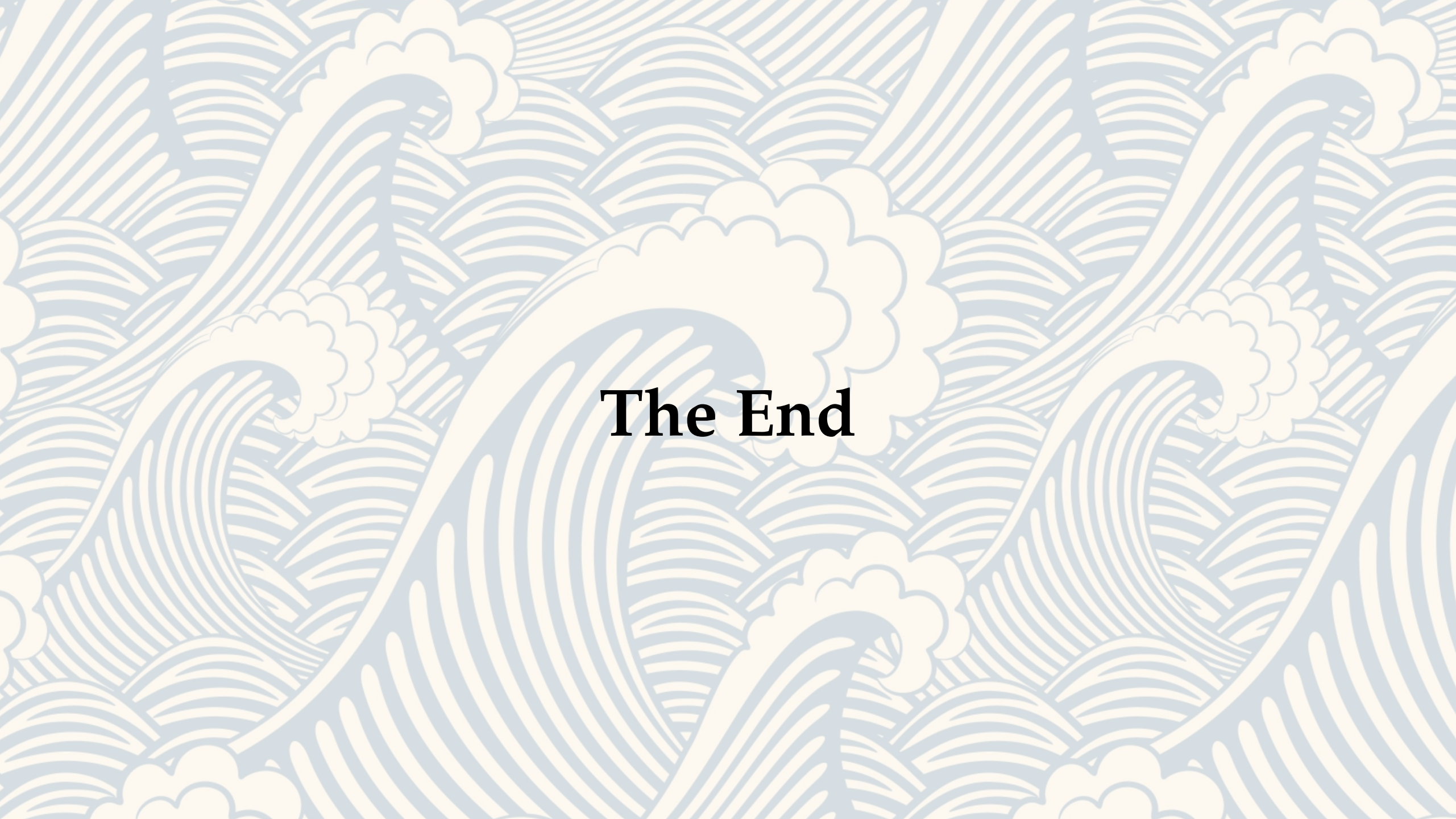
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How to think about optimization: *minimize f over a ball, step after step.*

Practical methods approximate this step: a ball + an approximation of f + a radius rule.



The End